

## ECA Table

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 8300 Hz - 9 kHz

|                                      |                             |                             |  |  |  |
|--------------------------------------|-----------------------------|-----------------------------|--|--|--|
| METEOROLOGICAL AIDS (5.54A)<br>5.54B | METEOROLOGICAL AIDS (5.54A) | Lightning detection systems |  |  |  |
|--------------------------------------|-----------------------------|-----------------------------|--|--|--|

### 9 kHz - 11.3 kHz

|                                                |                                                |                             |               |                                          |                             |
|------------------------------------------------|------------------------------------------------|-----------------------------|---------------|------------------------------------------|-----------------------------|
| METEOROLOGICAL AIDS (5.54A)<br>RADIONAVIGATION | METEOROLOGICAL AIDS (5.54A)<br>RADIONAVIGATION | Inductive applications      | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                |                                                | Lightning detection systems |               |                                          |                             |
|                                                |                                                | ULP-AMI                     | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 11.3 kHz - 14 kHz

|                 |                 |                        |               |                                          |                             |
|-----------------|-----------------|------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION | RADIONAVIGATION | Inductive applications | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                 |                 | ULP-AMI                | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 14 kHz - 19.95 kHz

|                                              |                                               |                           |               |                                          |                             |
|----------------------------------------------|-----------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE (5.57)<br>5.55 5.56 | FIXED<br>MARITIME MOBILE (5.57)<br>5.56 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                              |                                               | Land military systems     |               |                                          |                             |
|                                              |                                               | Maritime military systems |               |                                          |                             |
|                                              |                                               | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 19.95 kHz - 20.05 kHz

|                                                |                                                |                        |               |                                          |                             |
|------------------------------------------------|------------------------------------------------|------------------------|---------------|------------------------------------------|-----------------------------|
| STANDARD FREQUENCY AND TIME<br>SIGNAL (20 KHZ) | STANDARD FREQUENCY AND TIME<br>SIGNAL (20 KHZ) | Inductive applications | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                |                                                | ULP-AMI                | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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|-------------|-----------------------------|-------------|-------------------|----------|------|

## 20.05 kHz - 70 kHz

|                                                  |                                                   |                           |               |                                          |                             |
|--------------------------------------------------|---------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE (5.57)<br><br>5.56 5.58 | FIXED<br>MARITIME MOBILE (5.57)<br><br>5.56 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                  |                                                   | Land military systems     |               |                                          |                             |
|                                                  |                                                   | Maritime military systems |               |                                          |                             |
|                                                  |                                                   | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

## 70 kHz - 72 kHz

|                        |                                     |                           |               |                                          |                             |
|------------------------|-------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION (5.60) | RADIONAVIGATION (5.60)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                        |                                     | Land military systems     |               |                                          |                             |
|                        |                                     | Maritime military systems |               |                                          |                             |
|                        |                                     | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

## 72 kHz - 84 kHz

|                                                                       |                                                                             |                                    |               |                                          |                             |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE (5.57)<br>RADIONAVIGATION (5.60)<br><br>5.56 | FIXED<br>MARITIME MOBILE (5.57)<br>RADIONAVIGATION (5.60)<br><br>5.56 ECA36 | Inductive applications             | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                                       |                                                                             | Land military systems              |               |                                          |                             |
|                                                                       |                                                                             | Maritime military systems          |               |                                          |                             |
|                                                                       |                                                                             | Standard frequency and time signal |               |                                          | 77.5 kHz DCF time signal    |
|                                                                       |                                                                             | ULP-AMI                            | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

## 84 kHz - 86 kHz

|                        |                                     |                           |               |                                          |                             |
|------------------------|-------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION (5.60) | RADIONAVIGATION (5.60)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                        |                                     | Land military systems     |               |                                          |                             |
|                        |                                     | Maritime military systems |               |                                          |                             |
|                        |                                     | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

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|-------------|-----------------------------|-------------|-------------------|----------|------|

## 86 kHz - 90 kHz

|                                                                |                                                                      |                           |               |                                          |                             |
|----------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE (5.57)<br>RADIONAVIGATION<br><br>5.56 | FIXED<br>MARITIME MOBILE (5.57)<br>RADIONAVIGATION<br><br>5.56 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                                |                                                                      | Land military systems     |               |                                          |                             |
|                                                                |                                                                      | Maritime military systems |               |                                          |                             |
|                                                                |                                                                      | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

## 90 kHz - 110 kHz

|                                             |                                                   |                           |               |                                          |                             |
|---------------------------------------------|---------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION (5.62)<br>Fixed<br><br>5.64 | RADIONAVIGATION (5.62)<br>Fixed<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                             |                                                   | Land military systems     |               |                                          |                             |
|                                             |                                                   | Maritime military systems |               |                                          |                             |
|                                             |                                                   | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

## 110 kHz - 112 kHz

|                                                         |                                                               |                           |               |                                          |                             |
|---------------------------------------------------------|---------------------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE<br>RADIONAVIGATION<br><br>5.64 | FIXED<br>MARITIME MOBILE<br>RADIONAVIGATION<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                         |                                                               | Land military systems     |               |                                          |                             |
|                                                         |                                                               | Maritime military systems |               |                                          |                             |
|                                                         |                                                               | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

## 112 kHz - 115 kHz

|                        |                                     |                           |               |                                          |                             |
|------------------------|-------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION (5.60) | RADIONAVIGATION (5.60)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                        |                                     | Land military systems     |               |                                          |                             |
|                        |                                     | Maritime military systems |               |                                          |                             |
|                        |                                     | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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|-------------|-----------------------------|-------------|-------------------|----------|------|

### 115 kHz - 117.6 kHz

|                                                                     |                                                                      |                           |               |                                          |                             |
|---------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION (5.60)<br>Fixed<br>Maritime mobile<br><br>5.64 5.66 | RADIONAVIGATION (5.60)<br>Fixed<br>Maritime mobile<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                                     |                                                                      | Land military systems     |               |                                          |                             |
|                                                                     |                                                                      | Maritime military systems |               |                                          |                             |
|                                                                     |                                                                      | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 117.6 kHz - 126 kHz

|                                                                |                                                                      |                           |               |                                          |                             |
|----------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE<br>RADIONAVIGATION (5.60)<br><br>5.64 | FIXED<br>MARITIME MOBILE<br>RADIONAVIGATION (5.60)<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                                |                                                                      | Land military systems     |               |                                          |                             |
|                                                                |                                                                      | Maritime military systems |               |                                          |                             |
|                                                                |                                                                      | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 126 kHz - 129 kHz

|                        |                                     |                           |               |                                          |                             |
|------------------------|-------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| RADIONAVIGATION (5.60) | RADIONAVIGATION (5.60)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                        |                                     | Land military systems     |               |                                          |                             |
|                        |                                     | Maritime military systems |               |                                          |                             |
|                        |                                     | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 129 kHz - 130 kHz

|                                                                |                                                                      |                           |               |                                          |                             |
|----------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE<br>RADIONAVIGATION (5.60)<br><br>5.64 | FIXED<br>MARITIME MOBILE<br>RADIONAVIGATION (5.60)<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                                                |                                                                      | Land military systems     |               |                                          |                             |
|                                                                |                                                                      | Maritime military systems |               |                                          |                             |
|                                                                |                                                                      | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

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### 130 kHz - 135.7 kHz

|                                           |                                            |                           |               |                                          |                             |
|-------------------------------------------|--------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE<br><br>5.64 5.67 | FIXED<br>MARITIME MOBILE<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                           |                                            | Land military systems     |               |                                          |                             |
|                                           |                                            | Maritime military systems |               |                                          |                             |
|                                           |                                            | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 135.7 kHz - 137.8 kHz

|                                                                 |                                                                       |                           |               |                                          |                                 |
|-----------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------|---------------|------------------------------------------|---------------------------------|
| FIXED (5.64)<br>MARITIME MOBILE<br>Amateur (5.67A)<br><br>5.67B | FIXED (5.64)<br>MARITIME MOBILE<br>Amateur (5.67A)<br><br>5.67B ECA36 | Amateur                   |               | EN 301 783                               | Within the band 135.7-137.8 kHz |
|                                                                 |                                                                       | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz     |
|                                                                 |                                                                       | Land military systems     |               |                                          |                                 |
|                                                                 |                                                                       | Maritime military systems |               |                                          |                                 |
|                                                                 |                                                                       | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz       |

### 137.8 kHz - 148.5 kHz

|                                           |                                            |                           |               |                                          |                             |
|-------------------------------------------|--------------------------------------------|---------------------------|---------------|------------------------------------------|-----------------------------|
| FIXED<br>MARITIME MOBILE<br><br>5.64 5.67 | FIXED<br>MARITIME MOBILE<br><br>5.64 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330,<br>EN 303 447,<br>EN 303 454 | Within the band 9-148.5 kHz |
|                                           |                                            | Land military systems     |               |                                          |                             |
|                                           |                                            | Maritime military systems |               |                                          |                             |
|                                           |                                            | ULP-AMI                   | ERC/REC 70-03 | EN 302 195                               | Within the band 9-315 kHz   |

### 148.5 kHz - 255 kHz

|                                    |              |                        |               |                           |                                                                  |
|------------------------------------|--------------|------------------------|---------------|---------------------------|------------------------------------------------------------------|
| BROADCASTING<br><br>5.68 5.69 5.70 | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | Frequency Assignment plan GE75. Digital systems to be introduced |
|                                    |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                               |
|                                    |              | ULP-AMI                | ERC/REC 70-03 | EN 302 195                | Within the band 9-315 kHz                                        |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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|-------------|-----------------------------|-------------|-------------------|----------|------|

## 255 kHz - 283.5 kHz

|                                                          |                                                           |                               |               |                           |                                                                  |
|----------------------------------------------------------|-----------------------------------------------------------|-------------------------------|---------------|---------------------------|------------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>BROADCASTING<br><br>5.70 | AERONAUTICAL RADIONAVIGATION<br>BROADCASTING<br><br>ECA36 | Aeronautical military systems |               |                           |                                                                  |
|                                                          |                                                           | Beacons (aeronautical)        |               |                           | Frequency Assignment plan GE85                                   |
|                                                          |                                                           | Broadcasting                  |               | EN 302 017,<br>EN 302 245 | Frequency Assignment plan GE75. Digital systems to be introduced |
|                                                          |                                                           | Inductive applications        | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                               |
|                                                          |                                                           | Maritime military systems     |               |                           |                                                                  |
|                                                          |                                                           | ULP-AMI                       | ERC/REC 70-03 | EN 302 195                | Within the band 9-315 kHz                                        |

## 283.5 kHz - 315 kHz

|                                                                             |                                                                                   |                               |               |            |                                    |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>MARITIME RADIONAVIGATION (5.73)<br><br>5.74 | AERONAUTICAL RADIONAVIGATION<br>MARITIME RADIONAVIGATION (5.73)<br><br>5.74 ECA36 | Aeronautical military systems |               |            |                                    |
|                                                                             |                                                                                   | Beacons (aeronautical)        |               |            | Frequency Assignment plan GE85     |
|                                                                             |                                                                                   | Beacons (maritime)            |               |            | Frequency Assignment plan GE85     |
|                                                                             |                                                                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                             |                                                                                   | Maritime military systems     |               |            |                                    |
|                                                                             |                                                                                   | ULP-AMI                       | ERC/REC 70-03 | EN 302 195 | Within the band 9-315 kHz          |

## 315 kHz - 325 kHz

|                                                                             |                                                                              |                               |               |            |                                                                       |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>Maritime radionavigation (5.73)<br><br>5.75 | AERONAUTICAL RADIONAVIGATION<br>Maritime radionavigation (5.73)<br><br>ECA36 | Aeronautical military systems |               |            |                                                                       |
|                                                                             |                                                                              | Beacons (aeronautical)        |               |            | Frequency Assignment plan GE85                                        |
|                                                                             |                                                                              | Beacons (maritime)            |               |            | Frequency Assignment plan GE85. IALA - plan to allow differential GPS |
|                                                                             |                                                                              | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                                    |
|                                                                             |                                                                              | Maritime military systems     |               |            |                                                                       |
|                                                                             |                                                                              | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                                                       |

## 325 kHz - 405 kHz

|                              |                                           |                               |               |            |                                     |
|------------------------------|-------------------------------------------|-------------------------------|---------------|------------|-------------------------------------|
| AERONAUTICAL RADIONAVIGATION | AERONAUTICAL RADIONAVIGATION<br><br>ECA36 | Aeronautical military systems |               |            |                                     |
|                              |                                           | Beacons (aeronautical)        |               |            | Frequency Assignment plan GE85      |
|                              |                                           | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz. |
|                              |                                           | RFID                          | ERC/REC 70-03 | EN 300 330 | within frequency range 400-600 kHz  |
|                              |                                           | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                     |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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#### 405 kHz - 415 kHz

|                        |                                 |                               |               |            |                                                                       |
|------------------------|---------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------------------------|
| RADIONAVIGATION (5.76) | RADIONAVIGATION (5.76)<br>ECA36 | Aeronautical military systems |               |            |                                                                       |
|                        |                                 | Beacons (aeronautical)        |               |            | Frequency Assignment plan GE85                                        |
|                        |                                 | Beacons (maritime)            |               |            | Frequency Assignment plan GE85. IALA - plan to allow differential GPS |
|                        |                                 | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz.                                   |
|                        |                                 | Maritime military systems     |               |            |                                                                       |
|                        |                                 | RFID                          | ERC/REC 70-03 | EN 300 330 | within frequency range 400-600 kHz                                    |
|                        |                                 | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                                                       |

#### 415 kHz - 435 kHz

|                                                        |                                                                 |                               |               |            |                                     |
|--------------------------------------------------------|-----------------------------------------------------------------|-------------------------------|---------------|------------|-------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>MARITIME MOBILE (5.79) | AERONAUTICAL RADIONAVIGATION<br>MARITIME MOBILE (5.79)<br>ECA36 | Aeronautical military systems |               |            |                                     |
|                                                        |                                                                 | Beacons (aeronautical)        |               |            | Frequency Assignment plan GE85      |
|                                                        |                                                                 | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz. |
|                                                        |                                                                 | Maritime communications       |               | EN 300 338 | Frequency Assignment plan GE85      |
|                                                        |                                                                 | Maritime military systems     |               |            |                                     |
|                                                        |                                                                 | RFID                          | ERC/REC 70-03 | EN 300 330 | within frequency range 400-600 kHz  |
|                                                        |                                                                 | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                     |

#### 435 kHz - 472 kHz

|                                                                       |                                                                      |                               |               |                           |                                                           |
|-----------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------|---------------|---------------------------|-----------------------------------------------------------|
| MARITIME MOBILE (5.79)<br>Aeronautical radionavigation (5.77)<br>5.82 | MARITIME MOBILE (5.79)<br>Aeronautical radionavigation<br>5.82 ECA36 | Aeronautical military systems |               |                           |                                                           |
|                                                                       |                                                                      | Emergency detection           | ERC/REC 70-03 | EN 300 330,<br>EN 300 718 | Emergency detection is only with the band 456.9-457.1 kHz |
|                                                                       |                                                                      | Inductive applications        | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz.                       |
|                                                                       |                                                                      | Maritime communications       |               | EN 300 338                | Frequency Assignment plan GE85                            |
|                                                                       |                                                                      | Maritime military systems     |               |                           |                                                           |
|                                                                       |                                                                      | RFID                          | ERC/REC 70-03 | EN 300 330                | within frequency range 400-600 kHz                        |
|                                                                       |                                                                      | ULP-AID                       | ERC/REC 70-03 | EN 302 536                |                                                           |

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#### 472 kHz - 479 kHz

|                                                                                                         |                                                                                                   |                               |               |            |                                     |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------|---------------|------------|-------------------------------------|
| MARITIME MOBILE (5.79)<br>Aeronautical radionavigation (5.77 5.80)<br>Amateur (5.80A)<br><br>5.80B 5.82 | MARITIME MOBILE (5.79)<br>Aeronautical radionavigation<br>Amateur (5.80A)<br><br>5.80B 5.82 ECA36 | Aeronautical military systems |               |            |                                     |
|                                                                                                         |                                                                                                   | Amateur                       |               | EN 301 783 |                                     |
|                                                                                                         |                                                                                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz. |
|                                                                                                         |                                                                                                   | Maritime communications       |               | EN 300 338 | Frequency Assignment plan GE85      |
|                                                                                                         |                                                                                                   | Maritime military systems     |               |            |                                     |
|                                                                                                         |                                                                                                   | RFID                          | ERC/REC 70-03 | EN 300 330 | within frequency range 400-600 kHz  |
|                                                                                                         |                                                                                                   | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                     |

#### 479 kHz - 495 kHz

|                                                                                 |                                                                                |                               |               |            |                                                   |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------|---------------|------------|---------------------------------------------------|
| MARITIME MOBILE (5.79 5.79A)<br>Aeronautical radionavigation (5.77)<br><br>5.82 | MARITIME MOBILE (5.79 5.79A)<br>Aeronautical radionavigation<br><br>5.82 ECA36 | Aeronautical military systems |               |            |                                                   |
|                                                                                 |                                                                                | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz.               |
|                                                                                 |                                                                                | Maritime communications       |               | EN 300 338 | Frequency Assignment plan GE85                    |
|                                                                                 |                                                                                | Maritime military systems     |               |            |                                                   |
|                                                                                 |                                                                                | NAVTEX                        |               | EN 300 065 | 490 kHz: NAVTEX transmission in national language |
|                                                                                 |                                                                                | RFID                          | ERC/REC 70-03 | EN 300 330 | within frequency range 400-600 kHz                |
|                                                                                 |                                                                                | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                                   |

#### 495 kHz - 505 kHz

|                               |                     |                           |               |            |                                     |
|-------------------------------|---------------------|---------------------------|---------------|------------|-------------------------------------|
| MARITIME MOBILE (5.82C 5.82D) | MOBILE<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz. |
|                               |                     | Maritime military systems |               |            |                                     |
|                               |                     | RFID                      | ERC/REC 70-03 | EN 300 330 | within frequency range 400-600 kHz  |
|                               |                     | ULP-AID                   | ERC/REC 70-03 | EN 302 536 |                                     |

#### 505 kHz - 526.5 kHz

|                                                                   |                                                                                |                               |               |            |                                                   |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------|---------------|------------|---------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>MARITIME MOBILE (5.79 5.79A 5.84) | AERONAUTICAL RADIONAVIGATION<br>MARITIME MOBILE (5.79 5.79A 5.84)<br><br>ECA36 | Aeronautical military systems |               |            |                                                   |
|                                                                   |                                                                                | Beacons (aeronautical)        |               |            | Frequency Assignment plan GE85                    |
|                                                                   |                                                                                | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz.               |
|                                                                   |                                                                                | Maritime communications       |               | EN 300 338 | Frequency Assignment plan GE85                    |
|                                                                   |                                                                                | Maritime military systems     |               |            |                                                   |
|                                                                   |                                                                                | NAVTEX                        |               | EN 300 065 | 518 kHz: NAVTEX transmission in national language |
|                                                                   |                                                                                | ULP-AID                       | ERC/REC 70-03 | EN 302 536 |                                                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 526.5 kHz - 1606.5 kHz

|                            |              |                        |               |                           |                                                                  |
|----------------------------|--------------|------------------------|---------------|---------------------------|------------------------------------------------------------------|
| BROADCASTING<br>5.87 5.87A | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | Frequency Assignment plan GE75. Digital systems to be introduced |
|                            |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz.                              |
|                            |              | RFID                   | ERC/REC 70-03 | EN 300 330                | within frequency range 400-600 kHz                               |
|                            |              | ULP-AID                | ERC/REC 70-03 | EN 302 536                | within frequency range 315-600 kHz                               |

## 1606.5 kHz - 1625 kHz

|                                                        |                                                                          |                           |               |            |                                    |
|--------------------------------------------------------|--------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE<br>MARITIME MOBILE (5.90)<br>5.92 | FIXED<br>LAND MOBILE<br>MARITIME MOBILE (5.90)<br>Radiolocation<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                        |                                                                          | Land military systems     |               |            |                                    |
|                                                        |                                                                          | Maritime communications   |               | EN 303 402 | Frequency Assignment plan GE85     |
|                                                        |                                                                          | Maritime military systems |               |            |                                    |

## 1625 kHz - 1635 kHz

|                       |                             |                          |               |            |                                    |
|-----------------------|-----------------------------|--------------------------|---------------|------------|------------------------------------|
| RADIOLOCATION<br>5.93 | RADIOLOCATION<br>5.93 ECA36 | Inductive applications   | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                       |                             | Radiolocation (military) |               |            |                                    |

## 1635 kHz - 1800 kHz

|                                                             |                                                              |                           |               |            |                                    |
|-------------------------------------------------------------|--------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE<br>MARITIME MOBILE (5.90)<br>5.92 5.96 | FIXED<br>LAND MOBILE<br>MARITIME MOBILE (5.90)<br>5.96 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                             |                                                              | Land military systems     |               |            |                                    |
|                                                             |                                                              | Maritime communications   |               | EN 303 402 | Frequency Assignment plan GE85     |
|                                                             |                                                              | Maritime military systems |               |            |                                    |

## 1800 kHz - 1810 kHz

|                       |                             |                          |               |            |                                    |
|-----------------------|-----------------------------|--------------------------|---------------|------------|------------------------------------|
| RADIOLOCATION<br>5.93 | RADIOLOCATION<br>5.93 ECA36 | Inductive applications   | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                       |                             | Radiolocation (military) |               |            |                                    |

## 1810 kHz - 1850 kHz

|                            |                       |                        |               |            |                                    |
|----------------------------|-----------------------|------------------------|---------------|------------|------------------------------------|
| AMATEUR<br>5.98 5.99 5.100 | AMATEUR<br>5.98 5.100 | Amateur                |               | EN 301 783 | Within the band 1810-2000 kHz      |
|                            |                       | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 1850 kHz - 2000 kHz

|                                                                      |                                                                                  |                           |               |            |                                    |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.92 5.96 5.103 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Amateur<br><br>5.96 5.103 ECA36 | Amateur                   |               | EN 301 783 | Within the band 1810-2000 kHz      |
|                                                                      |                                                                                  | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                      |                                                                                  | Land military systems     |               |            |                                    |
|                                                                      |                                                                                  | Maritime communications   |               | EN 303 402 |                                    |
|                                                                      |                                                                                  | Maritime military systems |               |            |                                    |

## 2000 kHz - 2025 kHz

|                                                                     |                                                                      |                           |               |            |                                    |
|---------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.92 5.103 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.103 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                     |                                                                      | Land military systems     |               |            |                                    |
|                                                                     |                                                                      | Maritime communications   |               | EN 303 402 |                                    |
|                                                                     |                                                                      | Maritime military systems |               |            |                                    |

## 2025 kHz - 2045 kHz

|                                                                                                    |                                                                            |                           |               |            |                                    |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>Meteorological aids (5.104)<br><br>5.92 5.103 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.103 5.104 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                                                    |                                                                            | Land military systems     |               |            |                                    |
|                                                                                                    |                                                                            | Maritime communications   |               | EN 303 402 |                                    |
|                                                                                                    |                                                                            | Maritime military systems |               |            |                                    |
|                                                                                                    |                                                                            | Oceanographic buoys       |               |            | Meteorological                     |

## 2045 kHz - 2160 kHz

|                                                     |                                                           |                           |               |            |                                    |
|-----------------------------------------------------|-----------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE<br>MARITIME MOBILE<br><br>5.92 | FIXED<br>LAND MOBILE<br>MARITIME MOBILE<br><br>5.92 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                     |                                                           | Land military systems     |               |            |                                    |
|                                                     |                                                           | Maritime communications   |               | EN 303 402 | Frequency Assignment plan GE85     |
|                                                     |                                                           | Maritime military systems |               |            |                                    |

## 2160 kHz - 2170 kHz

|                                 |                                 |                          |               |            |                                    |
|---------------------------------|---------------------------------|--------------------------|---------------|------------|------------------------------------|
| RADIOLOCATION<br><br>5.93 5.107 | RADIOLOCATION<br><br>5.93 ECA36 | Inductive applications   | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                 |                                 | Radiolocation (military) |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 2170 kHz - 2173.5 kHz

|                 |                          |                           |               |            |                                    |
|-----------------|--------------------------|---------------------------|---------------|------------|------------------------------------|
| MARITIME MOBILE | MARITIME MOBILE<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                 |                          | Maritime communications   |               | EN 303 402 |                                    |
|                 |                          | Maritime military systems |               |            |                                    |

## 2173.5 kHz - 2190.5 kHz

|                                                          |                                                                |                         |               |                           |                                                                                     |
|----------------------------------------------------------|----------------------------------------------------------------|-------------------------|---------------|---------------------------|-------------------------------------------------------------------------------------|
| MOBILE (DISTRESS AND CALLING)<br>5.108 5.109 5.110 5.111 | MOBILE (DISTRESS AND CALLING)<br>5.108 5.109 5.110 5.111 ECA36 | DSC                     |               | EN 302 885,<br>EN 303 402 | 2187.5 kHz (DSC for distress and calling)                                           |
|                                                          |                                                                | Inductive applications  | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                  |
|                                                          |                                                                | Maritime communications |               | EN 303 402                | 2182 kHz (Radiotelephony distress and calling). 2174.5 kHz (Telex distress traffic) |

## 2190.5 kHz - 2194 kHz

|                 |                          |                           |               |            |                                    |
|-----------------|--------------------------|---------------------------|---------------|------------|------------------------------------|
| MARITIME MOBILE | MARITIME MOBILE<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                 |                          | Maritime communications   |               | EN 303 402 |                                    |
|                 |                          | Maritime military systems |               |            |                                    |

## 2194 kHz - 2300 kHz

|                                                                       |                                                                  |                           |               |            |                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.92 5.103 5.112 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.103 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                       |                                                                  | Land military systems     |               |            |                                    |
|                                                                       |                                                                  | Maritime communications   |               | EN 303 402 |                                    |
|                                                                       |                                                                  | Maritime military systems |               |            |                                    |

## 2300 kHz - 2498 kHz

|                                                                                    |                                                                  |                           |               |            |                                    |
|------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| BROADCASTING (5.113)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.103 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.103 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                                    |                                                                  | Land military systems     |               |            |                                    |
|                                                                                    |                                                                  | Maritime communications   |               | EN 303 402 |                                    |
|                                                                                    |                                                                  | Maritime military systems |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 2498 kHz - 2501 kHz

|                                                |                                                |                        |               |            |                                    |
|------------------------------------------------|------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL (2 500 KHZ) | STANDARD FREQUENCY AND TIME SIGNAL (2 500 KHZ) | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|------------------------------------------------|------------------------------------------------|------------------------|---------------|------------|------------------------------------|

## 2501 kHz - 2502 kHz

|                                                      |                                                      |                        |               |            |                                    |
|------------------------------------------------------|------------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL<br>Space research | STANDARD FREQUENCY AND TIME SIGNAL<br>Space research | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|------------------------------------------------------|------------------------------------------------------|------------------------|---------------|------------|------------------------------------|

## 2502 kHz - 2625 kHz

|                                                                           |                                                                           |                           |               |            |                                    |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.92 5.103 5.114 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.92 5.103 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                           |                                                                           | Land military systems     |               |            |                                    |
|                                                                           |                                                                           | Maritime military systems |               |            |                                    |

## 2625 kHz - 2650 kHz

|                                                         |                                                               |                           |               |            |                                    |
|---------------------------------------------------------|---------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| MARITIME MOBILE<br>MARITIME RADIONAVIGATION<br><br>5.92 | MARITIME MOBILE<br>MARITIME RADIONAVIGATION<br><br>5.92 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                         |                                                               | Maritime communications   |               | EN 303 402 |                                    |
|                                                         |                                                               | Maritime military systems |               |            |                                    |

## 2650 kHz - 2850 kHz

|                                                                     |                                                                           |                           |               |            |                                    |
|---------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.92 5.103 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.92 5.103 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                     |                                                                           | Land military systems     |               |            |                                    |
|                                                                     |                                                                           | Maritime military systems |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 2850 kHz - 3025 kHz

|                                        |                                               |                               |               |            |                                                                  |
|----------------------------------------|-----------------------------------------------|-------------------------------|---------------|------------|------------------------------------------------------------------|
| AERONAUTICAL MOBILE (R)<br>5.111 5.115 | AERONAUTICAL MOBILE (OR)<br>5.111 5.115 ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan                                       |
|                                        |                                               | Aeronautical military systems |               |            |                                                                  |
|                                        |                                               | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                               |
|                                        |                                               | SAR (communications)          |               | EN 303 402 | 3023 kHz (Aeronautical/Maritime radiotelephony SAR coordination) |

### 3025 kHz - 3155 kHz

|                          |                                   |                               |               |            |                                    |
|--------------------------|-----------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan         |
|                          |                                   | Aeronautical military systems |               |            |                                    |
|                          |                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

### 3155 kHz - 3200 kHz

|                                                                  |                                                                  |                           |               |            |                                                                       |
|------------------------------------------------------------------|------------------------------------------------------------------|---------------------------|---------------|------------|-----------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.116 5.117 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.116 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 3155-3400 kHz; and within the band 148.5 kHz - 30 MHz |
|                                                                  |                                                                  | Land military systems     |               |            |                                                                       |
|                                                                  |                                                                  | Maritime communications   |               | EN 303 402 |                                                                       |
|                                                                  |                                                                  | Maritime military systems |               |            |                                                                       |

### 3200 kHz - 3230 kHz

|                                                                                    |                                                                  |                           |               |            |                                                                       |
|------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------|---------------|------------|-----------------------------------------------------------------------|
| BROADCASTING (5.113)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.116 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.116 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 3155-3400 kHz; and within the band 148.5 kHz - 30 MHz |
|                                                                                    |                                                                  | Land military systems     |               |            |                                                                       |
|                                                                                    |                                                                  | Maritime communications   |               | EN 303 402 |                                                                       |
|                                                                                    |                                                                  | Maritime military systems |               |            |                                                                       |

### 3230 kHz - 3400 kHz

|                                                                                      |                                                              |                           |               |            |                                                                       |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------|---------------|------------|-----------------------------------------------------------------------|
| BROADCASTING (5.113)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>5.116 5.118 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>5.116 ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 3155-3400 kHz; and within the band 148.5 kHz - 30 MHz |
|                                                                                      |                                                              | Land military systems     |               |            |                                                                       |
|                                                                                      |                                                              | Maritime communications   |               | EN 303 402 |                                                                       |
|                                                                                      |                                                              | Maritime military systems |               |            |                                                                       |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 3400 kHz - 3500 kHz

|                         |                                  |                               |               |            |                                                     |
|-------------------------|----------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                         |                                  | Aeronautical military systems |               |            |                                                     |
|                         |                                  | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |

### 3500 kHz - 3800 kHz

|                                                                      |                                                                            |                           |               |            |                                    |
|----------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| AMATEUR<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.92 | AMATEUR<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.92 ECA36 | Amateur                   |               | EN 301 783 |                                    |
|                                                                      |                                                                            | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                      |                                                                            | Land military systems     |               |            |                                    |
|                                                                      |                                                                            | Maritime communications   |               | EN 303 402 |                                    |
|                                                                      |                                                                            | Maritime military systems |               |            |                                    |

### 3800 kHz - 3900 kHz

|                                                  |                                                               |                               |               |            |                                    |
|--------------------------------------------------|---------------------------------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR)<br>FIXED<br>LAND MOBILE | AERONAUTICAL MOBILE (OR)<br>FIXED<br>LAND MOBILE<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan         |
|                                                  |                                                               | Aeronautical military systems |               |            |                                    |
|                                                  |                                                               | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                  |                                                               | Land military systems         |               |            |                                    |

### 3900 kHz - 3950 kHz

|                                       |                                       |                               |               |            |                                    |
|---------------------------------------|---------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR)<br><br>5.123 | AERONAUTICAL MOBILE (OR)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan         |
|                                       |                                       | Aeronautical military systems |               |            |                                    |
|                                       |                                       | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

### 3950 kHz - 4000 kHz

|                       |                                    |                        |               |                           |                                    |
|-----------------------|------------------------------------|------------------------|---------------|---------------------------|------------------------------------|
| BROADCASTING<br>FIXED | BROADCASTING<br>FIXED<br><br>ECA36 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | Digital systems to be introduced   |
|                       |                                    | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz |
|                       |                                    | Land military systems  |               |                           |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 4000 kHz - 4063 kHz

|                                               |                                               |                           |               |                           |                                                          |
|-----------------------------------------------|-----------------------------------------------|---------------------------|---------------|---------------------------|----------------------------------------------------------|
| FIXED<br>MARITIME MOBILE (5.127)<br><br>5.126 | FIXED<br>MARITIME MOBILE (5.127)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                       |
|                                               |                                               | Land military systems     |               |                           |                                                          |
|                                               |                                               | Maritime communications   |               | EN 302 885,<br>EN 303 402 | Appendix 17 channelling plan. Appendix 25 allotment plan |
|                                               |                                               | Maritime military systems |               |                           |                                                          |

#### 4063 kHz - 4438 kHz

|                                                                          |                                                                          |                           |               |                           |                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------|---------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.79A 5.109 5.110 5.82D 5.130 5.131 5.132)<br><br>5.128 | MARITIME MOBILE (5.109 5.110 5.130 5.131 5.132 5.79A)<br><br>5.128 ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | centre frequency 4207.5 kHz (DSC distress traffic). Ship stations centre frequencies 4208, 4208.5, 4209 kHz. Coast stations 4219.5, 4220, 4220.5 kHz (DSC calling)                                                                                                                          |
|                                                                          |                                                                          | Eurobalise                | ERC/REC 70-03 | EN 302 608                | centre frequency at 4234 kHz                                                                                                                                                                                                                                                                |
|                                                                          |                                                                          | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                                                                                                                                                                                                          |
|                                                                          |                                                                          | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. centre frequency 4125 kHz (Radiotelephony distress and safety traffic. centre frequency 4177.5 kHz (Telex distress traffic). 4209.5 kHz (Meteorological and navigational warnings. centre frequency 4210 kHz (Safety Information) |
|                                                                          |                                                                          | Maritime military systems |               |                           |                                                                                                                                                                                                                                                                                             |
|                                                                          |                                                                          | NAVTEX                    |               | EN 300 065                | centre frequency 4209.5 kHz                                                                                                                                                                                                                                                                 |

#### 4438 kHz - 4488 kHz

|                                                                                           |                                                                                          |                           |               |            |                                    |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>Radiolocation (5.132A)<br><br>5.132B | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>Radiolocation (5.132A)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                                           |                                                                                          | Land military systems     |               |            |                                    |
|                                                                                           |                                                                                          | Maritime military systems |               |            |                                    |
|                                                                                           |                                                                                          | Radiolocation (military)  |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 4488 kHz - 4650 kHz

|                                                   |                                                                |                           |               |            |                                    |
|---------------------------------------------------|----------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R) | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                   |                                                                | Land military systems     |               |            |                                    |
|                                                   |                                                                | Maritime military systems |               |            |                                    |

#### 4650 kHz - 4700 kHz

|                         |                                      |                               |               |            |                                                     |
|-------------------------|--------------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                         |                                      | Aeronautical military systems |               |            |                                                     |
|                         |                                      | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |

#### 4700 kHz - 4750 kHz

|                          |                                       |                               |               |            |                                    |
|--------------------------|---------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan         |
|                          |                                       | Aeronautical military systems |               |            |                                    |
|                          |                                       | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

#### 4750 kHz - 4850 kHz

|                                                                          |                                                               |                               |               |            |                                    |
|--------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR)<br>BROADCASTING (5.113)<br>FIXED<br>LAND MOBILE | AERONAUTICAL MOBILE (OR)<br>FIXED<br>LAND MOBILE<br><br>ECA36 | Aeronautical communications   |               |            |                                    |
|                                                                          |                                                               | Aeronautical military systems |               |            |                                    |
|                                                                          |                                                               | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                          |                                                               | Land military systems         |               |            |                                    |

#### 4850 kHz - 4995 kHz

|                                              |                                   |                        |               |            |                                    |
|----------------------------------------------|-----------------------------------|------------------------|---------------|------------|------------------------------------|
| BROADCASTING (5.113)<br>FIXED<br>LAND MOBILE | FIXED<br>LAND MOBILE<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                              |                                   | Land military systems  |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 4995 kHz - 5003 kHz

|                                                |                                                |                        |               |            |                                    |
|------------------------------------------------|------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL (5 000 KHZ) | STANDARD FREQUENCY AND TIME SIGNAL (5 000 KHZ) | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|------------------------------------------------|------------------------------------------------|------------------------|---------------|------------|------------------------------------|

#### 5003 kHz - 5005 kHz

|                                                      |                                                      |                        |               |            |                                    |
|------------------------------------------------------|------------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL<br>Space research | STANDARD FREQUENCY AND TIME SIGNAL<br>Space research | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|------------------------------------------------------|------------------------------------------------------|------------------------|---------------|------------|------------------------------------|

#### 5005 kHz - 5060 kHz

|                               |       |                        |               |            |                                    |
|-------------------------------|-------|------------------------|---------------|------------|------------------------------------|
| BROADCASTING (5.113)<br>FIXED | FIXED | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                               | ECA36 | Land military systems  |               |            |                                    |

#### 5060 kHz - 5250 kHz

|                                                         |                                                         |                           |               |            |                                    |
|---------------------------------------------------------|---------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>Mobile except aeronautical mobile<br><br>5.133 | FIXED<br>Mobile except aeronautical mobile<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                         |                                                         | Land military systems     |               |            |                                    |
|                                                         |                                                         | Maritime military systems |               |            |                                    |

#### 5250 kHz - 5275 kHz

|                                                                                       |                                                                                      |                           |               |            |                                    |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation (5.132A)<br><br>5.133A | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation (5.132A)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                                       |                                                                                      | Land military systems     |               |            |                                    |
|                                                                                       |                                                                                      | Maritime military systems |               |            |                                    |
|                                                                                       |                                                                                      | Radiolocation (military)  |               |            |                                    |

#### 5275 kHz - 5351.5 kHz

|                                               |                                                            |                           |               |            |                                    |
|-----------------------------------------------|------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                               |                                                            | Land military systems     |               |            |                                    |
|                                               |                                                            | Maritime military systems |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 5351.5 kHz - 5366.5 kHz

|                                                                   |                                                                                |                           |               |            |                                    |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Amateur (5.133B) | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Amateur (5.133B)<br><br>ECA36 | Amateur                   |               | EN 301 783 |                                    |
|                                                                   |                                                                                | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                   |                                                                                | Land military systems     |               |            |                                    |
|                                                                   |                                                                                | Maritime military systems |               |            |                                    |

### 5366.5 kHz - 5450 kHz

|                                               |                                                            |                           |               |            |                                    |
|-----------------------------------------------|------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                               |                                                            | Land military systems     |               |            |                                    |
|                                               |                                                            | Maritime military systems |               |            |                                    |

### 5450 kHz - 5480 kHz

|                                                  |                                                               |                               |               |            |                                    |
|--------------------------------------------------|---------------------------------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR)<br>FIXED<br>LAND MOBILE | AERONAUTICAL MOBILE (OR)<br>FIXED<br>LAND MOBILE<br><br>ECA36 | Aeronautical communications   |               |            |                                    |
|                                                  |                                                               | Aeronautical military systems |               |            |                                    |
|                                                  |                                                               | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                  |                                                               | Land military systems         |               |            |                                    |

### 5480 kHz - 5680 kHz

|                                            |                                                   |                               |               |            |                                                                  |
|--------------------------------------------|---------------------------------------------------|-------------------------------|---------------|------------|------------------------------------------------------------------|
| AERONAUTICAL MOBILE (R)<br><br>5.111 5.115 | AERONAUTICAL MOBILE (OR)<br><br>5.111 5.115 ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan Including HF Data Links               |
|                                            |                                                   | Aeronautical military systems |               |            |                                                                  |
|                                            |                                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                               |
|                                            |                                                   | SAR (communications)          |               | EN 303 402 | 5680 kHz (Aeronautical/Maritime radiotelephony SAR coordination) |

### 5680 kHz - 5730 kHz

|                                             |                                                   |                               |               |            |                                                                  |
|---------------------------------------------|---------------------------------------------------|-------------------------------|---------------|------------|------------------------------------------------------------------|
| AERONAUTICAL MOBILE (OR)<br><br>5.111 5.115 | AERONAUTICAL MOBILE (OR)<br><br>5.111 5.115 ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan                                       |
|                                             |                                                   | Aeronautical military systems |               |            |                                                                  |
|                                             |                                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                               |
|                                             |                                                   | SAR (communications)          |               | EN 303 402 | 5680 kHz (Aeronautical/Maritime radiotelephony SAR coordination) |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 5730 kHz - 5900 kHz

|                      |                                   |                        |               |            |                                    |
|----------------------|-----------------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE | FIXED<br>LAND MOBILE<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                      |                                   | Land military systems  |               |            |                                    |

### 5900 kHz - 5950 kHz

|                                   |                                   |                        |               |                           |                                                                     |
|-----------------------------------|-----------------------------------|------------------------|---------------|---------------------------|---------------------------------------------------------------------|
| BROADCASTING (5.134)<br><br>5.136 | BROADCASTING (5.134)<br><br>5.136 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. |
|                                   |                                   | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                  |

### 5950 kHz - 6200 kHz

|              |              |                        |               |                           |                                                                    |
|--------------|--------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |

### 6200 kHz - 6525 kHz

|                                                                  |                                                                        |                           |               |                           |                                                                                                                                                                                             |
|------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------|---------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.109 5.110 5.130<br>5.132 5.137A)<br><br>5.137 | MARITIME MOBILE (5.109 5.110 5.130<br>5.132 5.137A)<br><br>5.137 ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | 6312 kHz (DSC distress traffic). 6312.5, 6313, 6313.5, 6331, 6331.5, 6332 kHz (DSC calling)                                                                                                 |
|                                                                  |                                                                        | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                                                                                                          |
|                                                                  |                                                                        | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. 6215 kHz. (Radiotelephony distress and safety traffic). 6268 kHz (Telex distress traffic). 6314 kHz (Maritime Safety Information) |
|                                                                  |                                                                        | Maritime military systems |               |                           |                                                                                                                                                                                             |

### 6525 kHz - 6685 kHz

|                         |                                      |                               |               |            |                                                     |
|-------------------------|--------------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                         |                                      | Aeronautical military systems |               |            |                                                     |
|                         |                                      | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 6685 kHz - 6765 kHz

|                          |                                   |                               |               |            |                                    |
|--------------------------|-----------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan         |
|                          |                                   | Aeronautical military systems |               |            |                                    |
|                          |                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

## 6765 kHz - 7000 kHz

|                                                                |                                                                      |                           |               |            |                                                                       |
|----------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|---------------|------------|-----------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.138 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br><br>5.138 ECA36 | ISM                       |               |            | Within the band 6765-6795 kHz                                         |
|                                                                |                                                                      | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 6765-6795 kHz; and within the band 148.5 kHz - 30 MHz |
|                                                                |                                                                      | Land military systems     |               |            |                                                                       |
|                                                                |                                                                      | Maritime military systems |               |            |                                                                       |

## 7000 kHz - 7100 kHz

|                                                        |                              |                        |               |            |                                    |
|--------------------------------------------------------|------------------------------|------------------------|---------------|------------|------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE<br><br>5.140 5.141 5.141A | AMATEUR<br>AMATEUR-SATELLITE | Amateur                |               | EN 301 783 | Within the band 7000-7200 kHz      |
|                                                        |                              | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

## 7100 kHz - 7200 kHz

|                              |         |                        |               |            |                                    |
|------------------------------|---------|------------------------|---------------|------------|------------------------------------|
| AMATEUR<br><br>5.141A 5.141B | AMATEUR | Amateur                |               | EN 301 783 | Within the band 7000-7200 kHz      |
|                              |         | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

## 7200 kHz - 7300 kHz

|              |              |                        |               |                           |                                                                                                      |
|--------------|--------------|------------------------|---------------|---------------------------|------------------------------------------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR - Article 12 planning procedure. Digital systems to be introduced. Within the band 7200-7450 kHz. |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 7300 kHz - 7400 kHz

|                                                           |                                      |                        |               |                           |                                                                                                    |
|-----------------------------------------------------------|--------------------------------------|------------------------|---------------|---------------------------|----------------------------------------------------------------------------------------------------|
| BROADCASTING (5.134)<br>5.143 5.143A 5.143B 5.143C 5.143D | BROADCASTING (5.134)<br>5.143 5.143B | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. Within the band 7200-7450 kHz. |
|                                                           |                                      | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                 |

### 7400 kHz - 7450 kHz

|                               |                        |                        |               |                           |                                                                                                    |
|-------------------------------|------------------------|------------------------|---------------|---------------------------|----------------------------------------------------------------------------------------------------|
| BROADCASTING<br>5.143B 5.143C | BROADCASTING<br>5.143B | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. Within the band 7200-7450 kHz. |
|                               |                        | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz                              |

### 7450 kHz - 8100 kHz

|                                                            |                                                            |                           |               |            |                                                                       |
|------------------------------------------------------------|------------------------------------------------------------|---------------------------|---------------|------------|-----------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.144 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz |
|                                                            |                                                            | Land military systems     |               |            |                                                                       |
|                                                            |                                                            | Maritime military systems |               |            |                                                                       |

### 8100 kHz - 8195 kHz

|                          |                                   |                           |               |            |                                                                       |
|--------------------------|-----------------------------------|---------------------------|---------------|------------|-----------------------------------------------------------------------|
| FIXED<br>MARITIME MOBILE | FIXED<br>MARITIME MOBILE<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz |
|                          |                                   | Land military systems     |               |            |                                                                       |
|                          |                                   | Maritime communications   |               | EN 303 402 | Appendix 17 channelling plan                                          |
|                          |                                   | Maritime military systems |               |            |                                                                       |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 8195 kHz - 8815 kHz

|                                                               |                                                                     |                           |               |                           |                                                                                                                                                                                               |
|---------------------------------------------------------------|---------------------------------------------------------------------|---------------------------|---------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.109 5.110 5.132 5.145 5.137A)<br><br>5.111 | MARITIME MOBILE (5.109 5.110 5.132 5.137A 5.145)<br><br>5.111 ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | 8414.5 kHz (DSC distress traffic). 8415, 8415.5, 8416, 8436.5, 8437, 8437.5 kHz (DSC calling)                                                                                                 |
|                                                               |                                                                     | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 7400-8800 kHz; and within the band 148.5 kHz - 30 MHz                                                                                                                         |
|                                                               |                                                                     | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. 8291 kHz (Radiotelephony distress and safety traffic).8376.5 kHz (Telex distress traffic). 8416.5 kHz (Maritime Safety Information) |
|                                                               |                                                                     | Maritime military systems |               |                           |                                                                                                                                                                                               |

### 8815 kHz - 8965 kHz

|                         |                                      |                               |               |            |                                                     |
|-------------------------|--------------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan1 Including HF Data Links |
|                         |                                      | Aeronautical military systems |               |            |                                                     |
|                         |                                      | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |

### 8965 kHz - 9040 kHz

|                          |                                       |                               |               |            |                                    |
|--------------------------|---------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan         |
|                          |                                       | Aeronautical military systems |               |            |                                    |
|                          |                                       | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

### 9040 kHz - 9305 kHz

|       |                    |                        |               |            |                                    |
|-------|--------------------|------------------------|---------------|------------|------------------------------------|
| FIXED | FIXED<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|       |                    | Land military systems  |               |            |                                    |

### 9305 kHz - 9355 kHz

|                                               |                                              |                        |               |            |                                    |
|-----------------------------------------------|----------------------------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>Radiolocation (5.145A)<br><br>5.145B | FIXED<br>Radiolocation (5.145A)<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                               |                                              | Land military systems  |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 9355 kHz - 9400 kHz

|       |       |                        |               |            |                                    |
|-------|-------|------------------------|---------------|------------|------------------------------------|
| FIXED | FIXED | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|       | ECA36 | Land military systems  |               |            |                                    |

### 9400 kHz - 9500 kHz

|                               |                               |                        |               |                           |                                                         |
|-------------------------------|-------------------------------|------------------------|---------------|---------------------------|---------------------------------------------------------|
| BROADCASTING (5.134)<br>5.146 | BROADCASTING (5.134)<br>5.146 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Within 9400-9900 kHz. |
|                               |                               | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                      |

### 9500 kHz - 9900 kHz

|                       |                       |                        |               |                           |                                                           |
|-----------------------|-----------------------|------------------------|---------------|---------------------------|-----------------------------------------------------------|
| BROADCASTING<br>5.147 | BROADCASTING<br>5.147 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. . Within 9400-9900 kHz. |
|                       |                       | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                        |

### 9900 kHz - 9995 kHz

|       |       |                        |               |            |                                    |
|-------|-------|------------------------|---------------|------------|------------------------------------|
| FIXED | FIXED | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|       | ECA36 | Land military systems  |               |            |                                    |

### 9995 kHz - 10003 kHz

|                                                          |                                                          |                        |               |            |                                    |
|----------------------------------------------------------|----------------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL (10 000 KHZ)<br>5.111 | STANDARD FREQUENCY AND TIME SIGNAL (10 000 KHZ)<br>5.111 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|----------------------------------------------------------|----------------------------------------------------------|------------------------|---------------|------------|------------------------------------|

### 10003 kHz - 10005 kHz

|                                                               |                                                               |                        |               |            |                                                       |
|---------------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------|------------|-------------------------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL<br>Space research<br>5.111 | STANDARD FREQUENCY AND TIME SIGNAL<br>Space research<br>5.111 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                    |
|                                                               |                                                               | SAR (communications)   |               |            | 10003 kHz (+/-3 kHz) concerning manned space vehicles |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 10005 kHz - 10100 kHz

|                                  |                                        |                               |               |            |                                                     |
|----------------------------------|----------------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R)<br>5.111 | AERONAUTICAL MOBILE (R)<br>5.111 ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                                  |                                        | Aeronautical military systems |               |            |                                                     |
|                                  |                                        | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |

### 10100 kHz - 10150 kHz

|                  |                           |                        |               |            |                                    |
|------------------|---------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>Amateur | FIXED<br>Amateur<br>ECA36 | Amateur                |               | EN 301 783 |                                    |
|                  |                           | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                  |                           | Land military systems  |               |            |                                    |

### 10150 kHz - 11175 kHz

|                                                |                                                         |                           |               |            |                                                                         |
|------------------------------------------------|---------------------------------------------------------|---------------------------|---------------|------------|-------------------------------------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R) | FIXED<br>Mobile except aeronautical mobile (R)<br>ECA36 | Euroloop                  | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz                                  |
|                                                |                                                         | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 10200-11000 kHz; and within the band 148.5 kHz - 30 MHz |
|                                                |                                                         | Land military systems     |               |            |                                                                         |
|                                                |                                                         | Maritime military systems |               |            |                                                                         |

### 11175 kHz - 11275 kHz

|                          |                                   |                               |               |            |                                        |
|--------------------------|-----------------------------------|-------------------------------|---------------|------------|----------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan             |
|                          |                                   | Aeronautical military systems |               |            |                                        |
|                          |                                   | Euroloop                      | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz |
|                          |                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |

### 11275 kHz - 11400 kHz

|                         |                                  |                               |               |            |                                                     |
|-------------------------|----------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                         |                                  | Aeronautical military systems |               |            |                                                     |
|                         |                                  | Euroloop                      | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz              |
|                         |                                  | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 11400 kHz - 11600 kHz

|       |                    |                        |               |            |                                        |
|-------|--------------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br><br>ECA36 | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz |
|       |                    | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                    | Land military systems  |               |            |                                        |

## 11600 kHz - 11650 kHz

|                                   |                                   |                        |               |                           |                                                                                                            |
|-----------------------------------|-----------------------------------|------------------------|---------------|---------------------------|------------------------------------------------------------------------------------------------------------|
| BROADCASTING (5.134)<br><br>5.146 | BROADCASTING (5.134)<br><br>5.146 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. Within frequency range 11600-12100 kHz |
|                                   |                                   | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                                                                     |
|                                   |                                   | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                         |

## 11650 kHz - 12050 kHz

|                           |                           |                        |               |                           |                                                                                                            |
|---------------------------|---------------------------|------------------------|---------------|---------------------------|------------------------------------------------------------------------------------------------------------|
| BROADCASTING<br><br>5.147 | BROADCASTING<br><br>5.147 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. Within frequency range 11600-12100 kHz |
|                           |                           | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                                                                     |
|                           |                           | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                         |
|                           |                           | RFID                   |               | EN 300 330                | Within frequency range 11.810-15.310 MHz                                                                   |

## 12050 kHz - 12100 kHz

|                                   |                                   |                        |               |                           |                                                                                                            |
|-----------------------------------|-----------------------------------|------------------------|---------------|---------------------------|------------------------------------------------------------------------------------------------------------|
| BROADCASTING (5.134)<br><br>5.146 | BROADCASTING (5.134)<br><br>5.146 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. Within frequency range 11600-12100 kHz |
|                                   |                                   | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                                                                     |
|                                   |                                   | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                         |
|                                   |                                   | RFID                   |               | EN 300 330                | Within frequency range 11.810-15.310 MHz                                                                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 12100 kHz - 12230 kHz

|       |                |                        |               |            |                                          |
|-------|----------------|------------------------|---------------|------------|------------------------------------------|
| FIXED | FIXED<br>ECA36 | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|       |                | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|       |                | Land military systems  |               |            |                                          |
|       |                | RFID                   |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |

## 12230 kHz - 13200 kHz

|                                                  |                                                           |                           |               |                           |                                                                                                                                                                                                                                 |
|--------------------------------------------------|-----------------------------------------------------------|---------------------------|---------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.109 5.110 5.132 5.145 5.137A) | MARITIME MOBILE (5.109 5.110 5.132 5.137A 5.145)<br>ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | Centre frequency 12577 kHz (DSC distress traffic). Centre frequencies 12577.5, 12578, 12578.5, 12657, 12657.5, 12658 kHz (DSC calling)                                                                                          |
|                                                  |                                                           | Euroloop                  | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                                                                                                                                                                                          |
|                                                  |                                                           | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                                                                                                                                              |
|                                                  |                                                           | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. Centre frequency 12290 kHz (Radiotelephony distress and safety traffic). centre frequency 12520 kHz (Telex distress traffic). 12579 kHz (Maritime Safety Information) |
|                                                  |                                                           | Maritime military systems |               |                           |                                                                                                                                                                                                                                 |
|                                                  |                                                           | RFID                      |               | EN 300 330                | Within frequency range 11.810-15.310 MHz                                                                                                                                                                                        |
|                                                  |                                                           | ULP-AID                   | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                                                                                                                                                                                          |

## 13200 kHz - 13260 kHz

|                          |                                   |                               |               |            |                                          |
|--------------------------|-----------------------------------|-------------------------------|---------------|------------|------------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan               |
|                          |                                   | Aeronautical military systems |               |            |                                          |
|                          |                                   | Euroloop                      | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                          |                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                          |                                   | RFID                          |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                          |                                   | ULP-AID                       | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 13260 kHz - 13360 kHz

|                         |                                  |                               |               |            |                                                     |
|-------------------------|----------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                         |                                  | Aeronautical military systems |               |            |                                                     |
|                         |                                  | Euroloop                      | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz              |
|                         |                                  | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |
|                         |                                  | RFID                          |               | EN 300 330 | within frequency range 11.810-15.310 MHz            |
|                         |                                  | ULP-AID                       | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz              |

### 13360 kHz - 13410 kHz

|                                       |                                             |                        |               |            |                                          |
|---------------------------------------|---------------------------------------------|------------------------|---------------|------------|------------------------------------------|
| FIXED<br>RADIO ASTRONOMY<br><br>5.149 | FIXED<br>RADIO ASTRONOMY<br><br>5.149 ECA36 | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                                       |                                             | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                                       |                                             | Land military systems  |               |            |                                          |
|                                       |                                             | RFID                   |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                                       |                                             | Radio astronomy        |               |            | Continuum observations                   |
|                                       |                                             | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

### 13410 kHz - 13450 kHz

|                                                |                                                             |                           |               |            |                                          |
|------------------------------------------------|-------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R) | FIXED<br>Mobile except aeronautical mobile (R)<br><br>ECA36 | Euroloop                  | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                                                |                                                             | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                                                |                                                             | Land military systems     |               |            |                                          |
|                                                |                                                             | Maritime military systems |               |            |                                          |
|                                                |                                                             | RFID                      |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                                                |                                                             | ULP-AID                   | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 13450 kHz - 13550 kHz

|                                                                                        |                                                                                       |                           |               |            |                                                                        |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R)<br>Radiolocation (5.132A)<br><br>5.149A | FIXED<br>Mobile except aeronautical mobile (R)<br>Radiolocation (5.132A)<br><br>ECA36 | Euroloop                  | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz.<br>Centre frequency 13.547 MHz |
|                                                                                        |                                                                                       | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                                     |
|                                                                                        |                                                                                       | Land military systems     |               |            |                                                                        |
|                                                                                        |                                                                                       | Maritime military systems |               |            |                                                                        |
|                                                                                        |                                                                                       | RFID                      |               | EN 300 330 | Within frequency range 11.810-15.310 MHz                               |
|                                                                                        |                                                                                       | ULP-AID                   | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz                                 |

### 13550 kHz - 13570 kHz

|                                                             |                                                                   |                           |               |            |                                                                          |
|-------------------------------------------------------------|-------------------------------------------------------------------|---------------------------|---------------|------------|--------------------------------------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R)<br><br>5.150 | FIXED<br>Mobile except aeronautical mobile (R)<br><br>5.150 ECA36 | Euroloop                  | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz                                   |
|                                                             |                                                                   | ISM                       |               |            | Within the band 13553-13567 kHz                                          |
|                                                             |                                                                   | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 13553-13567 kHz; and within the band 148.5 kHz - 30 MHz  |
|                                                             |                                                                   | Land military systems     |               |            |                                                                          |
|                                                             |                                                                   | Maritime military systems |               |            |                                                                          |
|                                                             |                                                                   | Non-specific SRDs         | ERC/REC 70-03 | EN 300 330 | Within the band 13553-13567 kHz                                          |
|                                                             |                                                                   | RFID                      |               | EN 300 330 | Centre frequency is 13.56 MHz. Within frequency range 11.810-15.310 MHz. |
|                                                             |                                                                   | ULP-AID                   | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz                                   |

### 13570 kHz - 13600 kHz

|                                   |                                   |                        |               |                           |                                                                    |
|-----------------------------------|-----------------------------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING (5.134)<br><br>5.151 | BROADCASTING (5.134)<br><br>5.151 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|                                   |                                   | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                             |
|                                   |                                   | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|                                   |                                   | RFID                   |               | EN 300 330                | Within frequency range 11.810-15.310 MHz                           |
|                                   |                                   | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 13600 kHz - 13800 kHz

|              |              |                        |               |                           |                                                                    |
|--------------|--------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|              |              | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                             |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|              |              | RFID                   |               | EN 300 330                | Within frequency range 11.810-15.310 MHz                           |
|              |              | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

### 13800 kHz - 13870 kHz

|                                   |                                   |                        |               |                           |                                                                    |
|-----------------------------------|-----------------------------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING (5.134)<br><br>5.151 | BROADCASTING (5.134)<br><br>5.151 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|                                   |                                   | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                             |
|                                   |                                   | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|                                   |                                   | RFID                   |               | EN 300 330                | Within frequency range 11.810-15.310 MHz                           |
|                                   |                                   | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

### 13870 kHz - 14000 kHz

|                                                |                                                             |                           |               |            |                                          |
|------------------------------------------------|-------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R) | FIXED<br>Mobile except aeronautical mobile (R)<br><br>ECA36 | Euroloop                  | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                                                |                                                             | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                                                |                                                             | Land military systems     |               |            |                                          |
|                                                |                                                             | Maritime military systems |               |            |                                          |
|                                                |                                                             | RFID                      |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                                                |                                                             | ULP-AID                   | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 14000 kHz - 14250 kHz

|                              |                              |                        |               |            |                                          |
|------------------------------|------------------------------|------------------------|---------------|------------|------------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE | AMATEUR<br>AMATEUR-SATELLITE | Amateur                |               | EN 301 783 | Within the band 14000-14350 kHz          |
|                              |                              | Amateur-satellite      |               |            |                                          |
|                              |                              | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                              |                              | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                              |                              | RFID                   |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                              |                              | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

### 14250 kHz - 14350 kHz

|                      |         |                        |               |            |                                          |
|----------------------|---------|------------------------|---------------|------------|------------------------------------------|
| AMATEUR<br><br>5.152 | AMATEUR | Amateur                |               | EN 301 783 | Within the band 14000-14350 kHz          |
|                      |         | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                      |         | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                      |         | RFID                   |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                      |         | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

### 14350 kHz - 14990 kHz

|                                                |                                                             |                           |               |            |                                          |
|------------------------------------------------|-------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R) | FIXED<br>Mobile except aeronautical mobile (R)<br><br>ECA36 | Euroloop                  | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                                                |                                                             | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                                                |                                                             | Land military systems     |               |            |                                          |
|                                                |                                                             | Maritime military systems |               |            |                                          |
|                                                |                                                             | RFID                      |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                                                |                                                             | ULP-AID                   | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

### 14990 kHz - 15005 kHz

|                                                                 |                                                                 |                        |               |            |                                                       |
|-----------------------------------------------------------------|-----------------------------------------------------------------|------------------------|---------------|------------|-------------------------------------------------------|
| STANDARD FREQUENCY AND TIME<br>SIGNAL (15 000 KHZ)<br><br>5.111 | STANDARD FREQUENCY AND TIME<br>SIGNAL (15 000 KHZ)<br><br>5.111 | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz                |
|                                                                 |                                                                 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                    |
|                                                                 |                                                                 | RFID                   |               | EN 300 330 | Within frequency range 11.810-15.310 MHz              |
|                                                                 |                                                                 | SAR (communications)   |               |            | 14993 kHz (+/-3 kHz) concerning manned space vehicles |
|                                                                 |                                                                 | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 15005 kHz - 15010 kHz

|                                                         |                                                         |                        |               |            |                                          |
|---------------------------------------------------------|---------------------------------------------------------|------------------------|---------------|------------|------------------------------------------|
| STANDARD FREQUENCY AND TIME<br>SIGNAL<br>Space research | STANDARD FREQUENCY AND TIME<br>SIGNAL<br>Space research | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                                                         |                                                         | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                                                         |                                                         | RFID                   |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                                                         |                                                         | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

### 15010 kHz - 15100 kHz

|                          |                                       |                               |               |            |                                          |
|--------------------------|---------------------------------------|-------------------------------|---------------|------------|------------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br><br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan               |
|                          |                                       | Aeronautical military systems |               |            |                                          |
|                          |                                       | Euroloop                      | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz   |
|                          |                                       | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz       |
|                          |                                       | RFID                          |               | EN 300 330 | Within frequency range 11.810-15.310 MHz |
|                          |                                       | ULP-AID                       | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz   |

### 15100 kHz - 15600 kHz

|              |              |                        |               |                           |                                                                    |
|--------------|--------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|              |              | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                             |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|              |              | RFID                   |               | EN 300 330                | within frequency range 11.810-15.310 MHz                           |
|              |              | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

### 15600 kHz - 15800 kHz

|                                   |                                   |                        |               |                           |                                                                    |
|-----------------------------------|-----------------------------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING (5.134)<br><br>5.146 | BROADCASTING (5.134)<br><br>5.146 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|                                   |                                   | Euroloop               | ERC/REC 70-03 | EN 302 609                | Mainly within the band 11100-16000 kHz                             |
|                                   |                                   | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|                                   |                                   | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 15800 kHz - 16100 kHz

|       |                |                        |               |            |                                        |
|-------|----------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br>ECA36 | Euroloop               | ERC/REC 70-03 | EN 302 609 | Mainly within the band 11100-16000 kHz |
|       |                | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                | Land military systems  |               |            |                                        |
|       |                | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

## 16100 kHz - 16200 kHz

|                                               |                                              |                        |               |            |                                        |
|-----------------------------------------------|----------------------------------------------|------------------------|---------------|------------|----------------------------------------|
| FIXED<br>Radiolocation (5.145A)<br><br>5.145B | FIXED<br>Radiolocation (5.145A)<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|                                               |                                              | Land military systems  |               |            |                                        |
|                                               |                                              | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

## 16200 kHz - 16360 kHz

|       |                |                        |               |            |                                        |
|-------|----------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                | Land military systems  |               |            |                                        |
|       |                | ULP-AID                |               | EN 300 330 | Within frequency range 12500-20000 kHz |

## 16360 kHz - 17410 kHz

|                                                  |                                                               |                           |               |                           |                                                                                                                                                                                              |
|--------------------------------------------------|---------------------------------------------------------------|---------------------------|---------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.109 5.110 5.132 5.145 5.137A) | MARITIME MOBILE (5.109 5.110 5.132 5.137A 5.145)<br><br>ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | 16804.5 kHz (DSC distress traffic).16805, 16805.5, 16806, 16903, 16903.5, 16904 kHz (DSC calling)                                                                                            |
|                                                  |                                                               | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                                                                                                           |
|                                                  |                                                               | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan.16420 kHz (Radiotelephony distress and safety traffic).16695 kHz (Telex distress traffic).16806.5 kHz (Maritime Safety Information) |
|                                                  |                                                               | Maritime military systems |               |                           |                                                                                                                                                                                              |
|                                                  |                                                               | ULP-AID                   | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                                                                                                                                                       |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 17410 kHz - 17480 kHz

|       |                |                        |               |            |                                        |
|-------|----------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                | Land military systems  |               |            |                                        |
|       |                | ULP-AID                |               | EN 300 330 | Within frequency range 12500-20000 kHz |

### 17480 kHz - 17550 kHz

|                               |                               |                        |               |                           |                                        |
|-------------------------------|-------------------------------|------------------------|---------------|---------------------------|----------------------------------------|
| BROADCASTING (5.134)<br>5.146 | BROADCASTING (5.134)<br>5.146 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | Digital systems to be introduced       |
|                               |                               | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz     |
|                               |                               | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz |

### 17550 kHz - 17900 kHz

|              |              |                        |               |                           |                                                                    |
|--------------|--------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|              |              | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

### 17900 kHz - 17970 kHz

|                         |                                  |                               |               |            |                                                     |
|-------------------------|----------------------------------|-------------------------------|---------------|------------|-----------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links |
|                         |                                  | Aeronautical military systems |               |            |                                                     |
|                         |                                  | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                  |
|                         |                                  | ULP-AID                       | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz              |

### 17970 kHz - 18030 kHz

|                          |                                   |                               |               |            |                                        |
|--------------------------|-----------------------------------|-------------------------------|---------------|------------|----------------------------------------|
| AERONAUTICAL MOBILE (OR) | AERONAUTICAL MOBILE (OR)<br>ECA36 | Aeronautical communications   |               |            | Appendix 26 Allotment Plan             |
|                          |                                   | Aeronautical military systems |               |            |                                        |
|                          |                                   | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|                          |                                   | ULP-AID                       | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 18030 kHz - 18052 kHz

|       |                |                        |               |            |                                        |
|-------|----------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                | Land military systems  |               |            |                                        |
|       |                | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

### 18052 kHz - 18068 kHz

|                         |                                  |                        |               |            |                                        |
|-------------------------|----------------------------------|------------------------|---------------|------------|----------------------------------------|
| FIXED<br>Space research | FIXED<br>Space research<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|                         |                                  | Land military systems  |               |            |                                        |
|                         |                                  | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

### 18068 kHz - 18168 kHz

|                                           |                              |                        |               |            |                                        |
|-------------------------------------------|------------------------------|------------------------|---------------|------------|----------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE<br><br>5.154 | AMATEUR<br>AMATEUR-SATELLITE | Amateur                |               | EN 301 783 |                                        |
|                                           |                              | Amateur-satellite      |               |            |                                        |
|                                           |                              | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|                                           |                              | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

### 18168 kHz - 18780 kHz

|                                            |                                                     |                           |               |                           |                                                                                    |
|--------------------------------------------|-----------------------------------------------------|---------------------------|---------------|---------------------------|------------------------------------------------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile | FIXED<br>Mobile except aeronautical mobile<br>ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | Centre frequencies at 18898.5, 18899. 18899.5 kHz (DSC) digital selective calling) |
|                                            |                                                     | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                 |
|                                            |                                                     | Land military systems     |               |                           |                                                                                    |
|                                            |                                                     | Maritime military systems |               |                           |                                                                                    |
|                                            |                                                     | ULP-AID                   | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                                             |

### 18780 kHz - 18900 kHz

|                 |                          |                           |               |            |                                        |
|-----------------|--------------------------|---------------------------|---------------|------------|----------------------------------------|
| MARITIME MOBILE | MARITIME MOBILE<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|                 |                          | Maritime communications   |               | EN 303 402 | Appendix 17 channelling plan           |
|                 |                          | Maritime military systems |               |            |                                        |
|                 |                          | ULP-AID                   | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 18900 kHz - 19020 kHz

|                               |                               |                        |               |                           |                                                                    |
|-------------------------------|-------------------------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING (5.134)<br>5.146 | BROADCASTING (5.134)<br>5.146 | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|                               |                               | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |
|                               |                               | ULP-AID                | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                             |

### 19020 kHz - 19680 kHz

|       |                |                        |               |            |                                        |
|-------|----------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                | Land military systems  |               |            |                                        |
|       |                | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

### 19680 kHz - 19800 kHz

|                         |                                  |                           |               |                           |                                                                                                     |
|-------------------------|----------------------------------|---------------------------|---------------|---------------------------|-----------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.132) | MARITIME MOBILE (5.132)<br>ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | 19703.5, 19704, 19704.5 kHz (DSC calling)                                                           |
|                         |                                  | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                  |
|                         |                                  | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. 19680.5 kHz (Maritime Safety Information) |
|                         |                                  | Maritime military systems |               |                           |                                                                                                     |
|                         |                                  | ULP-AID                   | ERC/REC 70-03 | EN 300 330                | Within frequency range 12500-20000 kHz                                                              |

### 19800 kHz - 19990 kHz

|       |                |                        |               |            |                                        |
|-------|----------------|------------------------|---------------|------------|----------------------------------------|
| FIXED | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|       |                | Land military systems  |               |            |                                        |
|       |                | ULP-AID                | ERC/REC 70-03 | EN 300 330 | Within frequency range 12500-20000 kHz |

### 19990 kHz - 19995 kHz

|                                                                  |                                                                  |                        |               |            |                                                       |
|------------------------------------------------------------------|------------------------------------------------------------------|------------------------|---------------|------------|-------------------------------------------------------|
| STANDARD FREQUENCY AND TIME<br>SIGNAL<br>Space research<br>5.111 | STANDARD FREQUENCY AND TIME<br>SIGNAL<br>Space research<br>5.111 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                    |
|                                                                  |                                                                  | SAR (communications)   |               |            | 19993 kHz (+/-3 kHz) concerning manned space vehicles |
|                                                                  |                                                                  | ULP-AID                |               | EN 300 330 | Within frequency range 12500-20000 kHz                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 19995 kHz - 20010 kHz

|                                                          |                                                          |                        |               |            |                                        |
|----------------------------------------------------------|----------------------------------------------------------|------------------------|---------------|------------|----------------------------------------|
| STANDARD FREQUENCY AND TIME SIGNAL (20 000 KHZ)<br>5.111 | STANDARD FREQUENCY AND TIME SIGNAL (20 000 KHZ)<br>5.111 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz     |
|                                                          |                                                          | ULP-AID                |               | EN 300 330 | Within frequency range 12500-20000 kHz |

## 20010 kHz - 21000 kHz

|                 |                              |                               |               |            |                                    |
|-----------------|------------------------------|-------------------------------|---------------|------------|------------------------------------|
| FIXED<br>Mobile | FIXED<br>Mobile<br><br>ECA36 | Aeronautical military systems |               |            |                                    |
|                 |                              | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                 |                              | Land military systems         |               |            |                                    |
|                 |                              | Maritime military systems     |               |            |                                    |

## 21000 kHz - 21450 kHz

|                              |                              |                        |               |            |                                    |
|------------------------------|------------------------------|------------------------|---------------|------------|------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE | AMATEUR<br>AMATEUR-SATELLITE | Amateur                |               | EN 301 783 |                                    |
|                              |                              | Amateur-satellite      |               |            |                                    |
|                              |                              | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

## 21450 kHz - 21850 kHz

|              |              |                        |               |                           |                                                                    |
|--------------|--------------|------------------------|---------------|---------------------------|--------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                 |

## 21850 kHz - 21870 kHz

|                         |                |                        |               |            |                                    |
|-------------------------|----------------|------------------------|---------------|------------|------------------------------------|
| FIXED (5.155A)<br>5.155 | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                         |                | Land military systems  |               |            |                                    |

## 21870 kHz - 21924 kHz

|                |                         |                        |               |            |                                    |
|----------------|-------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED (5.155B) | FIXED (5.155B)<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                |                         | Land military systems  |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 21924 kHz - 22000 kHz

|                         |                                  |                               |               |            |                                                      |
|-------------------------|----------------------------------|-------------------------------|---------------|------------|------------------------------------------------------|
| AERONAUTICAL MOBILE (R) | AERONAUTICAL MOBILE (R)<br>ECA36 | Aeronautical communications   |               |            | Appendix 27 Allotment Plan. Including HF Data Links. |
|                         |                                  | Aeronautical military systems |               |            |                                                      |
|                         |                                  | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                   |

## 22000 kHz - 22855 kHz

|                                         |                                         |                           |               |                           |                                                                                         |
|-----------------------------------------|-----------------------------------------|---------------------------|---------------|---------------------------|-----------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.132 5.137A)<br>5.156 | MARITIME MOBILE (5.132 5.137A)<br>ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | 22374.5, 22375, 22375.5, 22444, 22444.5, 22445 kHz (DSC calling)                        |
|                                         |                                         | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                      |
|                                         |                                         | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. 22376 kHz safety information. |
|                                         |                                         | Maritime military systems |               |                           |                                                                                         |

## 22855 kHz - 23000 kHz

|                |                |                        |               |            |                                    |
|----------------|----------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>5.156 | FIXED<br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                |                | Land military systems  |               |            |                                    |

## 23000 kHz - 23200 kHz

|                                                         |                                                         |                           |               |            |                                    |
|---------------------------------------------------------|---------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>Mobile except aeronautical mobile (R)<br>5.156 | FIXED<br>Mobile except aeronautical mobile (R)<br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                         |                                                         | Land military systems     |               |            |                                    |
|                                                         |                                                         | Maritime military systems |               |            |                                    |

## 23200 kHz - 23350 kHz

|                                            |                                                     |                               |               |            |                                    |
|--------------------------------------------|-----------------------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| AERONAUTICAL MOBILE (OR)<br>FIXED (5.156A) | AERONAUTICAL MOBILE (OR)<br>FIXED (5.156A)<br>ECA36 | Aeronautical communications   |               |            |                                    |
|                                            |                                                     | Aeronautical military systems |               |            |                                    |
|                                            |                                                     | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                            |                                                     | Land military systems         |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 23350 kHz - 24000 kHz

|                                                       |                                                                    |                           |               |            |                                    |
|-------------------------------------------------------|--------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.157) | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.157)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                       |                                                                    | Land military systems     |               |            |                                    |
|                                                       |                                                                    | Maritime military systems |               |            |                                    |

## 24000 kHz - 24450 kHz

|                      |                                   |                        |               |            |                                    |
|----------------------|-----------------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE | FIXED<br>LAND MOBILE<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                      |                                   | Land military systems  |               |            |                                    |

## 24450 kHz - 24600 kHz

|                                                             |                                                             |                        |               |            |                                    |
|-------------------------------------------------------------|-------------------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE<br>Radiolocation (5.132A)<br><br>5.158 | FIXED<br>LAND MOBILE<br>Radiolocation (5.132A)<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                             |                                                             | Land military systems  |               |            |                                    |

## 24600 kHz - 24890 kHz

|                      |                                   |                        |               |            |                                    |
|----------------------|-----------------------------------|------------------------|---------------|------------|------------------------------------|
| FIXED<br>LAND MOBILE | FIXED<br>LAND MOBILE<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                      |                                   | Land military systems  |               |            |                                    |

## 24890 kHz - 24990 kHz

|                              |                              |                        |               |            |                                    |
|------------------------------|------------------------------|------------------------|---------------|------------|------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE | AMATEUR<br>AMATEUR-SATELLITE | Amateur                |               | EN 301 783 |                                    |
|                              |                              | Amateur-satellite      |               |            |                                    |
|                              |                              | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

## 24990 kHz - 25005 kHz

|                                                    |                                                    |                        |               |            |                                    |
|----------------------------------------------------|----------------------------------------------------|------------------------|---------------|------------|------------------------------------|
| STANDARD FREQUENCY AND TIME<br>SIGNAL (25 000 KHZ) | STANDARD FREQUENCY AND TIME<br>SIGNAL (25 000 KHZ) | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|----------------------------------------------------|----------------------------------------------------|------------------------|---------------|------------|------------------------------------|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 25005 kHz - 25010 kHz

|                                                         |                                                                      |                        |               |            |                                       |
|---------------------------------------------------------|----------------------------------------------------------------------|------------------------|---------------|------------|---------------------------------------|
| STANDARD FREQUENCY AND TIME<br>SIGNAL<br>Space research | STANDARD FREQUENCY AND TIME<br>SIGNAL<br>Space research<br><br>ECA36 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz    |
|                                                         |                                                                      | Space research         |               |            | Scientific and medical space research |

## 25010 kHz - 25070 kHz

|                                               |                                                            |                           |               |            |                                    |
|-----------------------------------------------|------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                               |                                                            | Land military systems     |               |            |                                    |
|                                               |                                                            | Maritime military systems |               |            |                                    |

## 25070 kHz - 25210 kHz

|                 |                 |                           |               |                           |                                           |
|-----------------|-----------------|---------------------------|---------------|---------------------------|-------------------------------------------|
| MARITIME MOBILE | MARITIME MOBILE | DSC                       |               | EN 302 885,<br>EN 303 402 | 25208.5, 25209, 25209.5 kHz (DSC calling) |
|                 |                 | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz        |
|                 |                 | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan              |
|                 |                 | Maritime military systems |               |                           |                                           |

## 25210 kHz - 25550 kHz

|                                               |                                                            |                           |               |            |                                    |
|-----------------------------------------------|------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                               |                                                            | Land military systems     |               |            |                                    |
|                                               |                                                            | Maritime military systems |               |            |                                    |

## 25550 kHz - 25670 kHz

|                              |                              |                        |               |            |                                    |
|------------------------------|------------------------------|------------------------|---------------|------------|------------------------------------|
| RADIO ASTRONOMY<br><br>5.149 | RADIO ASTRONOMY<br><br>5.149 | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                              |                              | Radio astronomy        |               |            | Continuum observations             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 25670 kHz - 26100 kHz

|              |              |                        |               |                           |                                                                     |
|--------------|--------------|------------------------|---------------|---------------------------|---------------------------------------------------------------------|
| BROADCASTING | BROADCASTING | Broadcasting           |               | EN 302 017,<br>EN 302 245 | RR-Article 12 planning procedure. Digital systems to be introduced. |
|              |              | Inductive applications | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                  |

## 26100 kHz - 26175 kHz

|                         |                                      |                           |               |                           |                                                                                                    |
|-------------------------|--------------------------------------|---------------------------|---------------|---------------------------|----------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (5.132) | MARITIME MOBILE (5.132)<br><br>ECA36 | DSC                       |               | EN 302 885,<br>EN 303 402 | 26121, 26121.5, 26122 kHz (DSC calling)                                                            |
|                         |                                      | Inductive applications    | ERC/REC 70-03 | EN 300 330                | Within the band 148.5 kHz - 30 MHz                                                                 |
|                         |                                      | Maritime communications   |               | EN 303 402                | Appendix 17 channelling plan. Appendix 25 allotment plan. 26100.5 kHz Maritime Safety Information. |
|                         |                                      | Maritime military systems |               |                           |                                                                                                    |

## 26175 kHz - 26200 kHz

|                                               |                                                            |                           |               |            |                                    |
|-----------------------------------------------|------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                               |                                                            | Land military systems     |               |            |                                    |
|                                               |                                                            | Maritime military systems |               |            |                                    |

## 26200 kHz - 26350 kHz

|                                                                                       |                                                                                      |                           |               |            |                                    |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|---------------|------------|------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation (5.132A)<br><br>5.133A | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation (5.132A)<br><br>ECA36 | Inductive applications    | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                                                                       |                                                                                      | Land military systems     |               |            |                                    |
|                                                                                       |                                                                                      | Maritime military systems |               |            |                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 26350 kHz - 27500 kHz

|                                                            |                                                                  |                           |                                  |                           |                                                 |
|------------------------------------------------------------|------------------------------------------------------------------|---------------------------|----------------------------------|---------------------------|-------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.150 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.150 ECA36 | CB radio                  | ECC/DEC/(11)03,<br>ERC/REC 70-03 | EN 300 433                | (CEPT PR 27). Within the band 26.960-27.410 MHz |
|                                                            |                                                                  | Eurobalise                | ERC/REC 70-03                    | EN 302 608                | Centre frequency 27.095 MHz                     |
|                                                            |                                                                  | ISM                       |                                  |                           | Within the band 26.957-27.283 MHz               |
|                                                            |                                                                  | Inductive applications    | ERC/REC 70-03                    | EN 300 330                | Within the band 148.5 kHz - 30 MHz              |
|                                                            |                                                                  | Land military systems     |                                  |                           |                                                 |
|                                                            |                                                                  | Maritime military systems |                                  |                           |                                                 |
|                                                            |                                                                  | Model control             | ERC/REC 70-03                    | EN 300 220                | 26.995, 27.045, 27.095, 27.145, 27.195 MHz      |
|                                                            |                                                                  | Non-specific SRDs         | ERC/REC 70-03                    | EN 300 220,<br>EN 300 330 | Within the band 26.957-27.283 MHz               |

## 27500 kHz - 28 MHz

|                                        |                                                     |                               |               |            |                                    |
|----------------------------------------|-----------------------------------------------------|-------------------------------|---------------|------------|------------------------------------|
| FIXED<br>METEOROLOGICAL AIDS<br>MOBILE | FIXED<br>METEOROLOGICAL AIDS<br>MOBILE<br><br>ECA36 | Aeronautical military systems |               |            |                                    |
|                                        |                                                     | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |
|                                        |                                                     | Land military systems         |               |            |                                    |
|                                        |                                                     | Maritime military systems     |               |            |                                    |

## 28 MHz - 29.7 MHz

|                              |                              |                        |               |            |                                    |
|------------------------------|------------------------------|------------------------|---------------|------------|------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE | AMATEUR<br>AMATEUR-SATELLITE | Amateur                |               | EN 301 783 |                                    |
|                              |                              | Amateur-satellite      |               |            |                                    |
|                              |                              | Inductive applications | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz |

## 29.7 MHz - 30.005 MHz

|                 |                     |                               |               |            |                                                                                                                |
|-----------------|---------------------|-------------------------------|---------------|------------|----------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE | MOBILE<br><br>ECA36 | Aeronautical military systems |               |            |                                                                                                                |
|                 |                     | Inductive applications        | ERC/REC 70-03 | EN 300 330 | Within the band 148.5 kHz - 30 MHz                                                                             |
|                 |                     | Land military systems         |               |            |                                                                                                                |
|                 |                     | Maritime military systems     |               |            |                                                                                                                |
|                 |                     | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422 | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis. |
|                 |                     | ULP-MMI                       | ERC/REC 70-03 | EN 302 510 | Within the band 30.0-30.005 MHz                                                                                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 30.005 MHz - 30.01 MHz

|                                                                                    |                     |                               |               |            |                                                                                                               |
|------------------------------------------------------------------------------------|---------------------|-------------------------------|---------------|------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>SPACE OPERATION (SATELLITE<br>IDENTIFICATION)<br>SPACE RESEARCH | MOBILE<br><br>ECA36 | Aeronautical military systems |               |            |                                                                                                               |
|                                                                                    |                     | Land military systems         |               |            |                                                                                                               |
|                                                                                    |                     | Maritime military systems     |               |            |                                                                                                               |
|                                                                                    |                     | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422 | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |
|                                                                                    |                     | ULP-MMI                       | ERC/REC 70-03 | EN 302 510 |                                                                                                               |

### 30.01 MHz - 37.5 MHz

|                 |                     |                               |                                  |                                                                                                                                                   |                                                                                                                |
|-----------------|---------------------|-------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE | MOBILE<br><br>ECA36 | Aeronautical military systems |                                  |                                                                                                                                                   |                                                                                                                |
|                 |                     | Land military systems         |                                  |                                                                                                                                                   |                                                                                                                |
|                 |                     | Maritime military systems     |                                  |                                                                                                                                                   |                                                                                                                |
|                 |                     | Model control                 | ERC/DEC/(01)11,<br>ERC/REC 70-03 | EN 300 220                                                                                                                                        | Within the band 34.995-35.225 MHz only for flying models                                                       |
|                 |                     | PMR                           | T/R 25-08                        | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                                |
|                 |                     | Radio microphones and ALD     | ERC/REC 70-03                    | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis. |
|                 |                     | ULP-MMI                       | ERC/REC 70-03                    | EN 302 510                                                                                                                                        |                                                                                                                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 37.5 MHz - 38.25 MHz

|                                                 |                                              |                               |               |                                                                                                                                                   |                                                                                                                |
|-------------------------------------------------|----------------------------------------------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Radio astronomy<br><br>5.149 | MOBILE<br>Radio astronomy<br><br>5.149 ECA36 | Aeronautical military systems |               |                                                                                                                                                   |                                                                                                                |
|                                                 |                                              | Land military systems         |               |                                                                                                                                                   |                                                                                                                |
|                                                 |                                              | Maritime military systems     |               |                                                                                                                                                   |                                                                                                                |
|                                                 |                                              | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                                |
|                                                 |                                              | Radio astronomy               |               |                                                                                                                                                   | Continuum observations                                                                                         |
|                                                 |                                              | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis. |

### 38.25 MHz - 39 MHz

|                 |                     |                               |               |                                                                                                                                    |                                                                                                               |
|-----------------|---------------------|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE | MOBILE<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                    |                                                                                                               |
|                 |                     | Land military systems         |               |                                                                                                                                    |                                                                                                               |
|                 |                     | Maritime military systems     |               |                                                                                                                                    |                                                                                                               |
|                 |                     | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                 |                     | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                         | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |
|                 |                     |                               |               |                                                                                                                                    |                                                                                                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 39 MHz - 39.5 MHz

|                                                        |                                               |                               |                |                                                                                                                                    |                                                                                                                |
|--------------------------------------------------------|-----------------------------------------------|-------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Radiolocation (5.132A)<br><br>5.159 | MOBILE<br>Radiolocation (5.132A)<br><br>ECA36 | Aeronautical military systems |                |                                                                                                                                    |                                                                                                                |
|                                                        |                                               | Land military systems         |                |                                                                                                                                    |                                                                                                                |
|                                                        |                                               | Maritime military systems     |                |                                                                                                                                    |                                                                                                                |
|                                                        |                                               | Meteor scatter communications | ERC/REC/(00)04 |                                                                                                                                    | Within the band 39.0-39.2 MHz                                                                                  |
|                                                        |                                               | PMR                           | T/R 25-08      | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                                |
|                                                        |                                               | Radio microphones and ALD     | ERC/REC 70-03  | EN 300 422                                                                                                                         | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis. |

### 39.5 MHz - 39.986 MHz

|                 |                     |                               |               |                                                                                                                                    |                                                                                                               |
|-----------------|---------------------|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE | MOBILE<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                    |                                                                                                               |
|                 |                     | Land military systems         |               |                                                                                                                                    |                                                                                                               |
|                 |                     | Maritime military systems     |               |                                                                                                                                    |                                                                                                               |
|                 |                     | Meteor scatter communications |               |                                                                                                                                    | Within the band 39.0-39.2 MHz                                                                                 |
|                 |                     | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                 |                     | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                         | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 39.986 MHz - 40 MHz

|                                   |                                       |                               |               |                                                                                                                                    |                                                                                                               |
|-----------------------------------|---------------------------------------|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Space research | MOBILE<br>Space research<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                    |                                                                                                               |
|                                   |                                       | Land military systems         |               |                                                                                                                                    |                                                                                                               |
|                                   |                                       | Maritime military systems     |               |                                                                                                                                    |                                                                                                               |
|                                   |                                       | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                   |                                       | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                         | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

### 40 MHz - 40.02 MHz

|                                                                           |                                                                               |                               |               |                                                                                                                                    |                                                                                                               |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br>Space research | MOBILE<br>Earth exploration-satellite (5.159A)<br>Space research<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                    |                                                                                                               |
|                                                                           |                                                                               | Land military systems         |               |                                                                                                                                    |                                                                                                               |
|                                                                           |                                                                               | Maritime military systems     |               |                                                                                                                                    |                                                                                                               |
|                                                                           |                                                                               | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                                                           |                                                                               | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                         | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 40.02 MHz - 40.98 MHz

|                                                                      |                                                                   |                               |                                  |                                                                                                                                                   |                                                                                                                     |
|----------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br><br>5.150 | MOBILE<br>Earth exploration-satellite (5.159A)<br><br>5.150 ECA36 | Aeronautical military systems |                                  |                                                                                                                                                   |                                                                                                                     |
|                                                                      |                                                                   | ISM                           |                                  |                                                                                                                                                   | Within the band 40.66-40.7 MHz                                                                                      |
|                                                                      |                                                                   | Land military systems         |                                  |                                                                                                                                                   |                                                                                                                     |
|                                                                      |                                                                   | Maritime military systems     |                                  |                                                                                                                                                   |                                                                                                                     |
|                                                                      |                                                                   | Model control                 | ERC/DEC/(01)12,<br>ERC/REC 70-03 | EN 300 220                                                                                                                                        | Centre frequencies 40.665, 40.675, 40.685,<br>40.695 MHz                                                            |
|                                                                      |                                                                   | Non-specific SRDs             | ERC/REC 70-03                    | EN 300 220                                                                                                                                        |                                                                                                                     |
|                                                                      |                                                                   | PMR                           | T/R 25-08                        | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                                     |
|                                                                      |                                                                   | Radio microphones and ALD     | ERC/REC 70-03                    | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band<br>audio systems including tour guide systems on<br>a tuning range basis |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 40.98 MHz - 41.015 MHz

|                                                                                              |                                       |                               |               |                                                                                                                                                   |                                                                                                               |
|----------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br>Space research<br><br>5.160 5.161 | MOBILE<br>Space research<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                                   |                                                                                                               |
|                                                                                              |                                       | Land military systems         |               |                                                                                                                                                   |                                                                                                               |
|                                                                                              |                                       | Maritime military systems     |               |                                                                                                                                                   |                                                                                                               |
|                                                                                              |                                       | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                                                                              |                                       | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

#### 41.015 MHz - 42 MHz

|                                                                                   |                     |                               |               |                                                                                                                                                   |                                                                                                               |
|-----------------------------------------------------------------------------------|---------------------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br><br>5.160 5.161 5.161A | MOBILE<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                                   |                                                                                                               |
|                                                                                   |                     | Land military systems         |               |                                                                                                                                                   |                                                                                                               |
|                                                                                   |                     | Maritime military systems     |               |                                                                                                                                                   |                                                                                                               |
|                                                                                   |                     | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                                                                   |                     | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 42 MHz - 42.5 MHz

|                                                                                                       |                                                               |                               |               |                                                                                                                                                   |                                                                                                               |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br>Radiolocation (5.132A)<br><br>5.160 5.161B | FIXED<br>MOBILE<br>Radiolocation (5.132A)<br><br>5.161B ECA36 | Aeronautical military systems |               |                                                                                                                                                   |                                                                                                               |
|                                                                                                       |                                                               | Land military systems         |               |                                                                                                                                                   |                                                                                                               |
|                                                                                                       |                                                               | Maritime military systems     |               |                                                                                                                                                   |                                                                                                               |
|                                                                                                       |                                                               | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                                                                                       |                                                               | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

## 42.5 MHz - 44 MHz

|                                                                                   |                     |                               |               |                                                                                                                                                   |                                                                                                               |
|-----------------------------------------------------------------------------------|---------------------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br><br>5.160 5.161 5.161A | MOBILE<br><br>ECA36 | Aeronautical military systems |               |                                                                                                                                                   |                                                                                                               |
|                                                                                   |                     | Land military systems         |               |                                                                                                                                                   |                                                                                                               |
|                                                                                   |                     | Maritime military systems     |               |                                                                                                                                                   |                                                                                                               |
|                                                                                   |                     | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                                                                   |                     | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 44 MHz - 47 MHz

|                                                                             |                            |                               |               |                                                                                                                                                   |                                                                                                               |
|-----------------------------------------------------------------------------|----------------------------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Earth exploration-satellite (5.159A)<br><br>5.162 5.162A | MOBILE<br><br>5.162A ECA36 | Aeronautical military systems |               |                                                                                                                                                   |                                                                                                               |
|                                                                             |                            | Land military systems         |               |                                                                                                                                                   |                                                                                                               |
|                                                                             |                            | Maritime military systems     |               |                                                                                                                                                   |                                                                                                               |
|                                                                             |                            | PMR                           | T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                                                               |
|                                                                             |                            | Radio microphones and ALD     | ERC/REC 70-03 | EN 300 422                                                                                                                                        | Within the band 29.7-47.0 MHz. Narrow band audio systems including tour guide systems on a tuning range basis |
|                                                                             |                            | Wind profilers                |               |                                                                                                                                                   | In the range 46-68 MHz, geographical sharing with other services                                              |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 47 MHz - 50 MHz

|                                                                                      |                                       |                             |           |                                                                                                                                                   |                                                                  |
|--------------------------------------------------------------------------------------|---------------------------------------|-----------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| BROADCASTING<br>Earth exploration-satellite (5.159A)<br><br>5.162A 5.163 5.164 5.165 | LAND MOBILE<br><br>5.162A 5.164 ECA36 | Earth exploration-satellite |           |                                                                                                                                                   | In the range 48.5-50 MHz. Space Research/EESS                    |
|                                                                                      |                                       | Land military systems       |           |                                                                                                                                                   |                                                                  |
|                                                                                      |                                       | On-site paging              |           | EN 300 224                                                                                                                                        | On site paging in the band 47.0-47.25 MHz                        |
|                                                                                      |                                       | PMR                         | T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                  |
|                                                                                      |                                       | Wind profilers              |           |                                                                                                                                                   | In the range 46-68 MHz, geographical sharing with other services |

## 50 MHz - 52 MHz

|                                                                                               |                                                                |                       |           |                                                                                                                                                   |                                                                  |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| BROADCASTING<br>Amateur (5.166C 5.166E 5.166B)<br><br>5.162A 5.164 5.165 5.166A 5.169A 5.169B | LAND MOBILE<br>Amateur<br><br>5.162A 5.164 5.166A 5.169B ECA36 | Amateur               |           | EN 301 783                                                                                                                                        |                                                                  |
|                                                                                               |                                                                | Land military systems |           |                                                                                                                                                   |                                                                  |
|                                                                                               |                                                                | PMR                   | T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                                  |
|                                                                                               |                                                                | Wind profilers        |           |                                                                                                                                                   | In the range 46-68 MHz, geographical sharing with other services |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 52 MHz - 68 MHz

|                                                            |                                             |                       |           |                                                                                                                                                   |                                                                                                     |
|------------------------------------------------------------|---------------------------------------------|-----------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| BROADCASTING<br><br>5.162A 5.163 5.163 5.164 5.169A 5.169B | LAND MOBILE<br><br>5.162A 5.163 5.164 ECA36 | Land military systems |           |                                                                                                                                                   |                                                                                                     |
|                                                            |                                             | PMR                   | T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit band in 68-74.8 MHz paired with base station transmit band in 77.8-84.6 MHz |
|                                                            |                                             | Wind profilers        |           |                                                                                                                                                   | In the range 46-68 MHz, geographical sharing with other services                                    |

## 68 MHz - 70.45 MHz

|                                                            |                                     |                           |                              |                                                                                                                                                   |                                                                                                     |
|------------------------------------------------------------|-------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.175 | MOBILE<br>Amateur<br><br>ECA9 ECA36 | Amateur                   |                              | EN 301 783                                                                                                                                        | Within the band 69.9-70.5 MHz                                                                       |
|                                                            |                                     | Land military systems     |                              |                                                                                                                                                   |                                                                                                     |
|                                                            |                                     | Maritime military systems |                              |                                                                                                                                                   |                                                                                                     |
|                                                            |                                     | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit band in 68-74.8 MHz paired with base station transmit band in 77.8-84.6 MHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 70.45 MHz - 74.8 MHz

|                                                                                    |                                                                                            |                           |                              |                                                                                                                                                   |                                                                                                           |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.149 5.175 5.177 5.178 5.179 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Amateur<br>Radio astronomy<br><br>5.149 ECA9 ECA36 | Amateur                   |                              | EN 301 783                                                                                                                                        | Within the band 69.9-70.5 MHz                                                                             |
|                                                                                    |                                                                                            | Land military systems     |                              |                                                                                                                                                   |                                                                                                           |
|                                                                                    |                                                                                            | Maritime military systems |                              |                                                                                                                                                   |                                                                                                           |
|                                                                                    |                                                                                            | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit band in 68-74.8 MHz<br>paired with base station transmit band in<br>77.8-84.6 MHz |
|                                                                                    |                                                                                            | Radio astronomy           |                              |                                                                                                                                                   | Continuum observations (inter alia solar wind<br>monitoring in 73-74.6 MHz)                               |

## 74.8 MHz - 75.2 MHz

|                                           |                                           |     |  |  |                |
|-------------------------------------------|-------------------------------------------|-----|--|--|----------------|
| AERONAUTICAL RADIONAVIGATION<br><br>5.180 | AERONAUTICAL RADIONAVIGATION<br><br>5.180 | ILS |  |  | Marker beacons |
|-------------------------------------------|-------------------------------------------|-----|--|--|----------------|

## 75.2 MHz - 87.5 MHz

|                                                                     |                     |                           |                              |                                                                                                                                                   |                                                                                                                                                                                                                       |
|---------------------------------------------------------------------|---------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>Mobile except aeronautical mobile<br><br>5.175 5.179 5.187 | MOBILE<br><br>ECA36 | Land military systems     |                              |                                                                                                                                                   |                                                                                                                                                                                                                       |
|                                                                     |                     | Maritime military systems |                              |                                                                                                                                                   |                                                                                                                                                                                                                       |
|                                                                     |                     | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit band in 75.2-77.7 MHz<br>paired with base station transmit band in<br>85.0-87.5 MHz Base station transmit band in<br>77.8-84.6 MHz paired with mobile station<br>transmit band in 68-74.8 MHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 87.5 MHz - 100 MHz

|                       |              |                           |               |                           |                                |
|-----------------------|--------------|---------------------------|---------------|---------------------------|--------------------------------|
| BROADCASTING<br>5.190 | BROADCASTING | FM sound analogue         |               | EN 302 018,<br>EN 303 345 | Geneva Agreement GE84          |
|                       |              | Wireless audio/multimedia | ERC/REC 70-03 | EN 301 357                | Within the band 87.5-108.0 MHz |

## 100 MHz - 108 MHz

|                             |              |                           |               |                           |                                |
|-----------------------------|--------------|---------------------------|---------------|---------------------------|--------------------------------|
| BROADCASTING<br>5.192 5.194 | BROADCASTING | FM sound analogue         |               | EN 302 018,<br>EN 303 345 | Geneva Agreement GE84          |
|                             |              | Wireless audio/multimedia | ERC/REC 70-03 | EN 301 357                | Within the band 87.5-108.0 MHz |

## 108 MHz - 117.975 MHz

|                                              |                                                                   |                             |  |            |                                                                                                |
|----------------------------------------------|-------------------------------------------------------------------|-----------------------------|--|------------|------------------------------------------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>5.197 5.197A | AERONAUTICAL MOBILE (R)<br>AERONAUTICAL RADIONAVIGATION<br>5.197A | Aeronautical communications |  | EN 301 842 | Safety and regularity of flights, below 112 MHz limited to ground based data link transmitters |
|                                              |                                                                   | GBAS                        |  | EN 303 084 | GBAS/VDB within 112-117.975 MHz                                                                |
|                                              |                                                                   | ILS                         |  |            | Localiser within the band 108-112 MHz                                                          |
|                                              |                                                                   | VOR                         |  |            | Within the band 108-117.975 MHz                                                                |

## 117.975 MHz - 137 MHz

|                                                                                                        |                                                         |                             |  |                                          |                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------|--|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| AERONAUTICAL MOBILE (R)<br>AERONAUTICAL MOBILE-SATELLITE<br>(5.198A 5.198B)<br>5.111 5.200 5.201 5.202 | AERONAUTICAL MOBILE (R)<br>5.111 5.200 5.201 5.202 ECA5 | -                           |  | EN 300 676,<br>EN 301 841,<br>EN 302 961 | Maritime Personal Homing Beacon for search and rescue purposes. 123.1 MHz.                                                           |
|                                                                                                        |                                                         | Aeronautical communications |  | EN 300 676,<br>EN 301 841,<br>EN 301 842 | Safety and regularity of flights. EN 301 841-3 is for ground-based equipment. 121.5 MHz. Aeronautical mobile distress communication. |
|                                                                                                        |                                                         | EPIRBs                      |  | EN 300 152                               | Band only available for distress and safety.                                                                                         |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 137 MHz - 137.025 MHz

|                                                                                                                                                                                                                                           |                                                                                                                                                                                            |                               |                |            |                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|------------|--------------------------------------------------------------------|
| METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (5.208A 5.208B 5.209)<br>SPACE OPERATION (5.203C)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile (R)<br><br>5.204 5.205 5.206 5.207 5.208 | METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (5.208A 5.208B 5.209)<br>SPACE OPERATION (SPACE-TO-EARTH)<br>SPACE RESEARCH (5.203C)<br><br>5.206 5.208 ECA6 ECA36 | Aeronautical military systems |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Land military systems         |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Land mobile                   |                |            | Mobile restricted to Aeronautical Mobile (OR), including air sport |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | MSS Earth stations            | ERC/DEC/(99)06 | EN 301 721 | Non-geostationary                                                  |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Maritime military systems     |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Satellite systems (military)  |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Weather satellites            |                |            |                                                                    |

### 137.025 MHz - 137.175 MHz

|                                                                                                                                                                                                                                           |                                                                                                                                                                                            |                               |                |            |                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|------------|--------------------------------------------------------------------|
| METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>SPACE OPERATION (5.203C)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile (R)<br>Mobile-Satellite (5.208A 5.208B 5.209)<br><br>5.204 5.205 5.206 5.207 5.208 | METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>SPACE OPERATION (SPACE-TO-EARTH)<br>SPACE RESEARCH (5.203C)<br>Mobile-Satellite (5.208A 5.208B 5.209)<br><br>5.206 5.208 ECA6 ECA36 | Aeronautical military systems |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Land military systems         |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Land mobile                   |                |            | Mobile restricted to Aeronautical Mobile (OR), including air sport |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | MSS Earth stations            | ERC/DEC/(99)06 | EN 301 721 | Non-geostationary                                                  |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Maritime military systems     |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Satellite systems (military)  |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Weather satellites            |                |            |                                                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 137.175 MHz - 137.825 MHz

|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   |                               |                |            |                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|------------|--------------------------------------------------------------------|
| METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (5.208A 5.208B 5.209)<br>SPACE OPERATION (5.209A 5.203C)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile (R)<br><br>5.204 5.205 5.206 5.207 5.208 | METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (5.208A 5.208B 5.209)<br>SPACE OPERATION (5.203C 5.209A)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br><br>5.206 5.208 ECA6 ECA36 | Aeronautical military systems |                |            |                                                                    |
|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   | Land military systems         |                |            |                                                                    |
|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   | Land mobile                   |                |            | Mobile restricted to Aeronautical Mobile (OR), including air sport |
|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   | MSS Earth stations            | ERC/DEC/(99)06 | EN 301 721 | Non-geostationary                                                  |
|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   | Maritime military systems     |                |            |                                                                    |
|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   | Satellite systems (military)  |                |            |                                                                    |
|                                                                                                                                                                                                                                                  |                                                                                                                                                                                                   | Weather satellites            |                |            |                                                                    |

### 137.825 MHz - 138 MHz

|                                                                                                                                                                                                                                           |                                                                                                                                                                                            |                               |                |            |                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|------------|--------------------------------------------------------------------|
| METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>SPACE OPERATION (5.203C)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile (R)<br>Mobile-Satellite (5.208A 5.208B 5.209)<br><br>5.204 5.205 5.206 5.207 5.208 | METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>SPACE OPERATION (5.203C)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Mobile-Satellite (5.208A 5.208B 5.209)<br><br>5.206 5.208 ECA6 ECA36 | Aeronautical military systems |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Land military systems         |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Land mobile                   |                |            | Mobile restricted to Aeronautical Mobile (OR), including air sport |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | MSS Earth stations            | ERC/DEC/(99)06 | EN 301 721 | Non-geostationary                                                  |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Maritime military systems     |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Satellite systems (military)  |                |            |                                                                    |
|                                                                                                                                                                                                                                           |                                                                                                                                                                                            | Weather satellites            |                |            |                                                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 138 MHz - 143.6 MHz

|                                                     |                                                                                                    |                               |               |            |                                   |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------|---------------|------------|-----------------------------------|
| AERONAUTICAL MOBILE (OR)<br>5.210 5.211 5.212 5.214 | AERONAUTICAL MOBILE (OR)<br>LAND MOBILE<br>Space research (space-to-Earth)<br><br>5.211 ECA5 ECA36 | Aeronautical military systems |               |            |                                   |
|                                                     |                                                                                                    | Land military systems         |               |            |                                   |
|                                                     |                                                                                                    | Land mobile                   |               |            |                                   |
|                                                     |                                                                                                    | Maritime military systems     |               |            |                                   |
|                                                     |                                                                                                    | Non-specific SRDs             | ERC/REC 70-03 | EN 300 220 | Within the band 138.20-138.45 MHz |

### 143.6 MHz - 143.65 MHz

|                                                                                      |                                                                                                    |                               |  |  |  |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------|--|--|--|
| AERONAUTICAL MOBILE (OR)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br><br>5.211 5.212 5.214 | AERONAUTICAL MOBILE (OR)<br>LAND MOBILE<br>SPACE RESEARCH (SPACE-TO-EARTH)<br><br>5.211 ECA5 ECA36 | Aeronautical military systems |  |  |  |
|                                                                                      |                                                                                                    | Land military systems         |  |  |  |
|                                                                                      |                                                                                                    | Land mobile                   |  |  |  |
|                                                                                      |                                                                                                    | Maritime military systems     |  |  |  |

### 143.65 MHz - 144 MHz

|                                                         |                                                                 |                               |  |  |  |
|---------------------------------------------------------|-----------------------------------------------------------------|-------------------------------|--|--|--|
| AERONAUTICAL MOBILE (OR)<br><br>5.210 5.211 5.212 5.214 | AERONAUTICAL MOBILE (OR)<br>LAND MOBILE<br><br>5.211 ECA5 ECA36 | Aeronautical military systems |  |  |  |
|                                                         |                                                                 | Land military systems         |  |  |  |
|                                                         |                                                                 | Land mobile                   |  |  |  |
|                                                         |                                                                 | Maritime military systems     |  |  |  |

### 144 MHz - 146 MHz

|                                           |                              |                   |  |            |  |
|-------------------------------------------|------------------------------|-------------------|--|------------|--|
| AMATEUR<br>AMATEUR-SATELLITE<br><br>5.216 | AMATEUR<br>AMATEUR-SATELLITE | Amateur           |  | EN 301 783 |  |
|                                           |                              | Amateur-satellite |  |            |  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 146 MHz - 148 MHz

|                                                |                          |                           |                              |                                                                                                                                                   |  |
|------------------------------------------------|--------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--|
| FIXED<br>Mobile except aeronautical mobile (R) | MOBILE<br><br>ECA7 ECA36 | Maritime military systems |                              |                                                                                                                                                   |  |
|                                                |                          | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |  |

## 148 MHz - 149.9 MHz

|                                                                                                            |                                                                                       |                           |                              |                                                                                                                                                   |                   |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| FIXED<br>MOBILE-SATELLITE (5.209)<br>Mobile except aeronautical mobile (R)<br><br>5.218 5.218A 5.219 5.221 | MOBILE<br>MOBILE-SATELLITE (5.209)<br><br>5.218 5.218A 5.219 5.221 ECA6 ECA7<br>ECA36 | MSS Earth stations        | ERC/DEC/(99)06               | EN 301 721                                                                                                                                        | Non-geostationary |
|                                                                                                            |                                                                                       | Maritime military systems |                              |                                                                                                                                                   |                   |
|                                                                                                            |                                                                                       | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 149.9 MHz - 150.05 MHz

|                                |                                                            |                           |                              |                                                                                                                                                   |                               |
|--------------------------------|------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| MOBILE-SATELLITE (5.209 5.220) | MOBILE<br>MOBILE-SATELLITE (5.209 5.220)<br><br>ECA6 ECA36 | MSS Earth stations        | ERC/DEC/(99)06               | EN 301 721                                                                                                                                        | Non-geostationary             |
|                                |                                                            | Maritime military systems |                              |                                                                                                                                                   |                               |
|                                |                                                            | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Single frequency applications |

### 150.05 MHz - 153 MHz

|                                                                               |                                                                                 |                           |                              |                                                                                                                                                   |                                                    |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 ECA7 ECA36 | Maritime military systems |                              |                                                                                                                                                   |                                                    |
|                                                                               |                                                                                 | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                    |
|                                                                               |                                                                                 | Radio astronomy           |                              |                                                                                                                                                   | Continuum observations (inter-alia solar research) |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 153 MHz - 154 MHz

|                                                                          |                                                        |                           |                              |                                                                                                                                                   |                                                   |
|--------------------------------------------------------------------------|--------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>Meteorological aids | MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>ECA7 ECA36 | Maritime military systems |                              |                                                                                                                                                   |                                                   |
|                                                                          |                                                        | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Base station transmit paired with 148.4-149.4 MHz |

## 154 MHz - 156.4875 MHz

|                                                                   |                                                                   |                           |                              |                                                                                                                                                   |                |
|-------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.225A 5.226 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE (R)<br>5.226 ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03               | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929                                                                            | RR Appendix 18 |
|                                                                   |                                                                   | Maritime military systems |                              |                                                                                                                                                   |                |
|                                                                   |                                                                   | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 156.4875 MHz - 156.5125 MHz

|                                                               |                                                                               |                           |                |                                                                        |                |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------|----------------|
| MARITIME MOBILE (DISTRESS AND CALLING VIA DSC)<br>5.226 5.227 | MARITIME MOBILE (DISTRESS AND CALLING VIA DSC)<br>5.226 5.227 ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929 | RR Appendix 18 |
|                                                               |                                                                               | Maritime military systems |                |                                                                        |                |

### 156.5125 MHz - 156.5375 MHz

|                                                               |                                                                     |                           |                |                                                         |                                                           |
|---------------------------------------------------------------|---------------------------------------------------------------------|---------------------------|----------------|---------------------------------------------------------|-----------------------------------------------------------|
| MARITIME MOBILE (DISTRESS AND CALLING VIA DSC)<br>5.111 5.226 | MARITIME MOBILE (DISTRESS AND CALLING VIA DSC)<br>5.111 5.226 ECA36 | AMRD Group A              | ECC/DEC/(22)02 |                                                         |                                                           |
|                                                               |                                                                     | DSC                       | ECC/DEC/(19)03 | EN 301 025,<br>EN 301 929,<br>EN 302 885,<br>EN 303 132 | RR Appendix 18. Distress, safety and calling 156.525 MHz. |
|                                                               |                                                                     | Maritime military systems |                |                                                         |                                                           |

### 156.5375 MHz - 156.5625 MHz

|                                                               |                                                                                                                        |                           |                |                                                                        |                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------|----------------|
| MARITIME MOBILE (DISTRESS AND CALLING VIA DSC)<br>5.226 5.227 | MARITIME MOBILE (DISTRESS AND CALLING VIA DSC)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (R)<br>5.226 5.227 ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929 | RR Appendix 18 |
|                                                               |                                                                                                                        | Maritime military systems |                |                                                                        |                |

### 156.5625 MHz - 156.7625 MHz

|                                                         |                                                                |                           |                |                                                                        |                |
|---------------------------------------------------------|----------------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------|----------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL MOBILE (R)<br>5.226 | MOBILE EXCEPT AERONAUTICAL MOBILE (R)<br>5.226 ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929 | RR Appendix 18 |
|                                                         |                                                                | Maritime military systems |                |                                                                        |                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 156.7625 MHz - 156.7875 MHz

|                                                                           |                                                                      |                           |                |            |                                              |
|---------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------|----------------|------------|----------------------------------------------|
| MARITIME MOBILE<br>Mobile-Satellite (Earth-to-space)<br>5.111 5.226 5.228 | MARITIME MOBILE (DISTRESS AND<br>CALLING)<br>5.111 5.226 5.228 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 301 929 | RR Appendix 18. Satellite AIS Earth-to-space |
|                                                                           |                                                                      | Maritime military systems |                |            |                                              |

### 156.7875 MHz - 156.8125 MHz

|                                                          |                                                                |                           |                |            |                                                                                                                  |
|----------------------------------------------------------|----------------------------------------------------------------|---------------------------|----------------|------------|------------------------------------------------------------------------------------------------------------------|
| MARITIME MOBILE (DISTRESS AND<br>CALLING)<br>5.111 5.226 | MARITIME MOBILE (DISTRESS AND<br>CALLING)<br>5.111 5.226 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 300 162 | RR Appendix 18. Distress, safety and calling<br>156.8 MHz for the maritime mobile VHF<br>radiotelephone service. |
|                                                          |                                                                | Maritime military systems |                |            |                                                                                                                  |

### 156.8125 MHz - 156.8375 MHz

|                                                                           |                                            |                           |                |            |                                               |
|---------------------------------------------------------------------------|--------------------------------------------|---------------------------|----------------|------------|-----------------------------------------------|
| MARITIME MOBILE<br>Mobile-Satellite (Earth-to-space)<br>5.111 5.226 5.228 | MARITIME MOBILE<br>5.111 5.226 5.228 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 301 929 | RR Appendix 18. Satellite AIS Earth-to-space. |
|                                                                           |                                            | Maritime military systems |                |            |                                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 156.8375 MHz - 157.1875 MHz

|                                                       |                                                                     |                           |                              |                                                                                                                                                   |                |
|-------------------------------------------------------|---------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226) | MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226 ECA7 ECA8)<br><br>ECA36 | Maritime communications   | ECC/DEC/(19)03               | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929                                                                            | RR Appendix 18 |
|                                                       |                                                                     | Maritime military systems |                              |                                                                                                                                                   |                |
|                                                       |                                                                     | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                |

## 157.1875 MHz - 157.3375 MHz

|                                                                                                                       |                                                                                                                                  |                           |                              |                                                                                                                                                   |                |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226)<br>Maritime mobile-satellite (5.208A 5.208B<br>5.228AC 5.228AB) | Maritime mobile-satellite (5.208A 5.208B<br>5.228AB 5.228AC)<br>Mobile except aeronautical mobile (5.226)<br><br>ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03               | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929                                                                            | RR Appendix 18 |
|                                                                                                                       |                                                                                                                                  | Maritime military systems |                              |                                                                                                                                                   |                |
|                                                                                                                       |                                                                                                                                  | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 157.3375 MHz - 161.7875 MHz

|                                                       |                                                                     |                           |                              |                                                                                                                                                   |                                              |
|-------------------------------------------------------|---------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226) | MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226)<br><br>ECA7 ECA8 ECA36 | AMRD Group B              | ECC/DEC/(22)02               |                                                                                                                                                   | Within frequency range 160.8875-160.9125 MHz |
|                                                       |                                                                     | Maritime communications   | ECC/DEC/(19)03               | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929                                                                            | RR Appendix 18                               |
|                                                       |                                                                     | Maritime military systems |                              |                                                                                                                                                   |                                              |
|                                                       |                                                                     | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                              |

### 161.7875 MHz - 161.9375 MHz

|                                                                                                                       |                                                                                                                                     |                           |                              |                                                                                                                                                   |                |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226)<br>Maritime mobile-satellite (5.208A<br>5.228AC 5.228AB 5.208B) | MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226)<br>Maritime mobile-satellite (5.208A 5.208B<br>5.228AB 5.228AC)<br><br>ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03               | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929                                                                            | RR Appendix 18 |
|                                                                                                                       |                                                                                                                                     | Maritime military systems |                              |                                                                                                                                                   |                |
|                                                                                                                       |                                                                                                                                     | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 161.9375 MHz - 161.9625 MHz

|                                                                                                      |                                                                                                          |                           |                              |                                                                                                                                                   |                |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FIXED<br>MARITIME MOBILE-SATELLITE<br>(5.228AA)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.226 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Maritime mobile-satellite (5.228AA)<br><br>5.226 ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03               | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929                                                                            | RR Appendix 18 |
|                                                                                                      |                                                                                                          | Maritime military systems |                              |                                                                                                                                                   |                |
|                                                                                                      |                                                                                                          | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                |

### 161.9625 MHz - 161.9875 MHz

|                                                                                                               |                                                                                          |                           |                |                                                                        |                              |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------|------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.226)<br>Mobile-Satellite (5.228F)<br><br>5.226 5.228A 5.228B | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Mobile-Satellite (5.228F)<br><br>5.226 ECA7 ECA8 | AIS                       |                | EN 303 098                                                             | Centre frequency 161.975 MHz |
|                                                                                                               |                                                                                          | AMRD Group A              | ECC/DEC/(22)02 |                                                                        |                              |
|                                                                                                               |                                                                                          | Maritime communications   | ECC/DEC/(19)03 | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929 | RR Appendix 18               |
|                                                                                                               |                                                                                          | Maritime military systems |                |                                                                        |                              |

### 161.9875 MHz - 162.0125 MHz

|                                                                                                      |                                                                                                          |                           |                |                                                                        |                |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------|----------------|
| FIXED<br>MARITIME MOBILE-SATELLITE<br>(5.228AA)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.226 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Maritime mobile-satellite (5.228AA)<br><br>5.226 ECA7 ECA8 ECA36 | Maritime communications   | ECC/DEC/(19)03 | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929 | RR Appendix 18 |
|                                                                                                      |                                                                                                          | Maritime military systems |                |                                                                        |                |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 162.0125 MHz - 162.0375 MHz

|                                                                                                       |                                                                   |                           |                |                                                                        |                              |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------|------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Mobile-Satellite (5.228F)<br><br>5.226 5.228A 5.228B | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.226 ECA7 ECA8 ECA36 | AIS                       |                | EN 303 098                                                             | Centre frequency 162.025 MHz |
|                                                                                                       |                                                                   | AMRD Group A              | ECC/DEC/(22)02 |                                                                        |                              |
|                                                                                                       |                                                                   | Maritime communications   | ECC/DEC/(19)03 | EN 300 162,<br>EN 300 698,<br>EN 301 025,<br>EN 301 178,<br>EN 301 929 | RR Appendix 18               |
|                                                                                                       |                                                                   | Maritime military systems |                |                                                                        |                              |

## 162.0375 MHz - 174 MHz

|                                                            |                                                           |                           |                                  |                                                                                                                                                   |                                                             |
|------------------------------------------------------------|-----------------------------------------------------------|---------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.226 | Mobile except aeronautical mobile<br><br>5.226 ECA7 ECA36 | ALD                       | ECC/DEC/(05)02,<br>ERC/REC 70-03 | EN 300 422                                                                                                                                        | The bands 169.400-169.475 MHz and<br>169.4875-169.5875 MHz. |
|                                                            |                                                           | Maritime military systems |                                  |                                                                                                                                                   |                                                             |
|                                                            |                                                           | Meter reading             | ECC/DEC/(05)02,<br>ERC/REC 70-03 | EN 300 220                                                                                                                                        | Within the band 169.400-169.475 MHz                         |
|                                                            |                                                           | Non-specific SRDs         | ECC/DEC/(05)02,<br>ERC/REC 70-03 | EN 300 220                                                                                                                                        |                                                             |
|                                                            |                                                           | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 |                                                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 174 MHz - 223 MHz

|                       |                                      |                            |                                 |                           |                                                                                  |
|-----------------------|--------------------------------------|----------------------------|---------------------------------|---------------------------|----------------------------------------------------------------------------------|
| BROADCASTING<br>5.235 | BROADCASTING<br>LAND MOBILE<br>5.235 | Audio PMSE                 | ERC/REC 25-10                   | EN 300 454                | Radio microphones and In-ear monitors on a tuning range basis within 174-216 MHz |
|                       |                                      | Broadcasting (terrestrial) |                                 | EN 302 077,<br>EN 302 296 | Geneva Agreement 2006. TV Broadcasting T-DAB.                                    |
|                       |                                      | Radio microphones and ALD  | ERC/REC 25-10,<br>ERC/REC 70-03 | EN 300 422                | On a tuning range basis within 174-216 MHz                                       |

## 223 MHz - 225 MHz

|                                                      |              |                            |  |                           |                                               |
|------------------------------------------------------|--------------|----------------------------|--|---------------------------|-----------------------------------------------|
| BROADCASTING<br>Fixed<br>Mobile<br>5.243 5.246 5.247 | BROADCASTING | Broadcasting (terrestrial) |  | EN 302 077,<br>EN 302 296 | Geneva Agreement 2006. TV Broadcasting, T-DAB |
|------------------------------------------------------|--------------|----------------------------|--|---------------------------|-----------------------------------------------|

## 225 MHz - 230 MHz

|                                                |                                            |                            |  |                           |                                                                                                                                                   |
|------------------------------------------------|--------------------------------------------|----------------------------|--|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| BROADCASTING<br>Fixed<br>Mobile<br>5.246 5.247 | BROADCASTING<br>Land mobile<br>ECA10 ECA36 | Broadcasting (terrestrial) |  | EN 302 077,<br>EN 302 296 | Geneva Agreement 2006. This band is within the military tuning range 225-400 MHz. Sharing with defence on national basis. TV Broadcasting, T-DAB. |
|                                                |                                            | Defence systems            |  |                           |                                                                                                                                                   |

## 230 MHz - 235 MHz

|                                      |                       |                 |  |  |                                                                                                                                                            |
|--------------------------------------|-----------------------|-----------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>5.247 5.251 5.252 | MOBILE<br>ECA10 ECA36 | Defence systems |  |  |                                                                                                                                                            |
|                                      |                       | T-DAB           |  |  | T-DAB sharing with defence on a national basis. Wiesbaden 1995 Special Arrangement, as revised in Constanta, 2007. Within the frequency range 230-240 MHz. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 235 MHz - 240 MHz

|                                    |                                 |                 |  |  |                                                                                                                                                           |
|------------------------------------|---------------------------------|-----------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br><br>5.252 5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |  |  |                                                                                                                                                           |
|                                    |                                 | T-DAB           |  |  | T-DAB sharing with defence on a national basis. Wiesbaden 1995 Special Arrangement, as revised in Constanta, 2007. Within the frequency range 230-240 MHz |

## 240 MHz - 242.95 MHz

|                                          |                                 |                 |  |            |  |
|------------------------------------------|---------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br><br>5.111 5.254 5.256 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|------------------------------------------|---------------------------------|-----------------|--|------------|--|

## 242.95 MHz - 243.05 MHz

|                                          |                                              |        |  |            |                                                                |
|------------------------------------------|----------------------------------------------|--------|--|------------|----------------------------------------------------------------|
| FIXED<br>MOBILE<br><br>5.111 5.254 5.256 | AERONAUTICAL MOBILE<br><br>5.111 5.254 5.256 | EPIRBs |  | EN 300 152 | Band only available for distress and safety purposes 243.0 MHz |
|------------------------------------------|----------------------------------------------|--------|--|------------|----------------------------------------------------------------|

## 243.05 MHz - 267 MHz

|                                                                                     |                                 |                 |  |            |  |
|-------------------------------------------------------------------------------------|---------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.111 5.252 5.254 5.256 5.256A | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|-------------------------------------------------------------------------------------|---------------------------------|-----------------|--|------------|--|

## 267 MHz - 272 MHz

|                                                                        |                                       |                 |  |            |  |
|------------------------------------------------------------------------|---------------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br>Space Operation (space-to-Earth)<br><br>5.254 5.257 | MOBILE<br><br>5.254 5.257 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|------------------------------------------------------------------------|---------------------------------------|-----------------|--|------------|--|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 272 MHz - 273 MHz

|                                                                  |                                 |                 |  |            |  |
|------------------------------------------------------------------|---------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br>SPACE OPERATION (SPACE-TO-EARTH)<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|------------------------------------------------------------------|---------------------------------|-----------------|--|------------|--|

## 273 MHz - 312 MHz

|                              |                                 |                 |  |            |  |
|------------------------------|---------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|------------------------------|---------------------------------|-----------------|--|------------|--|

## 312 MHz - 315 MHz

|                                                   |                                       |                 |  |            |  |
|---------------------------------------------------|---------------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br>Mobile-Satellite (5.254 5.255) | MOBILE<br><br>5.254 5.255 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|---------------------------------------------------|---------------------------------------|-----------------|--|------------|--|

## 315 MHz - 322 MHz

|                              |                                 |                 |  |            |  |
|------------------------------|---------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|------------------------------|---------------------------------|-----------------|--|------------|--|

## 322 MHz - 328.6 MHz

|                                                 |                                                    |                 |  |  |                                                                 |
|-------------------------------------------------|----------------------------------------------------|-----------------|--|--|-----------------------------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 | MOBILE<br>RADIO ASTRONOMY<br><br>5.149 ECA10 ECA36 | Defence systems |  |  |                                                                 |
|                                                 |                                                    | Radio astronomy |  |  | Continuum and spectral line observations (e.g. deuterium), VLBI |

## 328.6 MHz - 335.4 MHz

|                                                 |                                           |     |  |  |            |
|-------------------------------------------------|-------------------------------------------|-----|--|--|------------|
| AERONAUTICAL RADIONAVIGATION<br><br>5.258 5.259 | AERONAUTICAL RADIONAVIGATION<br><br>5.258 | ILS |  |  | Glide path |
|-------------------------------------------------|-------------------------------------------|-----|--|--|------------|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 335.4 MHz - 380 MHz

|                              |                                      |                 |  |            |  |
|------------------------------|--------------------------------------|-----------------|--|------------|--|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA7 ECA10 ECA36 | Defence systems |  | EN 302 617 |  |
|------------------------------|--------------------------------------|-----------------|--|------------|--|

### 380 MHz - 385 MHz

|                              |                                 |                 |                                                                    |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------|---------------------------------|-----------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |                                                                    |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                         |
|                              |                                 | PPDR            | ECC/DEC/(06)05,<br>ECC/DEC/(08)05,<br>ERC/DEC/(01)19,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 302 426,<br>EN 302 561 | Within the bands 384.8-385.0 and 394.8-395.0 MHz for AGA, 384.750-384.800 MHz and 394.750-394.800 MHz may be used as preferred extension bands for AGA. Within the bands 380-380.15 and 390-390.15 MHz for DMO. Mobile station transmit paired with 390-395 MHz. PPDR sharing with defence applications. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

### 385 MHz - 387 MHz

|                              |                                 |                 |                              |                                                                                                      |                                                                                                                                 |
|------------------------------|---------------------------------|-----------------|------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |                              |                                                                                                      |                                                                                                                                 |
|                              |                                 | PPDR            | ECC/DEC/(08)05,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 302 426,<br>EN 302 561 | Mobile station transmit paired with 395-397 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 387 MHz - 390 MHz

|                                                                    |                           |                 |                              |                                                                                                      |                                                                                                                                                                                     |
|--------------------------------------------------------------------|---------------------------|-----------------|------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Mobile-Satellite (5.208A 5.254 5.255<br>5.208B) | MOBILE<br><br>ECA10 ECA36 | Defence systems |                              |                                                                                                      |                                                                                                                                                                                     |
|                                                                    |                           | PPDR            | ECC/DEC/(08)05,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 302 426,<br>EN 302 561 | Single frequency applications in 389.9-390 MHz. Mobile station transmit paired with 397.0-399.9 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

### 390 MHz - 395 MHz

|                              |                                 |                 |                                                                    |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------|---------------------------------|-----------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |                                                                    |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                               |
|                              |                                 | PPDR            | ECC/DEC/(06)05,<br>ECC/DEC/(08)05,<br>ERC/DEC/(01)19,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 302 426,<br>EN 302 561 | Within the bands 384.8-385.0 and 394.8-395.0 MHz for AGA, 384.750-384.800 MHz and 394.750-394.800 MHz may be used as preferred extension bands. Within the bands 380-380.15 and 390-390.15 MHz for DMO. Base station transmit paired with 380-385 MHz. PPDR sharing with defence applications. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

### 395 MHz - 399.9 MHz

|                              |                                 |                 |                              |                                                                                                      |                                                                                                                                   |
|------------------------------|---------------------------------|-----------------|------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br><br>5.254 | MOBILE<br><br>5.254 ECA10 ECA36 | Defence systems |                              |                                                                                                      |                                                                                                                                   |
|                              |                                 | PPDR            | ECC/DEC/(08)05,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 302 426,<br>EN 302 561 | Base station transmit paired with 385.0-389.9 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 399.9 MHz - 400.05 MHz

|                                              |                                |                    |                                   |            |  |
|----------------------------------------------|--------------------------------|--------------------|-----------------------------------|------------|--|
| MOBILE-SATELLITE (5.220 5.209 5.260A 5.260B) | MOBILE-SATELLITE (5.209 5.220) | MSS Earth stations | ERC/DEC/(99)05,<br>ERC/DEC/(99)06 | EN 301 721 |  |
|----------------------------------------------|--------------------------------|--------------------|-----------------------------------|------------|--|

### 400.15 MHz - 401 MHz

|                                                                                                                                                                                             |                                                                                                                                                                                             |                    |                                   |            |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|------------|-------------------|
| METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (5.208A 5.208B 5.209)<br>SPACE RESEARCH (5.263)<br>Space Operation (space-to-Earth)<br><br>5.262 5.264 | METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (5.208A 5.208B 5.209)<br>SPACE OPERATION (SPACE-TO-EARTH)<br>SPACE RESEARCH (5.263)<br><br>5.262 5.264 | MSS Earth stations | ERC/DEC/(99)05,<br>ERC/DEC/(99)06 | EN 301 721 | Non-geostationary |
|                                                                                                                                                                                             |                                                                                                                                                                                             | Sondes             |                                   | EN 302 054 |                   |
|                                                                                                                                                                                             |                                                                                                                                                                                             | Weather satellites |                                   |            |                   |

### 401 MHz - 402 MHz

|                                                                                                                                                                                                                         |                                                                                                                                       |                    |                                  |            |                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------|------------|------------------------------------|
| EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE)<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE)<br>SPACE OPERATION (SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile<br><br>5.264A 5.264B | EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE)<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE)<br><br>5.264A 5.264B | Sondes             |                                  | EN 302 054 |                                    |
|                                                                                                                                                                                                                         |                                                                                                                                       | ULP-AMI            | ERC/DEC/(01)17,<br>ERC/REC 70-03 | EN 302 537 |                                    |
|                                                                                                                                                                                                                         |                                                                                                                                       | Weather satellites |                                  |            | Data collection platform telemetry |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 402 MHz - 403 MHz

|                                                                                                                                                                                 |                                                                                                                                   |                    |                                  |            |                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------|------------|------------------------------------|
| EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE)<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE)<br>Fixed<br>Mobile except aeronautical mobile<br>5.264A 5.264B | EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE)<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE)<br>5.264A 5.264B | Sondes             |                                  | EN 302 054 |                                    |
|                                                                                                                                                                                 |                                                                                                                                   | ULP-AMI            | ERC/DEC/(01)17,<br>ERC/REC 70-03 | EN 301 839 |                                    |
|                                                                                                                                                                                 |                                                                                                                                   | Weather satellites |                                  |            | Data collection platform telemetry |

#### 403 MHz - 406 MHz

|                                                                            |                              |         |                                  |                           |  |
|----------------------------------------------------------------------------|------------------------------|---------|----------------------------------|---------------------------|--|
| METEOROLOGICAL AIDS<br>Fixed<br>Mobile except aeronautical mobile<br>5.265 | METEOROLOGICAL AIDS<br>5.265 | Sondes  |                                  | EN 302 054                |  |
|                                                                            |                              | ULP-AMI | ERC/DEC/(01)17,<br>ERC/REC 70-03 | EN 301 839,<br>EN 302 537 |  |

#### 406 MHz - 406.1 MHz

|                                                        |                                                        |        |  |                           |                                                      |
|--------------------------------------------------------|--------------------------------------------------------|--------|--|---------------------------|------------------------------------------------------|
| MOBILE-SATELLITE (EARTH-TO-SPACE)<br>5.265 5.266 5.267 | MOBILE-SATELLITE (EARTH-TO-SPACE)<br>5.265 5.266 5.267 | EPIRBs |  | EN 300 066,<br>EN 302 152 | Band only available for distress and safety purposes |
|--------------------------------------------------------|--------------------------------------------------------|--------|--|---------------------------|------------------------------------------------------|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 406.1 MHz - 410 MHz

|                                                                                     |                                                         |                           |                              |                                                                                                                                                   |                                                                                                              |
|-------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.265 | LAND MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.265 ECA36 | Land military systems     |                              |                                                                                                                                                   |                                                                                                              |
|                                                                                     |                                                         | Maritime military systems |                              |                                                                                                                                                   |                                                                                                              |
|                                                                                     |                                                         | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08 | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Single frequency applications. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05 |
|                                                                                     |                                                         | Radio astronomy           |                              |                                                                                                                                                   | Continuum observations, VLBI                                                                                 |

## 410 MHz - 420 MHz

|                                                                         |                                                   |                           |                                                 |                                                                                                                                                                  |                                                                                                                                 |
|-------------------------------------------------------------------------|---------------------------------------------------|---------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE RESEARCH (5.268) | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>ECA36 | Land military systems     |                                                 |                                                                                                                                                                  |                                                                                                                                 |
|                                                                         |                                                   | Maritime military systems |                                                 |                                                                                                                                                                  |                                                                                                                                 |
|                                                                         |                                                   | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08                    | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit paired with 420-430 MHz.                                                                                |
|                                                                         |                                                   | PPDR                      | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Mobile station transmit paired with 420-430 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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#### 420 MHz - 430 MHz

|                                                                                         |                                                                         |                           |                                                 |                                                                                                                                                                  |                                                                                                                                  |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation<br><br>5.269 5.270 5.271 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation<br><br>ECA7 ECA36 | Land military systems     |                                                 |                                                                                                                                                                  |                                                                                                                                  |
|                                                                                         |                                                                         | Maritime military systems |                                                 |                                                                                                                                                                  |                                                                                                                                  |
|                                                                                         |                                                                         | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08                    | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Base station transmit paired with 410-420 MHz.                                                                                   |
|                                                                                         |                                                                         | PPDR                      | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Base station transmit paired with 410-420 MHz.<br>PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |
|                                                                                         |                                                                         | Radiolocation (military)  |                                                 |                                                                                                                                                                  |                                                                                                                                  |

#### 430 MHz - 432 MHz

|                                                               |                                             |                          |               |            |                             |
|---------------------------------------------------------------|---------------------------------------------|--------------------------|---------------|------------|-----------------------------|
| AMATEUR<br>RADIOLOCATION<br><br>5.271 5.274 5.275 5.276 5.277 | AMATEUR<br>RADIOLOCATION<br><br>ECA12 ECA36 | Amateur                  |               | EN 301 783 | Within the band 430-440 MHz |
|                                                               |                                             | Radiolocation (military) |               |            |                             |
|                                                               |                                             | ULP-WMCE                 | ERC/REC 70-03 | EN 303 520 | Within the band 430-440 MHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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#### 432 MHz - 433.05 MHz

|                                                                                                       |                                                                                     |                            |               |            |                                                                                                                 |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------|---------------|------------|-----------------------------------------------------------------------------------------------------------------|
| AMATEUR<br>RADIOLOCATION<br>Earth exploration-satellite (5.279A)<br><br>5.138 5.271 5.276 5.277 5.280 | AMATEUR<br>RADIOLOCATION<br>Earth exploration-satellite (5.279A)<br><br>ECA12 ECA36 | Active sensors (satellite) |               |            | The use of this band by sensors in the EESS (active) shall be in accordance with Recommendation ITU-R SA 1260-1 |
|                                                                                                       |                                                                                     | Amateur                    |               | EN 301 783 | Within the band 430-440 MHz                                                                                     |
|                                                                                                       |                                                                                     | Radiolocation (military)   |               |            |                                                                                                                 |
|                                                                                                       |                                                                                     | ULP-WMCE                   | ERC/REC 70-03 | EN 303 520 | Within the band 430-440 MHz                                                                                     |

#### 433.05 MHz - 434.79 MHz

|                                                                                                             |                                                                                                                |                            |               |            |                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------|---------------|------------|-----------------------------------------------------------------------------------------------------------------|
| AMATEUR<br>RADIOLOCATION<br>Earth exploration-satellite (5.279A)<br><br>5.138 5.271 5.276 5.277 5.280 5.281 | AMATEUR<br>RADIOLOCATION<br>Earth exploration-satellite (5.279A)<br>Land mobile<br><br>5.138 5.280 ECA12 ECA36 | Active sensors (satellite) |               |            | The use of this band by sensors in the EESS (active) shall be in accordance with Recommendation ITU-R SA 1260-1 |
|                                                                                                             |                                                                                                                | Amateur                    |               | EN 301 783 | Within the band 430-440 MHz                                                                                     |
|                                                                                                             |                                                                                                                | ISM                        |               |            |                                                                                                                 |
|                                                                                                             |                                                                                                                | Non-specific SRDs          | ERC/REC 70-03 | EN 300 220 |                                                                                                                 |
|                                                                                                             |                                                                                                                | Radiolocation (military)   |               |            |                                                                                                                 |
|                                                                                                             |                                                                                                                | ULP-WMCE                   | ERC/REC 70-03 | EN 303 520 | Within the band 430-440 MHz                                                                                     |

#### 434.79 MHz - 438 MHz

|                                                                                                             |                                                                                                          |                            |               |            |                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------|---------------|------------|-----------------------------------------------------------------------------------------------------------------|
| AMATEUR<br>RADIOLOCATION<br>Earth exploration-satellite (5.279A)<br><br>5.138 5.271 5.276 5.277 5.280 5.282 | AMATEUR<br>AMATEUR-SATELLITE<br>RADIOLOCATION<br>Earth exploration-satellite (5.279A)<br><br>ECA12 ECA36 | Active sensors (satellite) |               |            | The use of this band by sensors in the EESS (active) shall be in accordance with Recommendation ITU-R SA 1260-1 |
|                                                                                                             |                                                                                                          | Amateur                    |               | EN 301 783 | Within the band 430-440 MHz                                                                                     |
|                                                                                                             |                                                                                                          | Amateur-satellite          |               |            | Amateur Satellite Service restricted to 435-438 MHz                                                             |
|                                                                                                             |                                                                                                          | Radiolocation (military)   |               |            |                                                                                                                 |
|                                                                                                             |                                                                                                          | ULP-WMCE                   | ERC/REC 70-03 | EN 303 520 | Within the band 430-440 MHz                                                                                     |

#### 438 MHz - 440 MHz

|                                                                     |                                             |                          |               |            |                             |
|---------------------------------------------------------------------|---------------------------------------------|--------------------------|---------------|------------|-----------------------------|
| AMATEUR<br>RADIOLOCATION<br><br>5.271 5.274 5.275 5.276 5.277 5.283 | AMATEUR<br>RADIOLOCATION<br><br>ECA12 ECA36 | Amateur                  |               | EN 301 783 | Within the band 430-440 MHz |
|                                                                     |                                             | Radiolocation (military) |               |            |                             |
|                                                                     |                                             | ULP-WMCE                 | ERC/REC 70-03 | EN 303 520 | Within the band 430-440 MHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 440 MHz - 450 MHz

|                                                                                                           |                                                                         |                           |                                  |                                                                                                                                                   |                                                                                                            |
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| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation<br><br>5.269 5.270 5.271 5.284 5.285 5.286 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation<br><br>ECA7 ECA36 | Land military systems     |                                  |                                                                                                                                                   |                                                                                                            |
|                                                                                                           |                                                                         | Maritime military systems |                                  |                                                                                                                                                   |                                                                                                            |
|                                                                                                           |                                                                         | On-site paging            |                                  | EN 300 224                                                                                                                                        | Call-out & answer-back                                                                                     |
|                                                                                                           |                                                                         | PMR 446                   | ECC/DEC/(15)05,<br>ERC/REC 70-03 | EN 303 405                                                                                                                                        | PMR446 in 446.0-446.2 MHz                                                                                  |
|                                                                                                           |                                                                         | PMR/PAMR                  | ECC/DEC/(19)02,<br>T/R 25-08     | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 302 561,<br>EN 303 039 | Single frequency operation. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |
|                                                                                                           |                                                                         | Radiolocation (military)  |                                  |                                                                                                                                                   |                                                                                                            |
|                                                                                                           |                                                                         | Wind profilers            |                                  |                                                                                                                                                   | Geographical sharing with other services                                                                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 450 MHz - 455 MHz

|                                                                                          |                          |                |                                                 |                                                                                                                                                                  |                                                                                                                                 |
|------------------------------------------------------------------------------------------|--------------------------|----------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.286AA)<br><br>5.209 5.271 5.286 5.286A 5.286B 5.286C<br>5.286D 5.286E | MOBILE<br><br>ECA7 ECA34 | On-site paging |                                                 | EN 300 224                                                                                                                                                       | Call-out & answer-back                                                                                                          |
|                                                                                          |                          | PMR/PAMR       | ECC/DEC/(19)02,<br>T/R 25-08                    | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit paired with 460-465 MHz.                                                                                |
|                                                                                          |                          | PPDR           | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Mobile station transmit paired with 460-465 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 455 MHz - 456 MHz

|                                                                          |                          |                |                                                 |                                                                                                                                                                  |                                                                                                                                 |
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| FIXED<br>MOBILE (5.286AA)<br><br>5.209 5.271 5.286A 5.286B 5.286C 5.286E | MOBILE<br><br>ECA7 ECA34 | Land mobile    |                                                 |                                                                                                                                                                  | Existing public cellular networks                                                                                               |
|                                                                          |                          | On-site paging |                                                 | EN 300 224                                                                                                                                                       | Call-out & answer-back                                                                                                          |
|                                                                          |                          | PMR/PAMR       | ECC/DEC/(19)02,<br>T/R 25-08                    | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit paired with 465-466 MHz.                                                                                |
|                                                                          |                          | PPDR           | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Mobile station transmit paired with 465-466 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 456 MHz - 459 MHz

|                                                    |                                |                         |                                                 |                                                                                                                                                                  |                                                                                                                                 |
|----------------------------------------------------|--------------------------------|-------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.286AA)<br><br>5.271 5.287 5.288 | MOBILE<br><br>5.287 ECA7 ECA34 | Land mobile             |                                                 |                                                                                                                                                                  | Existing public cellular networks                                                                                               |
|                                                    |                                | On-board communications |                                                 | EN 300 720                                                                                                                                                       | Within 457.5125-457.5875 MHz and 467.5125-467.5875 MHz                                                                          |
|                                                    |                                | On-site paging          |                                                 | EN 300 224                                                                                                                                                       | Call-out & answer-back                                                                                                          |
|                                                    |                                | PMR/PAMR                | ECC/DEC/(19)02,<br>T/R 25-08                    | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit paired with 466-469 MHz.                                                                                |
|                                                    |                                | PPDR                    | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Mobile station transmit paired with 466-469 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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#### 459 MHz - 460 MHz

|                                                                          |                    |                |                                                 |                                                                                                                                                                  |                                                                                                                                 |
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| FIXED<br>MOBILE (5.286AA)<br><br>5.209 5.271 5.286A 5.286B 5.286C 5.286E | MOBILE<br><br>ECA7 | Land mobile    |                                                 |                                                                                                                                                                  | Existing public cellular networks                                                                                               |
|                                                                          |                    | On-site paging |                                                 | EN 300 224                                                                                                                                                       | Call-out & answer-back                                                                                                          |
|                                                                          |                    | PMR/PAMR       | ECC/DEC/(19)02,<br>T/R 25-08                    | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Mobile station transmit paired with 469-470 MHz                                                                                 |
|                                                                          |                    | PPDR           | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Mobile station transmit paired with 469-470 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 460 MHz - 470 MHz

|                                                                                                       |                                      |                         |                                                 |                                                                                                                                                                  |                                                                                                                               |
|-------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.286AA)<br>Meteorological-satellite (space-to-Earth)<br><br>5.287 5.288 5.289 5.290 | MOBILE<br><br>5.287 5.289 ECA7 ECA34 | Land mobile             |                                                 |                                                                                                                                                                  | Existing public cellular networks                                                                                             |
|                                                                                                       |                                      | On-board communications |                                                 | EN 300 720                                                                                                                                                       | Within 457.5125-457.5875 MHz and 467.5125-467.5875 MHz                                                                        |
|                                                                                                       |                                      | On-site paging          |                                                 | EN 300 224                                                                                                                                                       | Call-out & answer-back                                                                                                        |
|                                                                                                       |                                      | PMR/PAMR                | ECC/DEC/(19)02, T/R 25-08                       | EN 300 086,<br>EN 300 113,<br>EN 300 219,<br>EN 300 296,<br>EN 300 341,<br>EN 300 390,<br>EN 300 471,<br>EN 301 166,<br>EN 301 908,<br>EN 302 561,<br>EN 303 039 | Base station transmit paired with 450-460 MHz.                                                                                |
|                                                                                                       |                                      | PPDR                    | ECC/DEC/(08)05,<br>ECC/DEC/(16)02,<br>T/R 25-08 | EN 300 113,<br>EN 301 449,<br>EN 301 502,<br>EN 301 511,<br>EN 301 526,<br>EN 301 908,<br>EN 302 426,<br>EN 302 561                                              | Base station transmit paired with 450-460 MHz. PPDR on a tuning range basis in 380-470 MHz range according to ECC/DEC/(08)05. |
|                                                                                                       |                                      | Space research          |                                                 |                                                                                                                                                                  | Allocation to EESS is via RR 5.289. Data collection platform telecommand. Geographical sharing with other services            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 470 MHz - 694 MHz

|                                                                                          |                                                |                            |                                 |                           |                                                                              |
|------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------|---------------------------------|---------------------------|------------------------------------------------------------------------------|
| BROADCASTING<br>5.149 5.291A 5.294 5.295A 5.296 5.300<br>5.304 5.306 5.307A 5.307B 5.312 | BROADCASTING<br>5.149 5.291A 5.296 5.306 ECA13 | Audio PMSE                 | ERC/REC 25-10                   | EN 300 422,<br>EN 300 454 | Audio links and Talkback on a tuning range basis                             |
|                                                                                          |                                                | Broadcasting (terrestrial) |                                 | EN 302 296,<br>EN 303 340 | Geneva Agreement 2006. TV Broadcasting                                       |
|                                                                                          |                                                | Radio astronomy            |                                 |                           | Continuum observations, VLBI                                                 |
|                                                                                          |                                                | Radio microphones and ALD  | ERC/REC 25-10,<br>ERC/REC 70-03 | EN 300 422                | Within the band 470-789 MHz on a tuning range basis                          |
|                                                                                          |                                                | Wind profilers             |                                 |                           | Limited to the band 470-494 MHz.<br>Geographical sharing with other services |

## 694 MHz - 790 MHz

|                                                                                                |                                                                                         |                            |                                                                         |                           |                                                                                             |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------|
| BROADCASTING<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.312A 5.317A 5.312B)<br><br>5.300 5.312 | BROADCASTING<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.312A 5.317A)<br><br>5.312 ECA38 | Audio PMSE                 | ERC/REC 25-10                                                           | EN 300 422,<br>EN 300 454 | Radio microphones and In-ear monitors on a tuning range basis within the band 733-757.5 MHz |
|                                                                                                |                                                                                         | Broadcasting (terrestrial) |                                                                         | EN 302 296,<br>EN 303 340 | Geneva Agreement 2006 TV Broadcasting                                                       |
|                                                                                                |                                                                                         | MFCN                       | ECC/DEC/(15)01,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(15)01 | EN 301 908                | Within the band 703-788 MHz                                                                 |
|                                                                                                |                                                                                         | PPDR                       | ECC/DEC/(16)02,<br>ECC/REC/(16)03                                       | EN 301 908                | BB-PPDR options in 698-703/753-758 MHz,<br>703-733/758-788 MHz and 733-736/788-791 MHz      |
|                                                                                                |                                                                                         | Radio microphones and ALD  | ERC/REC 25-10,<br>ERC/REC 70-03                                         | EN 300 422                | Within the band 470-703 MHz and 733-757.5 MHz on a tuning range basis                       |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 790 MHz - 862 MHz

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|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------|
| BROADCASTING<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.317A 5.316B 5.312B)<br><br>5.312 5.319 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.312 5.316B 5.317A ECA13 ECA38 | -                         |                                                                         |            | Geneva Agreement 2006                                                                               |
|                                                                                                         |                                                                             | MFCN                      | ECC/DEC/(09)03,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(11)04 | EN 301 908 | Within the band 791-862 MHz. Aerial UE are permitted in the band 832-862 MHz in ECC Decision (22)07 |
|                                                                                                         |                                                                             | PPDR                      | ECC/DEC/(16)02,<br>ECC/REC/(16)03                                       |            | BB-PPDR options in 698-703/753-758 MHz, 703-733/758-788 MHz and 733-736/788-791 MHz                 |
|                                                                                                         |                                                                             | Radio microphones and ALD | ERC/REC 25-10,<br>ERC/REC 70-03                                         | EN 300 422 | Within the band 821.5-832 MHz                                                                       |

| RR Region 1                                                                                                                              | European Common Allocations                    | Application                            | CEPT Deliverables                                                       | Standard                                 | Note                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>862 MHz - 890 MHz</b><br><br>BROADCASTING (5.322)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.312B 5.317A)<br><br>5.319 5.323 | MOBILE (5.317A)<br><br>5.323 ECA13 ECA29 ECA36 | -                                      |                                                                         |                                          | This band is identified for IMT in the RRs, but within CEPT this band is not planned for the harmonised introduction of IMT |
|                                                                                                                                          |                                                | Alarms                                 | ERC/REC 70-03                                                           | EN 300 220,<br>EN 303 406                | Within the bands 868.6-868.7, 869.25-869.4 and 869.65-869.7 MHz                                                             |
|                                                                                                                                          |                                                | Audio PMSE                             | ERC/REC 25-10                                                           | EN 300 422                               | Radio microphones and In-ear monitors within the band 863-865 MHz                                                           |
|                                                                                                                                          |                                                | GSM                                    | ECC/REC/(05)08,<br>ECC/REC/(08)02,<br>ERC/DEC/(97)02                    | EN 301 502,<br>EN 301 511,<br>EN 303 609 | Within the band 880-890 MHz paired with 925-935 MHz                                                                         |
|                                                                                                                                          |                                                | GSM-R                                  | ECC/REC/(05)08                                                          | EN 301 502,<br>EN 301 511                | Within the band 876-880 MHz paired with 921-925 MHz. Railway systems                                                        |
|                                                                                                                                          |                                                | Land military systems                  |                                                                         |                                          | The bands 870-876 MHz and 915-921 MHz are used for land military systems, specifically for unmanned systems.                |
|                                                                                                                                          |                                                | MCV                                    | ECC/DEC/(08)08                                                          |                                          | Within the band 880-915 MHz                                                                                                 |
|                                                                                                                                          |                                                | MFCN                                   | ECC/DEC/(06)13,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(08)02 | EN 301 908                               | Within the band 880-915 MHz. Aerial UE are permitted in ECC Decision (22)07.                                                |
|                                                                                                                                          |                                                | Maritime military systems              |                                                                         |                                          |                                                                                                                             |
|                                                                                                                                          |                                                | Non-specific SRDs                      | ERC/REC 70-03                                                           | EN 300 220                               | Within the band 862-876 MHz                                                                                                 |
|                                                                                                                                          |                                                | RFID                                   | ERC/REC 70-03                                                           | EN 302 208                               | Within the band 865-868 MHz                                                                                                 |
|                                                                                                                                          |                                                | RMR                                    | ECC/DEC/(20)02,<br>ECC/REC/(25)02                                       | EN 301 502,<br>EN 301 511                | Within the band 874.4-880.0 MHz                                                                                             |
|                                                                                                                                          |                                                | Radio microphones and ALD              | ERC/REC 25-10,<br>ERC/REC 70-03                                         | EN 300 422,<br>EN 301 357                | Within the band 863-865 MHz                                                                                                 |
|                                                                                                                                          |                                                | Social alarms                          | ERC/REC 70-03                                                           | EN 300 220                               | Within the band 869.2-869.25 MHz                                                                                            |
|                                                                                                                                          |                                                | Telemetry/Telecommand (military)       |                                                                         |                                          | Within the band 890-915 MHz                                                                                                 |
|                                                                                                                                          |                                                | Tracking, tracing and data acquisition | ERC/REC 70-03                                                           | EN 303 204,<br>EN 303 659                | Within the bands 865-868 MHz and 870-874.4 MHz                                                                              |
|                                                                                                                                          |                                                | Wideband data transmission systems     | ERC/REC 70-03                                                           | EN 304 220                               | Within the band 863-868 MHz                                                                                                 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 890 MHz - 942 MHz

|                                                                                                                     |                                                                                      |                                        |                                                                         |                                          |                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| BROADCASTING (5.322)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.317A 5.312B)<br>Radiolocation<br><br>5.323 | Mobile (5.317A)<br>Radiolocation<br><br>5.323 ECA13 ECA14 ECA29 ECA30<br>ECA32 ECA36 | -                                      |                                                                         |                                          | The band 915-925 MHz is identified for IMT in the RRs, but within CEPT this band is not planned for the harmonised introduction of IMT |
|                                                                                                                     |                                                                                      | GSM                                    | ECC/REC/(05)08,<br>ECC/REC/(08)02,<br>ERC/DEC/(94)01,<br>ERC/DEC/(97)02 | EN 301 502,<br>EN 301 511,<br>EN 303 609 | Within the band 890-915 MHz paired with 935-960 MHz                                                                                    |
|                                                                                                                     |                                                                                      | GSM-R                                  | ECC/REC/(05)08                                                          | EN 301 502,<br>EN 301 511                | Within the bands 876-880 MHz paired with 921-925 MHz                                                                                   |
|                                                                                                                     |                                                                                      | Land military systems                  |                                                                         |                                          | The bands 870-876 MHz and 915-921 MHz are used for land military systems, specifically for unmanned systems.                           |
|                                                                                                                     |                                                                                      | MCV                                    | ECC/DEC/(08)08                                                          |                                          | Within the band 880-915 MHz and 925-960 MHz                                                                                            |
|                                                                                                                     |                                                                                      | MFCN                                   | ECC/DEC/(06)13,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(08)02 | EN 301 908                               | Within the band 880-915 MHz and 925-960 MHz. Aerial UE are permitted in the band 880-915 MHz in ECC Decision (22)07.                   |
|                                                                                                                     |                                                                                      | Maritime military systems              |                                                                         |                                          |                                                                                                                                        |
|                                                                                                                     |                                                                                      | Non-specific SRDs                      | ERC/REC 70-03                                                           | EN 300 220                               | Within the band 915-919.4 MHz                                                                                                          |
|                                                                                                                     |                                                                                      | RFID                                   | ERC/REC 70-03                                                           | EN 302 208                               | Within the band 915-919.4 MHz                                                                                                          |
|                                                                                                                     |                                                                                      | RMR                                    | ECC/DEC/(20)02,<br>ECC/REC/(25)02                                       | EN 301 502,<br>EN 301 511                | Within the band 919.4-925 MHz                                                                                                          |
|                                                                                                                     |                                                                                      | Telemetry/Telecommand (military)       |                                                                         |                                          |                                                                                                                                        |
|                                                                                                                     |                                                                                      | Tracking, tracing and data acquisition | ERC/REC 70-03                                                           | EN 303 659                               | Within the band 915-919.4 MHz                                                                                                          |
|                                                                                                                     |                                                                                      | Wideband data transmission systems     | ERC/REC 70-03                                                           | EN 304 220                               | Within the band 915.8-919.4 MHz                                                                                                        |

## 942 MHz - 960 MHz

|                                                                                                    |                                                |      |                                                                         |                                          |                                               |
|----------------------------------------------------------------------------------------------------|------------------------------------------------|------|-------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| BROADCASTING (5.322)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.312B 5.317A)<br><br>5.323 | MOBILE (5.317A)<br><br>5.323 ECA13 ECA29 ECA32 | GSM  | ECC/REC/(05)08,<br>ECC/REC/(08)02,<br>ERC/DEC/(94)01,<br>ERC/DEC/(97)02 | EN 301 502,<br>EN 301 511,<br>EN 303 609 | Base station transmit paired with 897-915 MHz |
|                                                                                                    |                                                | MCV  | ECC/DEC/(08)08                                                          |                                          |                                               |
|                                                                                                    |                                                | MFCN | ECC/DEC/(06)13,<br>ECC/REC/(08)02                                       | EN 301 908                               |                                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 960 MHz - 1164 MHz

|                                                                                        |                                                                                                                                      |                               |  |  |                                                                    |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--|--|--------------------------------------------------------------------|
| AERONAUTICAL MOBILE (5.327A)<br>AERONAUTICAL RADIONAVIGATION<br>(5.328)<br><br>5.328AA | AERONAUTICAL MOBILE (5.327A)<br>AERONAUTICAL MOBILE-SATELLITE<br>(R)<br>AERONAUTICAL RADIONAVIGATION<br>(5.328)<br><br>5.328AA ECA36 | Aeronautical                  |  |  | Including DME and SSR                                              |
|                                                                                        |                                                                                                                                      | Aeronautical military systems |  |  | Military use includes JTIDS/MIDS and TACAN within 108.7-1092.3 MHz |

## 1164 MHz - 1215 MHz

|                                                                                                |                                                                                                      |                               |                |            |                                   |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|----------------|------------|-----------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>(5.328)<br>RADIONAVIGATION-SATELLITE<br>(5.328B)<br><br>5.328A | AERONAUTICAL RADIONAVIGATION<br>(5.328)<br>RADIONAVIGATION-SATELLITE<br>(5.328B)<br><br>5.328A ECA36 | Aeronautical military systems |                |            | Military use includes JTIDS/MIDS  |
|                                                                                                |                                                                                                      | Aeronautical navigation       |                |            |                                   |
|                                                                                                |                                                                                                      | GALILEO                       |                | EN 303 413 | Within the band 1164-1214 MHz     |
|                                                                                                |                                                                                                      | GLONASS                       |                | EN 303 413 | Within the band 1190.3-1213.8 MHz |
|                                                                                                |                                                                                                      | GNSS Repeater                 | ECC/REC/(10)02 | EN 302 645 | Within the band 1164-1300 MHz     |
|                                                                                                |                                                                                                      | Satellite systems (military)  |                |            |                                   |

## 1215 MHz - 1240 MHz

|                                                                                                                                                                    |                                                                                                                                                                    |                              |                |            |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|------------|-----------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION-SATELLITE<br>(5.328B 5.329 5.329A)<br>SPACE RESEARCH (ACTIVE)<br><br>5.330 5.331 5.332 | EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION-SATELLITE<br>(5.328B 5.329 5.329A)<br>SPACE RESEARCH (ACTIVE)<br><br>5.331 5.332 ECA36 | Active sensors (satellite)   |                |            |                                   |
|                                                                                                                                                                    |                                                                                                                                                                    | GLONASS                      |                | EN 303 413 | Within the band 1237.8-1253.8 MHz |
|                                                                                                                                                                    |                                                                                                                                                                    | GNSS Repeater                | ECC/REC/(10)02 | EN 302 645 | Within the band 1164-1300 MHz     |
|                                                                                                                                                                    |                                                                                                                                                                    | GPS                          |                | EN 303 413 | Within the band 1215.6-1239.6 MHz |
|                                                                                                                                                                    |                                                                                                                                                                    | Radiolocation (civil)        |                |            | Radar and Navigation systems      |
|                                                                                                                                                                    |                                                                                                                                                                    | Radiolocation (military)     |                |            |                                   |
|                                                                                                                                                                    |                                                                                                                                                                    | Satellite systems (military) |                |            |                                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 1240 MHz - 1300 MHz

|                                                                                                                                                                                                   |                                                                                                                                                                                                           |                              |                |            |                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|------------|-----------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION-SATELLITE (5.329 5.328B 5.329A)<br>SPACE RESEARCH (ACTIVE)<br>Amateur<br><br>5.282 5.330 5.331 5.332 5.332A 5.335 5.335A | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION-SATELLITE (5.228B 5.329 5.329A)<br>SPACE RESEARCH (ACTIVE)<br>Amateur<br>Amateur-satellite<br><br>5.282 5.331 5.332 5.335A ECA36 | Active sensors (satellite)   |                |            |                                   |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | Amateur                      | ECC/DEC/(25)01 | EN 301 783 | Within the band 1258-1300 MHz     |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | Amateur-satellite            | ECC/DEC/(25)01 |            | Within the band 1258-1300 MHz     |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | GALILEO                      | ECC/DEC/(25)01 | EN 303 413 | Within the band 1258-1300 MHz     |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | GLONASS                      |                | EN 303 413 | Within the band 1237.8-1253.8 MHz |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | GNSS Repeater                | ECC/REC/(10)02 | EN 302 645 | Within the band 1164-1300 MHz     |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | Radiolocation (civil)        |                |            | Radar and Navigation systems      |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | Radiolocation (military)     |                |            |                                   |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | Satellite systems (military) |                |            |                                   |
|                                                                                                                                                                                                   |                                                                                                                                                                                                           | Wind profilers               |                |            | Within the band 1270-1295 MHz     |

## 1300 MHz - 1350 MHz

|                                                                                                                         |                                                                                                                               |                              |  |  |                                                                              |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------|--|--|------------------------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION (5.337)<br>RADIOLOCATION<br>RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE)<br><br>5.149 5.337A | AERONAUTICAL RADIONAVIGATION (5.337)<br>RADIOLOCATION<br>RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE)<br><br>5.149 5.337A ECA36 | Radio astronomy              |  |  | Continuum and spectral line observations (e.g. neutral hydrogen line). VLBI. |
|                                                                                                                         |                                                                                                                               | Radiolocation (civil)        |  |  | Radar and Navigation systems                                                 |
|                                                                                                                         |                                                                                                                               | Radiolocation (military)     |  |  |                                                                              |
|                                                                                                                         |                                                                                                                               | Satellite navigation systems |  |  |                                                                              |
|                                                                                                                         |                                                                                                                               | Satellite systems (military) |  |  |                                                                              |

## 1350 MHz - 1400 MHz

|                                                                  |                                                                  |                               |                              |            |                                                                             |
|------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------|------------------------------|------------|-----------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>RADIOLOCATION<br><br>5.149 5.338 5.338A 5.339 | FIXED<br>MOBILE<br>RADIOLOCATION<br><br>5.149 5.338A 5.339 ECA36 | Aeronautical military systems |                              |            |                                                                             |
|                                                                  |                                                                  | Fixed                         | T/R 13-01                    | EN 302 217 | Low capacity fixed links                                                    |
|                                                                  |                                                                  | Land military systems         |                              |            |                                                                             |
|                                                                  |                                                                  | Maritime military systems     |                              |            |                                                                             |
|                                                                  |                                                                  | Radio astronomy               |                              |            | Continuum and spectral line observations (e.g. neutral hydrogen line). VLBI |
|                                                                  |                                                                  | Radio microphones and ALD     | ERC/REC 25-10, ERC/REC 70-03 | EN 300 422 | Radio microphones and In-ear monitors                                       |
|                                                                  |                                                                  | Radiolocation (military)      |                              |            |                                                                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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### 1400 MHz - 1427 MHz

|                                                                                                     |                                                                                                     |                             |                |  |                                                                                     |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------|----------------|--|-------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.341 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.341 | Passive sensors (satellite) | ECC/DEC/(11)01 |  | Measurement of soil moisture, salinity, ocean surface temperature, vegetation index |
|                                                                                                     |                                                                                                     | Radio astronomy             |                |  | Continuum and spectral line observations (e.g. neutral hydrogen line). VLBI.        |

### 1427 MHz - 1429 MHz

|                                                                                                            |                                                                                                         |                                  |                                                      |            |                          |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------|------------|--------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.341A)<br>SPACE OPERATION (EARTH-TO-SPACE)<br>5.338A 5.341 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE OPERATION (EARTH-TO-SPACE)<br>5.338A 5.341 ECA36 | Fixed                            | T/R 13-01                                            | EN 302 217 | Low capacity fixed links |
|                                                                                                            |                                                                                                         | Land military systems            |                                                      |            |                          |
|                                                                                                            |                                                                                                         | MFCN                             | ECC/DEC/(17)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(15)01 | EN 301 908 | Supplemental Downlink    |
|                                                                                                            |                                                                                                         | Maritime military systems        |                                                      |            |                          |
|                                                                                                            |                                                                                                         | Telemetry/Telecommand (military) |                                                      |            |                          |

### 1429 MHz - 1452 MHz

|                                                                              |                                                                     |                                  |                                                      |            |                          |
|------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------|------------------------------------------------------|------------|--------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.341A)<br>5.338A 5.341 5.342 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>5.338A 5.341 ECA36 | Fixed                            | T/R 13-01                                            | EN 302 217 | Low capacity fixed links |
|                                                                              |                                                                     | Land military systems            |                                                      |            |                          |
|                                                                              |                                                                     | MFCN                             | ECC/DEC/(17)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(15)01 | EN 301 908 | Supplemental Downlink    |
|                                                                              |                                                                     | Maritime military systems        |                                                      |            |                          |
|                                                                              |                                                                     | Telemetry/Telecommand (military) |                                                      |            |                          |

### 1452 MHz - 1492 MHz

|                                                                                                                               |                                                                                    |                                  |                                                      |                           |                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------|
| BROADCASTING<br>BROADCASTING-SATELLITE (5.208B)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.346)<br>5.341 5.342 5.345 | BROADCASTING<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Fixed<br>5.341 5.342 5.345 | MFCN                             | ECC/DEC/(13)03,<br>ECC/DEC/(22)01,<br>ECC/REC/(15)01 | EN 301 908                | Supplemental Downlink                                                                                  |
|                                                                                                                               |                                                                                    | T-DAB                            |                                                      | EN 302 077,<br>EN 303 345 | Within the band 1452.0-1479.5 MHz. Maastricht 2002 Special Arrangement, as revised in Constanta, 2007. |
|                                                                                                                               |                                                                                    | Telemetry/Telecommand (military) |                                                      |                           |                                                                                                        |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 1492 MHz - 1518 MHz

|                                                                           |                                                                  |                                  |                                                      |            |                          |
|---------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------|------------------------------------------------------|------------|--------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.341A)<br><br>5.341 5.342 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.341 ECA36 | Fixed                            | T/R 13-01                                            | EN 302 217 | Low capacity fixed links |
|                                                                           |                                                                  | Land military systems            |                                                      |            |                          |
|                                                                           |                                                                  | MFCN                             | ECC/DEC/(17)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(15)01 | EN 301 908 | Supplemental Downlink    |
|                                                                           |                                                                  | Maritime military systems        |                                                      |            |                          |
|                                                                           |                                                                  | Radio microphones and ALD        | ERC/REC 70-03                                        | EN 300 422 | On a tuning range basis  |
|                                                                           |                                                                  | Telemetry/Telecommand (military) |                                                      |            |                          |

## 1518 MHz - 1525 MHz

|                                                                                                                      |                                                                                                                      |                                  |                                   |                                          |                                                               |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------|------------------------------------------|---------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>MOBILE-SATELLITE (5.348 5.348A<br>5.348B 5.351A)<br><br>5.341 5.342 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>MOBILE-SATELLITE (5.348 5.348A<br>5.348B 5.351A)<br><br>5.341 ECA36 | Audio PMSE                       | ERC/REC 25-10                     | EN 300 422                               | Radio microphones and In-ear monitors on a tuning range basis |
|                                                                                                                      |                                                                                                                      | Fixed                            |                                   | EN 302 217                               | Unidirectional fixed links                                    |
|                                                                                                                      |                                                                                                                      | IMT-2000 satellite component     |                                   |                                          |                                                               |
|                                                                                                                      |                                                                                                                      | Land military systems            |                                   |                                          |                                                               |
|                                                                                                                      |                                                                                                                      | MSS Earth stations               | ECC/DEC/(04)09,<br>ECC/DEC/(12)01 | EN 301 444,<br>EN 301 473,<br>EN 301 681 |                                                               |
|                                                                                                                      |                                                                                                                      | Maritime military systems        |                                   |                                          |                                                               |
|                                                                                                                      |                                                                                                                      | Radio microphones and ALD        | ERC/REC 70-03                     | EN 300 422                               | On a tuning range basis                                       |
|                                                                                                                      |                                                                                                                      | Telemetry/Telecommand (military) |                                   |                                          |                                                               |

## 1525 MHz - 1530 MHz

|                                                                                                                                                                                                       |                                                                                                        |                              |                |                                                         |                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------|----------------|---------------------------------------------------------|----------------------------|
| FIXED<br>MOBILE-SATELLITE (5.208B 5.351A)<br>SPACE OPERATION (SPACE-TO-EARTH)<br>Earth exploration-satellite<br>Mobile except aeronautical mobile (5.349)<br><br>5.341 5.342 5.350 5.351 5.352A 5.354 | FIXED<br>MOBILE-SATELLITE (5.208B 5.351A)<br>SPACE OPERATION (SPACE-TO-EARTH)<br><br>5.341 5.351 5.354 | Fixed                        |                | EN 302 217                                              | Unidirectional fixed links |
|                                                                                                                                                                                                       |                                                                                                        | IMT-2000 satellite component |                |                                                         |                            |
|                                                                                                                                                                                                       |                                                                                                        | MSS Earth stations           | ECC/DEC/(12)01 | EN 301 426,<br>EN 301 444,<br>EN 301 473,<br>EN 301 681 |                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 1530 MHz - 1535 MHz

|                                                                                                                                                                                     |                                                                                                                                                                               |                              |                |                                                         |                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| MOBILE-SATELLITE (5.208B 5.353A 5.351A)<br>SPACE OPERATION (SPACE-TO-EARTH)<br>Earth exploration-satellite<br>Fixed<br>Mobile except aeronautical mobile<br>5.341 5.342 5.351 5.354 | MOBILE-SATELLITE (5.208B 5.351A 5.353A)<br>SPACE OPERATION (SPACE-TO-EARTH)<br>Earth exploration-satellite<br>Fixed<br>Mobile except aeronautical mobile<br>5.341 5.351 5.354 | IMT-2000 satellite component |                |                                                         |                                                                                                  |
|                                                                                                                                                                                     |                                                                                                                                                                               | MSS Earth stations           | ECC/DEC/(12)01 | EN 301 426,<br>EN 301 444,<br>EN 301 473,<br>EN 301 681 | Priority for GMDSS Distress, urgency and safety and for AMS(R)S categories 1 to 6 communications |

### 1535 MHz - 1559 MHz

|                                                                                             |                                                                                       |                              |                |                                                         |                                                                                                                                |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------|----------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| MOBILE-SATELLITE (5.208B 5.351A)<br>5.341 5.351 5.353A 5.354 5.355 5.356 5.357 5.357A 5.359 | MOBILE-SATELLITE (5.208B 5.351A)<br>5.341 5.351 5.353A 5.354 5.356 5.357 5.357A 5.359 | IMT-2000 satellite component |                |                                                         |                                                                                                                                |
|                                                                                             |                                                                                       | MSS Earth stations           | ECC/DEC/(12)01 | EN 301 426,<br>EN 301 444,<br>EN 301 473,<br>EN 301 681 | Priority for GMDSS Distress, urgency and safety and for AMS(R)S categories 1 to 6 communications within the band 1544-1545 MHz |

### 1559 MHz - 1610 MHz

|                                                                                                                          |                                                                                                                          |                              |                                   |                           |                                     |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------|---------------------------|-------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE (5.208B)<br>RADIONAVIGATION-SATELLITE (5.328B 5.329A)<br>5.341 | AERONAUTICAL RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE (5.208B)<br>RADIONAVIGATION-SATELLITE (5.328B 5.329A)<br>5.341 | GALILEO                      |                                   | EN 303 413                | Within the band 1559.42-1591.42 MHz |
|                                                                                                                          |                                                                                                                          | GLONASS                      |                                   | EN 303 413                | Within the band 1592.9-1610.5 MHz   |
|                                                                                                                          |                                                                                                                          | GNSS Pseudolites             | ECC/REC/(11)08                    |                           |                                     |
|                                                                                                                          |                                                                                                                          | GNSS Repeater                | ECC/REC/(10)02                    | EN 302 645                |                                     |
|                                                                                                                          |                                                                                                                          | GPS                          |                                   | EN 303 413                | Within the band 1563.42-1587.42 MHz |
|                                                                                                                          |                                                                                                                          | IMT-2000 satellite component |                                   |                           |                                     |
|                                                                                                                          |                                                                                                                          | MSS Earth stations           | ECC/DEC/(09)02,<br>ECC/DEC/(12)01 | EN 301 441,<br>EN 301 473 |                                     |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 1610 MHz - 1610.6 MHz

|                                                                                                                                 |                                                                                                                     |                              |                                   |                           |                                   |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------|---------------------------|-----------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>MOBILE-SATELLITE (5.351A)<br><br>5.341 5.355 5.359 5.364 5.366 5.367 5.368<br>5.369 5.371 5.372 | AERONAUTICAL RADIONAVIGATION<br>MOBILE-SATELLITE (5.351A)<br><br>5.341 5.359 5.364 5.366 5.367 5.368 5.371<br>5.372 | GLONASS                      |                                   |                           | Within the band 1592.9-1610.5 MHz |
|                                                                                                                                 |                                                                                                                     | IMT-2000 satellite component |                                   |                           |                                   |
|                                                                                                                                 |                                                                                                                     | MSS Earth stations           | ECC/DEC/(09)02,<br>ECC/DEC/(12)01 | EN 301 441,<br>EN 301 473 |                                   |

## 1610.6 MHz - 1613.8 MHz

|                                                                                                                                                          |                                                                                                                                              |                              |                                   |                           |                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------|---------------------------|----------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>MOBILE-SATELLITE (5.351A)<br>RADIO ASTRONOMY<br><br>5.149 5.341 5.355 5.359 5.364 5.366 5.367<br>5.368 5.369 5.371 5.372 | AERONAUTICAL RADIONAVIGATION<br>MOBILE-SATELLITE (5.351A)<br>RADIO ASTRONOMY<br><br>5.149 5.341 5.359 5.364 5.366 5.367 5.368<br>5.371 5.372 | IMT-2000 satellite component |                                   |                           |                                                          |
|                                                                                                                                                          |                                                                                                                                              | MSS Earth stations           | ECC/DEC/(09)02,<br>ECC/DEC/(12)01 | EN 301 441,<br>EN 301 473 |                                                          |
|                                                                                                                                                          |                                                                                                                                              | Radio astronomy              |                                   |                           | Spectral line observations (e.g. hydroxyl line).<br>VLBI |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 1613.8 MHz - 1621.35 MHz

|                                                                                                                                                                           |                                                                                                                                                        |                              |                                                      |                                          |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------|------------------------------------------|--|
| AERONAUTICAL RADIONAVIGATION<br>MOBILE-SATELLITE (5.351A)<br>Mobile-Satellite (5.208B)<br><br>5.341 5.355 5.359 5.364 5.365 5.366 5.367<br>5.368 5.369 5.371 5.372 5.372A | AERONAUTICAL RADIONAVIGATION<br>MOBILE-SATELLITE (5.351A)<br>Mobile-Satellite (5.208B)<br><br>5.341 5.359 5.364 5.365 5.366 5.367 5.368<br>5.371 5.372 | IMT-2000 satellite component |                                                      |                                          |  |
|                                                                                                                                                                           |                                                                                                                                                        | MSS Earth stations           | ECC/DEC/(09)02,<br>ECC/DEC/(09)04,<br>ECC/DEC/(12)01 | EN 301 426,<br>EN 301 441,<br>EN 301 473 |  |

### 1621.35 MHz - 1626.5 MHz

|                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                      |                              |                                                      |                                          |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------|------------------------------------------|--|
| AERONAUTICAL RADIONAVIGATION<br>MARITIME MOBILE-SATELLITE (5.373A<br>5.373)<br>MOBILE-SATELLITE (5.351A)<br>Mobile-Satellite (5.208B)<br>Mobile-satellite except maritime mobile<br>satellite (space-to-Earth)<br><br>5.341 5.355 5.359 5.364 5.365 5.366 5.367<br>5.368 5.369 5.371 5.372 | AERONAUTICAL RADIONAVIGATION<br>MARITIME MOBILE-SATELLITE (5.373<br>5.373A)<br>MOBILE-SATELLITE (5.351A)<br>Mobile-Satellite (5.208B)<br>Mobile-satellite except maritime mobile<br>satellite (space-to-Earth)<br><br>5.341 5.359 5.364 5.365 5.365 5.366 5.367<br>5.368 5.371 5.372 | IMT-2000 satellite component |                                                      |                                          |  |
|                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                      | MSS Earth stations           | ECC/DEC/(09)02,<br>ECC/DEC/(09)04,<br>ECC/DEC/(12)01 | EN 301 426,<br>EN 301 441,<br>EN 301 473 |  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 1626.5 MHz - 1660 MHz

|                                                                                               |                                                                                        |                              |                |                                          |                                                                                                                                    |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------|----------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| MOBILE-SATELLITE (5.351A)<br>5.341 5.351 5.353A 5.354 5.355 5.357A<br>5.359 5.374 5.375 5.376 | MOBILE-SATELLITE (5.351A)<br>5.100 5.341 5.351 5.353A 5.354 5.359<br>5.374 5.375 5.376 | ALS                          | ERC/REC 70-03  | EN 300 422                               | ALS is Assistive Listening Systems. Within 1656.5-1660.5 MHz                                                                       |
|                                                                                               |                                                                                        | IMT-2000 satellite component |                |                                          |                                                                                                                                    |
|                                                                                               |                                                                                        | MSS Earth stations           | ECC/DEC/(12)01 | EN 301 426,<br>EN 301 473,<br>EN 301 681 | Priority for GMDSS Distress, urgency and safety and for AMS(R)S categories 1 to 6 communications within the band 1645.5-1646.5 MHz |

## 1660 MHz - 1660.5 MHz

|                                                                                |                                                                                |                              |                |                                                         |                                                                     |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------|----------------|---------------------------------------------------------|---------------------------------------------------------------------|
| MOBILE-SATELLITE (5.351A)<br>RADIO ASTRONOMY<br>5.149 5.341 5.351 5.354 5.376A | MOBILE-SATELLITE (5.351A)<br>RADIO ASTRONOMY<br>5.149 5.341 5.351 5.354 5.376A | ALS                          | ERC/REC 70-03  | EN 300 422                                              | ALS is Assistive Listening Systems. Within 1656.5-1660.5 MHz        |
|                                                                                |                                                                                | IMT-2000 satellite component |                |                                                         |                                                                     |
|                                                                                |                                                                                | MSS Earth stations           | ECC/DEC/(12)01 | EN 301 426,<br>EN 301 444,<br>EN 301 473,<br>EN 301 681 |                                                                     |
|                                                                                |                                                                                | Radio astronomy              |                |                                                         | Continuum and spectral line observations (e.g. hydroxyl line), VLBI |

## 1660.5 MHz - 1668 MHz

|                                                                                                                       |                                                                                                                 |                 |  |  |                                                                     |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------|--|--|---------------------------------------------------------------------|
| RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Fixed<br>Mobile except aeronautical mobile<br>5.149 5.341 5.379 5.379A | RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Fixed<br>Mobile except aeronautical mobile<br>5.149 5.341 5.379A | Radio astronomy |  |  | Continuum and spectral line observations (e.g. hydroxyl line), VLBI |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------|--|--|---------------------------------------------------------------------|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 1668 MHz - 1668.4 MHz

|                                                                                                                                                                  |                                                                                                                                                            |                              |  |  |                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--|--|---------------------------------------------------------------------|
| MOBILE-SATELLITE (5.351A 5.379B 5.379C)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Fixed<br>Mobile except aeronautical mobile<br>5.149 5.341 5.379 5.379A | MOBILE-SATELLITE (5.351A 5.379B 5.379C)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Fixed<br>Mobile except aeronautical mobile<br>5.149 5.341 5.379A | IMT-2000 satellite component |  |  |                                                                     |
|                                                                                                                                                                  |                                                                                                                                                            | Radio astronomy              |  |  | Continuum and spectral line observations (e.g. hydroxyl line), VLBI |

### 1668.4 MHz - 1670 MHz

|                                                                                                                                                                 |                                                                                                                                                                 |                              |  |  |                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--|--|---------------------------------------------------------------------|
| FIXED<br>METEOROLOGICAL AIDS<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>MOBILE-SATELLITE (5.351A 5.379B 5.379C)<br>RADIO ASTRONOMY<br>5.149 5.341 5.379D 5.379E | FIXED<br>METEOROLOGICAL AIDS<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>MOBILE-SATELLITE (5.351A 5.379B 5.379C)<br>RADIO ASTRONOMY<br>5.149 5.341 5.379D 5.379E | IMT-2000 satellite component |  |  |                                                                     |
|                                                                                                                                                                 |                                                                                                                                                                 | Meteorology                  |  |  |                                                                     |
|                                                                                                                                                                 |                                                                                                                                                                 | Radio astronomy              |  |  | Continuum and spectral line observations (e.g. hydroxyl line), VLBI |

### 1670 MHz - 1675 MHz

|                                                                                                                                                       |                                                                                                                                                       |                              |                                   |                                          |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------|------------------------------------------|--|
| FIXED<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (5.351A 5.379B)<br>5.341 5.379D 5.379E 5.380A | METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (5.351A 5.379B)<br>Fixed<br>5.341 5.379D 5.379E 5.380A | IMT-2000 satellite component |                                   |                                          |  |
|                                                                                                                                                       |                                                                                                                                                       | MSS Earth stations           | ECC/DEC/(04)09,<br>ECC/DEC/(12)01 | EN 301 444,<br>EN 301 473,<br>EN 301 681 |  |
|                                                                                                                                                       |                                                                                                                                                       | Meteorology                  |                                   |                                          |  |
|                                                                                                                                                       |                                                                                                                                                       | Weather satellites           |                                   |                                          |  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 1675 MHz - 1690 MHz

|                                                                                                                               |                                                                                                                                     |                                |  |            |                            |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--|------------|----------------------------|
| FIXED<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>5.341 | FIXED<br>METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>5.341 ECA36 | Land military systems          |  |            |                            |
|                                                                                                                               |                                                                                                                                     | Maritime military systems      |  |            |                            |
|                                                                                                                               |                                                                                                                                     | Meteorological aids (military) |  |            |                            |
|                                                                                                                               |                                                                                                                                     | Sondes                         |  | EN 302 454 | Meteorological radiosondes |
|                                                                                                                               |                                                                                                                                     | Weather satellites             |  |            | Data collection platform   |

## 1690 MHz - 1700 MHz

|                                                                                                                                        |                                                                                                                                        |                                |  |  |                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--|--|--------------------------------------------------------------|
| METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile<br>5.289 5.341 5.382 | METEOROLOGICAL AIDS<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>Fixed<br>Mobile except aeronautical mobile<br>5.289 5.341 ECA36 | Land military systems          |  |  |                                                              |
|                                                                                                                                        |                                                                                                                                        | Maritime military systems      |  |  |                                                              |
|                                                                                                                                        |                                                                                                                                        | Meteorological aids (military) |  |  |                                                              |
|                                                                                                                                        |                                                                                                                                        | Weather satellites             |  |  | Data collection platform. Allocation to EESS is via RR 5.289 |

## 1700 MHz - 1710 MHz

|                                                                                                              |                                                                                                                                             |                                |  |  |                                                              |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--|--|--------------------------------------------------------------|
| FIXED<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>5.289 5.341 | FIXED<br>FIXED<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>MOBILE (5.384A)<br>Mobile except aeronautical mobile<br>5.289 5.341 ECA36 | Land military systems          |  |  |                                                              |
|                                                                                                              |                                                                                                                                             | Maritime military systems      |  |  |                                                              |
|                                                                                                              |                                                                                                                                             | Meteorological aids (military) |  |  |                                                              |
|                                                                                                              |                                                                                                                                             | Weather satellites             |  |  | Data collection platform. Allocation to EESS is via RR 5.289 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 1710 MHz - 1930 MHz

|                                                                            |                                                                                  |                           |                                                                                                              |                                          |                                                                  |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------|
| FIXED<br>MOBILE (5.384A 5.388A)<br><br>5.149 5.341 5.385 5.386 5.387 5.388 | FIXED<br>MOBILE (5.384A 5.388 5.388A ECA38)<br><br>5.149 5.341 5.385 ECA29 ECA36 | DECT                      | ERC/DEC/(94)03,<br>ERC/DEC/(98)22,<br>ERC/REC 70-03                                                          | EN 301 406                               | Within the band 1880-1900 MHz                                    |
|                                                                            |                                                                                  | GSM                       | ECC/REC/(05)08,<br>ECC/REC/(08)02,<br>ERC/DEC/(95)03                                                         | EN 301 502,<br>EN 301 511,<br>EN 303 609 |                                                                  |
|                                                                            |                                                                                  | Land military systems     |                                                                                                              |                                          |                                                                  |
|                                                                            |                                                                                  | Land mobile               |                                                                                                              |                                          | Mobile applications                                              |
|                                                                            |                                                                                  | MCA                       | ECC/DEC/(06)07                                                                                               | EN 302 480                               | Within the band 1920-1980 MHz                                    |
|                                                                            |                                                                                  | MCV                       | ECC/DEC/(08)08                                                                                               |                                          | Within the band 1920-1980 MHz                                    |
|                                                                            |                                                                                  | MFCN                      | ECC/DEC/(06)01,<br>ECC/DEC/(06)13,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(08)02,<br>ERC/REC 01-01 | EN 301 908                               | 1920-1980 MHz, Aerial UE are permitted – See ECC Decision (22)07 |
|                                                                            |                                                                                  | RMR                       | ECC/DEC/(20)02,<br>ECC/REC/(23)01                                                                            | EN 301 502,<br>EN 301 511                | Within the band 1900-1910 MHz                                    |
|                                                                            |                                                                                  | Radio microphones and ALD | ERC/REC 25-10,<br>ERC/REC 70-03                                                                              | EN 300 422                               | Within the band 1785-1805 MHz                                    |
|                                                                            |                                                                                  | UAS                       | ECC/REC/(24)02                                                                                               |                                          | Within the bands 1880-1900 MHz and 1910-1920 MHz                 |

## 1930 MHz - 1970 MHz

|                                       |                                                   |      |                                                                        |            |                                                                  |
|---------------------------------------|---------------------------------------------------|------|------------------------------------------------------------------------|------------|------------------------------------------------------------------|
| FIXED<br>MOBILE (5.388A)<br><br>5.388 | MOBILE (5.388 5.388A ECA38)<br>Fixed<br><br>ECA29 | -    |                                                                        |            | This band can also be used by fixed service on a national basis  |
|                                       |                                                   | MCA  | ECC/DEC/(06)07                                                         |            |                                                                  |
|                                       |                                                   | MCV  | ECC/DEC/(08)08                                                         |            |                                                                  |
|                                       |                                                   | MFCN | ECC/DEC/(06)01,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ERC/REC 01-01 | EN 301 908 | 1920-1980 MHz, Aerial UE are permitted – See ECC Decision (22)07 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 1970 MHz - 1980 MHz

|                                       |                                                   |      |                                                                        |            |                                                                  |
|---------------------------------------|---------------------------------------------------|------|------------------------------------------------------------------------|------------|------------------------------------------------------------------|
| FIXED<br>MOBILE (5.388A)<br><br>5.388 | MOBILE (5.388A ECA38)<br>Fixed<br><br>5.388 ECA29 | -    |                                                                        |            | This band can also be used by fixed service on a national basis  |
|                                       |                                                   | MCA  | ECC/DEC/(06)07                                                         |            |                                                                  |
|                                       |                                                   | MCV  | ECC/DEC/(08)08                                                         |            |                                                                  |
|                                       |                                                   | MFCN | ECC/DEC/(06)01,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ERC/REC 01-01 | EN 301 908 | 1920-1980 MHz, Aerial UE are permitted – See ECC Decision (22)07 |

## 1980 MHz - 2010 MHz

|                                                                                |                                                         |                    |                                                      |                                          |                                                                                                     |
|--------------------------------------------------------------------------------|---------------------------------------------------------|--------------------|------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>MOBILE-SATELLITE (5.351A)<br><br>5.388 5.389A 5.389B 5.389F | MOBILE<br>MOBILE-SATELLITE (5.351A)<br><br>5.388 5.389A | -                  |                                                      |                                          | This band can also be used by fixed service on a national basis                                     |
|                                                                                |                                                         | MSS Earth stations | ECC/DEC/(06)09,<br>ECC/DEC/(06)10,<br>ECC/DEC/(12)01 | EN 301 442,<br>EN 301 473,<br>EN 302 574 | The mobile satellite systems using this band may incorporate a complementary Ground Component (CGC) |

## 2010 MHz - 2025 MHz

|                                             |                 |            |               |            |                                                                 |
|---------------------------------------------|-----------------|------------|---------------|------------|-----------------------------------------------------------------|
| FIXED<br>MOBILE (5.388A 5.388)<br><br>5.388 | MOBILE<br>Fixed | -          |               |            | This band can also be used by fixed service on a national basis |
|                                             |                 | Video PMSE | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 2025 MHz - 2110 MHz

|                                                                                                                                                                                                                |                                                                                                                                                                                                                             |                                  |                |            |                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------|------------|------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) (SPACE-TO-SPACE)<br>FIXED<br>MOBILE (5.391)<br>SPACE OPERATION (EARTH-TO-SPACE) (SPACE-TO-SPACE)<br>SPACE RESEARCH (EARTH-TO-SPACE) (SPACE-TO-SPACE)<br><br>5.392 | EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE) (SPACE-TO-SPACE)<br>FIXED<br>MOBILE (5.391)<br>SPACE OPERATION (EARTH-TO-SPACE) (SPACE-TO-SPACE)<br>SPACE RESEARCH (EARTH-TO-SPACE) (SPACE-TO-SPACE)<br><br>5.392 ECA16A ECA36 | Aeronautical military systems    |                |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Fixed                            | T/R 13-01      | EN 302 217 |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Land military systems            |                |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | MSS Earth stations               | ECC/REC/(24)03 | EN 301 473 |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Maritime military systems        |                |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Space research                   |                |            | Satellite payload and platform telecommand                 |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Telemetry/Telecommand (military) |                |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Video PMSE                       | ERC/REC 25-10  | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links |

## 2110 MHz - 2120 MHz

|                                                                                       |                                                                                             |      |                                                     |            |                                                                                                                                             |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------|-----------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.388A)<br>SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE)<br><br>5.388 | MOBILE (5.388A)<br>SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE)<br>Fixed<br><br>5.388 ECA29 | -    |                                                     |            | Satellite payload and platform telecommand for space research (deep space). This band can also be used by fixed service on a national basis |
|                                                                                       |                                                                                             | MCA  | ECC/DEC/(06)07                                      |            |                                                                                                                                             |
|                                                                                       |                                                                                             | MCV  | ECC/DEC/(08)08                                      |            |                                                                                                                                             |
|                                                                                       |                                                                                             | MFCN | ECC/DEC/(06)01,<br>ECC/DEC/(22)01,<br>ERC/REC 01-01 | EN 301 908 |                                                                                                                                             |

## 2120 MHz - 2160 MHz

|                                       |                                             |      |                                                     |            |                                                                 |
|---------------------------------------|---------------------------------------------|------|-----------------------------------------------------|------------|-----------------------------------------------------------------|
| FIXED<br>MOBILE (5.388A)<br><br>5.388 | MOBILE (5.388A)<br>Fixed<br><br>5.388 ECA29 | -    |                                                     |            | This band can also be used by fixed service on a national basis |
|                                       |                                             | MCA  | ECC/DEC/(06)07                                      |            |                                                                 |
|                                       |                                             | MCV  | ECC/DEC/(08)08                                      |            |                                                                 |
|                                       |                                             | MFCN | ECC/DEC/(06)01,<br>ECC/DEC/(22)01,<br>ERC/REC 01-01 | EN 301 908 |                                                                 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 2160 MHz - 2170 MHz

|                                       |                                             |      |                                                     |            |                                                                 |
|---------------------------------------|---------------------------------------------|------|-----------------------------------------------------|------------|-----------------------------------------------------------------|
| FIXED<br>MOBILE (5.388A)<br><br>5.388 | MOBILE (5.388A)<br>Fixed<br><br>5.388 ECA29 | -    |                                                     |            | This band can also be used by fixed service on a national basis |
|                                       |                                             | MCA  | ECC/DEC/(06)07                                      |            |                                                                 |
|                                       |                                             | MCV  | ECC/DEC/(08)08                                      |            |                                                                 |
|                                       |                                             | MFCN | ECC/DEC/(06)01,<br>ECC/DEC/(22)01,<br>ERC/REC 01-01 | EN 301 908 |                                                                 |

## 2170 MHz - 2200 MHz

|                                                                         |                                                         |                    |                                                                         |                                          |                                                                                                     |
|-------------------------------------------------------------------------|---------------------------------------------------------|--------------------|-------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>MOBILE-SATELLITE (5.351A)<br><br>5.388 5.389A 5.389F | MOBILE<br>MOBILE-SATELLITE (5.351A)<br><br>5.388 5.389A | -                  |                                                                         |                                          | This band can also be used by fixed service on a national basis                                     |
|                                                                         |                                                         | MSS Earth stations | ECC/DEC/(06)09,<br>ECC/DEC/(06)10,<br>ECC/DEC/(12)01,<br>ECC/REC/(10)01 | EN 301 442,<br>EN 301 473,<br>EN 302 574 | The mobile satellite systems using this band may incorporate a Complementary Ground Component (CGC) |

## 2200 MHz - 2290 MHz

|                                                                                                                                                                                                                |                                                                                                                                                                                                                             |                                  |                                  |            |                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|------------|------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) (SPACE-TO-SPACE)<br>FIXED<br>MOBILE (5.391)<br>SPACE OPERATION (SPACE-TO-EARTH) (SPACE-TO-SPACE)<br>SPACE RESEARCH (SPACE-TO-EARTH) (SPACE-TO-SPACE)<br><br>5.392 | EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH) (SPACE-TO-SPACE)<br>FIXED<br>MOBILE (5.391)<br>SPACE OPERATION (SPACE-TO-EARTH) (SPACE-TO-SPACE)<br>SPACE RESEARCH (SPACE-TO-EARTH) (SPACE-TO-SPACE)<br><br>5.392 ECA16A ECA36 | Aeronautical military systems    |                                  |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Fixed                            | T/R 13-01                        | EN 302 217 |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Land military systems            |                                  |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | MSS Earth stations               | ECC/REC/(24)03                   |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Maritime military systems        |                                  |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Radio astronomy                  |                                  |            | Continuum observations, VLBI (used by SRS)                 |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Space research                   | ECC/REC/(10)01                   |            | EESS Satellite payload and platform telemetry              |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Telemetry/Telecommand (military) |                                  |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | UWB applications                 | ECC/DEC/(07)01,<br>ERC/REC 70-03 |            |                                                            |
|                                                                                                                                                                                                                |                                                                                                                                                                                                                             | Video PMSE                       | ERC/REC 25-10                    | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 2290 MHz - 2300 MHz

|                                                                                                  |                                                                                                  |                  |                                  |            |                                                                                                                      |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------|----------------------------------|------------|----------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE RESEARCH (DEEP SPACE)<br>(SPACE-TO-EARTH) | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE RESEARCH (DEEP SPACE)<br>(SPACE-TO-EARTH) | Land mobile      |                                  |            | Mobile applications                                                                                                  |
|                                                                                                  |                                                                                                  | Space research   |                                  |            | Satellite payload and platform telemetry for space research (deep space). Continuum observations, VLBI (used by SRS) |
|                                                                                                  |                                                                                                  | UWB applications | ECC/DEC/(07)01,<br>ERC/REC 70-03 |            |                                                                                                                      |
|                                                                                                  |                                                                                                  | Video PMSE       | ERC/REC 25-10                    | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links                                                           |

## 2300 MHz - 2400 MHz

|                                                                   |                                                                   |                                  |                                                                         |            |                                                                           |
|-------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------|------------|---------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.384A)<br>Amateur<br>Radiolocation<br><br>5.395 | FIXED<br>MOBILE (5.384A)<br>Amateur<br>Radiolocation<br><br>ECA36 | Aeronautical military systems    |                                                                         |            |                                                                           |
|                                                                   |                                                                   | Aeronautical telemetry           | ERC/REC 62-02                                                           |            | Parts of the band are used for aeronautical telemetry on a national basis |
|                                                                   |                                                                   | Amateur                          |                                                                         | EN 301 783 | Within the band 2300-2450 MHz                                             |
|                                                                   |                                                                   | Land military systems            |                                                                         |            |                                                                           |
|                                                                   |                                                                   | MFCN                             | ECC/DEC/(14)02,<br>ECC/DEC/(22)01,<br>ECC/REC/(14)04,<br>ECC/REC/(15)04 | EN 301 908 | Shared use of spectrum envisaged                                          |
|                                                                   |                                                                   | Maritime military systems        |                                                                         |            |                                                                           |
|                                                                   |                                                                   | Telemetry/Telecommand (military) |                                                                         |            |                                                                           |
|                                                                   |                                                                   | Video PMSE                       | ECC/REC/(15)04,<br>ERC/REC 25-10                                        | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links                |

## 2400 MHz - 2450 MHz

|                                                                |                                                                                     |                                    |               |            |                                                            |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------|---------------|------------|------------------------------------------------------------|
| FIXED<br>MOBILE<br>Amateur<br>Radiolocation<br><br>5.150 5.282 | FIXED<br>MOBILE<br>Amateur<br>Amateur-satellite<br>Radiolocation<br><br>5.150 5.282 | Amateur                            |               | EN 301 783 | Within the band 2300-2450 MHz                              |
|                                                                |                                                                                     | Amateur-satellite                  |               |            |                                                            |
|                                                                |                                                                                     | ISM                                |               |            |                                                            |
|                                                                |                                                                                     | Non-specific SRDs                  | ERC/REC 70-03 | EN 300 440 | Within the band 2400.0-2483.5 MHz                          |
|                                                                |                                                                                     | RFID                               | ERC/REC 70-03 | EN 300 440 | Within the band 2446-2454 MHz                              |
|                                                                |                                                                                     | Radiodetermination applications    | ERC/REC 70-03 | EN 300 440 | Within the band 2400.0-2483.5 MHz                          |
|                                                                |                                                                                     | Video PMSE                         | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links |
|                                                                |                                                                                     | Wideband data transmission systems | ERC/REC 70-03 | EN 300 328 | Within the band 2400-2483.5 MHz                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 2450 MHz - 2483.5 MHz

|                                               |                              |                                    |               |            |                                                            |
|-----------------------------------------------|------------------------------|------------------------------------|---------------|------------|------------------------------------------------------------|
| FIXED<br>MOBILE<br>Radiolocation<br><br>5.150 | FIXED<br>MOBILE<br><br>5.150 | ISM                                |               |            |                                                            |
|                                               |                              | Non-specific SRDs                  | ERC/REC 70-03 | EN 300 440 | Within the band 2400.0-2483.5 MHz                          |
|                                               |                              | RFID                               | ERC/REC 70-03 | EN 300 440 | Within the band 2446-2454 MHz                              |
|                                               |                              | Radiodetermination applications    | ERC/REC 70-03 | EN 300 440 | Within the band 2400.0-2483.5 MHz                          |
|                                               |                              | Video PMSE                         | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links |
|                                               |                              | Wideband data transmission systems | ERC/REC 70-03 | EN 300 328 | Within the band 2400-2483.5 MHz                            |

## 2483.5 MHz - 2500 MHz

|                                                                                                                                                            |                                                                       |                              |                                   |            |                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------|-----------------------------------|------------|--------------------------------------------------------------|
| FIXED<br>MOBILE<br>MOBILE-SATELLITE (5.351A)<br>RADIODETERMINATION-SATELLITE (5.398)<br>Radiolocation (5.398A)<br><br>5.150 5.368 5.372A 5.399 5.401 5.402 | FIXED<br>MOBILE<br>MOBILE-SATELLITE (5.351A)<br><br>5.150 5.399 5.402 | IMT-2000 satellite component |                                   |            |                                                              |
|                                                                                                                                                            |                                                                       | ISM                          |                                   |            |                                                              |
|                                                                                                                                                            |                                                                       | LP-AMI                       | ERC/REC 70-03                     | EN 301 559 | Low Power Active Medical Implants and associated peripherals |
|                                                                                                                                                            |                                                                       | Land mobile                  |                                   |            | Mobile applications                                          |
|                                                                                                                                                            |                                                                       | MBANS                        | ERC/REC 70-03                     | EN 303 203 |                                                              |
|                                                                                                                                                            |                                                                       | MSS Earth stations           | ECC/DEC/(09)02,<br>ECC/DEC/(12)01 | EN 301 441 |                                                              |
|                                                                                                                                                            |                                                                       | Video PMSE                   | ERC/REC 25-10                     | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links   |

## 2500 MHz - 2520 MHz

|                                                                                    |                                                        |      |                                                                         |            |                                                                  |
|------------------------------------------------------------------------------------|--------------------------------------------------------|------|-------------------------------------------------------------------------|------------|------------------------------------------------------------------|
| FIXED (5.410)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.384A 5.409A)<br><br>5.412 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.384A) | MCV  | ECC/DEC/(08)08                                                          |            |                                                                  |
|                                                                                    |                                                        | MFCN | ECC/DEC/(05)05,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(11)05 | EN 301 908 | 2500-2570 MHz, Aerial UE are permitted – See ECC Decision (22)07 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 2520 MHz - 2655 MHz

|                                                                                                                                             |                                                                                            |      |                                                                         |            |                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------|------------|------------------------------------------------------------------|
| BROADCASTING-SATELLITE (5.413 5.416)<br>FIXED (5.410)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A)<br><br>5.339 5.412 5.418B 5.418C | FIXED<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A ECA38)<br><br>5.339 5.418B 5.418C ECA16 | MCV  | ECC/DEC/(08)08                                                          |            |                                                                  |
|                                                                                                                                             |                                                                                            | MFCN | ECC/DEC/(05)05,<br>ECC/DEC/(22)01,<br>ECC/DEC/(22)07,<br>ECC/REC/(11)05 | EN 301 908 | 2500-2570 MHz, Aerial UE are permitted – See ECC Decision (22)07 |

## 2655 MHz - 2670 MHz

|                                                                                                                                                                                                                              |                                                                                                                                                                       |                 |                                                      |            |                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------|------------|------------------------------|
| BROADCASTING-SATELLITE (5.208B 5.413 5.416)<br>FIXED (5.410)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A)<br>Earth exploration-satellite (passive)<br>Radio astronomy<br>Space research (passive)<br><br>5.149 5.412 | FIXED<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A)<br>Earth exploration-satellite (passive)<br>Radio astronomy<br>Space research (passive)<br><br>5.149 5.208B ECA16 | MCV             | ECC/DEC/(08)08                                       |            |                              |
|                                                                                                                                                                                                                              |                                                                                                                                                                       | MFCN            | ECC/DEC/(05)05,<br>ECC/DEC/(22)01,<br>ECC/REC/(11)05 | EN 301 908 |                              |
|                                                                                                                                                                                                                              |                                                                                                                                                                       | Radio astronomy |                                                      |            | Continuum observations, VLBI |

## 2670 MHz - 2690 MHz

|                                                                                                                                                                               |                                                                                     |                 |                                                      |            |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------|------------------------------------------------------|------------|------------------------------|
| FIXED (5.410)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A 5.409A)<br>Earth exploration-satellite (passive)<br>Radio astronomy<br>Space research (passive)<br><br>5.149 5.412 | FIXED<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.384A)<br>Radio astronomy<br><br>5.149 | MCV             | ECC/DEC/(08)08                                       |            |                              |
|                                                                                                                                                                               |                                                                                     | MFCN            | ECC/DEC/(05)05,<br>ECC/DEC/(22)01,<br>ECC/REC/(11)05 | EN 301 908 |                              |
|                                                                                                                                                                               |                                                                                     | Radio astronomy |                                                      |            | Continuum observations, VLBI |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 2690 MHz - 2700 MHz

|                                                                                                         |                                                                                                   |                             |  |  |                              |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------|--|--|------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.422 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite) |  |  |                              |
|                                                                                                         |                                                                                                   | Radio astronomy             |  |  | Continuum observations, VLBI |

## 2700 MHz - 2900 MHz

|                                                                    |                                                                          |                          |                |            |                                                            |
|--------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------|----------------|------------|------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION (5.337)<br>Radiolocation<br><br>5.423 | AERONAUTICAL RADIONAVIGATION (5.337)<br>Radiolocation<br><br>5.423 ECA36 | Aeronautical navigation  | ECC/REC/(02)09 |            | Radar and navigation systems                               |
|                                                                    |                                                                          | Radiolocation (civil)    |                |            |                                                            |
|                                                                    |                                                                          | Radiolocation (military) |                |            |                                                            |
|                                                                    |                                                                          | Video PMSE               | ERC/REC 25-10  | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links |
|                                                                    |                                                                          | Weather radar            |                | EN 303 347 |                                                            |

## 2900 MHz - 3100 MHz

|                                                                      |                                                                            |                          |  |                           |                              |
|----------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------|--|---------------------------|------------------------------|
| RADIOLOCATION (5.424A)<br>RADIONAVIGATION (5.426)<br><br>5.425 5.427 | RADIOLOCATION (5.424A)<br>RADIONAVIGATION (5.426)<br><br>5.425 5.427 ECA36 | Radiolocation (civil)    |  | EN 302 248,<br>EN 302 752 | Radar and navigation systems |
|                                                                      |                                                                            | Radiolocation (military) |  |                           |                              |

## 3100 MHz - 3300 MHz

|                                                                                                     |                                                                                                     |                            |                                                                                           |            |                                                                                                 |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------|
| RADIOLOCATION<br>Earth exploration-satellite (active)<br>Space research (active)<br><br>5.149 5.428 | RADIOLOCATION<br>Earth exploration-satellite (active)<br>Space research (active)<br><br>5.149 ECA36 | Active sensors (satellite) |                                                                                           |            |                                                                                                 |
|                                                                                                     |                                                                                                     | Radio astronomy            |                                                                                           |            | Spectral line observations (e.g. methine line)                                                  |
|                                                                                                     |                                                                                                     | Radiolocation (civil)      |                                                                                           |            | Radars                                                                                          |
|                                                                                                     |                                                                                                     | Radiolocation (military)   |                                                                                           |            |                                                                                                 |
|                                                                                                     |                                                                                                     | UWB applications           | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES) |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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### 3300 MHz - 3400 MHz

|                                                  |                              |                          |                                                                                           |            |                                                                                                    |
|--------------------------------------------------|------------------------------|--------------------------|-------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------|
| RADIOLOCATION<br>5.149 5.429 5.429A 5.429B 5.430 | RADIOLOCATION<br>5.149 ECA36 | Radio astronomy          |                                                                                           |            | Spectral line observations (e.g. methine line)                                                     |
|                                                  |                              | Radiolocation (civil)    |                                                                                           |            | Upper limit for airborne radars 3410 MHz                                                           |
|                                                  |                              | Radiolocation (military) |                                                                                           |            | Upper limit for airborne radars is 3410 MHz                                                        |
|                                                  |                              | UWB applications         | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2).<br>Location Application for Emergency Services (LAES) |

### 3400 MHz - 3600 MHz

|                                                                                                                      |                                                                                                                                       |                          |                                                                                            |            |                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.430A)<br>Radiolocation<br>5.431 | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.430A ECA38)<br>Amateur<br>Radiolocation<br>ECA36 | Amateur                  |                                                                                            | EN 301 783 | Within the band 3400-3410 MHz                                                                      |
|                                                                                                                      |                                                                                                                                       | MFCN                     | ECC/DEC/(11)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(15)01,<br>ECC/REC/(20)03,<br>ECC/REC/(21)02 | EN 301 908 |                                                                                                    |
|                                                                                                                      |                                                                                                                                       | Radiolocation (civil)    |                                                                                            |            | Upper limit for airborne radars is 3410 MHz                                                        |
|                                                                                                                      |                                                                                                                                       | Radiolocation (military) |                                                                                            |            | Upper limit for airborne radars is 3410 MHz                                                        |
|                                                                                                                      |                                                                                                                                       | UWB applications         | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03  | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2).<br>Location Application for Emergency Services (LAES) |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 3600 MHz - 3800 MHz

|                                                                                                                        |                                                                                        |                  |                                                                                            |            |                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.433B 5.434A 5.434B)<br><br>5.435A | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE (5.434A 5.434B ECA38)<br><br>ECA37 | ESV              | ECC/DEC/(05)09                                                                             | EN 301 447 | Within the band 3700-3800 MHz                                                                      |
|                                                                                                                        |                                                                                        | Fixed            | ERC/REC 12-08                                                                              | EN 302 217 | Medium/high capacity fixed                                                                         |
|                                                                                                                        |                                                                                        | MFCN             | ECC/DEC/(11)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(15)01,<br>ECC/REC/(20)03,<br>ECC/REC/(21)02 | EN 301 908 |                                                                                                    |
|                                                                                                                        |                                                                                        | UWB applications | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03  | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2).<br>Location Application for Emergency Services (LAES) |
|                                                                                                                        |                                                                                        | VSAT             |                                                                                            | EN 301 443 | Within the band 3700-3800 MHz                                                                      |

### 3800 MHz - 4200 MHz

|                                                     |                                           |                  |                                                                                           |            |                                                                                                    |
|-----------------------------------------------------|-------------------------------------------|------------------|-------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>Mobile | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH) | ESV              | ECC/DEC/(05)09                                                                            | EN 301 447 | Within the band 3700-4200 MHz                                                                      |
|                                                     |                                           | Fixed            | ERC/REC 12-08                                                                             | EN 302 217 | Medium/high capacity fixed                                                                         |
|                                                     |                                           | MFCN             | ECC/DEC/(24)01                                                                            |            |                                                                                                    |
|                                                     |                                           | UWB applications | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2).<br>Location Application for Emergency Services (LAES) |
|                                                     |                                           | VSAT             |                                                                                           | EN 301 443 |                                                                                                    |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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#### 4200 MHz - 4400 MHz

|                                                                                              |                                                                                              |                               |                                                                                           |            |                                                                                                    |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------|
| AERONAUTICAL MOBILE (5.436)<br>AERONAUTICAL RADIONAVIGATION (5.438)<br><br>5.437 5.439 5.440 | AERONAUTICAL MOBILE (5.436)<br>AERONAUTICAL RADIONAVIGATION (5.438)<br><br>5.437 5.440 ECA36 | Aeronautical military systems |                                                                                           |            |                                                                                                    |
|                                                                                              |                                                                                              | Altimeters                    |                                                                                           |            |                                                                                                    |
|                                                                                              |                                                                                              | Passive sensors (satellite)   |                                                                                           |            | For sea surface temperature measurements                                                           |
|                                                                                              |                                                                                              | UWB applications              | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2).<br>Location Application for Emergency Services (LAES) |
|                                                                                              |                                                                                              | WAIC                          |                                                                                           |            | Wireless Avionics Intra-Communications                                                             |

#### 4400 MHz - 4500 MHz

|                 |                                    |                                  |                                                                                           |            |                                                                                                    |
|-----------------|------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE | FIXED<br>MOBILE<br><br>ECA20 ECA36 | Aeronautical military systems    |                                                                                           |            |                                                                                                    |
|                 |                                    | Land military systems            |                                                                                           |            |                                                                                                    |
|                 |                                    | Maritime military systems        |                                                                                           |            |                                                                                                    |
|                 |                                    | Telemetry/Telecommand (military) |                                                                                           |            |                                                                                                    |
|                 |                                    | UWB applications                 | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2).<br>Location Application for Emergency Services (LAES) |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 4500 MHz - 4800 MHz

|                                            |                                                               |                                  |                                                                                           |            |                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.441)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.441)<br>MOBILE<br><br>ECA20 ECA36 | Aeronautical military systems    |                                                                                           |            |                                                                                                 |
|                                            |                                                               | FSS Earth stations               |                                                                                           |            | FSS not to be implemented in NATO Europe. Fixed-Satellite frequency plan in 4500-4800 MHz       |
|                                            |                                                               | Land military systems            |                                                                                           |            |                                                                                                 |
|                                            |                                                               | Maritime military systems        |                                                                                           |            |                                                                                                 |
|                                            |                                                               | TLPR                             | ERC/REC 70-03                                                                             | EN 302 372 |                                                                                                 |
|                                            |                                                               | Telemetry/Telecommand (military) |                                                                                           |            |                                                                                                 |
|                                            |                                                               | UWB applications                 | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/REC/(11)09,<br>ECC/REC/(11)10,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. Location Tracking Type 2 (LT2). Location Application for Emergency Services (LAES) |

## 4800 MHz - 4990 MHz

|                                                                                          |                                                                                                |                                  |                |            |                                                                           |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------|----------------|------------|---------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.442 5.440A 5.441A 5.441B)<br>Radio astronomy<br><br>5.149 5.339 5.443 | FIXED<br>MOBILE (5.440A 5.441A 5.441B 5.442)<br>Radio astronomy<br><br>5.149 5.339 ECA20 ECA36 | Aeronautical military systems    |                |            |                                                                           |
|                                                                                          |                                                                                                | BBDR                             | ECC/REC/(08)04 | EN 302 625 | Within the band 4940-4990 MHz. Optimal band for BBDR within the PPDR uses |
|                                                                                          |                                                                                                | Land military systems            |                |            |                                                                           |
|                                                                                          |                                                                                                | Maritime military systems        |                |            |                                                                           |
|                                                                                          |                                                                                                | Passive sensors (satellite)      |                |            | Space Research and EESS (passive) above 4950 MHz in some countries        |
|                                                                                          |                                                                                                | Radio astronomy                  |                |            | Continuum and spectral line observations, (e.g. formaldehyde line), VLBI  |
|                                                                                          |                                                                                                | TLPR                             | ERC/REC 70-03  | EN 302 372 |                                                                           |
|                                                                                          |                                                                                                | Telemetry/Telecommand (military) |                |            |                                                                           |

## 4990 MHz - 5000 MHz

|                                                                                                           |                                                                                           |                                  |               |            |                              |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------|---------------|------------|------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br>Space research (passive)<br><br>5.149 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 ECA20 ECA36 | Aeronautical military systems    |               |            |                              |
|                                                                                                           |                                                                                           | Land military systems            |               |            |                              |
|                                                                                                           |                                                                                           | Maritime military systems        |               |            |                              |
|                                                                                                           |                                                                                           | Radio astronomy                  |               |            | Continuum observations, VLBI |
|                                                                                                           |                                                                                           | TLPR                             | ERC/REC 70-03 | EN 302 372 |                              |
|                                                                                                           |                                                                                           | Telemetry/Telecommand (military) |               |            |                              |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 5000 MHz - 5010 MHz

|                                                                                                                    |                                                                                                                                                                   |                              |               |            |                                                                  |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------|------------|------------------------------------------------------------------|
| AERONAUTICAL MOBILE-SATELLITE (5.443AA)<br>AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE) | AERONAUTICAL MOBILE-SATELLITE (5.443AA)<br>AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (EARTH-TO-SPACE)<br>Radio astronomy<br>Space research (passive) | GALILEO                      |               |            | For future use by Galileo                                        |
|                                                                                                                    |                                                                                                                                                                   | Radio astronomy              |               |            | Continuum observation, VLBI                                      |
|                                                                                                                    |                                                                                                                                                                   | Satellite navigation systems |               |            | Aeronautical Radionavigation and FSS envisaged in some countries |
|                                                                                                                    |                                                                                                                                                                   | TLPR                         | ERC/REC 70-03 | EN 302 372 |                                                                  |

## 5010 MHz - 5030 MHz

|                                                                                                                   |                                                                                                                                                                  |                              |               |            |                                                                  |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------|------------|------------------------------------------------------------------|
| AERONAUTICAL MOBILE-SATELLITE (5.443AA)<br>AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (5.328B 5.443B) | AERONAUTICAL MOBILE-SATELLITE (5.443AA)<br>AERONAUTICAL RADIONAVIGATION RADIONAVIGATION-SATELLITE (5.328B 5.443B)<br>Radio astronomy<br>Space research (passive) | GALILEO                      |               |            |                                                                  |
|                                                                                                                   |                                                                                                                                                                  | Radio astronomy              |               |            | Continuum observation, VLBI                                      |
|                                                                                                                   |                                                                                                                                                                  | Satellite navigation systems |               |            | Aeronautical Radionavigation and FSS envisaged in some countries |
|                                                                                                                   |                                                                                                                                                                  | TLPR                         | ERC/REC 70-03 | EN 302 372 |                                                                  |

## 5030 MHz - 5091 MHz

|                                                                                                                 |                                                                                                                 |      |               |            |                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------|---------------|------------|----------------------------------------------------------------------------------------|
| AERONAUTICAL MOBILE (5.443C)<br>AERONAUTICAL MOBILE-SATELLITE (5.443D)<br>AERONAUTICAL RADIONAVIGATION<br>5.444 | AERONAUTICAL MOBILE (5.443C)<br>AERONAUTICAL MOBILE-SATELLITE (5.443D)<br>AERONAUTICAL RADIONAVIGATION<br>5.444 | MLS  |               |            | Aeronautical Radionavigation envisaged in some countries. FSS in use in some countries |
|                                                                                                                 |                                                                                                                 | TLPR | ERC/REC 70-03 | EN 302 372 |                                                                                        |

## 5091 MHz - 5150 MHz

|                                                                                                                                           |                                                                                                                                           |      |               |            |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------|---------------|------------|------------------------------|
| AERONAUTICAL MOBILE (5.444B)<br>AERONAUTICAL MOBILE-SATELLITE (5.443AA)<br>AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.444A)<br>5.444 | AERONAUTICAL MOBILE (5.444B)<br>AERONAUTICAL MOBILE-SATELLITE (5.443AA)<br>AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.444A)<br>5.444 | -    |               |            | FSS in use in some countries |
|                                                                                                                                           |                                                                                                                                           | TLPR | ERC/REC 70-03 | EN 302 372 |                              |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 5150 MHz - 5250 MHz

|                                                                                                                                                                |                                                                                                                                                    |                        |                               |            |                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------|------------|----------------------------------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION (5.446D)<br>FIXED-SATELLITE (5.447A)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446B 5.446A)<br><br>5.446 5.446C 5.447 5.447B 5.447C | AERONAUTICAL RADIONAVIGATION FIXED-SATELLITE (5.447A)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.446B)<br><br>5.446 5.446C 5.447 5.447B 5.447C | Aeronautical telemetry |                               |            |                                                                                        |
|                                                                                                                                                                |                                                                                                                                                    | BBDR                   | ECC/REC/(08)04                | EN 302 625 | Temporary use by PPDR users                                                            |
|                                                                                                                                                                |                                                                                                                                                    | Feeder links           |                               |            | Feeder links for MSS. Aeronautical Radionavigation and FSS envisaged in some countries |
|                                                                                                                                                                |                                                                                                                                                    | RLAN                   | ECC/DEC/(04)08, ERC/REC 70-03 | EN 301 893 | WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz                             |
|                                                                                                                                                                |                                                                                                                                                    | TLPR                   | ERC/REC 70-03                 | EN 302 372 |                                                                                        |

## 5250 MHz - 5255 MHz

|                                                                                                                                                                  |                                                                                                                                                                 |                            |                               |            |                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------|------------|------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F)<br>RADIOLOCATION<br>SPACE RESEARCH (5.447D)<br><br>5.447E 5.448 5.448A | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F)<br>RADIOLOCATION<br>SPACE RESEARCH (5.447D)<br><br>5.448A ECA22 ECA36 | -                          |                               |            | Position fixing                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | Active sensors (satellite) |                               |            |                                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | Maritime radar             |                               |            | Shipborne and VTS radar                                    |
|                                                                                                                                                                  |                                                                                                                                                                 | RLAN                       | ECC/DEC/(04)08, ERC/REC 70-03 | EN 301 893 | WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz |
|                                                                                                                                                                  |                                                                                                                                                                 | Radiolocation (military)   |                               |            |                                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | TLPR                       | ERC/REC 70-03                 | EN 302 372 |                                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | Weather radar              |                               | EN 303 347 | Ground based and airborne                                  |

## 5255 MHz - 5350 MHz

|                                                                                                                                                                  |                                                                                                                                                                 |                            |                               |            |                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------|------------|------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br><br>5.447E 5.448 5.448A | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.447F)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br><br>5.448A ECA22 ECA36 | -                          |                               |            | Position fixing                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | Active sensors (satellite) |                               |            |                                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | Maritime radar             |                               |            | Shipborne and VTS radar                                    |
|                                                                                                                                                                  |                                                                                                                                                                 | RLAN                       | ECC/DEC/(04)08, ERC/REC 70-03 | EN 301 893 | WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz |
|                                                                                                                                                                  |                                                                                                                                                                 | Radiolocation (military)   |                               |            |                                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | TLPR                       | ERC/REC 70-03                 | EN 302 372 |                                                            |
|                                                                                                                                                                  |                                                                                                                                                                 | Weather radar              |                               | EN 303 347 | Ground based and airborne                                  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 5350 MHz - 5460 MHz

|                                                                                                                                   |                                                                                                                                                      |                            |               |            |                           |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|------------|---------------------------|
| AERONAUTICAL RADIONAVIGATION (5.449)<br>EARTH EXPLORATION-SATELLITE (5.448B)<br>RADIOLOCATION (5.448D)<br>SPACE RESEARCH (5.448C) | AERONAUTICAL RADIONAVIGATION (5.449)<br>EARTH EXPLORATION-SATELLITE (5.448B)<br>RADIOLOCATION (5.448D)<br>SPACE RESEARCH (5.448C)<br><br>ECA22 ECA36 | -                          |               |            | Position fixing           |
|                                                                                                                                   |                                                                                                                                                      | Active sensors (satellite) |               |            |                           |
|                                                                                                                                   |                                                                                                                                                      | Maritime radar             |               |            | Shipborne and VTS radar   |
|                                                                                                                                   |                                                                                                                                                      | Radiolocation (military)   |               |            |                           |
|                                                                                                                                   |                                                                                                                                                      | TLPR                       | ERC/REC 70-03 | EN 302 372 |                           |
|                                                                                                                                   |                                                                                                                                                      | Weather radar              |               | EN 303 347 | Ground based and airborne |

### 5460 MHz - 5470 MHz

|                                                                                                                                    |                                                                                                                                                |                            |               |            |                           |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|------------|---------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION (5.448D)<br>RADIONAVIGATION (5.449)<br>SPACE RESEARCH (ACTIVE)<br><br>5.448B | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION (5.448D)<br>RADIONAVIGATION (5.449)<br>SPACE RESEARCH (ACTIVE)<br><br>5.448B ECA22 ECA36 | -                          |               |            | Position fixing           |
|                                                                                                                                    |                                                                                                                                                | Active sensors (satellite) |               |            |                           |
|                                                                                                                                    |                                                                                                                                                | Maritime radar             |               |            | Shipborne and VTS radar   |
|                                                                                                                                    |                                                                                                                                                | Radiolocation (military)   |               |            |                           |
|                                                                                                                                    |                                                                                                                                                | TLPR                       | ERC/REC 70-03 | EN 302 372 |                           |
|                                                                                                                                    |                                                                                                                                                | Weather radar              |               | EN 303 347 | Ground based and airborne |

### 5470 MHz - 5570 MHz

|                                                                                                                                                                                                      |                                                                                                                                                                                                      |                            |                                  |            |                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------|------------|------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>MARITIME RADIONAVIGATION<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A)<br>RADIOLOCATION (5.450B)<br>SPACE RESEARCH (ACTIVE)<br><br>5.448B 5.450 5.451 | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>MARITIME RADIONAVIGATION<br>MOBILE EXCEPT AERONAUTICAL MOBILE (5.446A 5.450A)<br>RADIOLOCATION (5.450B)<br>SPACE RESEARCH (ACTIVE)<br><br>5.448B ECA22 ECA36 | -                          |                                  |            | Position fixing                                            |
|                                                                                                                                                                                                      |                                                                                                                                                                                                      | Active sensors (satellite) |                                  |            |                                                            |
|                                                                                                                                                                                                      |                                                                                                                                                                                                      | Maritime radar             |                                  |            | Shipborne and VTS radar                                    |
|                                                                                                                                                                                                      |                                                                                                                                                                                                      | RLAN                       | ECC/DEC/(04)08,<br>ERC/REC 70-03 | EN 301 893 | WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz |
|                                                                                                                                                                                                      |                                                                                                                                                                                                      | Radiolocation (military)   |                                  |            |                                                            |
|                                                                                                                                                                                                      |                                                                                                                                                                                                      | TLPR                       | ERC/REC 70-03                    | EN 302 372 |                                                            |
|                                                                                                                                                                                                      |                                                                                                                                                                                                      | Weather radar              |                                  | EN 303 347 | Ground based and airborne                                  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 5570 MHz - 5650 MHz

|                                                                                                                                     |                                                                                                                                     |                          |                                  |            |                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------|------------|------------------------------------------------------------|
| MARITIME RADIONAVIGATION<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.446A 5.450A)<br>RADIOLOCATION (5.450B)<br><br>5.450 5.451 5.452 | MARITIME RADIONAVIGATION<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.446A 5.450A)<br>RADIOLOCATION (5.450B)<br><br>5.452 ECA22 ECA36 | -                        |                                  |            | Position fixing                                            |
|                                                                                                                                     |                                                                                                                                     | Maritime radar           |                                  |            | Shipborne and VTS radar                                    |
|                                                                                                                                     |                                                                                                                                     | RLAN                     | ECC/DEC/(04)08,<br>ERC/REC 70-03 | EN 301 893 | WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz |
|                                                                                                                                     |                                                                                                                                     | Radiolocation (military) |                                  |            |                                                            |
|                                                                                                                                     |                                                                                                                                     | TLPR                     | ERC/REC 70-03                    | EN 302 372 |                                                            |
|                                                                                                                                     |                                                                                                                                     | Weather radar            |                                  | EN 303 347 | Ground based                                               |

## 5650 MHz - 5725 MHz

|                                                                                                                                                      |                                                                                                                                                       |                          |                                  |            |                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------|------------|------------------------------------------------------------|
| MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.446A 5.450A)<br>RADIOLOCATION<br>Amateur<br>Space research (deep space)<br><br>5.282 5.451 5.453 5.454 5.455 | MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.446A 5.450A)<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite (Earth-to-space)<br><br>5.282 ECA22 ECA23 ECA36 | -                        |                                  |            | Position fixing                                            |
|                                                                                                                                                      |                                                                                                                                                       | Amateur                  |                                  | EN 301 783 | Within the band 5650-5850 MHz                              |
|                                                                                                                                                      |                                                                                                                                                       | Amateur-satellite        |                                  |            | Within the band 5650-5670 MHz                              |
|                                                                                                                                                      |                                                                                                                                                       | Maritime radar           |                                  |            | Shipborne and VTS radar                                    |
|                                                                                                                                                      |                                                                                                                                                       | RLAN                     | ECC/DEC/(04)08,<br>ERC/REC 70-03 | EN 301 893 | WAS/RLANs within the bands 5150-5350 MHz and 5470-5725 MHz |
|                                                                                                                                                      |                                                                                                                                                       | Radiolocation (military) |                                  |            |                                                            |
|                                                                                                                                                      |                                                                                                                                                       | TLPR                     | ERC/REC 70-03                    | EN 302 372 |                                                            |
|                                                                                                                                                      |                                                                                                                                                       | Weather radar            |                                  | EN 303 347 | Ground based and airborne                                  |

## 5725 MHz - 5830 MHz

|                                                                                             |                                                                                                                |                          |                |            |                                                                                  |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------|----------------|------------|----------------------------------------------------------------------------------|
| FIXED-SATELLITE (EARTH-TO-SPACE)<br>RADIOLOCATION<br>Amateur<br><br>5.150 5.451 5.453 5.455 | FIXED-SATELLITE (EARTH-TO-SPACE)<br>RADIOLOCATION<br>Amateur<br>Fixed<br>Mobile<br><br>5.150 ECA17 ECA22 ECA36 | Amateur                  |                | EN 301 783 | Within the band 5650-5850 MHz                                                    |
|                                                                                             |                                                                                                                | BFWA                     | ECC/REC/(06)04 | EN 302 502 | Within the band 5725-5875 MHz                                                    |
|                                                                                             |                                                                                                                | ISM                      |                |            | Within the band 5725-5875 MHz                                                    |
|                                                                                             |                                                                                                                | Non-specific SRDs        | ERC/REC 70-03  | EN 300 440 | Within the band 5725-5875 MHz                                                    |
|                                                                                             |                                                                                                                | Radiolocation (military) |                |            |                                                                                  |
|                                                                                             |                                                                                                                | TLPR                     | ERC/REC 70-03  | EN 302 372 |                                                                                  |
|                                                                                             |                                                                                                                | TTT                      | ERC/REC 70-03  | EN 300 674 | Within the band 5795-5805 MHz. TTT in the band 5805-5815 MHz on a national basis |
|                                                                                             |                                                                                                                | WIA                      | ERC/REC 70-03  | EN 303 258 | Within the band 5725-5875 MHz                                                    |
|                                                                                             |                                                                                                                | Weather radar            |                | EN 303 347 | Ground based and airborne                                                        |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 5830 MHz - 5850 MHz

|                                                                                                                                   |                                                                                                                                                      |                          |               |            |                               |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|------------|-------------------------------|
| FIXED-SATELLITE (EARTH-TO-SPACE)<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite (space-to-Earth)<br><br>5.150 5.451 5.453 5.455 | FIXED-SATELLITE (EARTH-TO-SPACE)<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite (space-to-Earth)<br>Fixed<br>Mobile<br><br>5.150 ECA22 ECA23 ECA36 | -                        |               |            | Within the band 5725-5875 MHz |
|                                                                                                                                   |                                                                                                                                                      | Amateur                  |               | EN 301 783 | Within the band 5650-5850 MHz |
|                                                                                                                                   |                                                                                                                                                      | Amateur-satellite        |               |            | Within the band 5830-5850 MHz |
|                                                                                                                                   |                                                                                                                                                      | BFWA                     |               |            | Within the band 5725-5875 MHz |
|                                                                                                                                   |                                                                                                                                                      | ISM                      |               |            | Within the band 5725-5875 MHz |
|                                                                                                                                   |                                                                                                                                                      | Non-specific SRDs        | ERC/REC 70-03 | EN 300 440 | Within the band 5725-5875 MHz |
|                                                                                                                                   |                                                                                                                                                      | Radiolocation (military) |               |            |                               |
|                                                                                                                                   |                                                                                                                                                      | TLPR                     | ERC/REC 70-03 | EN 302 372 |                               |
|                                                                                                                                   |                                                                                                                                                      | WIA                      | ERC/REC 70-03 | EN 303 258 | Within the band 5725-5875 MHz |
|                                                                                                                                   |                                                                                                                                                      | Weather radar            |               | EN 303 347 | Ground based and airborne     |

## 5850 MHz - 5925 MHz

|                                                                  |                                                                  |                   |                                                     |            |                                                                                                                   |
|------------------------------------------------------------------|------------------------------------------------------------------|-------------------|-----------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br><br>5.150 | FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br><br>5.150 | BFWA              | ECC/REC/(06)04                                      | EN 302 502 | Within the band 5725-5875 MHz                                                                                     |
|                                                                  |                                                                  | ISM               |                                                     |            | Within the band 5725-5875 MHz                                                                                     |
|                                                                  |                                                                  | ITS               | ECC/DEC/(08)01,<br>ECC/REC/(08)01,<br>ERC/REC 70-03 | EN 302 571 | Safety related applications within the band 5875-5935 MHz. Non-Safety applications within the band 5855-5875 MHz. |
|                                                                  |                                                                  | MBR               | ECC/REC/(17)03                                      | EN 303 276 | Within 5852-5872 MHz and 5880-5900 MHz                                                                            |
|                                                                  |                                                                  | Non-specific SRDs | ERC/REC 70-03                                       | EN 300 440 | Within the band 5725-5875 MHz                                                                                     |
|                                                                  |                                                                  | TLPR              | ERC/REC 70-03                                       | EN 302 372 |                                                                                                                   |
|                                                                  |                                                                  | VSAT              |                                                     | EN 301 443 |                                                                                                                   |
|                                                                  |                                                                  | WIA               | ERC/REC 70-03                                       | EN 303 258 | Within the band 5725-5875 MHz                                                                                     |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 5925 MHz - 6700 MHz

|                                                                                                     |                                                                                                                        |                             |                                                                        |            |                                                                                                      |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------|
| FIXED (5.457)<br>FIXED-SATELLITE (5.457A 5.457B)<br>MOBILE (5.457C 5.457E)<br><br>5.149 5.440 5.458 | FIXED<br>FIXED-SATELLITE (5.457A)<br>MOBILE (5.457E)<br>Earth exploration-satellite (passive)<br><br>5.149 5.440 5.458 | -                           |                                                                        |            |                                                                                                      |
|                                                                                                     |                                                                                                                        | ESV                         | ECC/DEC/(05)09                                                         | EN 301 447 | Within the band 5925-6425 MHz                                                                        |
|                                                                                                     |                                                                                                                        | FSS Earth stations          |                                                                        |            |                                                                                                      |
|                                                                                                     |                                                                                                                        | Fixed                       | ECC/REC/(14)06,<br>ERC/REC 14-01,<br>ERC/REC 14-02                     | EN 302 217 | Point-to-point                                                                                       |
|                                                                                                     |                                                                                                                        | ITS                         | ECC/DEC/(08)01,<br>ERC/REC 70-03                                       |            | Urban rail systems only 5925–5935 MHz.<br>Safety related applications within the band 5875-5935 MHz. |
|                                                                                                     |                                                                                                                        | LPR                         | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                      |
|                                                                                                     |                                                                                                                        | Passive sensors (satellite) |                                                                        |            | For sea surface temperature, sea surface wind speed and soil moisture measurements                   |
|                                                                                                     |                                                                                                                        | RLAN                        | ECC/DEC/(20)01                                                         | EN 303 687 | Within the band 5945-6425 MHz                                                                        |
|                                                                                                     |                                                                                                                        | Radio astronomy             |                                                                        |            | Spectral line observations (e.g. methanol line), VLBI.                                               |
|                                                                                                     |                                                                                                                        | TLPR                        | ERC/REC 70-03                                                          | EN 302 372 |                                                                                                      |
|                                                                                                     |                                                                                                                        | UWB applications            | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as UWB on-board aircraft regulation within the band 6.0- 8.5 GHz                 |
|                                                                                                     |                                                                                                                        | VSAT                        |                                                                        | EN 301 443 |                                                                                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 6700 MHz - 7075 MHz

|                                                                                |                                                                                                                         |                             |                                                                        |            |                                                                                                                   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.441)<br>MOBILE (5.457E)<br><br>5.458 5.458A 5.458B | FIXED<br>FIXED-SATELLITE (5.441)<br>MOBILE (5.457E)<br>Earth exploration-satellite (passive)<br><br>5.458 5.458A 5.458B | Feeder links                |                                                                        |            |                                                                                                                   |
|                                                                                |                                                                                                                         | Fixed                       | ECC/REC/(14)06,<br>ERC/REC 14-02                                       | EN 302 217 | Point-to-point                                                                                                    |
|                                                                                |                                                                                                                         | LPR                         | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                                   |
|                                                                                |                                                                                                                         | Passive sensors (satellite) |                                                                        |            | For sea surface temperature, sea surface wind speed and soil moisture measurements                                |
|                                                                                |                                                                                                                         | TLPR                        | ERC/REC 70-03                                                          | EN 302 372 |                                                                                                                   |
|                                                                                |                                                                                                                         | UWB applications            | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                                   |
|                                                                                |                                                                                                                         | VSAT                        |                                                                        | EN 301 443 |                                                                                                                   |
|                                                                                |                                                                                                                         | Video PMSE                  | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link within 7-8.5 GHz. |

## 7075 MHz - 7145 MHz

|                                       |                                                                                |                             |                                                                        |            |                                                                                                  |
|---------------------------------------|--------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE (5.457E)<br><br>5.458 | FIXED<br>MOBILE (5.457E)<br>Earth exploration-satellite (passive)<br><br>5.458 | Fixed                       | ECC/REC/(02)06,<br>ECC/REC/(14)06,<br>ERC/REC 14-02                    | EN 302 217 | Point-to-point                                                                                   |
|                                       |                                                                                | LPR                         | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                       |                                                                                | Passive sensors (satellite) |                                                                        |            | For sea surface temperature, sea surface wind speed and soil moisture measurements               |
|                                       |                                                                                | UWB applications            | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                       |                                                                                | Video PMSE                  | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

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## 7145 MHz - 7190 MHz

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|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>SPACE RESEARCH (DEEP SPACE)<br>(EARTH-TO-SPACE)<br><br>5.458 | FIXED<br>MOBILE<br>SPACE RESEARCH (DEEP SPACE)<br>(EARTH-TO-SPACE)<br>Space Operation (Earth-to-space)<br><br>5.458 | Fixed            | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                 |                                                                                                                     | LPR              | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                 |                                                                                                                     | UWB applications | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                                 |                                                                                                                     | Video PMSE       | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

## 7190 MHz - 7235 MHz

|                                                                                                          |                                                                                                          |                             |                                                                        |            |                                                                                                  |
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| EARTH EXPLORATION-SATELLITE<br>(5.460A 5.460B)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (5.460)<br><br>5.458 | EARTH EXPLORATION-SATELLITE<br>(5.460A 5.460B)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (5.460)<br><br>5.458 | Fixed                       | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                          |                                                                                                          | LPR                         | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                          |                                                                                                          | Passive sensors (satellite) |                                                                        |            | For sea surface temperature, sea surface wind speed and soil moisture measurements               |
|                                                                                                          |                                                                                                          | UWB applications            | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                                                          |                                                                                                          | Video PMSE                  | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 7235 MHz - 7250 MHz

|                                                                         |                                                                                                  |                             |                                                                        |            |                                                                                                  |
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| EARTH EXPLORATION-SATELLITE<br>(5.460A)<br>FIXED<br>MOBILE<br><br>5.458 | EARTH EXPLORATION-SATELLITE<br>(5.460A)<br>FIXED<br>Space research (Earth-to-space)<br><br>5.458 | Fixed                       | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                         |                                                                                                  | LPR                         | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                         |                                                                                                  | Passive sensors (satellite) |                                                                        |            | For sea surface temperature, sea surface wind speed and soil moisture measurements               |
|                                                                         |                                                                                                  | UWB applications            | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                                                         |                                                                                                  | Video PMSE                  | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

## 7250 MHz - 7300 MHz

|                                                                  |                                                                        |                              |                                                                        |            |                                                                                                  |
|------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br><br>5.461 | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br><br>5.461 ECA36 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point. FIXED and MOBILE services not to be implemented in most NATO countries           |
|                                                                  |                                                                        | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                  |                                                                        | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                  |                                                                        | MSS Earth stations           |                                                                        |            | Mobile satellite applications within the band 7250-7375 MHz                                      |
|                                                                  |                                                                        | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                  |                                                                        | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                                                  |                                                                        | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 7300 MHz - 7375 MHz

|                                                                                                |                                                                                                      |                              |                                                                        |            |                                                                                                  |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461 | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461 ECA36 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                |                                                                                                      | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                |                                                                                                      | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                |                                                                                                      | MSS Earth stations           |                                                                        |            | Mobile satellite applications within the band 7250-7375 MHz                                      |
|                                                                                                |                                                                                                      | Maritime military systems    |                                                                        |            |                                                                                                  |
|                                                                                                |                                                                                                      | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                |                                                                                                      | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                                                |                                                                                                      | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

## 7375 MHz - 7450 MHz

|                                                                                                                                                    |                                                                                                                                                          |                              |                                                                        |            |                                                                                                  |
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| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MARITIME MOBILE-SATELLITE<br>(5.461AA 5.461AB)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461AC | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MARITIME MOBILE-SATELLITE<br>(5.461AA 5.461AB)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461AC ECA36 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                                                                    |                                                                                                                                                          | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | MSS Earth stations           |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | Maritime military systems    |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                                                                                                    |                                                                                                                                                          | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 7450 MHz - 7550 MHz

|                                                                                                                                                                                                           |                                                                                                                                                                                                                 |                              |                                                                        |            |                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MARITIME MOBILE-SATELLITE<br>(5.461AA 5.461AB)<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461A 5.461AC | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MARITIME MOBILE-SATELLITE<br>(5.461AA 5.461AB)<br>METEOROLOGICAL-SATELLITE<br>(SPACE-TO-EARTH)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461A 5.461AC ECA36 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | Maritime military systems    |                                                                        |            |                                                                                                  |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |
|                                                                                                                                                                                                           |                                                                                                                                                                                                                 | Weather satellites           |                                                                        |            | Limited to geostationary systems                                                                 |

## 7550 MHz - 7750 MHz

|                                                                                                                                                    |                                                                                                                                                          |                              |                                                                        |            |                                                                                                  |
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| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MARITIME MOBILE-SATELLITE<br>(5.461AA 5.461AB)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461AC | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MARITIME MOBILE-SATELLITE<br>(5.461AA 5.461AB)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.461AC ECA36 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                                                                    |                                                                                                                                                          | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | Maritime military systems    |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                                                                    |                                                                                                                                                          | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                                                                                                    |                                                                                                                                                          | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 7750 MHz - 7900 MHz

|                                                                                       |                                                                                       |                    |                                                                        |            |                                                                                                  |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>METEOROLOGICAL-SATELLITE<br>(5.461B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>METEOROLOGICAL-SATELLITE<br>(5.461B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | Fixed              | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                       |                                                                                       | LPR                | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                       |                                                                                       | UWB applications   | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                                                                       |                                                                                       | Video PMSE         | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |
|                                                                                       |                                                                                       | Weather satellites |                                                                        |            | Limited to non-geostationary systems                                                             |

## 7900 MHz - 8025 MHz

|                                                                  |                                                                        |                              |                                                                        |            |                                                                                                  |
|------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br><br>5.461 | FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br><br>5.461 ECA36 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                  |                                                                        | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                  |                                                                        | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                  |                                                                        | MSS Earth stations           |                                                                        |            | Mobile satellite applications                                                                    |
|                                                                  |                                                                        | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                  |                                                                        | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                  |                                                                        | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 8025 MHz - 8175 MHz

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| EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH)<br>FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE (5.463)<br><br>5.462A | EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH)<br>FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE (5.463)<br><br>5.462A ECA36 | Earth exploration-satellite  |                                                                        |            | Satellite payload telemetry                                                                      |
|                                                                                                                           |                                                                                                                                 | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                                           |                                                                                                                                 | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                                           |                                                                                                                                 | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                                           |                                                                                                                                 | Land mobile                  |                                                                        |            | Mobile applications within the band 8025-8200 MHz                                                |
|                                                                                                                           |                                                                                                                                 | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                                           |                                                                                                                                 | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                                                                                                           |                                                                                                                                 | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

## 8175 MHz - 8215 MHz

|                                                                                                                                                                        |                                                                                                                                                                              |                              |                                                                        |            |                                                                                                  |
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| EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH)<br>FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE)<br>MOBILE (5.463)<br><br>5.462A | EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH)<br>FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>METEOROLOGICAL-SATELLITE (EARTH-TO-SPACE)<br>MOBILE (5.463)<br><br>5.462A ECA36 | Earth exploration-satellite  |                                                                        |            | Satellite payload telemetry                                                                      |
|                                                                                                                                                                        |                                                                                                                                                                              | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                                                                                        |                                                                                                                                                                              | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                                                                                        |                                                                                                                                                                              | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                                                                                        |                                                                                                                                                                              | Land mobile                  |                                                                        |            | Mobile applications within the band 8025-8200 MHz                                                |
|                                                                                                                                                                        |                                                                                                                                                                              | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                                                                                        |                                                                                                                                                                              | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB. On-board aircraft regulation within the band 6.0-8.5 GHz                            |
|                                                                                                                                                                        |                                                                                                                                                                              | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 8215 MHz - 8400 MHz

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| EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH)<br>FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE (5.463)<br><br>5.462A | EARTH EXPLORATION-SATELLITE (SPACE-TO-EARTH)<br>FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br><br>5.462A 5.463 | Earth exploration-satellite  |                                                                        |            | Satellite payload telemetry                                                                      |
|                                                                                                                           |                                                                                                               | Fixed                        | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                   |
|                                                                                                                           |                                                                                                               | LPR                          | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                  |
|                                                                                                                           |                                                                                                               | Land military systems        |                                                                        |            |                                                                                                  |
|                                                                                                                           |                                                                                                               | Radio astronomy              |                                                                        |            | Continuum observations, VLBI (used by SRS)                                                       |
|                                                                                                                           |                                                                                                               | Satellite systems (military) |                                                                        |            |                                                                                                  |
|                                                                                                                           |                                                                                                               | UWB applications             | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                  |
|                                                                                                                           |                                                                                                               | Video PMSE                   | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link. |

## 8400 MHz - 8500 MHz

|                                                                               |                                                  |                  |                                                                        |            |                                                                                                                                       |
|-------------------------------------------------------------------------------|--------------------------------------------------|------------------|------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE RESEARCH (5.465 5.466) | FIXED<br>SPACE RESEARCH (5.465)<br>Radiolocation | Fixed            | ECC/REC/(02)06                                                         | EN 302 217 | Point-to-point                                                                                                                        |
|                                                                               |                                                  | LPR              | ECC/DEC/(11)02,<br>ERC/REC 70-03                                       | EN 302 729 |                                                                                                                                       |
|                                                                               |                                                  | Space research   |                                                                        |            | Satellite payload telemetry. The band 8400-8450 MHz is limited to deep space applications. Continuum observations, VLBI (used by SRS) |
|                                                                               |                                                  | UWB applications | ECC/DEC/(06)04,<br>ECC/DEC/(07)01,<br>ECC/DEC/(12)03,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB as well as on-board aircraft regulation within the band 6.0-8.5 GHz                                                       |
|                                                                               |                                                  | Video PMSE       | ERC/REC 25-10                                                          | EN 302 064 | Cordless Cameras; Portable video links; Mobile video links; Temporary point-to-point video link.                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 8500 MHz - 8550 MHz

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|------------------------------|------------------------------------|-------------------------------|----------------------------------|------------|-------------------------------------------|
| RADIOLOCATION<br>5.468 5.469 | RADIOLOCATION<br>5.469 ECA24 ECA36 | Aeronautical military systems |                                  |            |                                           |
|                              |                                    | Aeronautical navigation       |                                  | EN 303 064 | Civil and military e.g. airfield approach |
|                              |                                    | Radiolocation (civil)         |                                  | EN 303 135 | Shipborne, land and airborne surveillance |
|                              |                                    | Radiolocation (military)      |                                  |            | Shipborne, land and airborne surveillance |
|                              |                                    | TLPR                          | ERC/REC 70-03                    | EN 302 372 |                                           |
|                              |                                    | UWB applications              | ECC/DEC/(06)04,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB                               |

## 8550 MHz - 8650 MHz

|                                                                                                        |                                                                                                              |                               |                                  |            |                                           |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|------------|-------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>5.468 5.469 5.469A | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>5.469 5.469A ECA24 ECA36 | Active sensors (satellite)    |                                  |            |                                           |
|                                                                                                        |                                                                                                              | Aeronautical military systems |                                  |            |                                           |
|                                                                                                        |                                                                                                              | Aeronautical navigation       |                                  | EN 303 064 | Civil and military e.g. airfield approach |
|                                                                                                        |                                                                                                              | Radiolocation (civil)         |                                  | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                                                                        |                                                                                                              | Radiolocation (military)      |                                  |            | Shipborne, land and airborne surveillance |
|                                                                                                        |                                                                                                              | TLPR                          | ERC/REC 70-03                    | EN 302 372 |                                           |
|                                                                                                        |                                                                                                              | UWB applications              | ECC/DEC/(06)04,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 8650 MHz - 8750 MHz

|                              |                                    |                               |                                  |            |                                           |
|------------------------------|------------------------------------|-------------------------------|----------------------------------|------------|-------------------------------------------|
| RADIOLOCATION<br>5.468 5.469 | RADIOLOCATION<br>5.469 ECA24 ECA36 | Aeronautical military systems |                                  |            |                                           |
|                              |                                    | Aeronautical navigation       |                                  | EN 303 064 | Civil and military e.g. airfield approach |
|                              |                                    | Radiolocation (civil)         |                                  | EN 303 135 | Shipborne, land and airborne surveillance |
|                              |                                    | Radiolocation (military)      |                                  |            | Shipborne, land and airborne surveillance |
|                              |                                    | TLPR                          | ERC/REC 70-03                    | EN 302 372 |                                           |
|                              |                                    | UWB applications              | ECC/DEC/(06)04,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB                               |

## 8750 MHz - 8850 MHz

|                                                                |                                                                                              |                               |                                  |            |                                           |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|------------|-------------------------------------------|
| AERONAUTICAL RADIONAVIGATION (5.470)<br>RADIOLOCATION<br>5.471 | AERONAUTICAL RADIONAVIGATION (5.470)<br>RADIOLOCATION<br>Space research<br>5.471 ECA24 ECA36 | Aeronautical military systems |                                  |            |                                           |
|                                                                |                                                                                              | Aeronautical navigation       |                                  | EN 303 064 | Civil and military e.g. airfield approach |
|                                                                |                                                                                              | Radiolocation (civil)         |                                  | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                                |                                                                                              | Radiolocation (military)      |                                  |            | Shipborne, land and airborne surveillance |
|                                                                |                                                                                              | TLPR                          | ERC/REC 70-03                    | EN 302 372 |                                           |
|                                                                |                                                                                              | UWB applications              | ECC/DEC/(06)04,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB                               |

## 8850 MHz - 9000 MHz

|                                                            |                                                                                          |                               |                                  |            |                                           |
|------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|------------|-------------------------------------------|
| MARITIME RADIONAVIGATION (5.472)<br>RADIOLOCATION<br>5.473 | MARITIME RADIONAVIGATION (5.472)<br>RADIOLOCATION<br>Space research<br>5.473 ECA24 ECA36 | Aeronautical military systems |                                  |            |                                           |
|                                                            |                                                                                          | Aeronautical navigation       |                                  | EN 303 064 | Civil and military e.g. airfield approach |
|                                                            |                                                                                          | Radiolocation (civil)         |                                  | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                            |                                                                                          | Radiolocation (military)      |                                  |            | Shipborne, land and airborne surveillance |
|                                                            |                                                                                          | TLPR                          | ERC/REC 70-03                    | EN 302 372 |                                           |
|                                                            |                                                                                          | UWB applications              | ECC/DEC/(06)04,<br>ERC/REC 70-03 | EN 302 065 | Generic UWB                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 9000 MHz - 9200 MHz

|                                                                              |                                                                                                            |                               |               |                           |                                                                        |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------|---------------|---------------------------|------------------------------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>(5.337)<br>Radiolocation<br><br>5.471 5.473A | AERONAUTICAL RADIONAVIGATION<br>(5.337)<br>RADIOLOCATION<br>Space research<br><br>5.471 5.473A ECA24 ECA36 | Aeronautical military systems |               |                           |                                                                        |
|                                                                              |                                                                                                            | Aeronautical navigation       |               | EN 303 064                | Civil and military e.g. airfield approach                              |
|                                                                              |                                                                                                            | Radiolocation (civil)         |               | EN 303 135,<br>EN 303 213 | Shipborne, land and airborne surveillance. EN 303 213-1 X-band sensors |
|                                                                              |                                                                                                            | Radiolocation (military)      |               |                           | Shipborne, land and airborne surveillance                              |
|                                                                              |                                                                                                            | TLPR                          | ERC/REC 70-03 | EN 302 372                |                                                                        |

## 9200 MHz - 9300 MHz

|                                                                                                                                      |                                                                                                                                                                    |                                 |               |            |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|------------|-------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(5.474A 5.474B 5.474C)<br>MARITIME RADIONAVIGATION (5.472)<br>RADIOLOCATION<br><br>5.473 5.474 5.474D | EARTH EXPLORATION-SATELLITE<br>(5.474A 5.474B 5.474C)<br>MARITIME RADIONAVIGATION (5.472)<br>RADIOLOCATION<br>Space research<br><br>5.473 5.474 5.474D ECA24 ECA36 | Aeronautical military systems   |               |            |                                           |
|                                                                                                                                      |                                                                                                                                                                    | Aeronautical navigation         |               | EN 303 064 | Civil and military e.g. airfield approach |
|                                                                                                                                      |                                                                                                                                                                    | Radiodetermination applications | ERC/REC 70-03 | EN 300 440 | Within the band 9200-9975 MHz;            |
|                                                                                                                                      |                                                                                                                                                                    | Radiolocation (civil)           |               | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                                                                                                      |                                                                                                                                                                    | Radiolocation (military)        |               |            | Shipborne, land and airborne surveillance |
|                                                                                                                                      |                                                                                                                                                                    | Synthetic aperture radar        |               |            |                                           |
|                                                                                                                                      |                                                                                                                                                                    | TLPR                            | ERC/REC 70-03 | EN 302 372 |                                           |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 9300 MHz - 9500 MHz

|                                                                                                                                                           |                                                                                                                                                                          |                                 |               |                                                                        |                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION (5.475)<br>SPACE RESEARCH (ACTIVE)<br><br>5.427 5.474 5.475 5.475A 5.475B 5.476A | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION (5.475)<br>SPACE RESEARCH (ACTIVE)<br><br>5.427 5.474 5.475 5.475A 5.475B 5.476A<br>ECA24 ECA36 | Aeronautical military systems   |               |                                                                        |                                                                         |
|                                                                                                                                                           |                                                                                                                                                                          | Aeronautical navigation         |               | EN 303 064                                                             | Civil and military e.g. airfield approach                               |
|                                                                                                                                                           |                                                                                                                                                                          | Radiodetermination applications | ERC/REC 70-03 | EN 300 440                                                             | Within the band 9200-9975 MHz;                                          |
|                                                                                                                                                           |                                                                                                                                                                          | Radiolocation (civil)           |               | EN 302 194,<br>EN 302 248,<br>EN 302 752,<br>EN 303 135,<br>EN 303 213 | Shipborne, land and airborne surveillance EN 303 213-6-1 X-band sensors |
|                                                                                                                                                           |                                                                                                                                                                          | Radiolocation (military)        |               |                                                                        | Shipborne, land and airborne surveillance                               |
|                                                                                                                                                           |                                                                                                                                                                          | Satellite systems (military)    |               |                                                                        |                                                                         |
|                                                                                                                                                           |                                                                                                                                                                          | TLPR                            | ERC/REC 70-03 | EN 302 372                                                             |                                                                         |
|                                                                                                                                                           |                                                                                                                                                                          | Weather radar                   |               | EN 303 347                                                             | Shipborne, land and airborne surveillance                               |

## 9500 MHz - 9800 MHz

|                                                                                                                   |                                                                                                            |                                 |               |            |                                           |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|------------|-------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>RADIONAVIGATION<br>SPACE RESEARCH (ACTIVE)<br><br>5.476A | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br><br>5.476A ECA24 ECA36 | Active sensors (satellite)      |               |            |                                           |
|                                                                                                                   |                                                                                                            | Aeronautical military systems   |               |            |                                           |
|                                                                                                                   |                                                                                                            | Aeronautical navigation         |               | EN 303 064 | Civil and military e.g. airfield approach |
|                                                                                                                   |                                                                                                            | Radiodetermination applications | ERC/REC 70-03 | EN 300 440 | Within the band 9200-9975 MHz             |
|                                                                                                                   |                                                                                                            | Radiolocation (civil)           |               | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                                                                                   |                                                                                                            | Radiolocation (military)        |               |            | Shipborne, land and airborne surveillance |
|                                                                                                                   |                                                                                                            | Satellite systems (military)    |               |            |                                           |
|                                                                                                                   |                                                                                                            | TLPR                            | ERC/REC 70-03 | EN 302 372 |                                           |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 9800 MHz - 9900 MHz

|                                                                                                                            |                                                                                                                         |                                 |               |            |                                           |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|------------|-------------------------------------------|
| RADIOLOCATION<br>Earth exploration-satellite (active)<br>Fixed<br>Space research (active)<br><br>5.477 5.478 5.478A 5.478B | RADIOLOCATION<br>Earth exploration-satellite (active)<br>Space research (active)<br><br>5.478 5.478A 5.478B ECA24 ECA36 | Aeronautical military systems   |               |            |                                           |
|                                                                                                                            |                                                                                                                         | Aeronautical navigation         |               | EN 303 064 | Civil and military e.g. airfield approach |
|                                                                                                                            |                                                                                                                         | Radiodetermination applications | ERC/REC 70-03 | EN 300 440 | Within the band 9200-9975 MHz;            |
|                                                                                                                            |                                                                                                                         | Radiolocation (civil)           |               | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                                                                                            |                                                                                                                         | Radiolocation (military)        |               |            | Shipborne, land and airborne surveillance |
|                                                                                                                            |                                                                                                                         | Satellite systems (military)    |               |            |                                           |
|                                                                                                                            |                                                                                                                         | TLPR                            | ERC/REC 70-03 | EN 302 372 |                                           |

## 9900 MHz - 10000 MHz

|                                                                                                          |                                                                                                          |                                 |               |            |                                           |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------|---------------|------------|-------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(5.474A 5.474B 5.474C)<br>RADIOLOCATION<br>Fixed<br><br>5.477 5.478 5.479 | EARTH EXPLORATION-SATELLITE<br>(5.474A 5.474B 5.474C)<br>RADIOLOCATION<br>Fixed<br><br>5.477 5.478 5.479 | Aeronautical military systems   |               |            |                                           |
|                                                                                                          |                                                                                                          | Aeronautical navigation         |               | EN 303 064 | Civil and military e.g. Airfield approach |
|                                                                                                          |                                                                                                          | Radiodetermination applications | ERC/REC 70-03 | EN 300 440 | Within the band 9200-9975 MHz             |
|                                                                                                          |                                                                                                          | Radiolocation (civil)           |               | EN 303 135 | Shipborne, land and airborne surveillance |
|                                                                                                          |                                                                                                          | Radiolocation (military)        |               |            | Shipborne, land and airborne surveillance |
|                                                                                                          |                                                                                                          | Satellite systems (military)    |               |            |                                           |
|                                                                                                          |                                                                                                          | Synthetic aperture radar        |               |            |                                           |
|                                                                                                          |                                                                                                          | TLPR                            | ERC/REC 70-03 | EN 302 372 |                                           |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 10000 MHz - 10400 MHz

|                                                                                                                          |                                                                                                                                       |                               |               |            |                                                                                |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------|------------|--------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(5.474A 5.474B 5.474C)<br>FIXED<br>MOBILE<br>RADIOLOCATION<br>Amateur<br><br>5.474D 5.479 | EARTH EXPLORATION-SATELLITE<br>(5.474A 5.474B 5.474C)<br>FIXED<br>MOBILE<br>RADIOLOCATION<br>Amateur<br><br>5.474D 5.479 ECA17A ECA36 | Aeronautical military systems |               |            |                                                                                |
|                                                                                                                          |                                                                                                                                       | Amateur                       |               | EN 301 783 | Within the band 10-10.5 GHz                                                    |
|                                                                                                                          |                                                                                                                                       | FWA                           |               | EN 302 326 | Including Point-to-Multipoint                                                  |
|                                                                                                                          |                                                                                                                                       | Fixed                         | ERC/REC 12-05 | EN 302 217 |                                                                                |
|                                                                                                                          |                                                                                                                                       | Land military systems         |               |            |                                                                                |
|                                                                                                                          |                                                                                                                                       | Maritime military systems     |               |            |                                                                                |
|                                                                                                                          |                                                                                                                                       | Radiolocation (civil)         |               |            |                                                                                |
|                                                                                                                          |                                                                                                                                       | Radiolocation (military)      |               |            |                                                                                |
|                                                                                                                          |                                                                                                                                       | Synthetic aperture radar      |               |            |                                                                                |
|                                                                                                                          |                                                                                                                                       | TLPR                          | ERC/REC 70-03 | EN 302 372 |                                                                                |
|                                                                                                                          |                                                                                                                                       | Video PMSE                    | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Portable video links;<br>Temporary point-to-point video link |

## 10400 MHz - 10450 MHz

|                                             |                                                                       |                               |               |            |                                                                                |
|---------------------------------------------|-----------------------------------------------------------------------|-------------------------------|---------------|------------|--------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>RADIOLOCATION<br>Amateur | FIXED<br>RADIOLOCATION<br>Amateur<br>Mobile<br><br>ECA17 ECA17A ECA36 | Aeronautical military systems |               |            |                                                                                |
|                                             |                                                                       | Amateur                       |               | EN 301 783 | Within the band 10-10.5 GHz                                                    |
|                                             |                                                                       | Land military systems         |               |            |                                                                                |
|                                             |                                                                       | Maritime military systems     |               |            |                                                                                |
|                                             |                                                                       | Radiolocation (civil)         |               |            | Low power radars in certain subbands                                           |
|                                             |                                                                       | Radiolocation (military)      |               |            |                                                                                |
|                                             |                                                                       | TLPR                          | ERC/REC 70-03 | EN 302 372 |                                                                                |
|                                             |                                                                       | Video PMSE                    | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Portable video links;<br>Temporary point-to-point video link |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 10450 MHz - 10.5 GHz

|                                                            |                                                                                                        |                               |               |            |                                                                                |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------|---------------|------------|--------------------------------------------------------------------------------|
| RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.481 | FIXED<br>MOBILE<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.481 ECA17 ECA17A ECA23 ECA36 | Aeronautical military systems |               |            |                                                                                |
|                                                            |                                                                                                        | Amateur                       |               | EN 301 783 | Within the band 10-10.5 GHz                                                    |
|                                                            |                                                                                                        | Amateur-satellite             |               |            |                                                                                |
|                                                            |                                                                                                        | Land military systems         |               |            |                                                                                |
|                                                            |                                                                                                        | Maritime military systems     |               |            |                                                                                |
|                                                            |                                                                                                        | Radiolocation (civil)         |               |            |                                                                                |
|                                                            |                                                                                                        | Radiolocation (military)      |               |            |                                                                                |
|                                                            |                                                                                                        | TLPR                          | ERC/REC 70-03 | EN 302 372 |                                                                                |
|                                                            |                                                                                                        | Video PMSE                    | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Portable video links;<br>Temporary point-to-point video link |

## 10.5 GHz - 10.55 GHz

|                                  |                                                |                                 |               |                           |                                                                                |
|----------------------------------|------------------------------------------------|---------------------------------|---------------|---------------------------|--------------------------------------------------------------------------------|
| FIXED<br>MOBILE<br>Radiolocation | FIXED<br>MOBILE<br>Radiolocation<br><br>ECA17A | Fixed                           | ERC/REC 12-05 | EN 302 217,<br>EN 302 326 | Including Point-to-Multipoint                                                  |
|                                  |                                                | Radiodetermination applications | ERC/REC 70-03 | EN 300 440                | Within the band 10.5-10.6 GHz;                                                 |
|                                  |                                                | TLPR                            | ERC/REC 70-03 | EN 302 372                |                                                                                |
|                                  |                                                | Video PMSE                      | ERC/REC 25-10 | EN 302 064                | Cordless Cameras; Portable video links;<br>Temporary point-to-point video link |

## 10.55 GHz - 10.6 GHz

|                                                                |                                                                              |                                 |               |                           |                                                                                |
|----------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------|---------------|---------------------------|--------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Radiolocation<br><br>ECA17A | Fixed                           | ERC/REC 12-05 | EN 302 217,<br>EN 302 326 | Including Point-to-Multipoint                                                  |
|                                                                |                                                                              | Radiodetermination applications | ERC/REC 70-03 | EN 300 440                | Within the band 10.5-10.6 GHz,                                                 |
|                                                                |                                                                              | TLPR                            | ERC/REC 70-03 | EN 302 372                |                                                                                |
|                                                                |                                                                              | Video PMSE                      | ERC/REC 25-10 | EN 302 064                | Cordless Cameras; Portable video links;<br>Temporary point-to-point video link |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 10.6 GHz - 10.68 GHz

|                                                                                                                                                                           |                                                                                                                                                                                 |                             |                                  |                           |                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------|---------------------------|--------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Mobile except aeronautical mobile<br>Radiolocation<br>5.149 5.482 5.482A | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Mobile except aeronautical mobile<br>Radiolocation<br>5.149 5.482 5.482A ECA17 | Fixed                       | ECC/DEC/(10)01,<br>ERC/REC 12-05 | EN 302 217,<br>EN 302 326 | Including Point-to-Multipoint                                                  |
|                                                                                                                                                                           |                                                                                                                                                                                 | Passive sensors (satellite) | ECC/DEC/(10)01                   |                           | Surface emissivity and precipitation measurements                              |
|                                                                                                                                                                           |                                                                                                                                                                                 | Radio astronomy             |                                  |                           | Continuum observations, VLBI                                                   |
|                                                                                                                                                                           |                                                                                                                                                                                 | Video PMSE                  | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Portable video links;<br>Temporary point-to-point video link |

## 10.68 GHz - 10.7 GHz

|                                                                                                     |                                                                                               |                             |  |  |                                                  |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------|--|--|--------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.483 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 | Passive sensors (satellite) |  |  | Surface emissivity and precipitation measurement |
|                                                                                                     |                                                                                               | Radio astronomy             |  |  | Continuum observations, VLBI                     |

## 10.7 GHz - 10.95 GHz

|                                                                                                     |                                                                                                                                          |                    |                                   |                                                         |                                                                                   |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.441)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.441)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Mobile-Satellite (space-to-Earth) | AES                | ECC/DEC/(05)11,<br>ECC/DEC/(19)04 | EN 302 186                                              |                                                                                   |
|                                                                                                     |                                                                                                                                          | ESIM               | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                                                                   |
|                                                                                                     |                                                                                                                                          | ESV                | ECC/DEC/(05)10                    | EN 302 340                                              |                                                                                   |
|                                                                                                     |                                                                                                                                          | FSS Earth stations | ERC/DEC/(00)08                    | EN 301 427,<br>EN 301 430,<br>EN 302 448                | Within the band 10.7-10.95/11.2-11.45 GHz in accordance with App 30B of RR - VSAT |
|                                                                                                     |                                                                                                                                          | Fixed              | ERC/DEC/(00)08,<br>ERC/REC 12-06  | EN 302 217                                              | Limited to high capacity fixed links                                              |
|                                                                                                     |                                                                                                                                          | HEST               | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |                                                                                   |
|                                                                                                     |                                                                                                                                          | NGSO FSS           | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                                                                   |
|                                                                                                     |                                                                                                                                          | VSAT               | ECC/DEC/(03)04                    | EN 301 428                                              | SNG                                                                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 10.95 GHz - 11.2 GHz

|                                                                                                             |                                                                                                             |          |                                   |                                                         |                                      |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------|-----------------------------------|---------------------------------------------------------|--------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.484A 5.484B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.484A 5.484B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | AES      | ECC/DEC/(05)11,<br>ECC/DEC/(19)04 | EN 302 186                                              |                                      |
|                                                                                                             |                                                                                                             | ESIM     | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                      |
|                                                                                                             |                                                                                                             | ESV      | ECC/DEC/(05)10                    | EN 302 340                                              |                                      |
|                                                                                                             |                                                                                                             | Fixed    | ERC/DEC/(00)08,<br>ERC/REC 12-06  | EN 302 217                                              | Limited to high capacity fixed links |
|                                                                                                             |                                                                                                             | HEST     | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |                                      |
|                                                                                                             |                                                                                                             | NGSO FSS | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                      |
|                                                                                                             |                                                                                                             | VSAT     | ECC/DEC/(03)04                    | EN 301 428                                              | SNG                                  |

## 11.2 GHz - 11.45 GHz

|                                                                                                     |                                                                                                     |          |                                   |                                                         |                                      |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------|-----------------------------------|---------------------------------------------------------|--------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.441)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.441)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | AES      | ECC/DEC/(05)11,<br>ECC/DEC/(19)04 | EN 302 186                                              |                                      |
|                                                                                                     |                                                                                                     | ESIM     | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                      |
|                                                                                                     |                                                                                                     | ESV      | ECC/DEC/(05)10                    | EN 302 340                                              |                                      |
|                                                                                                     |                                                                                                     | Fixed    | ERC/DEC/(00)08,<br>ERC/REC 12-06  | EN 302 217                                              | Limited to high capacity fixed links |
|                                                                                                     |                                                                                                     | HEST     | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |                                      |
|                                                                                                     |                                                                                                     | NGSO FSS | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                      |
|                                                                                                     |                                                                                                     | VSAT     | ECC/DEC/(03)04                    | EN 301 428                                              | SNG                                  |

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## 11.45 GHz - 11.7 GHz

|                                                                                                             |                                                                                                             |          |                                   |                                                         |                                      |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------|-----------------------------------|---------------------------------------------------------|--------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.484A 5.484B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.484)<br>FIXED-SATELLITE (5.484A 5.484B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE | AES      | ECC/DEC/(05)11,<br>ECC/DEC/(19)04 | EN 302 186                                              |                                      |
|                                                                                                             |                                                                                                             | ESIM     | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                      |
|                                                                                                             |                                                                                                             | ESV      | ECC/DEC/(05)10                    | EN 302 340                                              |                                      |
|                                                                                                             |                                                                                                             | Fixed    | ERC/DEC/(00)08,<br>ERC/REC 12-06  | EN 302 217                                              | Limited to high capacity fixed links |
|                                                                                                             |                                                                                                             | HEST     | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |                                      |
|                                                                                                             |                                                                                                             | NGSO FSS | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                      |
|                                                                                                             |                                                                                                             | VSAT     | ECC/DEC/(03)04                    | EN 301 428                                              | SNG                                  |

## 11.7 GHz - 12.5 GHz

|                                                                                                                  |                                                                                                  |                          |                                   |                                                         |                                                                      |
|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------|---------------------------------------------------------|----------------------------------------------------------------------|
| BROADCASTING<br>BROADCASTING-SATELLITE (5.492)<br>FIXED<br>Mobile except aeronautical mobile<br><br>5.487 5.487A | BROADCASTING-SATELLITE (5.492)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.487 5.487A ECA28 | AES                      | ECC/DEC/(19)04                    | EN 302 186                                              |                                                                      |
|                                                                                                                  |                                                                                                  | Broadcasting (satellite) | ERC/DEC/(00)08                    | EN 302 340,<br>EN 302 448                               | In accordance with App 30 of RR. SIT within the band 12.4 - 12.5 GHz |
|                                                                                                                  |                                                                                                  | ESIM                     | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                                                      |
|                                                                                                                  |                                                                                                  | HEST                     | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |                                                                      |
|                                                                                                                  |                                                                                                  | NGSO FSS                 | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 12.5 GHz - 12.75 GHz

|                                                                                              |                                                                                  |          |                                   |                                                         |  |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------|-----------------------------------|---------------------------------------------------------|--|
| FIXED-SATELLITE (EARTH-TO-SPACE)<br>FIXED-SATELLITE (5.484A 5.484B)<br><br>5.494 5.495 5.496 | FIXED-SATELLITE (EARTH-TO-SPACE)<br>FIXED-SATELLITE (5.484A 5.484B)<br><br>5.496 | AES      | ECC/DEC/(05)11,<br>ECC/DEC/(19)04 | EN 302 186                                              |  |
|                                                                                              |                                                                                  | ESIM     | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |  |
|                                                                                              |                                                                                  | ESV      | ECC/DEC/(05)10                    | EN 302 340                                              |  |
|                                                                                              |                                                                                  | HEST     | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |  |
|                                                                                              |                                                                                  | NGSO FSS | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |  |

## 12.75 GHz - 13.25 GHz

|                                                                                                   |                                         |                    |                |            |  |
|---------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|----------------|------------|--|
| FIXED<br>FIXED-SATELLITE (5.441 5.496A)<br>MOBILE<br>Space research (deep space) (space-to-Earth) | FIXED<br>FIXED-SATELLITE (5.441 5.496A) | AES                | ECC/DEC/(19)04 | EN 302 186 |  |
|                                                                                                   |                                         | FSS Earth stations |                | EN 301 430 |  |
|                                                                                                   |                                         | Fixed              | ERC/REC 12-02  | EN 302 217 |  |

## 13.25 GHz - 13.4 GHz

|                                                                                                                             |                                                                                                                             |                                  |  |  |                                                  |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------|--|--|--------------------------------------------------|
| AERONAUTICAL RADIONAVIGATION (5.497)<br>EARTH EXPLORATION-SATELLITE (ACTIVE)<br>SPACE RESEARCH (ACTIVE)<br><br>5.498A 5.499 | AERONAUTICAL RADIONAVIGATION (5.497)<br>EARTH EXPLORATION-SATELLITE (ACTIVE)<br>SPACE RESEARCH (ACTIVE)<br><br>5.498A ECA26 | Active sensors (satellite)       |  |  | Altimeters, scatterometers, precipitation radars |
|                                                                                                                             |                                                                                                                             | Airborne doppler navigation aids |  |  |                                                  |
|                                                                                                                             |                                                                                                                             | Maritime radar                   |  |  | Ship berthing radars                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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### 13.4 GHz - 13.65 GHz

|                                                                                                                                                                                                                              |                                                                                                                                                      |                                  |               |            |                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------|------------|--------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>FIXED-SATELLITE (5.499A 5.499B)<br>RADIOLOCATION<br>SPACE RESEARCH (5.499C 5.499D)<br>Standard frequency and time signal-satellite (Earth-to-space)<br><br>5.499E 5.500 5.501 5.501B | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>FIXED-SATELLITE (5.499A 5.499B)<br>RADIOLOCATION<br>SPACE RESEARCH (5.499C 5.499D)<br><br>5.501B ECA26 ECA36 | -                                |               |            | Data relay satellites                            |
|                                                                                                                                                                                                                              |                                                                                                                                                      | Active sensors (satellite)       |               |            | Altimeters, scatterometers, precipitation radars |
|                                                                                                                                                                                                                              |                                                                                                                                                      | Airborne doppler navigation aids |               |            |                                                  |
|                                                                                                                                                                                                                              |                                                                                                                                                      | FSS Earth stations               |               |            |                                                  |
|                                                                                                                                                                                                                              |                                                                                                                                                      | Maritime radar                   |               |            | Ship berthing radars                             |
|                                                                                                                                                                                                                              |                                                                                                                                                      | Radiodetermination applications  | ERC/REC 70-03 | EN 300 440 | Within the band 13.4-14.0 GHz                    |
|                                                                                                                                                                                                                              |                                                                                                                                                      | Radiolocation (military)         |               |            |                                                  |

### 13.65 GHz - 13.75 GHz

|                                                                                                                                                                                   |                                                                                                            |                                  |               |            |                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------|---------------|------------|--------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (5.501A)<br>Standard frequency and time signal-satellite (Earth-to-space)<br><br>5.499 5.500 5.501 5.501B | EARTH EXPLORATION-SATELLITE (ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (5.501A)<br><br>5.501B ECA26 ECA36 | -                                |               |            | Data relay satellites                            |
|                                                                                                                                                                                   |                                                                                                            | Active sensors (satellite)       |               |            | Altimeters, scatterometers, precipitation radars |
|                                                                                                                                                                                   |                                                                                                            | Airborne doppler navigation aids |               |            |                                                  |
|                                                                                                                                                                                   |                                                                                                            | Maritime radar                   |               |            | Ship berthing radars                             |
|                                                                                                                                                                                   |                                                                                                            | Radiodetermination applications  | ERC/REC 70-03 | EN 300 440 | Within the band 13.4-14.0 GHz                    |
|                                                                                                                                                                                   |                                                                                                            | Radiolocation (military)         |               |            |                                                  |

### 13.75 GHz - 14 GHz

|                                                                                                                                                                                                  |                                                                                            |                                 |               |            |                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------|---------------|------------|-------------------------------------------------|
| FIXED-SATELLITE (5.484A)<br>RADIOLOCATION<br>Earth exploration-satellite<br>Space research<br>Standard frequency and time signal-satellite (Earth-to-space)<br><br>5.499 5.500 5.501 5.502 5.503 | FIXED-SATELLITE (5.484A)<br>RADIOLOCATION<br>Space research<br><br>5.502 5.503 ECA26 ECA36 | -                               |               |            | Data relay satellites                           |
|                                                                                                                                                                                                  |                                                                                            | FSS Earth stations              |               | EN 301 430 | minimum antenna size imposed according to 5.502 |
|                                                                                                                                                                                                  |                                                                                            | Maritime radar                  |               |            | Navigation radars, ship berthing radars         |
|                                                                                                                                                                                                  |                                                                                            | Passive sensors (satellite)     |               |            | Future VLBI measurements                        |
|                                                                                                                                                                                                  |                                                                                            | Radiodetermination applications | ERC/REC 70-03 | EN 300 440 | Within the band 13.4-14.0 GHz                   |
|                                                                                                                                                                                                  |                                                                                            | Radiolocation (military)        |               |            |                                                 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 14 GHz - 14.25 GHz

|                                                                                                                                                                           |                                                                                                                                         |                    |                                   |                                                         |                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|
| FIXED-SATELLITE (5.457A 5.457B<br>5.484A 5.506 5.506B 5.484B)<br>RADIONAVIGATION (5.504)<br>Mobile-Satellite (5.504B 5.504C 5.506A)<br>Space research<br><br>5.504A 5.505 | FIXED-SATELLITE (5.457A 5.457B<br>5.484A 5.484B 5.506 5.506B)<br>Mobile-Satellite (5.504B 5.504C 5.506A)<br>Space research<br><br>5.504 | AES                | ECC/DEC/(05)11                    | EN 302 186                                              |                                                                                        |
|                                                                                                                                                                           |                                                                                                                                         | ESIM               | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                                                                        |
|                                                                                                                                                                           |                                                                                                                                         | ESV                | ECC/DEC/(05)10                    | EN 302 340                                              |                                                                                        |
|                                                                                                                                                                           |                                                                                                                                         | HEST               | ECC/DEC/(06)03                    | EN 301 428,<br>EN 301 459                               |                                                                                        |
|                                                                                                                                                                           |                                                                                                                                         | MSS Earth stations |                                   | EN 301 427,<br>EN 302 977                               |                                                                                        |
|                                                                                                                                                                           |                                                                                                                                         | NGSO FSS           | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                                                                        |
|                                                                                                                                                                           |                                                                                                                                         | VSAT               | ECC/DEC/(03)04,<br>ERC/REC 13-03  | EN 301 428,<br>EN 301 430                               | Low density carriers, including VSATs and digital SNG, are encouraged to use this band |

## 14.25 GHz - 14.3 GHz

|                                                                                                                                                                                 |                                                                                                                                         |                    |                                   |                                                         |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|---------------------------------------------------------|-----|
| FIXED-SATELLITE (5.457A 5.457B<br>5.484A 5.484B 5.506 5.506B)<br>RADIONAVIGATION (5.504)<br>Mobile-Satellite (5.504B 5.506A 5.508A)<br>Space research<br><br>5.504A 5.505 5.508 | FIXED-SATELLITE (5.457A 5.457B<br>5.484A 5.484B 5.506 5.506B)<br>Mobile-Satellite (5.504B 5.506A 5.508A)<br>Space research<br><br>5.504 | AES                | ECC/DEC/(05)11                    | EN 302 186                                              |     |
|                                                                                                                                                                                 |                                                                                                                                         | ESIM               | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |     |
|                                                                                                                                                                                 |                                                                                                                                         | ESV                | ECC/DEC/(05)10                    | EN 302 340                                              |     |
|                                                                                                                                                                                 |                                                                                                                                         | MSS Earth stations |                                   | EN 301 427,<br>EN 302 977                               |     |
|                                                                                                                                                                                 |                                                                                                                                         | NGSO FSS           | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |     |
|                                                                                                                                                                                 |                                                                                                                                         | VSAT               | ECC/DEC/(03)04,<br>ERC/REC 13-03  | EN 301 428,<br>EN 301 430                               | SNG |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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### 14.3 GHz - 14.4 GHz

|                                                                                                                                                                                                   |                                                                                                       |                    |                                   |                                                         |                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.506 5.457A 5.506B 5.484A 5.457B 5.484B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Mobile-Satellite (5.504B 5.506A 5.509A)<br>Radionavigation-satellite<br><br>5.504A | FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B)<br>Mobile-Satellite (5.504B 5.506A 5.509A) | AES                | ECC/DEC/(05)11                    | EN 302 186                                              |                                                                                 |
|                                                                                                                                                                                                   |                                                                                                       | ESIM               | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                                                                 |
|                                                                                                                                                                                                   |                                                                                                       | ESV                | ECC/DEC/(05)10                    | EN 302 340                                              |                                                                                 |
|                                                                                                                                                                                                   |                                                                                                       | FSS Earth stations |                                   | EN 302 340                                              | Fixed links to be coordinated with Fixed Satellite Services on a national basis |
|                                                                                                                                                                                                   |                                                                                                       | MSS Earth stations |                                   | EN 301 427,<br>EN 302 977                               |                                                                                 |
|                                                                                                                                                                                                   |                                                                                                       | NGSO FSS           | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                                                                 |
|                                                                                                                                                                                                   |                                                                                                       | VSAT               | ECC/DEC/(03)04,<br>ERC/REC 13-03  | EN 301 428,<br>EN 301 430                               | SNG                                                                             |

### 14.4 GHz - 14.47 GHz

|                                                                                                                                                                                                         |                                                                                                                     |                    |                                   |                                                         |                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.457A 5.457B 5.484A 5.506 5.506B 5.484B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Mobile-Satellite (5.504B 5.506A 5.509A)<br>Space research (space-to-Earth)<br><br>5.504A | FIXED-SATELLITE (5.457A 5.457B 5.484A 5.484B 5.506 5.506B)<br>Mobile-Satellite (5.504B 5.506A 5.509A)<br><br>5.504A | AES                | ECC/DEC/(05)11                    | EN 302 186                                              |                                                                                 |
|                                                                                                                                                                                                         |                                                                                                                     | ESIM               | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                                                                 |
|                                                                                                                                                                                                         |                                                                                                                     | ESV                | ECC/DEC/(05)10                    | EN 302 340                                              |                                                                                 |
|                                                                                                                                                                                                         |                                                                                                                     | FSS Earth stations |                                   | EN 302 340                                              | Fixed links to be coordinated with Fixed Satellite Services on a national basis |
|                                                                                                                                                                                                         |                                                                                                                     | MSS Earth stations |                                   | EN 301 427,<br>EN 302 977                               |                                                                                 |
|                                                                                                                                                                                                         |                                                                                                                     | NGSO FSS           | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                                                                 |
|                                                                                                                                                                                                         |                                                                                                                     | VSAT               | ECC/DEC/(03)04,<br>ERC/REC 13-03  | EN 301 428,<br>EN 301 430                               | SNG                                                                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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### 14.47 GHz - 14.5 GHz

|                                                                                                                                                                                        |                                                                                                                         |                    |                                   |                                                         |                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.457A 5.457B 5.484A 5.506 5.506B)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Mobile-Satellite (5.504B 5.506A 5.509A)<br>Radio astronomy<br><br>5.149 5.504A | FIXED-SATELLITE (5.457A 5.484A 5.506)<br>Mobile-Satellite (5.504B 5.506A 5.509A)<br>Radio astronomy<br><br>5.149 5.504A | AES                | ECC/DEC/(05)11                    | EN 302 186                                              |                                                                                |
|                                                                                                                                                                                        |                                                                                                                         | ESIM               | ECC/DEC/(18)04,<br>ECC/DEC/(18)05 | EN 302 448,<br>EN 302 977,<br>EN 303 980,<br>EN 303 981 |                                                                                |
|                                                                                                                                                                                        |                                                                                                                         | ESV                | ECC/DEC/(05)10                    | EN 302 340                                              |                                                                                |
|                                                                                                                                                                                        |                                                                                                                         | FSS Earth stations |                                   | EN 302 340                                              | Fixed links to be coordinated with Fixed Satellite Service on a national basis |
|                                                                                                                                                                                        |                                                                                                                         | MSS Earth stations |                                   | EN 301 427,<br>EN 302 977                               |                                                                                |
|                                                                                                                                                                                        |                                                                                                                         | NGSO FSS           | ECC/DEC/(17)04                    | EN 303 980,<br>EN 303 981                               |                                                                                |
|                                                                                                                                                                                        |                                                                                                                         | Radio astronomy    |                                   |                                                         | Spectral line observations, VLBI                                               |
|                                                                                                                                                                                        |                                                                                                                         | VSAT               | ERC/REC 13-03                     | EN 301 428,<br>EN 301 430                               | SNG                                                                            |

### 14.5 GHz - 14.75 GHz

|                                                                                                          |                                                       |                               |               |            |                                         |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------|---------------|------------|-----------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.510 5.509B 5.509C 5.509D 5.509E 5.509F)<br>MOBILE<br>Space research (5.509G) | FIXED<br>MOBILE<br>Radio astronomy<br><br>ECA20 ECA36 | Aeronautical military systems |               |            |                                         |
|                                                                                                          |                                                       | Fixed                         | ERC/REC 12-07 | EN 302 217 |                                         |
|                                                                                                          |                                                       | Land military systems         |               |            |                                         |
|                                                                                                          |                                                       | Maritime military systems     |               |            |                                         |
|                                                                                                          |                                                       | Radio astronomy               |               |            | VLBI (when compatible with primary use) |

### 14.75 GHz - 14.8 GHz

|                                                                       |                                                       |                               |  |  |                                         |
|-----------------------------------------------------------------------|-------------------------------------------------------|-------------------------------|--|--|-----------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.510)<br>MOBILE<br>Space research (5.509G) | FIXED<br>MOBILE<br>Radio astronomy<br><br>ECA20 ECA36 | Aeronautical military systems |  |  |                                         |
|                                                                       |                                                       | Land military systems         |  |  |                                         |
|                                                                       |                                                       | Maritime military systems     |  |  |                                         |
|                                                                       |                                                       | Radio astronomy               |  |  | VLBI (when compatible with primary use) |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 14.8 GHz - 15.35 GHz

|                                                         |                                                             |                               |               |            |                                         |
|---------------------------------------------------------|-------------------------------------------------------------|-------------------------------|---------------|------------|-----------------------------------------|
| FIXED<br>MOBILE<br>SPACE RESEARCH (5.510A)<br><br>5.339 | FIXED<br>MOBILE<br>Radio astronomy<br><br>5.339 ECA20 ECA36 | Aeronautical military systems |               |            |                                         |
|                                                         |                                                             | Fixed                         | ERC/REC 12-07 | EN 302 217 |                                         |
|                                                         |                                                             | Land military systems         |               |            |                                         |
|                                                         |                                                             | Maritime military systems     |               |            |                                         |
|                                                         |                                                             | Radio astronomy               |               |            | VLBI (when compatible with primary use) |

## 15.35 GHz - 15.4 GHz

|                                                                                                            |                                                                                                      |                             |  |  |                              |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------|--|--|------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.511 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite) |  |  |                              |
|                                                                                                            |                                                                                                      | Radio astronomy             |  |  | Continuum observations, VLBI |

## 15.4 GHz - 15.41 GHz

|                                                               |                                                               |                                  |  |  |                                 |
|---------------------------------------------------------------|---------------------------------------------------------------|----------------------------------|--|--|---------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>RADIOLOCATION (5.511E 5.511F) | RADIOLOCATION (5.511E 5.511F)<br>Aeronautical radionavigation | Airborne doppler navigation aids |  |  | Doppler radar low power sensing |
|                                                               |                                                               | Radiolocation (civil)            |  |  | Ground movement radars          |

## 15.41 GHz - 15.43 GHz

|                                                                                               |                                                               |                                  |  |  |                                 |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------|--|--|---------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>RADIOLOCATION (5.511F 5.511E)<br>Aeronautical mobile (5.511G) | RADIOLOCATION (5.511E 5.511F)<br>Aeronautical radionavigation | Airborne doppler navigation aids |  |  | Doppler radar low power sensing |
|                                                                                               |                                                               | Radiolocation (civil)            |  |  | Ground movement radars          |

## 15.43 GHz - 15.63 GHz

|                                                                                                                                         |                                                                                                                 |                                  |  |  |                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------|--|--|---------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>FIXED-SATELLITE (5.511A)<br>RADIOLOCATION (5.511E 5.511F)<br>Aeronautical mobile (5.511G)<br><br>5.511C | AERONAUTICAL RADIONAVIGATION<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>RADIOLOCATION (5.511E 5.511F)<br><br>5.511C | Airborne doppler navigation aids |  |  | Doppler radar low power sensing |
|                                                                                                                                         |                                                                                                                 | FSS Earth stations               |  |  | MSS feeder links                |
|                                                                                                                                         |                                                                                                                 | Radiolocation (civil)            |  |  | Ground movement radars          |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 15.63 GHz - 15.7 GHz

|                                                                                               |                                                               |                                  |  |  |                                 |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------|--|--|---------------------------------|
| AERONAUTICAL RADIONAVIGATION<br>RADIOLOCATION (5.511E 5.511F)<br>Aeronautical mobile (5.511G) | AERONAUTICAL RADIONAVIGATION<br>RADIOLOCATION (5.511E 5.511F) | Airborne doppler navigation aids |  |  | Doppler radar low power sensing |
|                                                                                               |                                                               | Radiolocation (civil)            |  |  | Ground movement radars          |

### 15.7 GHz - 16.6 GHz

|                              |                        |                          |  |  |  |
|------------------------------|------------------------|--------------------------|--|--|--|
| RADIOLOCATION<br>5.512 5.513 | RADIOLOCATION<br>ECA36 | Radiolocation (military) |  |  |  |
|------------------------------|------------------------|--------------------------|--|--|--|

### 16.6 GHz - 17.1 GHz

|                                                                              |                                                                        |                          |  |  |  |
|------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------|--|--|--|
| RADIOLOCATION<br>Space research (deep space) (Earth-to-space)<br>5.512 5.513 | RADIOLOCATION<br>Space research (deep space) (Earth-to-space)<br>ECA36 | Radiolocation (military) |  |  |  |
|------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------|--|--|--|

### 17.1 GHz - 17.2 GHz

|                              |                                  |                          |               |            |  |
|------------------------------|----------------------------------|--------------------------|---------------|------------|--|
| RADIOLOCATION<br>5.512 5.513 | RADIOLOCATION<br>Mobile<br>ECA36 | GBSAR                    | ERC/REC 70-03 | EN 303 661 |  |
|                              |                                  | Radiolocation (military) |               |            |  |

### 17.2 GHz - 17.3 GHz

|                                                                                                           |                                                                                                               |                          |               |            |  |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------|---------------|------------|--|
| EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>5.512 5.513 5.513A | EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>MOBILE<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>5.513A ECA36 | GBSAR                    | ERC/REC 70-03 | EN 303 661 |  |
|                                                                                                           |                                                                                                               | Radiolocation (military) |               |            |  |

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### 17.3 GHz - 17.7 GHz

|                                                                                          |                                                                                          |                          |                |            |                                                      |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------|----------------|------------|------------------------------------------------------|
| FIXED-SATELLITE (5.516)<br>FIXED-SATELLITE (5.516A 5.516B)<br>Radiolocation<br><br>5.514 | FIXED-SATELLITE (5.516)<br>FIXED-SATELLITE (5.516A 5.516B)<br>Radiolocation<br><br>ECA36 | FSS Earth stations       | ECC/DEC/(05)08 |            | High Density FSS                                     |
|                                                                                          |                                                                                          | Feeder links             |                |            | Feeder links for the BSS service. Appendix 30A of RR |
|                                                                                          |                                                                                          | GSO ESOMPs               | ECC/DEC/(13)01 | EN 303 978 |                                                      |
|                                                                                          |                                                                                          | NGSO ESOMPs              | ECC/DEC/(15)04 | EN 303 979 |                                                      |
|                                                                                          |                                                                                          | Radiolocation (military) |                |            |                                                      |

### 17.7 GHz - 18.1 GHz

|                                                                                      |                                                                            |                    |                               |            |                                                      |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------|-------------------------------|------------|------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.516)<br>FIXED-SATELLITE (5.484A 5.517A 5.517B)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.516 5.517A 5.517B)<br>FIXED-SATELLITE (5.484A) | FSS Earth stations | ERC/DEC/(00)07                |            |                                                      |
|                                                                                      |                                                                            | Feeder links       |                               |            | Feeder links for the BSS service. Appendix 30A of RR |
|                                                                                      |                                                                            | Fixed              | ERC/DEC/(00)07, ERC/REC 12-03 | EN 302 217 |                                                      |
|                                                                                      |                                                                            | GSO ESOMPs         | ECC/DEC/(13)01                | EN 303 978 |                                                      |
|                                                                                      |                                                                            | NGSO ESOMPs        | ECC/DEC/(15)04                | EN 303 979 |                                                      |

### 18.1 GHz - 18.4 GHz

|                                                                                                                                            |                                                                                                                                                                  |                    |                               |            |                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------|------------|----------------------------------|
| FIXED<br>FIXED-SATELLITE (5.520)<br>FIXED-SATELLITE (5.484A 5.517A 5.517B 5.516B)<br>INTER-SATELLITE (5.521A)<br>MOBILE<br><br>5.519 5.521 | FIXED<br>FIXED-SATELLITE (5.520)<br>FIXED-SATELLITE (5.484A 5.517A 5.517B)<br>INTER-SATELLITE (5.521A)<br>METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH)<br><br>5.519 | FSS Earth stations | ERC/DEC/(00)07                |            |                                  |
|                                                                                                                                            |                                                                                                                                                                  | Feeder links       |                               |            | Feeder links for the BSS service |
|                                                                                                                                            |                                                                                                                                                                  | Fixed              | ERC/DEC/(00)07, ERC/REC 12-03 | EN 302 217 |                                  |
|                                                                                                                                            |                                                                                                                                                                  | GSO ESOMPs         | ECC/DEC/(13)01                | EN 303 978 |                                  |
|                                                                                                                                            |                                                                                                                                                                  | NGSO ESOMPs        | ECC/DEC/(15)04                | EN 303 979 |                                  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 18.4 GHz - 18.6 GHz

|                                                                                                 |                                                                                |                    |                                  |            |  |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------|----------------------------------|------------|--|
| FIXED<br>FIXED-SATELLITE (5.484A 5.517A<br>5.517B 5.516B)<br>INTER-SATELLITE (5.521A)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.484A 5.517A<br>5.517B)<br>INTER-SATELLITE (5.521A) | FSS Earth stations | ERC/DEC/(00)07                   |            |  |
|                                                                                                 |                                                                                | Fixed              | ERC/DEC/(00)07,<br>ERC/REC 12-03 | EN 302 217 |  |
|                                                                                                 |                                                                                | GSO ESOMPs         | ECC/DEC/(13)01                   | EN 303 978 |  |
|                                                                                                 |                                                                                | NGSO ESOMPs        | ECC/DEC/(15)04                   | EN 303 979 |  |

## 18.6 GHz - 18.8 GHz

|                                                                                                                                                                               |                                                                                             |                             |                                  |            |                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------|----------------------------------|------------|------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED<br>FIXED-SATELLITE (5.522B 5.517A)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>Space research (passive)<br><br>5.522A 5.522C | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED<br>FIXED-SATELLITE (5.522B)<br><br>5.522A | FSS Earth stations          | ERC/DEC/(00)07                   |            |                                                      |
|                                                                                                                                                                               |                                                                                             | Fixed                       | ERC/DEC/(00)07,<br>ERC/REC 12-03 | EN 302 217 |                                                      |
|                                                                                                                                                                               |                                                                                             | GSO ESOMPs                  | ECC/DEC/(13)01                   | EN 303 978 |                                                      |
|                                                                                                                                                                               |                                                                                             | NGSO ESOMPs                 | ECC/DEC/(15)04                   | EN 303 979 |                                                      |
|                                                                                                                                                                               |                                                                                             | Passive sensors (satellite) |                                  |            | Surface emissivity, snow, sea, ice and precipitation |

## 18.8 GHz - 19.3 GHz

|                                                                                                 |                                                                                |                    |                                  |            |  |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------|----------------------------------|------------|--|
| FIXED<br>FIXED-SATELLITE (5.523A 5.517A<br>5.517B 5.516B)<br>INTER-SATELLITE (5.521A)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.517A 5.517B<br>5.523A)<br>INTER-SATELLITE (5.521A) | FSS Earth stations | ERC/DEC/(00)07                   |            |  |
|                                                                                                 |                                                                                | Fixed              | ERC/DEC/(00)07,<br>ERC/REC 12-03 | EN 302 217 |  |
|                                                                                                 |                                                                                | GSO ESOMPs         | ECC/DEC/(13)01                   | EN 303 978 |  |
|                                                                                                 |                                                                                | NGSO ESOMPs        | ECC/DEC/(15)04                   | EN 303 979 |  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 19.3 GHz - 19.7 GHz

|                                                                                                                |                                                                  |                    |                                  |            |  |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------|----------------------------------|------------|--|
| FIXED<br>FIXED-SATELLITE (5.523B 5.523C<br>5.523D 5.523E 5.517A)<br>INTER-SATELLITE (5.521A 5.523DA)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.517A 5.523B<br>5.523C 5.523D 5.523E) | FSS Earth stations | ERC/DEC/(00)07                   |            |  |
|                                                                                                                |                                                                  | Fixed              | ERC/DEC/(00)07,<br>ERC/REC 12-03 | EN 302 217 |  |
|                                                                                                                |                                                                  | GSO ESOMPs         | ECC/DEC/(13)01                   | EN 303 978 |  |
|                                                                                                                |                                                                  | NGSO ESOMPs        | ECC/DEC/(15)04                   | EN 303 979 |  |

### 19.7 GHz - 20.1 GHz

|                                                                                                                                       |                                                                                                                          |                    |                |                           |                                      |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|---------------------------|--------------------------------------|
| FIXED-SATELLITE (5.484A 5.516B<br>5.527A 5.484B 5.517B)<br>INTER-SATELLITE (5.521A)<br>Mobile-Satellite (space-to-Earth)<br><br>5.524 | FIXED-SATELLITE (5.484A 5.484B<br>5.516B 5.517B 5.527A)<br>INTER-SATELLITE (5.521A)<br>Mobile-Satellite (space-to-Earth) | FSS Earth stations | ECC/DEC/(05)08 |                           | High Density FSS                     |
|                                                                                                                                       |                                                                                                                          | GSO ESOMPs         | ECC/DEC/(13)01 | EN 303 978                |                                      |
|                                                                                                                                       |                                                                                                                          | HEST               | ECC/DEC/(06)03 | EN 301 428,<br>EN 301 459 |                                      |
|                                                                                                                                       |                                                                                                                          | MSS Earth stations |                |                           | For uncoordinated Earth stations SUT |
|                                                                                                                                       |                                                                                                                          | NGSO ESOMPs        | ECC/DEC/(15)04 | EN 303 979                |                                      |

### 20.1 GHz - 20.2 GHz

|                                                                                                                                                                   |                                                                                                                                                             |                    |                |                           |                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|---------------------------|--------------------------------------|
| FIXED-SATELLITE (5.484A 5.516B<br>5.527A 5.484B 5.517B)<br>INTER-SATELLITE (5.521A)<br>MOBILE-SATELLITE (SPACE-TO-<br>EARTH)<br><br>5.524 5.525 5.526 5.527 5.528 | FIXED-SATELLITE (5.484A 5.484B<br>5.516B 5.517B 5.527A)<br>INTER-SATELLITE (5.521A)<br>MOBILE-SATELLITE (SPACE-TO-<br>EARTH)<br><br>5.525 5.526 5.527 5.528 | FSS Earth stations | ECC/DEC/(05)08 |                           | High Density FSS                     |
|                                                                                                                                                                   |                                                                                                                                                             | GSO ESOMPs         | ECC/DEC/(13)01 | EN 303 978                |                                      |
|                                                                                                                                                                   |                                                                                                                                                             | HEST               | ECC/DEC/(06)03 | EN 301 428,<br>EN 301 459 |                                      |
|                                                                                                                                                                   |                                                                                                                                                             | MSS Earth stations |                |                           | For uncoordinated Earth stations SUT |
|                                                                                                                                                                   |                                                                                                                                                             | NGSO ESOMPs        | ECC/DEC/(15)04 | EN 303 979                |                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 20.2 GHz - 21.2 GHz

|                                                                                                                                                            |                                                                                    |                              |  |  |                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------|--|--|----------------------------------|
| FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>Standard frequency and time signal-satellite (space-to-Earth)<br><br>5.524 5.529A | FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br><br>ECA36 | MSS Earth stations           |  |  | For uncoordinated Earth stations |
|                                                                                                                                                            |                                                                                    | Satellite systems (military) |  |  |                                  |

## 21.2 GHz - 21.4 GHz

|                                                                                      |                                                                                      |            |               |            |                                                       |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|---------------|------------|-------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (PASSIVE) | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (PASSIVE) | Video PMSE | ERC/REC 25-10 | EN 302 064 | Cordless Cameras; Temporary point-to-point video link |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|---------------|------------|-------------------------------------------------------|

## 21.4 GHz - 22 GHz

|                                                                         |                                                      |                          |                                  |            |                                                       |
|-------------------------------------------------------------------------|------------------------------------------------------|--------------------------|----------------------------------|------------|-------------------------------------------------------|
| BROADCASTING-SATELLITE (5.208B)<br>FIXED<br>MOBILE<br><br>5.530A 5.530B | BROADCASTING-SATELLITE (5.208B)<br><br>5.530A 5.530B | Broadcasting (satellite) |                                  |            |                                                       |
|                                                                         |                                                      | SRR                      | ECC/DEC/(04)10,<br>ERC/REC 70-03 |            | Within the frequency band 21.65-22 GHz                |
|                                                                         |                                                      | Video PMSE               | ERC/REC 25-10                    | EN 302 064 | Cordless Cameras; Temporary point-to-point video link |

## 22 GHz - 22.2 GHz

|                                                                                                    |                                                                                                                  |                 |                                  |                           |                                                                  |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------|---------------------------|------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.531D 5.531A 5.531B<br>5.531F 5.531C)<br><br>5.149 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.149 ECA17A | Fixed           | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                                  |
|                                                                                                    |                                                                                                                  | Radio astronomy |                                  |                           | Continuum and spectral line observations (e.g. water line), VLBI |
|                                                                                                    |                                                                                                                  | SRR             | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                                  |
|                                                                                                    |                                                                                                                  | Video PMSE      | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 22.2 GHz - 22.21 GHz

|                                                            |       |                 |                                  |                           |                                                                  |
|------------------------------------------------------------|-------|-----------------|----------------------------------|---------------------------|------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.149 | FIXED | Fixed           | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                                  |
|                                                            |       | Radio astronomy |                                  |                           | Continuum and spectral line observations (e.g. water line), VLBI |
|                                                            |       | SRR             | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                                  |
|                                                            |       | Video PMSE      | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link            |

## 22.21 GHz - 22.5 GHz

|                                                                                                                                                          |                                                                                                                                                                                   |                 |                                  |                           |                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------|---------------------------|------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.149 5.532 | FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Earth exploration-satellite (passive)<br>Mobile (ECA39)<br><br>5.149 5.532 ECA17A | Fixed           | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                                  |
|                                                                                                                                                          |                                                                                                                                                                                   | Radio astronomy |                                  |                           | Continuum and spectral line observations (e.g. water line), VLBI |
|                                                                                                                                                          |                                                                                                                                                                                   | SRR             | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                                  |
|                                                                                                                                                          |                                                                                                                                                                                   | Video PMSE      | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link            |

## 22.5 GHz - 22.55 GHz

|                 |                                                                                      |                 |                                  |                           |                                                                  |
|-----------------|--------------------------------------------------------------------------------------|-----------------|----------------------------------|---------------------------|------------------------------------------------------------------|
| FIXED<br>MOBILE | FIXED<br>MOBILE (ECA39)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>ECA17A | Fixed           | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                                  |
|                 |                                                                                      | Radio astronomy |                                  |                           | Continuum and spectral line observations (e.g. water line), VLBI |
|                 |                                                                                      | SRR             | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                                  |
|                 |                                                                                      | Video PMSE      | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 22.55 GHz - 23.15 GHz

|                                                                                     |                                                                                                                  |                 |                                  |                           |                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------|---------------------------|------------------------------------------------------------------|
| FIXED<br>INTER-SATELLITE (5.338A)<br>MOBILE<br>SPACE RESEARCH (5.532A)<br><br>5.149 | FIXED<br>INTER-SATELLITE (5.338A)<br>MOBILE (ECA39)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>ECA17A | Fixed           | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                                  |
|                                                                                     |                                                                                                                  | Radio astronomy |                                  |                           | Continuum and spectral line observations (e.g. water line), VLBI |
|                                                                                     |                                                                                                                  | SRR             | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                                  |
|                                                                                     |                                                                                                                  | Video PMSE      | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link            |

## 23.15 GHz - 23.55 GHz

|                                             |                                                     |            |                                  |                           |                                                       |
|---------------------------------------------|-----------------------------------------------------|------------|----------------------------------|---------------------------|-------------------------------------------------------|
| FIXED<br>INTER-SATELLITE (5.338A)<br>MOBILE | FIXED<br>INTER-SATELLITE (5.338A)<br>MOBILE (ECA39) | Fixed      | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                       |
|                                             |                                                     | SRR        | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                       |
|                                             |                                                     | Video PMSE | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link |

## 23.55 GHz - 23.6 GHz

|                 |                                            |            |                                  |                           |                                                       |
|-----------------|--------------------------------------------|------------|----------------------------------|---------------------------|-------------------------------------------------------|
| FIXED<br>MOBILE | FIXED<br>INTER-SATELLITE<br>MOBILE (ECA39) | Fixed      | T/R 13-02                        | EN 302 217,<br>EN 302 326 |                                                       |
|                 |                                            | SRR        | ECC/DEC/(04)10,<br>ERC/REC 70-03 |                           |                                                       |
|                 |                                            | Video PMSE | ERC/REC 25-10                    | EN 302 064                | Cordless Cameras; Temporary point-to-point video link |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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|-------------|-----------------------------|-------------|-------------------|----------|------|

## 23.6 GHz - 24 GHz

|                                                                                                   |                                                                                                   |                             |                                  |            |                                                                            |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------|------------|----------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite) |                                  |            | Measurement of water vapour, liquid water, clouds for atmospheric sounding |
|                                                                                                   |                                                                                                   | Radio astronomy             |                                  |            | Continuum and spectral line observations (e.g. ammonia line). VLBI         |
|                                                                                                   |                                                                                                   | SRR                         | ECC/DEC/(04)10,<br>ERC/REC 70-03 |            |                                                                            |
|                                                                                                   |                                                                                                   | Video PMSE                  | ERC/REC 25-10                    | EN 302 064 | Cordless Cameras; Temporary point-to-point video link                      |

## 24 GHz - 24.05 GHz

|                                           |                                           |                   |                                  |            |                                                       |
|-------------------------------------------|-------------------------------------------|-------------------|----------------------------------|------------|-------------------------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE<br><br>5.150 | AMATEUR<br>AMATEUR-SATELLITE<br><br>5.150 | Amateur           |                                  | EN 301 783 | Within the band 24-24.25 GHz                          |
|                                           |                                           | Amateur-satellite |                                  |            |                                                       |
|                                           |                                           | ISM               |                                  |            | Within the band 24-24.25 GHz                          |
|                                           |                                           | Non-specific SRDs | ERC/REC 70-03                    | EN 300 440 | Within the band 24-24.25 GHz                          |
|                                           |                                           | SRR               | ECC/DEC/(04)10,<br>ERC/REC 70-03 |            |                                                       |
|                                           |                                           | Video PMSE        | ERC/REC 25-10                    | EN 302 064 | Cordless Cameras; Temporary point-to-point video link |

## 24.05 GHz - 24.25 GHz

|                                                                               |                                                                                                        |                                 |                                  |            |                                                       |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-------------------------------------------------------|
| RADIOLOCATION<br>Amateur<br>Earth exploration-satellite (active)<br><br>5.150 | RADIOLOCATION<br>Amateur<br>Earth exploration-satellite (active)<br>Fixed<br>Mobile<br><br>5.150 ECA36 | Active sensors (satellite)      |                                  |            | Rain radars from satellites                           |
|                                                                               |                                                                                                        | Amateur                         |                                  | EN 301 783 | Within the band 24-24.25 GHz                          |
|                                                                               |                                                                                                        | ISM                             |                                  |            | Within the band 24-24.25 GHz                          |
|                                                                               |                                                                                                        | LPR                             | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                                                       |
|                                                                               |                                                                                                        | Non-specific SRDs               | ERC/REC 70-03                    | EN 300 440 | Within the band 24-24.25 GHz                          |
|                                                                               |                                                                                                        | Radiodetermination applications |                                  | EN 300 440 |                                                       |
|                                                                               |                                                                                                        | Radiolocation (military)        |                                  |            |                                                       |
|                                                                               |                                                                                                        | SRR                             | ECC/DEC/(04)10,<br>ERC/REC 70-03 |            |                                                       |
|                                                                               |                                                                                                        | TLPR                            | ERC/REC 70-03                    | EN 302 372 |                                                       |
|                                                                               |                                                                                                        | TTT                             | ERC/REC 70-03                    | EN 302 858 | Automotive radars                                     |
|                                                                               |                                                                                                        | Video PMSE                      | ERC/REC 25-10                    | EN 302 064 | Cordless Cameras; Temporary point-to-point video link |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 24.25 GHz - 24.45 GHz

|                                                                |                                                |            |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------|------------------------------------------------|------------|------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.338A 5.532AB) | FIXED<br>MOBILE (5.338A 5.532AB)<br><br>ECA17A | Fixed      | T/R 13-02                                            | EN 302 217,<br>EN 302 326 | Unidirectional fixed links                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                |                                                | LPR        | ECC/DEC/(11)02,<br>ERC/REC 70-03                     | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                |                                                | MFCN       | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                |                                                | SRR        | ECC/DEC/(04)10,<br>ERC/REC 70-03                     | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 24.25-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                |                                                | TLPR       | ERC/REC 70-03                                        | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                |                                                | Video PMSE | ERC/REC 25-10                                        | EN 302 064                | Cordless Cameras; Temporary point-to-point video link                                                                                                                                                                                                                                                                                                                                                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 24.45 GHz - 24.5 GHz

|                                                                                   |                                                |            |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|------------------------------------------------|------------|------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.338A 5.532AB) | FIXED<br>MOBILE (5.338A 5.532AB)<br><br>ECA17A | Fixed      | T/R 13-02                                            | EN 302 217,<br>EN 302 326 | Unidirectional fixed links                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                   |                                                | LPR        | ECC/DEC/(11)02,<br>ERC/REC 70-03                     | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                   |                                                | MFCN       | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                   |                                                | SRR        | ECC/DEC/(04)10,<br>ERC/REC 70-03                     | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                   |                                                | TLPR       | ERC/REC 70-03                                        | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                   |                                                | Video PMSE | ERC/REC 25-10                                        | EN 302 064                | Cordless Cameras; Temporary point-to-point video link                                                                                                                                                                                                                                                                                                                                                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 24.5 GHz - 24.65 GHz

|                                                                                   |                                  |       |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|----------------------------------|-------|------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.338A 5.532AB) | FIXED<br>MOBILE (5.338A 5.532AB) | FWA   | ECC/REC/(11)01                                       | EN 302 326                | CRS paired with 25.5-26.5 GHz for FDD systems                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                   |                                  | Fixed | T/R 13-02                                            | EN 302 217,<br>EN 302 326 |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                   |                                  | LPR   | ECC/DEC/(11)02,<br>ERC/REC 70-03                     | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                   |                                  | MFCN  | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                   |                                  | SRR   | ECC/DEC/(04)10,<br>ERC/REC 70-03                     | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                   |                                  | TLPR  | ERC/REC 70-03                                        | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 24.65 GHz - 24.75 GHz

|                                                                                                               |                                                              |       |                                                                         |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| FIXED<br>FIXED-SATELLITE (5.532B)<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.338A 5.532AB) | FIXED<br>FIXED-SATELLITE (5.532B)<br>MOBILE (5.338A 5.532AB) | FWA   | ECC/REC/(11)01                                                          | EN 302 326                | CRS paired with 25.5-26.5 GHz for FDD systems                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                               |                                                              | Fixed | T/R 13-02                                                               | EN 302 217,<br>EN 302 326 |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                               |                                                              | LPR   | ECC/DEC/(11)02,<br>ERC/REC 70-03                                        | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                               |                                                              | MFCN  | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(20)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                               |                                                              | SRR   | ECC/DEC/(04)10,<br>ERC/REC 70-03                                        | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                                               |                                                              | TLPR  | ERC/REC 70-03                                                           | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 24.75 GHz - 25.25 GHz

|                                                                                                               |                                                                                 |       |                                                                         |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| FIXED<br>FIXED-SATELLITE (5.532B)<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.338A 5.532AB) | FIXED<br>FIXED-SATELLITE (5.532B)<br>INTER-SATELLITE<br>MOBILE (5.338A 5.532AB) | FWA   | ECC/REC/(11)01                                                          | EN 302 326                | CRS paired with 25.5-26.5 GHz for FDD systems                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                               |                                                                                 | Fixed | T/R 13-02                                                               | EN 302 217,<br>EN 302 326 |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                               |                                                                                 | LPR   | ECC/DEC/(11)02,<br>ERC/REC 70-03                                        | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                               |                                                                                 | MFCN  | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(20)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                               |                                                                                 | SRR   | ECC/DEC/(04)10,<br>ERC/REC 70-03                                        | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                                               |                                                                                 | TLPR  | ERC/REC 70-03                                                           | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 25.25 GHz - 25.5 GHz

|                                                                                                                                       |                                                                          |                               |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------|------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED (5.534A)<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br>Standard frequency and time signal-satellite (Earth-to-space) | FIXED<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br><br>ECA36 | Aeronautical military systems |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                       |                                                                          | FWA                           | ECC/REC/(11)01                                       | EN 302 326                | CRS paired with 25.5-26.5 GHz for FDD systems                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                                                       |                                                                          | Fixed                         | T/R 13-02                                            | EN 302 217,<br>EN 302 326 |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                       |                                                                          | LPR                           | ECC/DEC/(11)02,<br>ERC/REC 70-03                     | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                       |                                                                          | Land military systems         |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                       |                                                                          | MFCN                          | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                       |                                                                          | Maritime military systems     |                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                       |                                                                          | SRR                           | ECC/DEC/(04)10,<br>ERC/REC 70-03                     | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                                                                       |                                                                          | TLPR                          | ERC/REC 70-03                                        | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 25.5 GHz - 26.5 GHz

|                                                                                                                                                                                                                        |                                                                                                                                                    |                               |                                                                         |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (5.536B)<br>FIXED (5.534A)<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br>SPACE RESEARCH (5.536C)<br>Standard frequency and time signal-satellite (Earth-to-space)<br><br>5.536A | FIXED<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br>SPACE RESEARCH (5.536C)<br>Earth exploration-satellite (5.536B)<br><br>5.536A ECA36 | Aeronautical military systems |                                                                         |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | FWA                           | ECC/REC/(11)01                                                          | EN 302 326                | TS should be paired with 24.5-25.5 GHz for FDD systems                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                                                                        |                                                                                                                                                    | Fixed                         | T/R 13-02                                                               | EN 302 217,<br>EN 302 326 |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | LPR                           | ECC/DEC/(11)02,<br>ERC/REC 70-03                                        | EN 302 729                |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | Land military systems         |                                                                         |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | MFCN                          | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(19)01,<br>ECC/REC/(23)02 |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | Maritime military systems     |                                                                         |                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | SRR                           | ECC/DEC/(04)10,<br>ERC/REC 70-03                                        | EN 302 288                | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 24.25-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type-approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                                                                                                                                                        |                                                                                                                                                    | Space research                | ECC/REC/(19)01                                                          |                           | Satellite payload telemetry                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                                                                                                                                        |                                                                                                                                                    | TLPR                          | ERC/REC 70-03                                                           | EN 302 372                |                                                                                                                                                                                                                                                                                                                                                                                                                            |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 26.5 GHz - 27 GHz

|                                                                                                                                                                                                                        |                                                                                                                                                    |                       |                                                                         |            |                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| EARTH EXPLORATION-SATELLITE (5.536B)<br>FIXED (5.534A)<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br>SPACE RESEARCH (5.536C)<br>Standard frequency and time signal-satellite (Earth-to-space)<br><br>5.536A | FIXED<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br>SPACE RESEARCH (5.536C)<br>Earth exploration-satellite (5.536B)<br><br>5.536A ECA36 | Land military systems |                                                                         |            |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | MFCN                  | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(19)01,<br>ECC/REC/(23)02 |            |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                        |                                                                                                                                                    | SRR                   | ECC/DEC/(04)10,<br>ERC/REC 70-03                                        | EN 302 288 | New SRR systems shall not be introduced in CEPT countries in the frequency bands 21.65-26.65 GHz as of 1 July 2013. New SRR systems may only be introduced in CEPT countries in the frequency bands 24.25-26.65 GHz until 1 January 2018; this date is extended by 4 years for SRR equipment mounted on motor vehicles for which a type approval application has been submitted and has been granted before 1 January 2018 |
|                                                                                                                                                                                                                        |                                                                                                                                                    | Space research        | ECC/REC/(19)01                                                          |            | Satellite payload telemetry                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                                                                                                                                        |                                                                                                                                                    | TLPR                  | ERC/REC 70-03                                                           | EN 302 372 |                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 27 GHz - 27.5 GHz

|                                                             |                                                                                                                          |                       |                                                                         |  |  |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------|--|--|
| FIXED<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB) | FIXED<br>INTER-SATELLITE (5.536)<br>MOBILE (5.338A 5.532AB)<br>Earth exploration-satellite (space-to-Earth)<br><br>ECA36 | Land military systems |                                                                         |  |  |
|                                                             |                                                                                                                          | MFCN                  | ECC/DEC/(18)06,<br>ECC/DEC/(22)01,<br>ECC/REC/(19)01,<br>ECC/REC/(23)02 |  |  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 27.5 GHz - 28.5 GHz

|                                                                                                                                |                                                                                                             |                    |                                |                        |                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXED (5.537A)<br>FIXED-SATELLITE (5.484A 5.516B 5.539 5.517A 5.517B)<br>INTER-SATELLITE (5.521A)<br>MOBILE<br><br>5.538 5.540 | FIXED<br>FIXED-SATELLITE (5.484A 5.516B 5.517A 5.517B 5.539)<br>INTER-SATELLITE (5.521A)<br><br>5.538 5.540 | FSS Earth stations | ECC/DEC/(05)01                 |                        | The Earth-to-Space direction for uncoordinated Earth stations within the band 27.5-27.8285 GHz. The Space-to-Earth direction is limited to beacons for uplink power control 27.5-27.501 GHz |
|                                                                                                                                |                                                                                                             | FWA                | ECC/DEC/(05)01, ECC/REC/(11)01 | EN 302 326             | CRS paired with 28.5-29.5 GHz for FDD systems.                                                                                                                                              |
|                                                                                                                                |                                                                                                             | Feeder links       |                                |                        | Feeder links to be used for Broadcasting satellites (HDTV) 27.5-29.5 GHz                                                                                                                    |
|                                                                                                                                |                                                                                                             | Fixed              | ECC/DEC/(05)01, T/R 13-02      | EN 302 217, EN 302 326 | For frequency arrangement between FS and FSS see ECC/DEC/(05)01. CRS paired with 28.5-29.5 GHz for FDD systems.                                                                             |
|                                                                                                                                |                                                                                                             | GSO ESOMPs         | ECC/DEC/(13)01                 | EN 303 978             |                                                                                                                                                                                             |
|                                                                                                                                |                                                                                                             | NGSO ESOMPs        | ECC/DEC/(15)04                 | EN 303 979             |                                                                                                                                                                                             |
|                                                                                                                                |                                                                                                             | NGSO FSS           | ECC/DEC/(05)01                 | EN 303 699             |                                                                                                                                                                                             |

## 28.5 GHz - 29.1 GHz

|                                                                                                                                                               |                                                                                                                                                     |                    |                                |                        |                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.484A 5.516B 5.523A 5.539 5.517A 5.517B)<br>INTER-SATELLITE (5.521A)<br>MOBILE<br>Earth exploration-satellite (5.541)<br><br>5.540 | FIXED<br>FIXED-SATELLITE (5.484A 5.516B 5.517A 5.517B 5.523A 5.539)<br>INTER-SATELLITE (5.521A)<br>Earth exploration-satellite (5.541)<br><br>5.540 | FSS Earth stations | ECC/DEC/(05)01                 |                        | Uncoordinated Earth stations within the band 28.4445-28.8365 GHz                                               |
|                                                                                                                                                               |                                                                                                                                                     | FWA                | ECC/DEC/(05)01, ECC/REC/(11)01 | EN 302 326             | TS paired with 27.5-28.5 GHz for FDD systems.                                                                  |
|                                                                                                                                                               |                                                                                                                                                     | Feeder links       |                                |                        | Feeder links to be used for Broadcasting satellites (HDTV) 27.5-29.5 GHz                                       |
|                                                                                                                                                               |                                                                                                                                                     | Fixed              | ECC/DEC/(05)01, T/R 13-02      | EN 302 217, EN 302 326 | For frequency arrangement between FS and FSS see ECC/DEC/(05)01. TS paired with 27.5-28.5 GHz for FDD systems. |
|                                                                                                                                                               |                                                                                                                                                     | GSO ESOMPs         | ECC/DEC/(13)01                 | EN 303 978             |                                                                                                                |
|                                                                                                                                                               |                                                                                                                                                     | NGSO ESOMPs        | ECC/DEC/(15)04                 | EN 303 979             |                                                                                                                |
|                                                                                                                                                               |                                                                                                                                                     | NGSO FSS           | ECC/DEC/(05)01                 | EN 303 699             |                                                                                                                |

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## 29.1 GHz - 29.5 GHz

|                                                                                                                                                                         |                                                                                                                                                               |                    |                                   |                           |                                                                                    |
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| FIXED<br>FIXED-SATELLITE (5.516B 5.523C<br>5.523E 5.535A 5.539 5.541A 5.517A)<br>INTER-SATELLITE (5.521A)<br>MOBILE<br>Earth exploration-satellite (5.541)<br><br>5.540 | FIXED<br>FIXED-SATELLITE (5.516B 5.517A<br>5.523C 5.523E 5.535A 5.539 5.541A)<br>INTER-SATELLITE (5.521A)<br>Earth exploration-satellite (5.541)<br><br>5.540 | FSS Earth stations | ECC/DEC/(05)01                    |                           | Uncoordinated Earth stations within the band 29.4525-29.5 GHz                      |
|                                                                                                                                                                         |                                                                                                                                                               | FWA                | ECC/DEC/(05)01,<br>ECC/REC/(11)01 | EN 302 326                | TS paired with 27.5-28.5 GHz for FDD systems.                                      |
|                                                                                                                                                                         |                                                                                                                                                               | Feeder links       |                                   |                           | Feeder links to be used for Broadcasting satellites (HDTV) 27.5-29.5 GHz           |
|                                                                                                                                                                         |                                                                                                                                                               | Fixed              | ECC/DEC/(05)01,<br>T/R 13-02      | EN 302 217,<br>EN 302 326 | Within the band 29.0605-29.4525 GHz. TS paired with 27.5-28.5 GHz for FDD systems. |
|                                                                                                                                                                         |                                                                                                                                                               | GSO ESOMPs         | ECC/DEC/(13)01                    | EN 303 978                |                                                                                    |

## 29.5 GHz - 29.9 GHz

|                                                                                                                                                                                          |                                                                                                                                                                                    |                    |                |            |                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|------------|------------------|
| FIXED-SATELLITE (5.484A 5.516B<br>5.539 5.484B 5.527A 5.517B)<br>INTER-SATELLITE (5.521A)<br>Earth exploration-satellite (5.541)<br>Mobile-Satellite (Earth-to-space)<br><br>5.540 5.542 | FIXED-SATELLITE (5.484A 5.484B<br>5.516B 5.517B 5.527A 5.539)<br>INTER-SATELLITE (5.521A)<br>Earth exploration-satellite (5.541)<br>Mobile-Satellite (Earth-to-space)<br><br>5.540 | FSS Earth stations | ECC/DEC/(05)08 |            | High Density FSS |
|                                                                                                                                                                                          |                                                                                                                                                                                    | GSO ESOMPs         | ECC/DEC/(13)01 | EN 303 978 |                  |
|                                                                                                                                                                                          |                                                                                                                                                                                    | HEST               | ECC/DEC/(06)03 | EN 301 459 |                  |
|                                                                                                                                                                                          |                                                                                                                                                                                    | MSS Earth stations |                |            |                  |
|                                                                                                                                                                                          |                                                                                                                                                                                    | NGSO ESOMPs        | ECC/DEC/(15)04 | EN 303 979 |                  |

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## 29.9 GHz - 30 GHz

|                                                                                                                                                                                                                    |                                                                                                                                                                                                           |                    |                |            |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|------------|------------------|
| FIXED-SATELLITE (5.484A 5.516B<br>5.539 5.484B 5.527A 5.517B)<br>INTER-SATELLITE (5.521A)<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>Earth exploration-satellite (5.541 5.543)<br>5.525 5.526 5.527 5.538 5.540 5.542 | EARTH EXPLORATION-SATELLITE (5.541 5.543)<br>FIXED-SATELLITE (5.484A 5.484B 5.516B 5.517B 5.527A 5.539)<br>INTER-SATELLITE (5.521A)<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>5.525 5.526 5.527 5.538 5.540 | FSS Earth stations | ECC/DEC/(05)08 |            | High Density FSS |
|                                                                                                                                                                                                                    |                                                                                                                                                                                                           | GSO ESOMPs         | ECC/DEC/(13)01 | EN 303 978 |                  |
|                                                                                                                                                                                                                    |                                                                                                                                                                                                           | HEST               | ECC/DEC/(06)03 | EN 301 459 |                  |
|                                                                                                                                                                                                                    |                                                                                                                                                                                                           | MSS Earth stations |                |            |                  |
|                                                                                                                                                                                                                    |                                                                                                                                                                                                           | NGSO ESOMPs        | ECC/DEC/(15)04 | EN 303 979 |                  |

## 30 GHz - 31 GHz

|                                                                                                                                                |                                                                            |                              |  |  |                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------|--|--|----------------------------------|
| FIXED-SATELLITE (5.338A)<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>Standard frequency and time signal-satellite (space-to-Earth)<br>5.529A 5.542 | FIXED-SATELLITE (5.338A)<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br><br>ECA36 | FSS Earth stations           |  |  | For uncoordinated Earth stations |
|                                                                                                                                                |                                                                            | MSS Earth stations           |  |  |                                  |
|                                                                                                                                                |                                                                            | Satellite systems (military) |  |  |                                  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 31 GHz - 31.3 GHz

|                                                                                                                                                      |                                              |                 |                |                           |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------|----------------|---------------------------|------------------------|
| FIXED (5.338A 5.543B 5.543A)<br>MOBILE<br>Space research (5.544 5.545)<br>Standard frequency and time signal-satellite (space-to-Earth)<br><br>5.149 | FIXED (5.338A 5.543B)<br>MOBILE<br><br>5.149 | Fixed           | ECC/REC/(02)02 | EN 302 217,<br>EN 302 326 |                        |
|                                                                                                                                                      |                                              | Radio astronomy |                |                           | Continuum observations |

### 31.3 GHz - 31.5 GHz

|                                                                                                   |                                                                                                   |                             |                |  |                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------|----------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite) | ECC/DEC/(10)02 |  | Measurement of sea ice, water vapour, oil spills, liquid water, clouds, surface temperature, emissivity and atmospheric attenuation. Reference window for the 50-60 GHz range |
|                                                                                                   |                                                                                                   | Radio astronomy             |                |  | Continuum observations                                                                                                                                                        |

### 31.5 GHz - 31.8 GHz

|                                                                                                                                                       |                                                                                                                                                       |                             |  |  |                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Fixed<br>Mobile except aeronautical mobile<br><br>5.149 5.546 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>Fixed<br>Mobile except aeronautical mobile<br><br>5.149 5.546 | Fixed                       |  |  |                                                                                                                                                                               |
|                                                                                                                                                       |                                                                                                                                                       | Passive sensors (satellite) |  |  | Measurement of sea ice, water vapour, oil spills, liquid water, clouds, surface temperature. Emissivity and atmospheric attenuation. Reference window for the 50-60 GHz range |
|                                                                                                                                                       |                                                                                                                                                       | Radio astronomy             |  |  | Continuum observations                                                                                                                                                        |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 31.8 GHz - 32 GHz

|                                                                                                                |                                                                                                         |       |                |            |                                        |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------|----------------|------------|----------------------------------------|
| FIXED (5.547A)<br>RADIONAVIGATION<br>SPACE RESEARCH (DEEP SPACE)<br>(SPACE-TO-EARTH)<br><br>5.547 5.547B 5.548 | FIXED (5.547A)<br>RADIONAVIGATION<br>SPACE RESEARCH (DEEP SPACE)<br>(SPACE-TO-EARTH)<br><br>5.547 5.548 | FWA   | ECC/REC/(11)01 | EN 302 326 | Point-to-Point and Point-to-Multipoint |
|                                                                                                                |                                                                                                         | Fixed | ERC/REC/(01)02 | EN 302 217 | High Density FS                        |

### 32 GHz - 32.3 GHz

|                                                                                                                |                                                                                                         |       |                |            |                                        |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------|----------------|------------|----------------------------------------|
| FIXED (5.547A)<br>RADIONAVIGATION<br>SPACE RESEARCH (DEEP SPACE)<br>(SPACE-TO-EARTH)<br><br>5.547 5.547C 5.548 | FIXED (5.547A)<br>RADIONAVIGATION<br>SPACE RESEARCH (DEEP SPACE)<br>(SPACE-TO-EARTH)<br><br>5.547 5.548 | FWA   | ECC/REC/(11)01 | EN 302 326 | Point-to-Point and Point-to-Multipoint |
|                                                                                                                |                                                                                                         | Fixed | ERC/REC/(01)02 | EN 302 217 | High Density FS                        |

### 32.3 GHz - 33 GHz

|                                                                                |                                                                         |       |                |            |                                        |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------|----------------|------------|----------------------------------------|
| FIXED (5.547A)<br>INTER-SATELLITE<br>RADIONAVIGATION<br><br>5.547 5.547D 5.548 | FIXED (5.547A)<br>INTER-SATELLITE<br>RADIONAVIGATION<br><br>5.547 5.548 | FWA   | ECC/REC/(11)01 | EN 302 326 | Point-to-Point and Point-to-Multipoint |
|                                                                                |                                                                         | Fixed | ERC/REC/(01)02 | EN 302 217 | High Density FS                        |

### 33 GHz - 33.4 GHz

|                                                       |                                                                   |                          |                |            |                                        |
|-------------------------------------------------------|-------------------------------------------------------------------|--------------------------|----------------|------------|----------------------------------------|
| FIXED (5.547A)<br>RADIONAVIGATION<br><br>5.547 5.547E | FIXED (5.547A)<br>INTER-SATELLITE<br>RADIONAVIGATION<br><br>5.547 | FWA                      | ECC/REC/(11)01 | EN 302 326 | Point-to-Point and Point-to-Multipoint |
|                                                       |                                                                   | Fixed                    | ERC/REC/(01)02 | EN 302 217 | High Density FS                        |
|                                                       |                                                                   | Radiolocation (military) |                |            |                                        |

### 33.4 GHz - 34.2 GHz

|                            |                            |                          |  |  |  |
|----------------------------|----------------------------|--------------------------|--|--|--|
| RADIOLOCATION<br><br>5.549 | RADIOLOCATION<br><br>ECA36 | Radiolocation (military) |  |  |  |
|----------------------------|----------------------------|--------------------------|--|--|--|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 34.2 GHz - 34.7 GHz

|                                                                           |                                                                           |                          |  |  |  |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------|--|--|--|
| RADIOLOCATION<br>SPACE RESEARCH (DEEP SPACE)<br>(EARTH-TO-SPACE)<br>5.549 | RADIOLOCATION<br>SPACE RESEARCH (DEEP SPACE)<br>(EARTH-TO-SPACE)<br>ECA36 | Radiolocation (military) |  |  |  |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------|--|--|--|

### 34.7 GHz - 35.2 GHz

|                                          |                                          |                          |  |  |  |
|------------------------------------------|------------------------------------------|--------------------------|--|--|--|
| RADIOLOCATION<br>Space research<br>5.549 | RADIOLOCATION<br>Space research<br>ECA36 | Radiolocation (military) |  |  |  |
|------------------------------------------|------------------------------------------|--------------------------|--|--|--|

### 35.2 GHz - 35.5 GHz

|                                               |                                               |                            |  |  |                            |
|-----------------------------------------------|-----------------------------------------------|----------------------------|--|--|----------------------------|
| METEOROLOGICAL AIDS<br>RADIOLOCATION<br>5.549 | METEOROLOGICAL AIDS<br>RADIOLOCATION<br>ECA36 | Active sensors (satellite) |  |  | Rain radar from satellites |
|                                               |                                               | Radiolocation (military)   |  |  |                            |

### 35.5 GHz - 36 GHz

|                                                                                                                            |                                                                                                                            |                            |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------|--|--|--|
| EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>METEOROLOGICAL AIDS<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>5.549 5.549A | EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>METEOROLOGICAL AIDS<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>5.549A ECA36 | Active sensors (satellite) |  |  |  |
|                                                                                                                            |                                                                                                                            | Radiolocation (military)   |  |  |  |

### 36 GHz - 37 GHz

|                                                                                                         |                                                                                                                            |                             |  |  |                                                                                  |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------|--|--|----------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (PASSIVE)<br>5.149 5.550A | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (PASSIVE)<br>Radio astronomy<br>5.149 5.550A | Passive sensors (satellite) |  |  | EESS surface emissivity, snow, sea ice and precipitation                         |
|                                                                                                         |                                                                                                                            | Radio astronomy             |  |  | Spectral line observations (Hydrogen cyanide and Hydroxyl lines) 36.43-36.50 GHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 37 GHz - 37.5 GHz

|                                                                                                         |                                                       |       |           |  |                                                                    |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------|-----------|--|--------------------------------------------------------------------|
| FIXED<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.550B)<br>SPACE OPERATION (SPACE-TO-EARTH)<br><br>5.547 | FIXED<br>SPACE RESEARCH (SPACE-TO-EARTH)<br><br>5.547 | Fixed | T/R 12-01 |  | Major use by civil Fixed Service systems. High Density fixed links |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------|-----------|--|--------------------------------------------------------------------|

### 37.5 GHz - 38 GHz

|                                                                                                                                                                                            |                                                                                                                                           |                    |                |  |                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|--|--------------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.550C 5.550CA)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.550B)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.547 | FIXED<br>FIXED-SATELLITE (5.550C 5.550CA)<br>SPACE RESEARCH (SPACE-TO-EARTH)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.547 | FSS Earth stations | ERC/DEC/(00)02 |  | Uncoordinated Earth stations shall not claim protection from the Fixed Service |
|                                                                                                                                                                                            |                                                                                                                                           | Fixed              | T/R 12-01      |  | Major use by civil Fixed Service systems. High Density fixed links             |

### 38 GHz - 39.5 GHz

|                                                                                                                            |                                                                                                         |                    |                |  |                                                                                |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------|----------------|--|--------------------------------------------------------------------------------|
| FIXED (5.550D)<br>FIXED-SATELLITE (5.550C)<br>MOBILE (5.550B)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.547 | FIXED (5.550D)<br>FIXED-SATELLITE (5.550C)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.547 | FSS Earth stations | ERC/DEC/(00)02 |  | Uncoordinated Earth stations shall not claim protection from the Fixed Service |
|                                                                                                                            |                                                                                                         | Fixed              | T/R 12-01      |  | Major use by civil Fixed Service systems. High Density fixed links             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 39.5 GHz - 40 GHz

|                                                                                                                                                                      |                                                                                                                                                             |                    |                                   |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|--|--|
| FIXED<br>FIXED-SATELLITE (5.516B 5.550C)<br>MOBILE (5.550B)<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.547 5.550E | FIXED<br>FIXED-SATELLITE (5.516B 5.550C)<br>MOBILE<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.547 5.550E | FSS Earth stations | ECC/REC/(22)02,<br>ERC/DEC/(00)02 |  |  |
|                                                                                                                                                                      |                                                                                                                                                             | MSS Earth stations | ERC/DEC/(00)02                    |  |  |

### 40 GHz - 40.5 GHz

|                                                                                                                                                                                                                                                   |                                                                                                                                                                                   |                    |                                   |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|--|--|
| EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE)<br>FIXED<br>FIXED-SATELLITE (5.516B 5.550C)<br>MOBILE (5.550B)<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>SPACE RESEARCH (EARTH-TO-SPACE)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.550E | FIXED<br>FIXED-SATELLITE (5.516B)<br>MOBILE<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>SPACE RESEARCH (EARTH-TO-SPACE)<br>Earth exploration-satellite (space-to-Earth)<br><br>5.550E | FSS Earth stations | ECC/REC/(22)02,<br>ERC/DEC/(00)02 |  |  |
|                                                                                                                                                                                                                                                   |                                                                                                                                                                                   | MSS Earth stations | ERC/DEC/(00)02                    |  |  |

### 40.5 GHz - 41 GHz

|                                                                                                                                                            |                                                                                                                                |                    |                                                      |            |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------|------------|--|
| BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (5.550C)<br>LAND MOBILE (5.550B)<br>Aeronautical mobile<br>Maritime mobile<br><br>5.547 | BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>LAND MOBILE (5.550B)<br>Aeronautical mobile<br>Maritime mobile<br><br>5.547 | FSS Earth stations | ECC/DEC/(23)01,<br>ECC/REC/(22)01                    |            |  |
|                                                                                                                                                            |                                                                                                                                | MFCN               | ECC/DEC/(22)06,<br>ECC/REC/(22)01,<br>ECC/REC/(22)02 |            |  |
|                                                                                                                                                            |                                                                                                                                | Point-to-Point     | ECC/REC/(01)04                                       | EN 302 217 |  |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 41 GHz - 42.5 GHz

|                                                                                                                                                                          |                                                                                                                                                                          |                    |                                                      |            |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------|------------|--|
| BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (5.550C)<br>LAND MOBILE (5.550B)<br>Aeronautical mobile<br>Maritime mobile<br><br>5.547 5.551H 5.551I | BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (5.550C)<br>LAND MOBILE (5.550B)<br>Aeronautical mobile<br>Maritime mobile<br><br>5.547 5.551H 5.551I | FSS Earth stations | ECC/DEC/(23)01,<br>ECC/REC/(22)01                    |            |  |
|                                                                                                                                                                          |                                                                                                                                                                          | MFCN               | ECC/DEC/(22)06,<br>ECC/REC/(22)01,<br>ECC/REC/(22)02 |            |  |
|                                                                                                                                                                          |                                                                                                                                                                          | Point-to-Point     | ECC/REC/(01)04                                       | EN 302 217 |  |

#### 42.5 GHz - 43.5 GHz

|                                                                                                                         |                                                                                                                |                    |                                                      |            |                                                                             |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------|------------|-----------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE (5.550B)<br>RADIO ASTRONOMY<br><br>5.149 5.547 | FIXED<br>FIXED-SATELLITE (5.552)<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.547 | FSS Earth stations | ECC/DEC/(23)01,<br>ECC/REC/(22)01                    |            |                                                                             |
|                                                                                                                         |                                                                                                                | MFCN               | ECC/DEC/(22)06,<br>ECC/REC/(22)01,<br>ECC/REC/(22)02 |            |                                                                             |
|                                                                                                                         |                                                                                                                | Point-to-Point     | ECC/REC/(01)04                                       | EN 302 217 |                                                                             |
|                                                                                                                         |                                                                                                                | Radio astronomy    |                                                      |            | Continuum and spectral line observations (e.g. silicon monoxide line), VLBI |

#### 43.5 GHz - 45.5 GHz

|                                                                                                        |                                                                          |                               |  |  |  |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------|--|--|--|
| MOBILE (5.553 5.553A)<br>MOBILE-SATELLITE<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.554 | MOBILE (5.553)<br>MOBILE-SATELLITE<br>Fixed-satellite<br><br>5.554 ECA36 | Aeronautical military systems |  |  |  |
|                                                                                                        |                                                                          | Land military systems         |  |  |  |
|                                                                                                        |                                                                          | Maritime military systems     |  |  |  |
|                                                                                                        |                                                                          | Satellite systems (military)  |  |  |  |

#### 45.5 GHz - 47 GHz

|                                                                                                        |                                                                                                 |   |  |  |  |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---|--|--|--|
| MOBILE (5.553 5.553A)<br>MOBILE-SATELLITE<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.554 | MOBILE (5.553)<br>MOBILE-SATELLITE<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.554 | - |  |  |  |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---|--|--|--|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 47 GHz - 47.2 GHz

|                              |                              |                   |  |  |  |
|------------------------------|------------------------------|-------------------|--|--|--|
| AMATEUR<br>AMATEUR-SATELLITE | AMATEUR<br>AMATEUR-SATELLITE | Amateur           |  |  |  |
|                              |                              | Amateur-satellite |  |  |  |

## 47.2 GHz - 47.5 GHz

|                                                                          |                                                                          |                    |                |  |                                    |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------|----------------|--|------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552 5.550C)<br>MOBILE (5.553B)<br><br>5.552A | FIXED<br>FIXED-SATELLITE (5.550C 5.552)<br>MOBILE (5.553B)<br><br>5.552A | FSS Earth stations | ECC/DEC/(21)01 |  | Coordinated gateway Earth stations |
|                                                                          |                                                                          | Feeder links       |                |  | For 40 GHz Broadcasting satellites |

## 47.5 GHz - 47.9 GHz

|                                                                                               |                                                                                               |                    |                                   |  |                                    |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------|-----------------------------------|--|------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552 5.550C)<br>FIXED-SATELLITE (5.516B 5.554A)<br>MOBILE (5.553B) | FIXED<br>FIXED-SATELLITE (5.550C 5.552)<br>FIXED-SATELLITE (5.516B 5.554A)<br>MOBILE (5.553B) | FSS Earth stations | ECC/DEC/(05)08,<br>ECC/DEC/(21)01 |  | High Density FSS                   |
|                                                                                               |                                                                                               | Feeder links       |                                   |  | For 40 GHz Broadcasting satellites |

## 47.9 GHz - 48.2 GHz

|                                                                          |                                                                          |                    |                |  |                                    |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------|----------------|--|------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552 5.550C)<br>MOBILE (5.553B)<br><br>5.552A | FIXED<br>FIXED-SATELLITE (5.550C 5.552)<br>MOBILE (5.553B)<br><br>5.552A | FSS Earth stations | ECC/DEC/(21)01 |  | Coordinated gateway Earth stations |
|                                                                          |                                                                          | Feeder links       |                |  | For 40 GHz Broadcasting satellites |

## 48.2 GHz - 48.54 GHz

|                                                                                                |                                                                                                |                    |                                   |            |                                                 |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|------------|-------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552 5.550C)<br>FIXED-SATELLITE (5.516B 5.554A<br>5.555B)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.550C 5.552)<br>FIXED-SATELLITE (5.516B 5.554A<br>5.555B)<br>MOBILE | FSS Earth stations | ECC/DEC/(05)08,<br>ECC/DEC/(21)01 |            | High Density FSS                                |
|                                                                                                |                                                                                                | Feeder links       |                                   |            | For 40 GHz Broadcasting satellites              |
|                                                                                                |                                                                                                | Fixed              | ERC/REC 12-11                     | EN 302 217 | Within the band 48.5-50.2 GHz and 50.9-52.6 GHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

#### 48.54 GHz - 49.44 GHz

|                                                                            |                                                                                                      |                    |                |            |                                                            |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------|----------------|------------|------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552 5.550C)<br>MOBILE<br><br>5.149 5.340 5.555 | FIXED<br>FIXED-SATELLITE (5.550C 5.552)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.340 5.555 ECA17A | FSS Earth stations | ECC/DEC/(21)01 |            |                                                            |
|                                                                            |                                                                                                      | Feeder links       |                |            | 48.5-49.2 GHz for 40 GHz Broadcasting satellites           |
|                                                                            |                                                                                                      | Fixed              | ERC/REC 12-11  | EN 302 217 | Within the band 48.5-50.2 GHz and 50.9-52.6 GHz            |
|                                                                            |                                                                                                      | Radio astronomy    |                |            | Spectral line observations (e.g. carbon monosulphide line) |

#### 49.44 GHz - 50.2 GHz

|                                                                                                    |                                                                                                                  |                    |                                   |            |                                                 |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------|------------|-------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.552 5.338A 5.550C)<br>FIXED-SATELLITE (5.516B 5.554A 5.555B)<br>MOBILE | FIXED<br>FIXED-SATELLITE (5.338A 5.550C 5.552)<br>FIXED-SATELLITE (5.516B 5.554A 5.555B)<br>MOBILE<br><br>ECA17A | FSS Earth stations | ECC/DEC/(05)08,<br>ECC/DEC/(21)01 |            | High Density FSS                                |
|                                                                                                    |                                                                                                                  | Fixed              | ERC/REC 12-11                     | EN 302 217 | Within the band 48.5-50.2 GHz and 50.9-52.6 GHz |

#### 50.2 GHz - 50.4 GHz

|                                                                                |                                                                                |                             |  |  |                                                                                                                |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------|--|--|----------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite) |  |  | Atmospheric temperature sounding. Terrestrial passive radiometers. Reference window for the 52.6-59.3 GHz band |
|                                                                                |                                                                                | Radio astronomy             |  |  | Continuum and spectral line observations                                                                       |

#### 50.4 GHz - 51.4 GHz

|                                                                                         |                                                                               |                    |                |            |                                                 |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------|----------------|------------|-------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (5.338A 5.550C)<br>MOBILE<br>Mobile-Satellite (Earth-to-space) | FIXED<br>FIXED-SATELLITE (5.338A 5.550C)<br>Mobile-Satellite (Earth-to-space) | FSS Earth stations | ECC/DEC/(21)01 |            | Coordinated gateway Earth stations              |
|                                                                                         |                                                                               | Fixed              | ERC/REC 12-11  | EN 302 217 | Within the band 48.5-50.2 GHz and 50.9-52.6 GHz |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 51.4 GHz - 52.4 GHz

|                                                                         |                                                                                            |                    |                |            |                                                 |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------|----------------|------------|-------------------------------------------------|
| FIXED (5.338A)<br>FIXED-SATELLITE (5.555C)<br>MOBILE<br><br>5.547 5.556 | FIXED (5.338A)<br>FIXED-SATELLITE (5.555C)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.547 5.556 | FSS Earth stations | ECC/DEC/(21)01 |            | Coordinated gateway Earth stations              |
|                                                                         |                                                                                            | Fixed              | ERC/REC 12-11  | EN 302 217 | Within the band 48.5-50.2 GHz and 50.9-52.6 GHz |
|                                                                         |                                                                                            | Radio astronomy    |                |            | Continuum and spectral line observations        |

## 52.4 GHz - 52.6 GHz

|                                             |                                                           |                 |               |            |                                                 |
|---------------------------------------------|-----------------------------------------------------------|-----------------|---------------|------------|-------------------------------------------------|
| FIXED (5.338A)<br>MOBILE<br><br>5.547 5.556 | FIXED (5.338A)<br>MOBILE<br>RADIO ASTRONOMY (5.547 5.556) | Fixed           | ERC/REC 12-11 | EN 302 217 | Within the band 48.5-50.2 GHz and 50.9-52.6 GHz |
|                                             |                                                           | Radio astronomy |               |            | Continuum and spectral line observations        |

## 52.6 GHz - 54.25 GHz

|                                                                                      |                                                                                      |                             |  |  |                                                                   |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------|--|--|-------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.556 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.556 | Passive sensors (satellite) |  |  | Atmospheric temperature sounding. Terrestrial passive radiometers |
|                                                                                      |                                                                                      | Radio astronomy             |  |  | Continuum and spectral line observations                          |

## 54.25 GHz - 55.78 GHz

|                                                                                                             |                                                                   |                             |  |  |                                                                   |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------|--|--|-------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.556A)<br>SPACE RESEARCH (PASSIVE)<br><br>5.556B | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE) | Passive sensors (satellite) |  |  | Atmospheric temperature sounding. Terrestrial passive radiometers |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------|--|--|-------------------------------------------------------------------|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 55.78 GHz - 56.9 GHz

|                                                                                                                                                      |                                                                                                                                    |                             |               |            |                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------|------------|----------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED (5.557A)<br>INTER-SATELLITE (5.556A)<br>MOBILE (5.558)<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 5.557 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED (5.557A)<br>INTER-SATELLITE (5.556A)<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 5.558 | Fixed                       | ERC/REC 12-12 | EN 302 217 | High density fixed links         |
|                                                                                                                                                      |                                                                                                                                    | Passive sensors (satellite) |               |            | Atmospheric temperature sounding |

## 56.9 GHz - 57 GHz

|                                                                                                                                             |                                                                                                                                       |                             |               |            |                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------|------------|----------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>INTER-SATELLITE (5.558A)<br>MOBILE (5.558)<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 5.557 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>INTER-SATELLITE (5.558A)<br>MOBILE (5.558)<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 | Fixed                       | ERC/REC 12-12 | EN 302 217 | High density fixed links         |
|                                                                                                                                             |                                                                                                                                       | Passive sensors (satellite) |               |            | Atmospheric temperature sounding |

## 57 GHz - 58.2 GHz

|                                                                                                                                             |                                                                                                                                       |                                    |                                  |                                          |                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------|------------------------------------------|-----------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>INTER-SATELLITE (5.556A)<br>MOBILE (5.558)<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 5.557 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>INTER-SATELLITE (5.556A)<br>MOBILE (5.558)<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 | Fixed                              |                                  | EN 302 217                               | Un-coordinated deployment. High density fixed links |
|                                                                                                                                             |                                                                                                                                       | LPR                                | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729                               |                                                     |
|                                                                                                                                             |                                                                                                                                       | Non-specific SRDs                  | ERC/REC 70-03                    | EN 305 550                               | Within the band 57-64 GHz                           |
|                                                                                                                                             |                                                                                                                                       | Passive sensors (satellite)        |                                  |                                          | Atmospheric temperature sounding                    |
|                                                                                                                                             |                                                                                                                                       | TLPR                               | ERC/REC 70-03                    | EN 302 372                               |                                                     |
|                                                                                                                                             |                                                                                                                                       | Wideband data transmission systems | ERC/REC 70-03                    | EN 302 567,<br>EN 303 722,<br>EN 303 753 |                                                     |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 58.2 GHz - 59 GHz

|                                                                                                         |                                                                                                                             |                                    |                               |                                    |                                                                   |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|------------------------------------|-------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>MOBILE<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 5.556 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.547 5.556 ECA6 ECA19 | Fixed                              |                               | EN 302 217                         | Un-coordinated deployment. High density fixed links               |
|                                                                                                         |                                                                                                                             | LPR                                | ECC/DEC/(11)02, ERC/REC 70-03 | EN 302 729                         |                                                                   |
|                                                                                                         |                                                                                                                             | Non-specific SRDs                  | ERC/REC 70-03                 | EN 305 550                         | Within the band 57-64 GHz                                         |
|                                                                                                         |                                                                                                                             | Passive sensors (satellite)        |                               |                                    | Atmospheric temperature sounding. Terrestrial passive radiometers |
|                                                                                                         |                                                                                                                             | Radio astronomy                    |                               |                                    | Continuum and spectral line observations                          |
|                                                                                                         |                                                                                                                             | TLPR                               | ERC/REC 70-03                 | EN 302 372                         |                                                                   |
|                                                                                                         |                                                                                                                             | Wideband data transmission systems | ERC/REC 70-03                 | EN 302 567, EN 303 722, EN 303 753 |                                                                   |

## 59 GHz - 59.3 GHz

|                                                                                                                                                   |                                                                                                                                                   |                                    |                               |                                    |                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|------------------------------------|-------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>INTER-SATELLITE (5.556A)<br>MOBILE (5.558)<br>RADIOLOCATION (5.559)<br>SPACE RESEARCH (PASSIVE) | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>FIXED<br>INTER-SATELLITE (5.556A)<br>MOBILE (5.558)<br>RADIOLOCATION (5.559)<br>SPACE RESEARCH (PASSIVE) | Fixed                              |                               | EN 302 217                         | High density fixed links                                          |
|                                                                                                                                                   |                                                                                                                                                   | LPR                                | ECC/DEC/(11)02, ERC/REC 70-03 | EN 302 729                         |                                                                   |
|                                                                                                                                                   |                                                                                                                                                   | Non-specific SRDs                  | ERC/REC 70-03                 | EN 305 550                         | Within the band 57-64 GHz                                         |
|                                                                                                                                                   |                                                                                                                                                   | Passive sensors (satellite)        |                               |                                    | Atmospheric temperature sounding. Terrestrial passive radiometers |
|                                                                                                                                                   |                                                                                                                                                   | TLPR                               | ERC/REC 70-03                 | EN 302 372                         |                                                                   |
|                                                                                                                                                   |                                                                                                                                                   | Wideband data transmission systems | ERC/REC 70-03                 | EN 302 567, EN 303 722, EN 303 753 |                                                                   |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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### 59.3 GHz - 64 GHz

|                                                                                  |                                                                                  |                                    |                                  |                                          |                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------|----------------------------------|------------------------------------------|----------------------------------|
| FIXED<br>INTER-SATELLITE<br>MOBILE (5.558)<br>RADIOLOCATION (5.559)<br><br>5.138 | FIXED<br>INTER-SATELLITE<br>MOBILE (5.558)<br>RADIOLOCATION (5.559)<br><br>5.138 | Fixed                              |                                  | EN 302 217                               | High density fixed links         |
|                                                                                  |                                                                                  | ISM                                |                                  |                                          | Within the band 61.0-61.5 GHz    |
|                                                                                  |                                                                                  | ITS                                | ECC/DEC/(09)01                   | EN 302 686                               | Within the band 63.72- 65.88 GHz |
|                                                                                  |                                                                                  | LPR                                | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729                               |                                  |
|                                                                                  |                                                                                  | Non-specific SRDs                  | ERC/REC 70-03                    | EN 305 550                               | Within the band 57-64 GHz        |
|                                                                                  |                                                                                  | TLPR                               | ERC/REC 70-03                    | EN 302 372                               |                                  |
|                                                                                  |                                                                                  | Wideband data transmission systems | ERC/REC 70-03                    | EN 302 567,<br>EN 303 722,<br>EN 303 753 |                                  |

### 64 GHz - 65 GHz

|                                                                                     |                                                                                     |                                    |                                  |                                          |                                          |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------|----------------------------------|------------------------------------------|------------------------------------------|
| FIXED<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.547 5.556 | FIXED<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br><br>5.547 5.556 | Fixed                              |                                  | EN 302 217                               | High density fixed links                 |
|                                                                                     |                                                                                     | ITS                                | ECC/DEC/(09)01,<br>ERC/REC 70-03 | EN 302 686                               | Within the band 63.72 - 65.88 GHz        |
|                                                                                     |                                                                                     | Radio astronomy                    |                                  |                                          | Continuum and spectral line observations |
|                                                                                     |                                                                                     | Wideband data transmission systems | ERC/REC 70-03                    | EN 302 567,<br>EN 303 722,<br>EN 303 753 |                                          |

### 65 GHz - 66 GHz

|                                                                                                                                |                                                                                                                                |                                    |                                  |                                          |                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------|------------------------------------------|-----------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>FIXED<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE RESEARCH<br><br>5.547 | EARTH EXPLORATION-SATELLITE<br>FIXED<br>INTER-SATELLITE<br>MOBILE EXCEPT AERONAUTICAL<br>MOBILE<br>SPACE RESEARCH<br><br>5.547 | Fixed                              |                                  | EN 302 217                               | High density fixed links                                              |
|                                                                                                                                |                                                                                                                                | ITS                                | ECC/DEC/(09)01,<br>ERC/REC 70-03 | EN 302 686                               | Within the band 63.72 - 65.88 GHz                                     |
|                                                                                                                                |                                                                                                                                | Land mobile                        |                                  |                                          | Broadband mobile systems for connection to IBCN paired with 62-63 GHz |
|                                                                                                                                |                                                                                                                                | Wideband data transmission systems | ERC/REC 70-03                    | EN 302 567,<br>EN 303 722,<br>EN 303 753 |                                                                       |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 66 GHz - 71 GHz

|                                                                                                                                  |                                                                                                                          |                                    |               |                                          |                      |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------|------------------------------------------|----------------------|
| INTER-SATELLITE<br>MOBILE (5.553 5.558)<br>MOBILE-SATELLITE<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.554 5.559AA | INTER-SATELLITE<br>MOBILE (5.553 5.558)<br>MOBILE-SATELLITE<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.554 | -                                  |               |                                          | Future civil systems |
|                                                                                                                                  |                                                                                                                          | Wideband data transmission systems | ERC/REC 70-03 | EN 302 567,<br>EN 303 722,<br>EN 303 753 |                      |

## 71 GHz - 74 GHz

|                                                                                          |                                                                                          |       |                |            |  |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------|----------------|------------|--|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (SPACE-TO-EARTH) | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (SPACE-TO-EARTH) | Fixed | ECC/REC/(05)07 | EN 302 217 |  |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------|----------------|------------|--|

## 74 GHz - 75.5 GHz

|                                                                                                                                               |                                                                                                                                               |                |                                  |            |                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|------------|---------------------------------------------|
| BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>Space research (space-to-Earth)<br><br>5.561 | BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>Space research (space-to-Earth)<br><br>5.561 | Fixed          | ECC/REC/(05)07                   | EN 302 217 |                                             |
|                                                                                                                                               |                                                                                                                                               | LPR            | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                                             |
|                                                                                                                                               |                                                                                                                                               | Space research |                                  |            | VLBI measurements within the band 74-84 GHz |
|                                                                                                                                               |                                                                                                                                               | TLPR           | ERC/REC 70-03                    | EN 302 372 |                                             |

## 75.5 GHz - 76 GHz

|                                                                                                                                               |                                                                                                                                        |                   |                                  |            |                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------|------------|-------------------------------|
| BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>Space research (space-to-Earth)<br><br>5.561 | BROADCASTING<br>BROADCASTING-SATELLITE<br>FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>Amateur<br>Amateur-satellite<br><br>5.561 ECA35 | Amateur           |                                  |            | Within the band 75.5-81.5 GHz |
|                                                                                                                                               |                                                                                                                                        | Amateur-satellite |                                  |            | Within the band 75.5-81.5 GHz |
|                                                                                                                                               |                                                                                                                                        | Fixed             | ECC/REC/(05)07                   | EN 302 217 |                               |
|                                                                                                                                               |                                                                                                                                        | LPR               | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                               |
|                                                                                                                                               |                                                                                                                                        | Space research    |                                  |            | VLBI                          |
|                                                                                                                                               |                                                                                                                                        | TLPR              | ERC/REC 70-03                    | EN 302 372 |                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 76 GHz - 77.5 GHz

|                                                                                                                  |                                                                                                                  |                       |                                  |                           |                                                  |
|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------|---------------------------|--------------------------------------------------|
| RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>Space research (space-to-Earth)<br><br>5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>Space research (space-to-Earth)<br><br>5.149 | Amateur               |                                  |                           | Within the band 75.5-81.5 GHz                    |
|                                                                                                                  |                                                                                                                  | Amateur-satellite     |                                  |                           | Within the band 75.5-81.5 GHz                    |
|                                                                                                                  |                                                                                                                  | GBSAR                 | ECC/DEC/(21)02,<br>ERC/REC 70-03 | EN 303 661                | Within frequency band 76-77 GHz                  |
|                                                                                                                  |                                                                                                                  | LPR                   | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729                |                                                  |
|                                                                                                                  |                                                                                                                  | Radio astronomy       |                                  |                           | Continuum and spectral line observations         |
|                                                                                                                  |                                                                                                                  | Radiolocation (civil) |                                  |                           |                                                  |
|                                                                                                                  |                                                                                                                  | Railway applications  | ERC/REC 70-03                    | EN 301 091                | Obstruction/vehicle detection at level crossings |
|                                                                                                                  |                                                                                                                  | SRR                   | ECC/DEC/(04)03,<br>ERC/REC 70-03 | EN 302 264                |                                                  |
|                                                                                                                  |                                                                                                                  | TLPR                  | ERC/REC 70-03                    | EN 302 372                |                                                  |
|                                                                                                                  |                                                                                                                  | TTT                   | ECC/DEC/(16)01,<br>ERC/REC 70-03 | EN 301 091,<br>EN 303 360 | Within the band 76-77 GHz. Rotorcraft Radar      |

## 77.5 GHz - 78 GHz

|                                                                                                                           |                                                                                                        |                   |                                  |            |                                          |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------|----------------------------------|------------|------------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE<br>RADIOLOCATION (5.559B)<br>Radio astronomy<br>Space research (space-to-Earth)<br><br>5.149 | AMATEUR<br>AMATEUR-SATELLITE<br>RADIOLOCATION (5.559B)<br>Space research (space-to-Earth)<br><br>5.149 | Amateur           |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                           |                                                                                                        | Amateur-satellite |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                           |                                                                                                        | LPR               | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                                          |
|                                                                                                                           |                                                                                                        | Radio astronomy   |                                  |            | Continuum and spectral line observations |
|                                                                                                                           |                                                                                                        | SRR               | ECC/DEC/(04)03,<br>ERC/REC 70-03 | EN 302 264 |                                          |
|                                                                                                                           |                                                                                                        | TLPR              | ERC/REC 70-03                    | EN 302 372 |                                          |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 78 GHz - 79 GHz

|                                                                                                                        |                                                                                                                        |                       |                                  |            |                                          |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------|------------|------------------------------------------|
| RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>Radio astronomy<br>Space research (space-to-Earth)<br><br>5.149 5.560 | RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>Radio astronomy<br>Space research (space-to-Earth)<br><br>5.149 5.560 | Amateur               |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                        |                                                                                                                        | Amateur-satellite     |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                        |                                                                                                                        | LPR                   | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                                          |
|                                                                                                                        |                                                                                                                        | Radio astronomy       |                                  |            | Continuum and spectral line observations |
|                                                                                                                        |                                                                                                                        | Radiolocation (civil) |                                  |            |                                          |
|                                                                                                                        |                                                                                                                        | SRR                   | ECC/DEC/(04)03,<br>ERC/REC 70-03 | EN 302 264 |                                          |
|                                                                                                                        |                                                                                                                        | TLPR                  | ERC/REC 70-03                    | EN 302 372 |                                          |

## 79 GHz - 81 GHz

|                                                                                                                  |                                                                               |                       |                                  |            |                                          |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------|----------------------------------|------------|------------------------------------------|
| RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>Space research (space-to-Earth)<br><br>5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.149 | Amateur               |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                  |                                                                               | Amateur-satellite     |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                  |                                                                               | LPR                   | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                                          |
|                                                                                                                  |                                                                               | Radio astronomy       |                                  |            | Continuum and spectral line observations |
|                                                                                                                  |                                                                               | Radiolocation (civil) |                                  |            |                                          |
|                                                                                                                  |                                                                               | SRR                   | ECC/DEC/(04)03,<br>ERC/REC 70-03 | EN 302 264 |                                          |
|                                                                                                                  |                                                                               | TLPR                  | ERC/REC 70-03                    | EN 302 372 |                                          |

## 81 GHz - 84 GHz

|                                                                                                                                                                             |                                                                                                                                                                             |                   |                                  |            |                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------|------------|------------------------------------------|
| FIXED (5.338A)<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>RADIO ASTRONOMY<br>Space research (space-to-Earth)<br><br>5.149 5.561A | FIXED (5.338A)<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>RADIO ASTRONOMY<br>Space research (space-to-Earth)<br><br>5.149 5.561A | Amateur           |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                                                                             |                                                                                                                                                                             | Amateur-satellite |                                  |            | Within the band 75.5-81.5 GHz            |
|                                                                                                                                                                             |                                                                                                                                                                             | Fixed             | ECC/REC/(05)07                   | EN 302 217 |                                          |
|                                                                                                                                                                             |                                                                                                                                                                             | LPR               | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 |                                          |
|                                                                                                                                                                             |                                                                                                                                                                             | Radio astronomy   |                                  |            | Continuum and spectral line observations |
|                                                                                                                                                                             |                                                                                                                                                                             | TLPR              | ERC/REC 70-03                    | EN 302 372 |                                          |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 84 GHz - 86 GHz

|                                                                                              |                                                                                              |                 |                                  |            |                                          |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------|----------------------------------|------------|------------------------------------------|
| FIXED (5.338A)<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 | FIXED (5.338A)<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 | Fixed           | ECC/REC/(05)07                   | EN 302 217 |                                          |
|                                                                                              |                                                                                              | LPR             | ECC/DEC/(11)02,<br>ERC/REC 70-03 | EN 302 729 | Within the band 84-85 GHz                |
|                                                                                              |                                                                                              | Radio astronomy |                                  |            | Continuum and spectral line observations |
|                                                                                              |                                                                                              | TLPR            | ERC/REC 70-03                    | EN 302 372 | Within the band 84-85 GHz                |

## 86 GHz - 92 GHz

|                                                                                                      |                                                                                                      |                             |  |  |                                                                                                                |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------|--|--|----------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite) |  |  | Measurement of clouds, oil spills, ice, snow, rain, reference window for the temperature sounding near 118 GHz |
|                                                                                                      |                                                                                                      | Radio astronomy             |  |  | Continuum and spectral line observations. VLBI                                                                 |

## 92 GHz - 94 GHz

|                                                                           |                                                                           |                 |                                   |  |                                          |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------|-----------------------------------|--|------------------------------------------|
| FIXED (5.338A)<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | FIXED (5.338A)<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | Fixed           | ECC/REC/(14)01,<br>ECC/REC/(18)02 |  |                                          |
|                                                                           |                                                                           | Radio astronomy |                                   |  | Continuum and spectral line observations |

## 94 GHz - 94.1 GHz

|                                                                                                                            |                                                                                                                            |                            |  |  |                                          |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------|--|--|------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>Radio astronomy<br><br>5.562 5.562A | EARTH EXPLORATION-SATELLITE<br>(ACTIVE)<br>RADIOLOCATION<br>SPACE RESEARCH (ACTIVE)<br>Radio astronomy<br><br>5.562 5.562A | Active sensors (satellite) |  |  | Cloud radars                             |
|                                                                                                                            |                                                                                                                            | Radio astronomy            |  |  | Continuum and spectral line observations |
|                                                                                                                            |                                                                                                                            | Space research             |  |  |                                          |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
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## 94.1 GHz - 95 GHz

|                                                                  |                                                                  |                 |                                   |  |                                          |
|------------------------------------------------------------------|------------------------------------------------------------------|-----------------|-----------------------------------|--|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | Fixed           | ECC/REC/(14)01,<br>ECC/REC/(18)02 |  |                                          |
|                                                                  |                                                                  | Radio astronomy |                                   |  | Continuum and spectral line observations |

## 95 GHz - 100 GHz

|                                                                                                                        |                                                                                                                        |                 |                |  |                                          |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|--|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.149 5.554 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.149 5.554 | Fixed           | ECC/REC/(18)02 |  |                                          |
|                                                                                                                        |                                                                                                                        | Radio astronomy |                |  | Continuum and spectral line observations |

## 100 GHz - 102 GHz

|                                                                                                            |                                                                                                            |                             |  |  |                                           |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------|--|--|-------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.341 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.341 | Passive sensors (satellite) |  |  | Limb sounding of atmospheric constituents |
|                                                                                                            |                                                                                                            | Radio astronomy             |  |  | Continuum and spectral line observations  |

## 102 GHz - 105 GHz

|                                                       |                                                       |                 |                |  |                                          |
|-------------------------------------------------------|-------------------------------------------------------|-----------------|----------------|--|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.341 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.341 | Fixed           | ECC/REC/(18)02 |  |                                          |
|                                                       |                                                       | Radio astronomy |                |  | Continuum and spectral line observations |

## 105 GHz - 109.5 GHz

|                                                                                  |                                                                                  |                 |                |  |                                          |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------|----------------|--|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br><br>5.149 5.341 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br><br>5.149 5.341 | Fixed           | ECC/REC/(18)02 |  |                                          |
|                                                                                  |                                                                                  | Radio astronomy |                |  | Continuum and spectral line observations |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

### 109.5 GHz - 111.8 GHz

|                                                                                                     |                                                                                                     |                 |  |  |                                          |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------|--|--|------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.341 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.341 | Radio astronomy |  |  | Continuum and spectral line observations |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------|--|--|------------------------------------------|

### 111.8 GHz - 114.25 GHz

|                                                                              |                                                                              |                 |                |  |                                          |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------|----------------|--|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br>5.149 5.341 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br>5.149 5.341 | Fixed           | ECC/REC/(18)02 |  |                                          |
|                                                                              |                                                                              | Radio astronomy |                |  | Continuum and spectral line observations |

### 114.25 GHz - 116 GHz

|                                                                                                     |                                                                                                     |                 |  |  |                                          |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------|--|--|------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.341 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br>5.340 5.341 | Radio astronomy |  |  | Continuum and spectral line observations |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------|--|--|------------------------------------------|

### 116 GHz - 119.98 GHz

|                                                                                                        |                                                                            |                                 |                                  |            |                                                                               |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.562C)<br>SPACE RESEARCH (PASSIVE)<br>5.341 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.562C)<br>5.341 | Passive sensors (satellite)     |                                  |            | Passive sensing as part of the oxygen absorption band with peak at 118.75 GHz |
|                                                                                                        |                                                                            | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                               |

### 119.98 GHz - 120.02 GHz

|                                                                                                        |                                                                            |                                 |                                  |            |                                                                               |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.562C)<br>SPACE RESEARCH (PASSIVE)<br>5.341 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.562C)<br>5.341 | Passive sensors (satellite)     |                                  |            | Passive sensing as part of the oxygen absorption band with peak at 118.75 GHz |
|                                                                                                        |                                                                            | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                               |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 120.02 GHz - 122.25 GHz

|                                                                                                            |                                                                                                            |                                 |                                  |            |                                                                               |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.562C)<br>SPACE RESEARCH (PASSIVE)<br><br>5.138 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>INTER-SATELLITE (5.562C)<br>SPACE RESEARCH (PASSIVE)<br><br>5.138 | Non-specific SRDs               | ERC/REC 70-03                    | EN 305 550 | Within the band 122-123 GHz                                                   |
|                                                                                                            |                                                                                                            | Passive sensors (satellite)     |                                  |            | Passive sensing as part of the oxygen absorption band with peak at 118.75 GHz |
|                                                                                                            |                                                                                                            | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                               |

## 122.25 GHz - 123 GHz

|                                                                    |                                                                                         |                                 |                                  |            |                             |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-----------------------------|
| FIXED<br>INTER-SATELLITE<br>MOBILE (5.558)<br>Amateur<br><br>5.138 | FIXED<br>INTER-SATELLITE<br>MOBILE (5.558)<br>Amateur<br>Amateur-satellite<br><br>5.138 | Amateur                         |                                  |            |                             |
|                                                                    |                                                                                         | Amateur-satellite               |                                  |            |                             |
|                                                                    |                                                                                         | Non-specific SRDs               | ERC/REC 70-03                    | EN 305 550 | Within the band 122-123 GHz |
|                                                                    |                                                                                         | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                             |

## 123 GHz - 130 GHz

|                                                                                                                                                             |                                                                                                                                                             |                                 |                                  |            |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br>Radio astronomy<br><br>5.149 5.554 | FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE-SATELLITE (SPACE-TO-EARTH)<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br>Radio astronomy<br><br>5.149 5.554 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                                                                                             |                                                                                                                                                             | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 130 GHz - 134 GHz

|                                                                                                                           |                                                                                                                           |                                 |                                  |            |                                          |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| EARTH EXPLORATION-SATELLITE (5.562E)<br>FIXED<br>INTER-SATELLITE<br>MOBILE (5.558)<br>RADIO ASTRONOMY<br><br>5.149 5.562A | EARTH EXPLORATION-SATELLITE (5.562E)<br>FIXED<br>INTER-SATELLITE<br>MOBILE (5.558)<br>RADIO ASTRONOMY<br><br>5.149 5.562A | Fixed                           | ECC/REC/(18)01                   |            |                                          |
|                                                                                                                           |                                                                                                                           | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                                                           |                                                                                                                           | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 134 GHz - 136 GHz

|                                                 |                                                 |                                 |                                  |            |                                          |
|-------------------------------------------------|-------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE<br>Radio astronomy | AMATEUR<br>AMATEUR-SATELLITE<br>Radio astronomy | Amateur                         |                                  |            | Within the band 134-141 GHz              |
|                                                 |                                                 | Amateur-satellite               |                                  |            | Within the band 134-141 GHz              |
|                                                 |                                                 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                 |                                                 | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 136 GHz - 141 GHz

|                                                                               |                                                                               |                                 |                                  |            |                                          |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.149 | Amateur                         |                                  |            | Within the band 134-141 GHz              |
|                                                                               |                                                                               | Amateur-satellite               |                                  |            | Within the band 134-141 GHz              |
|                                                                               |                                                                               | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                               |                                                                               | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 141 GHz - 148.5 GHz

|                                                                  |                                                                  |                                 |                                  |            |                                          |
|------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | Fixed                           | ECC/REC/(18)01                   |            |                                          |
|                                                                  |                                                                  | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                  |                                                                  | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 148.5 GHz - 151.5 GHz

|                                                                                                      |                                                                                                      |                                 |                                  |            |                                                             |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite)     |                                  |            | Harmonised reference window for passive sensor observations |
|                                                                                                      |                                                                                                      | Radio astronomy                 |                                  |            | Continuum and spectral line observations                    |
|                                                                                                      |                                                                                                      | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                             |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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### 151.5 GHz - 155.5 GHz

|                                                                  |                                                                  |                                 |                                  |            |                                          |
|------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>RADIOLOCATION<br><br>5.149 | Fixed                           | ECC/REC/(18)01                   |            |                                          |
|                                                                  |                                                                  | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                  |                                                                  | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

### 155.5 GHz - 158.5 GHz

|                                                                                                                        |                                                                                                                        |                                 |                                  |            |                                                    |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|----------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br><br>5.149 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br><br>5.149 | Fixed                           | ECC/REC/(18)01                   |            |                                                    |
|                                                                                                                        |                                                                                                                        | Passive sensors (satellite)     |                                  |            | Protection until 1.1.2018                          |
|                                                                                                                        |                                                                                                                        | Radio astronomy                 |                                  |            | Spectral line and wide band continuum observations |
|                                                                                                                        |                                                                                                                        | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                    |

### 158.5 GHz - 164 GHz

|                                                                                          |                                                                                          |                                 |                                  |            |  |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (SPACE-TO-EARTH) | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>MOBILE-SATELLITE (SPACE-TO-EARTH) | Fixed                           | ECC/REC/(18)01                   |            |  |
|                                                                                          |                                                                                          | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |

### 164 GHz - 167 GHz

|                                                                                                      |                                                                                                      |                                 |                                  |            |                                                                                                                                      |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite)     |                                  |            | Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz. Atmospheric limb sounding of the 164.38 GHz CO line |
|                                                                                                      |                                                                                                      | Radio astronomy                 |                                  |            | Continuum and spectral line observations                                                                                             |
|                                                                                                      |                                                                                                      | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                                                                                      |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 167 GHz - 174.5 GHz

|                                                                                             |                                                                                             |                                 |                                  |            |                                                                         |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-------------------------------------------------------------------------|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>INTER-SATELLITE<br>MOBILE (5.558)<br><br>5.149 | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>INTER-SATELLITE<br>MOBILE (5.558)<br><br>5.149 | Fixed                           | ECC/REC/(18)01                   |            |                                                                         |
|                                                                                             |                                                                                             | Radio astronomy                 |                                  |            | Within the band 168-174.5 GHz. Continuum and spectral line observations |
|                                                                                             |                                                                                             | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                         |

## 174.5 GHz - 174.8 GHz

|                                            |                                            |                                 |                                  |            |  |
|--------------------------------------------|--------------------------------------------|---------------------------------|----------------------------------|------------|--|
| FIXED<br>INTER-SATELLITE<br>MOBILE (5.558) | FIXED<br>INTER-SATELLITE<br>MOBILE (5.558) | Fixed                           | ECC/REC/(18)01                   |            |  |
|                                            |                                            | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |

## 174.8 GHz - 182 GHz

|                                                                                                  |                                                                                                  |                                 |                                  |            |                                                                                 |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|---------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>INTER-SATELLITE (5.562H)<br>SPACE RESEARCH (PASSIVE) | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>INTER-SATELLITE (5.562H)<br>SPACE RESEARCH (PASSIVE) | Passive sensors (satellite)     |                                  |            | Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz |
|                                                                                                  |                                                                                                  | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                                 |

## 182 GHz - 185 GHz

|                                                                                                      |                                                                                                      |                                 |                                  |            |                                                                                 |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|---------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite)     |                                  |            | Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz |
|                                                                                                      |                                                                                                      | Radio astronomy                 |                                  |            | Continuum and spectral line observations                                        |
|                                                                                                      |                                                                                                      | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                                 |

## 185 GHz - 190 GHz

|                                                                                                  |                                                                                                  |                                 |                                  |            |                                                                                 |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|---------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>INTER-SATELLITE (5.562H)<br>SPACE RESEARCH (PASSIVE) | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>INTER-SATELLITE (5.562H)<br>SPACE RESEARCH (PASSIVE) | Passive sensors (satellite)     |                                  |            | Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz |
|                                                                                                  |                                                                                                  | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                                 |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 190 GHz - 191.8 GHz

|                                                                                |                                                                                |                                 |                               |            |                                                                                 |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------|---------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite)     |                               |            | Passive sensing of the water vapour absorption line whose peak is at 183.31 GHz |
|                                                                                |                                                                                | Radio astronomy                 |                               |            | Continuum and spectral line observations                                        |
|                                                                                |                                                                                | Radiodetermination applications | ECC/DEC/(22)03, ERC/REC 70-03 | EN 305 550 |                                                                                 |

## 191.8 GHz - 200 GHz

|                                                                                                                                |                                                                                                                                |                                 |                               |            |                                          |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------|------------------------------------------|
| FIXED INTER-SATELLITE MOBILE (5.558)<br>MOBILE-SATELLITE RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.149 5.341 5.554 | FIXED INTER-SATELLITE MOBILE (5.558)<br>MOBILE-SATELLITE RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.149 5.341 5.554 | Radio astronomy                 |                               |            | Continuum and spectral line observations |
|                                                                                                                                |                                                                                                                                | Radiodetermination applications | ECC/DEC/(22)03, ERC/REC 70-03 | EN 305 550 |                                          |

## 200 GHz - 202 GHz

|                                                                                                                |                                                                                                                |                                 |                               |            |                                                                                             |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------|---------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.341 5.563A | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.341 5.563A | Earth exploration-satellite     |                               |            | (EESS) Atmospheric limb sounding and atmospheric remote sensing of nitrous oxide at 201 GHz |
|                                                                                                                |                                                                                                                | Radio astronomy                 |                               |            | Continuum and spectral line observations                                                    |
|                                                                                                                |                                                                                                                | Radiodetermination applications | ECC/DEC/(22)03, ERC/REC 70-03 | EN 305 550 |                                                                                             |

## 202 GHz - 209 GHz

|                                                                                                                |                                                                                                                |                                 |                               |            |                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------|---------------------------------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.341 5.563A | EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.341 5.563A | Earth exploration-satellite     |                               |            | (EESS) Atmospheric limb sounding and atmospheric remote sensing of water vapour at 203.4 GHz and ozone at 208.5 GHz |
|                                                                                                                |                                                                                                                | Radio astronomy                 |                               |            | Continuum and spectral line observations                                                                            |
|                                                                                                                |                                                                                                                | Radiodetermination applications | ECC/DEC/(22)03, ERC/REC 70-03 | EN 305 550 |                                                                                                                     |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 209 GHz - 217 GHz

|                                                                                           |                                                                                           |                                 |                                  |            |                                          |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.341 | FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.341 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                           |                                                                                           | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 217 GHz - 226 GHz

|                                                                                                                      |                                                                                                                      |                                 |                                  |            |                                          |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br><br>5.149 5.341 | FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br>SPACE RESEARCH (5.562B)<br><br>5.149 5.341 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                                                      |                                                                                                                      | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 226 GHz - 231.5 GHz

|                                                                                                      |                                                                                                      |                                 |                                  |            |                                                                                            |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--------------------------------------------------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 | Passive sensors (satellite)     |                                  |            | Atmospheric limb sounding. Reference window for higher frequency water vapour measurements |
|                                                                                                      |                                                                                                      | Radio astronomy                 |                                  |            | Continuum and spectral line observations (e.g. CO line), VLBI                              |
|                                                                                                      |                                                                                                      | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                                                            |

## 231.5 GHz - 232 GHz

|                                  |                                  |                                 |                                  |            |  |
|----------------------------------|----------------------------------|---------------------------------|----------------------------------|------------|--|
| FIXED<br>MOBILE<br>Radiolocation | FIXED<br>MOBILE<br>Radiolocation | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |
|----------------------------------|----------------------------------|---------------------------------|----------------------------------|------------|--|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 232 GHz - 235 GHz

|                                                                      |                                                                      |                                 |                                  |            |  |
|----------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>Radiolocation | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>Radiolocation | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |
|----------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|

## 235 GHz - 238 GHz

|                                                                                                                                                  |                                                                                                                               |                                 |                                  |            |                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|-----------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(5.563AA)<br>FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>SPACE RESEARCH (PASSIVE)<br><br>5.563A 5.563B | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>SPACE RESEARCH (PASSIVE)<br><br>5.563A 5.563B | Passive sensors (satellite)     |                                  |            | Passive sensing limited to microwave sounding |
|                                                                                                                                                  |                                                                                                                               | Radio astronomy                 |                                  |            | Continuum and spectral line observations      |
|                                                                                                                                                  |                                                                                                                               | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                               |

## 238 GHz - 239.2 GHz

|                                                                                                                      |                                                                                                                      |                                 |                                  |            |  |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|
| FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>RADIOLOCATION<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>RADIOLOCATION<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|

## 239.2 GHz - 240 GHz

|                                                                                                                                               |                                                                                                                      |                                 |                                  |            |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>RADIOLOCATION<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE | FIXED<br>FIXED-SATELLITE (SPACE-TO-EARTH)<br>MOBILE<br>RADIOLOCATION<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|--|

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
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## 240 GHz - 241 GHz

|                                                        |                                  |                                 |                                  |            |  |
|--------------------------------------------------------|----------------------------------|---------------------------------|----------------------------------|------------|--|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIOLOCATION | FIXED<br>MOBILE<br>RADIOLOCATION | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |  |
|--------------------------------------------------------|----------------------------------|---------------------------------|----------------------------------|------------|--|

## 241 GHz - 242.2 GHz

|                                                                                                                    |                                                                                 |                                 |                                  |            |                                          |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>5.138 5.149 | Amateur                         |                                  |            |                                          |
|                                                                                                                    |                                                                                 | Amateur-satellite               |                                  |            | Within the band 241-250 GHz              |
|                                                                                                                    |                                                                                 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                                                    |                                                                                 | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 242.2 GHz - 244.2 GHz

|                                                                                 |                                                                                 |                                 |                                  |            |                                          |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>5.138 5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>5.138 5.149 | Amateur                         |                                  |            |                                          |
|                                                                                 |                                                                                 | Amateur-satellite               |                                  |            |                                          |
|                                                                                 |                                                                                 | Non-specific SRDs               | ERC/REC 70-03                    | EN 305 550 | Within the band 244-246 GHz              |
|                                                                                 |                                                                                 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                 |                                                                                 | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 244.2 GHz - 247.2 GHz

|                                                                                                                                        |                                                                                 |                                 |                                  |            |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| EARTH EXPLORATION-SATELLITE (PASSIVE)<br>RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite (5.138 5.149)<br>5.138 5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br>5.138 5.149 | Amateur                         |                                  |            |                                          |
|                                                                                                                                        |                                                                                 | Amateur-satellite               |                                  |            |                                          |
|                                                                                                                                        |                                                                                 | Non-specific SRDs               | ERC/REC 70-03                    | EN 305 550 | Within the band 244-246 GHz              |
|                                                                                                                                        |                                                                                 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                                                                        |                                                                                 | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 247.2 GHz - 248 GHz

|                                                                               |                                                                                     |                                 |                                  |            |                                          |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.149 | RADIO ASTRONOMY<br>RADIOLOCATION<br>Amateur<br>Amateur-satellite<br><br>5.138 5.149 | Amateur                         |                                  |            |                                          |
|                                                                               |                                                                                     | Amateur-satellite               |                                  |            |                                          |
|                                                                               |                                                                                     | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                               |                                                                                     | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 248 GHz - 250 GHz

|                                                              |                                                              |                                 |                                  |            |                                          |
|--------------------------------------------------------------|--------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| AMATEUR<br>AMATEUR-SATELLITE<br>Radio astronomy<br><br>5.149 | AMATEUR<br>AMATEUR-SATELLITE<br>Radio astronomy<br><br>5.149 | Amateur                         |                                  |            |                                          |
|                                                              |                                                              | Amateur-satellite               |                                  |            | Within the band 241-250 GHz              |
|                                                              |                                                              | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                              |                                                              | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                          |

## 250 GHz - 252 GHz

|                                                                                                             |                                                                                                             |                                 |                                  |            |                                                    |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|----------------------------------------------------|
| EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.563A | EARTH EXPLORATION-SATELLITE<br>(PASSIVE)<br>RADIO ASTRONOMY<br>SPACE RESEARCH (PASSIVE)<br><br>5.340 5.563A | Earth exploration-satellite     |                                  |            | (EESS) Limb sounding of nitrous oxide near 251 GHz |
|                                                                                                             |                                                                                                             | Radio astronomy                 |                                  |            | Continuum and spectral line observations           |
|                                                                                                             |                                                                                                             | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 |                                                    |

## 252 GHz - 265 GHz

|                                                                                                                                            |                                                                                                                                            |                                 |                                  |            |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------------|------------------------------------------|
| FIXED<br>MOBILE<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>RADIO ASTRONOMY<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.149 5.554 | FIXED<br>MOBILE<br>MOBILE-SATELLITE (EARTH-TO-SPACE)<br>RADIO ASTRONOMY<br>RADIONAVIGATION<br>RADIONAVIGATION-SATELLITE<br><br>5.149 5.554 | Radio astronomy                 |                                  |            | Continuum and spectral line observations |
|                                                                                                                                            |                                                                                                                                            | Radiodetermination applications | ECC/DEC/(22)03,<br>ERC/REC 70-03 | EN 305 550 | within frequency range 116-260 GHz       |

| RR Region 1 | European Common Allocations | Application | CEPT Deliverables | Standard | Note |
|-------------|-----------------------------|-------------|-------------------|----------|------|
|-------------|-----------------------------|-------------|-------------------|----------|------|

## 265 GHz - 275 GHz

|                                                                                            |                                                                                            |                 |  |  |                                          |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------|--|--|------------------------------------------|
| FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.563A | FIXED<br>FIXED-SATELLITE (EARTH-TO-SPACE)<br>MOBILE<br>RADIO ASTRONOMY<br><br>5.149 5.563A | Radio astronomy |  |  | Continuum and spectral line observations |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------|--|--|------------------------------------------|

## 275 GHz - 3000 GHz

|                                   |                                   |             |  |  |                                                                                  |
|-----------------------------------|-----------------------------------|-------------|--|--|----------------------------------------------------------------------------------|
| Not allocated<br><br>5.564A 5.565 | Not allocated<br><br>5.564A 5.565 | Fixed       |  |  | Within the frequency bands 275-296 GHz, 306-313 GHz, 318-333 GHz and 356-450 GHz |
|                                   |                                   | Land mobile |  |  | Within the frequency bands 275-296 GHz, 306-313 GHz, 318-333 GHz and 356-450 GHz |

## ECA Footnotes

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECA5            | In parts of this band aeronautical stations and aircraft stations utilise the preferred 8.33 kHz channel spacing for non secure communications requirements.                                                                                                                                                                                                                                                                                                                                          |
| ECA6            | The mobile-satellite service is limited to low earth orbiting satellites.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ECA7            | This band can also be used by low capacity fixed links in rural areas on a national basis. These links need to be coordinated with mobile service and require full protection.                                                                                                                                                                                                                                                                                                                        |
| ECA8            | Any use of low capacity fixed links shall be avoided in areas where such use might cause harmful interference to the maritime mobile VHF radiocommunication service.                                                                                                                                                                                                                                                                                                                                  |
| ECA9            | CEPT administrations may authorise all or parts of the band 69.9-70.5 MHz to the amateur service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                |
| ECA10           | The range 225-399.9 MHz is essential to NATO and is in military use for land mobile, mobile-satellite, Air/Ground/Air and specific maritime and terrestrial communications, including ITU Region 2. This NATO UHF band 225-400 MHz is the only harmonised and commonly available resource managed by NATO on a daily basis in and for NATO nations. It is recognised that 380-385 MHz and 390-395 MHz are currently shared with narrowband Public Protection and Disaster Relief (PPDR) applications. |
| ECA12           | The applicable RR 5 footnotes in column 1 remain in force. Administrations are however urged to aim for the fullest possible harmonisation with the ITU Table of Allocations and ECA.                                                                                                                                                                                                                                                                                                                 |
| ECA13           | CEPT administrations are urged to take all practical steps to clear the band 645-960 MHz of the assignments to the aeronautical radionavigation service.                                                                                                                                                                                                                                                                                                                                              |
| ECA14           | Radiolocation limited to military requirements for naval ship borne radars.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ECA16           | Use of the band by the mobile service is limited to tactical radio relay and Video links applications.                                                                                                                                                                                                                                                                                                                                                                                                |
| ECA16A          | Use of the band by the mobile service is limited to tactical radio relay and SAP/SAB applications.                                                                                                                                                                                                                                                                                                                                                                                                    |
| ECA17           | In the sub-bands 5755-5765 MHz, 10.36-10.37 GHz, 10.45-10.46 GHz the amateur service operates on a secondary basis. In making assignments to other services, CEPT administrations are requested wherever possible to maintain these sub-bands in such a way as to facilitate the reception of amateur emissions with minimal power flux densities.                                                                                                                                                    |
| ECA17A          | Use of the band by the mobile service is limited to Video links.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ECA19           | This band is allocated to the radio astronomy service. CEPT administrations are urged to take all practicable steps to protect the radio astronomy service from harmful interference. Emissions from space or airborne stations in this and adjacent bands can cause serious harmful interference.                                                                                                                                                                                                    |
| ECA20           | This fixed service band is designated for common use by civil and non civil users. Any user priorities in respect of preferred channels or sub-bands are to be determined after discussions between interested parties.                                                                                                                                                                                                                                                                               |
| ECA22           | The band 5250-5850 MHz is utilised for a variety of radiodetermination applications falling within the radionavigation and radiolocation services. This band will be subject to further detailed consideration.                                                                                                                                                                                                                                                                                       |
| ECA23           | In the sub-bands 5660-5670 MHz (earth to space), 5830-5850 MHz (space to earth) and 10.45-10.50 GHz the amateur-satellite additionally operates on a secondary and non interference basis to other services. In making assignments to other services, CEPT administrations are requested wherever possible to maintain these allocations in such a way as to facilitate the reception of amateur emissions with minimal power flux densities.                                                         |
| ECA24           | The band 8500-10000 MHz is utilised for a variety of radiodetermination applications falling within the radionavigation and radiolocation services. This band will be subject to further detailed consideration in conjunction with the band 5250-5850 MHz (see ECA22).                                                                                                                                                                                                                               |
| ECA26           | The band 13.25-14.0 GHz is utilised for a variety of radiodetermination applications falling within the radionavigation and radiolocation services. This band will be subject to further detailed consideration.                                                                                                                                                                                                                                                                                      |
| ECA28           | CEPT administrations shall not deploy new fixed service systems in the band 11.7-12.5 GHz (ERC/DEC(00)08).                                                                                                                                                                                                                                                                                                                                                                                            |
| ECA29           | The frequency bands 890-915 / 935-960 MHz, 880-890 / 925-935 MHz, 1710-1785 / 1805-1880 MHz, 1920-1980 MHz and 2110-2170 MHz are reserved for public cellular mobile use only. Other services such as the fixed service should only be allowed in the above bands where coexistence with public mobile systems is possible i.e. in sparsely populated or rural areas where the frequency band is not needed for mobile cellular systems.                                                              |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                        |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECA30           | National administrations should consider co-ordination zones around the EISCAT sites when using the band 925-935 MHz for mobile services including international planning for military services. Short Range Devices should not use this band.                                          |
| ECA32           | The bands 880-915 MHz and 925-960 MHz are currently used for GSM (2nd generation terrestrial mobile system) in most CEPT member countries and by IMT, depending on the market demands and national licensing schemes.                                                                   |
| ECA34           | Parts of the bands 450-457.5/460-467.5 MHz may also be used for existing and evolving public cellular networks on a national basis.                                                                                                                                                     |
| ECA35           | In Europe the band 75.5-76 GHz is also allocated to the Amateur and Amateur Satellite services.                                                                                                                                                                                         |
| ECA36           | A frequency band, which has been harmonised by NATO and NATO member nations for military use as defined in the NATO Joint Civil/Military Frequency Agreement (NJFA) 2014. Note: NATO Joint Civil/Military Frequency Agreement (NJFA) - Extract for Public Disclosure – 14 February 2017 |
| ECA37           | In Europe the allocation to the mobile service is limited to the band 3400-3800 MHz.                                                                                                                                                                                                    |
| ECA38           | Administrations may choose at national level to allow MFCN for the command and control and payload links of UAS within the current MFCN bands. Administrations are requested to ensure protection of other existing systems and services in these frequency bands                       |
| ECA39           | Administrations shall avoid deployment of high-density mobile systems incl. high-density fixed wireless access in the 22.0-23.6 GHz frequency band (ECC/DEC/(18)06)                                                                                                                     |

# Radio Regulations Footnotes

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.54A           | Use of the 8.3-11.3 kHz frequency band by stations in the meteorological aids service is limited to passive use only. In the band 9-11.3 kHz, meteorological aids stations shall not claim protection from stations of the radionavigation service submitted for notification to the Bureau prior to 1 January 2013. For sharing between stations of the meteorological aids service and stations in the radionavigation service submitted for notification after this date, the most recent version of Recommendation ITU-R RS.1881 should be applied. (WRC-12) |
| 5.54B           | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Egypt, the United Arab Emirates, the Russian Federation, Iran (Islamic Republic of), Iraq, Kuwait, Lebanon, Morocco, Qatar, the Syrian Arab Republic, Sudan and Tunisia, the frequency band 8.3-9 kHz is also allocated to the radionavigation, fixed and mobile services on a primary basis. (WRC-15)                                                                                                                                                                                                 |
| 5.55            | Additional allocation: in Armenia, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the frequency band 14-17 kHz is also allocated to the radionavigation service on a primary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                          |
| 5.56            | The stations of services to which the frequency bands 14-19.95 kHz and 20.05-70 kHz and in Region 1 also the frequency bands 72-84 kHz and 86-90 kHz are allocated may transmit standard frequency and time signals. Such stations shall be afforded protection from harmful interference. In Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the frequencies 25 kHz and 50 kHz will be used for this purpose under the same conditions. (WRC-23)                                                        |
| 5.57            | The use of the bands 14-19.95 kHz, 20.05-70 kHz and 70-90 kHz (72-84 kHz and 86-90 kHz in Region 1) by the maritime mobile service is limited to coast radiotelegraph stations (A1A and F1B only). Exceptionally, the use of class J2B or J7B emissions is authorized subject to the necessary bandwidth not exceeding that normally used for class A1A or F1B emissions in the band concerned.                                                                                                                                                                  |
| 5.58            | Additional allocation: in Armenia, Azerbaijan, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the frequency band 67-70 kHz is also allocated to the radionavigation service on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                              |
| 5.60            | In the bands 70-90 kHz (70-86 kHz in Region 1) and 110-130 kHz (112-130 kHz in Region 1), pulsed radionavigation systems may be used on condition that they do not cause harmful interference to other services to which these bands are allocated.                                                                                                                                                                                                                                                                                                              |
| 5.62            | Administrations which operate stations in the radionavigation service in the band 90-110 kHz are urged to coordinate technical and operating characteristics in such a way as to avoid harmful interference to the services provided by these stations.                                                                                                                                                                                                                                                                                                          |
| 5.64            | Only classes A1A or F1B, A2C, A3C, F1C or F3C emissions are authorized for stations of the fixed service in the bands allocated to this service between 90 kHz and 160 kHz (148.5 kHz in Region 1) and for stations of the maritime mobile service in the bands allocated to this service between 110 kHz and 160 kHz (148.5 kHz in Region 1). Exceptionally, class J2B or J7B emissions are also authorized in the bands between 110 kHz and 160 kHz (148.5 kHz in Region 1) for stations of the maritime mobile service.                                       |
| 5.66            | Different category of service: in Germany, the allocation of the band 115-117.6 kHz to the fixed and maritime mobile services is on a primary basis (see No. 5.33) and to the radionavigation service on a secondary basis (see No. 5.32).                                                                                                                                                                                                                                                                                                                       |
| 5.67            | Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 130-148.5 kHz is also allocated to the radionavigation service on a secondary basis. Within and between these countries this service shall have an equal right to operate. (WRC-19)                                                                                                                                                                                                                                                                                                    |
| 5.67A           | Stations in the amateur service using frequencies in the band 135.7-137.8 kHz shall not exceed a maximum radiated power of 1 W (e.i.r.p.) and shall not cause harmful interference to stations of the radionavigation service operating in countries listed in No. 5.67. WRC-07)                                                                                                                                                                                                                                                                                 |
| 5.67B           | The use of the frequency band 135.7-137.8 kHz in Algeria, Egypt, Iraq, Lebanon, Syrian Arab Republic, Sudan, South Sudan and Tunisia is limited to the fixed and maritime mobile services. The amateur service shall not be used in the above-mentioned countries in the frequency band 135.7-137.8 kHz, and this should be taken into account by the countries authorizing such use. (WRC-19)                                                                                                                                                                   |
| 5.68            | Alternative allocation: in Congo (Rep of the), the Dem. Rep. of the Congo and South Africa, the frequency band 160-200 kHz is allocated to the fixed service on a primary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                        |
| 5.69            | Additional allocation: in Somalia, the band 200-255 kHz is also allocated to the aeronautical radionavigation service on a primary basis.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.70            | Alternative allocation: in Angola, Botswana, Burundi, the Central African Rep., Congo (Rep. of the), Eswatini, Ethiopia, Kenya, Lesotho, Madagascar, Malawi, Mozambique, Namibia, Nigeria, Oman, the Dem. Rep. of the Congo, South Africa, Tanzania, Chad, Zambia and Zimbabwe, the frequency band 200-283.5 kHz is allocated to the aeronautical radionavigation service on a primary basis. (WRC-19)                                                                                                                                                           |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.73            | The band 285-325 kHz (283.5-325 kHz in Region 1) in the maritime radionavigation service may be used to transmit supplementary navigational information using narrow-band techniques, on condition that no harmful interference is caused to radiobeacon stations operating in the radionavigation service. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.74            | Additional allocation: in Region 1, the frequency band 285.3-285.7 kHz is also allocated to the maritime radionavigation service (other than radiobeacons) on a primary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.75            | Different category of service: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Moldova, Kyrgyzstan, Tajikistan, Turkmenistan, Ukraine and the Black Sea areas of Romania, the allocation of the band 315-325 kHz to the maritime radionavigation service is on a primary basis under the condition that in the Baltic Sea area, the assignment of frequencies in this band to new stations in the maritime or aeronautical radionavigation services shall be subject to prior consultation between the administrations concerned. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.76            | The frequency 410 kHz is designated for radio direction-finding in the maritime radionavigation service. The other radionavigation services to which the band 405-415 kHz is allocated shall not cause harmful interference to radio direction-finding in the band 406.5-413.5 kHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.77            | Different category of service: in Australia, China, the French overseas communities of Region 3, Korea (Rep. of), India, Iran (Islamic Republic of), Japan, Pakistan, Papua New Guinea, the Dem. People's Rep. of Korea and Sri Lanka, the allocation of the frequency band 415-495 kHz to the aeronautical radionavigation service is on a primary basis. In Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Latvia, Uzbekistan and Kyrgyzstan, the allocation of the frequency band 435-495 kHz to the aeronautical radionavigation service is on a primary basis. Administrations in all the aforementioned countries shall take all practical steps necessary to ensure that aeronautical radionavigation stations in the frequency band 435-495 kHz do not cause interference to reception by coast stations of transmissions from ship stations on frequencies designated for ship stations on a worldwide basis. (WRC-19) |
| 5.79            | In the maritime mobile service, the frequency bands 415-495 kHz and 505-526.5 kHz are limited to radiotelegraphy and may also be used for the NAVDAT system in accordance with the most recent version of Recommendation ITU-R M.2010, subject to agreement between interested and affected administrations. NAVDAT transmitting stations are limited to coast stations. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.79A           | When establishing coast stations in the NAVTEX service on the frequencies 490 kHz, 518 kHz and 4209.5 kHz, administrations are strongly recommended to coordinate the operating characteristics in accordance with the procedures of the International Maritime Organization (IMO) (see Resolution 339 (Rev.WRC-07)). (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.80            | In Region 2, the use of the band 435-495 kHz by the aeronautical radionavigation service is limited to non-directional beacons not employing voice transmission                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.80A           | The maximum equivalent isotropically radiated power (e.i.r.p.) of stations in the amateur service using frequencies in the band 472-479 kHz shall not exceed 1 W. Administrations may increase this limit of e.i.r.p. to 5 W in portions of their territory which are at a distance of over 800 km from the borders of Algeria, Saudi Arabia, Azerbaijan, Bahrain, Belarus, China, Comoros, Djibouti, Egypt, United Arab Emirates, the Russian Federation, Iran (Islamic Republic of), Iraq, Jordan, Kazakhstan, Kuwait, Lebanon, Libya, Morocco, Mauritania, Oman, Uzbekistan, Qatar, Syrian Arab Republic, Kyrgyzstan, Somalia, Sudan, Tunisia, Ukraine and Yemen. In this frequency band, stations in the amateur service shall not cause harmful interference to, or claim protection from, stations of the aeronautical radionavigation service. (WRC-12)                                                                                     |
| 5.80B           | The use of the frequency band 472-479 kHz in Algeria, Saudi Arabia, Azerbaijan, Bahrain, Belarus, China, Comoros, Djibouti, Egypt, United Arab Emirates, the Russian Federation, Iraq, Jordan, Kazakhstan, Kuwait, Lebanon, Libya, Mauritania, Oman, Uzbekistan, Qatar, Syrian Arab Republic, Kyrgyzstan, Somalia, Sudan, Tunisia and Yemen is limited to the maritime mobile and aeronautical radionavigation services. The amateur service shall not be used in the above-mentioned countries in this frequency band, and this should be taken into account by the countries authorizing such use. (WRC-12)                                                                                                                                                                                                                                                                                                                                      |
| 5.82            | In the maritime mobile service, the frequency 490 kHz is to be used exclusively for the transmission by coast stations of navigational and meteorological warnings and urgent information to ships, by means of narrow-band direct-printing telegraphy. The conditions for use of the frequency 490 kHz are prescribed in Articles 31 and 52. In using the band 415-495 kHz for the aeronautical radionavigation service, administrations are requested to ensure that no harmful interference is caused to the frequency 490 kHz. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.82C           | The frequency band 495-505 kHz is used for the international NAVDAT system as described in the most recent version of Recommendation ITU-R M.2010. NAVDAT transmitting stations are limited to coast stations. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.82D           | When establishing coast stations in the NAVDAT system on the frequencies 500 kHz and 4 226 kHz, the conditions for the use of the frequencies 500 kHz and 4 226 kHz are prescribed in Articles 31 and 52. Administrations are strongly recommended to coordinate the NAVDAT systems operating characteristics in accordance with the procedures of the International Maritime Organization (IMO) (see Resolution 364 (WRC-23)). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.84            | The conditions for the use of the frequency 518 kHz by the maritime mobile service are prescribed in Articles 31 and 52. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.87            | Additional allocation: in Angola, Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia and Niger, the frequency band 526.5-535 kHz is also allocated to the mobile service on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.87A           | Additional allocation: in Uzbekistan, the band 526.5-1606.5 kHz is also allocated to the radionavigation service on a primary basis. Such use is subject to agreement obtained under No. 9.21 with administrations concerned and limited to ground-based radiobeacons in operation on 27 October 1997 until the end of their lifetime. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.90            | In the band 1605-1705 kHz, in cases where a broadcasting station of Region 2 is concerned, the service area of the maritime mobile stations in Region 1 shall be limited to that provided by ground-wave propagation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.92            | Some countries of Region 1 use radiodetermination systems in the bands 1606.5-1625 kHz, 1635-1800 kHz, 1850-2160 kHz, 2194-2300 kHz, 2502-2850 kHz and 3500-3800 kHz, subject to agreement obtained under No. 9.21. The radiated mean power of these stations shall not exceed 50 W.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.93            | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Hungary, Kazakhstan, Latvia, Lithuania, Mongolia, Nigeria, Uzbekistan, Poland, Kyrgyzstan, Slovakia, Tajikistan, Chad, Turkmenistan and Ukraine, the frequency bands 1625-1635 kHz, 1800-1810 kHz and 2160-2170 kHz are also allocated to the fixed and land mobile services on a primary basis, subject to agreement obtained under No. 9.21. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.96            | In Germany, Armenia, Austria, Azerbaijan, Belarus, Croatia, Denmark, Estonia, the Russian Federation, Finland, Georgia, Hungary, Iceland, Ireland, Israel, Kazakhstan, Latvia, Liechtenstein, Lithuania, Malta, Moldova, Norway, Uzbekistan, Poland, Kyrgyzstan, Slovakia, the Czech Rep., the United Kingdom, Sweden, Switzerland, Tajikistan, Turkmenistan and Ukraine, administrations may allocate up to 200 kHz to their amateur service in the frequency bands 1715-1800 kHz and 1850-2000 kHz. However, when allocating the frequency bands within this range to their amateur service, administrations shall, after prior consultation with administrations of neighbouring countries, take such steps as may be necessary to prevent harmful interference from their amateur service to the fixed and mobile services of other countries. The mean power of any amateur station shall not exceed 10 W. (WRC-15) |
| 5.98            | Alternative allocation: in Armenia, Azerbaijan, Belarus, Belgium, Cameroon, Congo (Rep. of the), Denmark, Eritrea, Spain, Ethiopia, the Russian Federation, Georgia, Greece, Italy, Kazakhstan, Lebanon, Lithuania, the Syrian Arab Republic, Türkiye, Kyrgyzstan, Somalia, Tajikistan, Tunisia and Turkmenistan, the frequency band 1 810- 1 830 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.99            | Additional allocation: in Saudi Arabia, Austria, Egypt, Iraq, Libya, Uzbekistan, Romania, Slovakia, Slovenia, Chad, and Togo, the frequency band 1 810-1 830 kHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.100           | In Region 1, the authorization to use the band 1810-1830 kHz by the amateur service in countries situated totally or partially north of 40° N shall be given only after consultation with the countries mentioned in Nos. 5.98 and 5.99 to define the necessary steps to be taken to prevent harmful interference between amateur stations and stations of other services operating in accordance with Nos. 5.98 and 5.99.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.103           | In Region 1, in making assignments to stations in the fixed and mobile services in the bands 1850-2045 kHz, 2194-2498 kHz, 2502-2625 kHz and 2650-2850 kHz, administrations should bear in mind the special requirements of the maritime mobile service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.104           | In Region 1, the use of the band 2025-2045 kHz by the meteorological aids service is limited to oceanographic buoy stations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.107           | Additional allocation: in Saudi Arabia, Eritrea, Eswatini, Ethiopia, Iraq, Libya and Somalia, the frequency band 2 160-2 170 kHz is also allocated to the fixed and mobile, except aeronautical mobile (R), services on a primary basis. The mean power of stations in these services shall not exceed 50 W. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.108           | The carrier frequency 2182 kHz is an international distress and calling frequency for radiotelephony. The conditions for the use of the band 2173.5-2190.5 kHz are prescribed in Articles 31 and 52. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.109           | The frequencies 2187.5 kHz, 4207.5 kHz, 6312 kHz, 8414.5 kHz, 12577 kHz and 16804.5 kHz are international distress frequencies for digital selective calling. The conditions for the use of these frequencies are prescribed in Article 31.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.110           | The frequencies 2 174.5 kHz, 4 177.5 kHz, 6 268 kHz, 8 376.5 kHz, 12 520 kHz and 16 695 kHz are used for the automatic connection system (ACS), as described in the most recent version of Recommendation ITU-R M.541. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.111           | The carrier frequencies 2182 kHz, 3023 kHz, 5680 kHz, 8364 kHz and the frequencies 121.5 MHz, 156.525 MHz, 156.8 MHz and 243 MHz may also be used, in accordance with the procedures in force for terrestrial radiocommunication services, for search and rescue operations concerning manned space vehicles. The conditions for the use of the frequencies are prescribed in Article 31. The same applies to the frequencies 10003 kHz, 14993 kHz and 19993 kHz, but in each of these cases emissions must be confined in a band of $\pm 3$ kHz about the frequency. (WRC-07)                                                                                                                                                                                                                                                                                                                                           |
| 5.112           | Alternative allocation: in Sri Lanka, the frequency band 2 194-2 300 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.113           | For the conditions for the use of the bands 2300-2495 kHz (2498 kHz in Region 1), 3200-3400 kHz, 4750-4995 kHz and 5005-5060 kHz by the broadcasting service, see Nos. 5.16 to 5.20, 5.21 and 23.3 to 23.10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.114           | Alternative allocation: in Iraq, the frequency band 2 502-2 625 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.115           | The carrier (reference) frequencies 3023 kHz and 5680 kHz may also be used, in accordance with Article 31 by stations of the maritime mobile service engaged in coordinated search and rescue operations. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.116           | Administrations are urged to authorize the use of the band 3155-3195 kHz to provide a common worldwide channel for low power wireless hearing aids. Additional channels for these devices may be assigned by administrations in the bands between 3155 kHz and 3400 kHz to suit local needs. It should be noted that frequencies in the range 3000 kHz to 4000 kHz are suitable for hearing aid devices which are designed to operate over short distances within the induction field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.117           | Alternative allocation: in Liberia, Sri Lanka and Togo, the frequency band 3 155-3 200 kHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.118           | Additional allocation: in the United States, Mexico and Peru, the frequency band 3 230-3 400 kHz is also allocated to the radiolocation service on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.123           | Additional allocation: in Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia, South Africa, Zambia and Zimbabwe, the frequency band 3 900-3 950 kHz is also allocated to the broadcasting service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.126           | In Region 3, the stations of those services to which the band 3 995-4 005 kHz is allocated may transmit standard frequency and time signals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.127           | The use of the band 4000-4063 kHz by the maritime mobile service is limited to ship stations using radiotelephony (see No. 52.220 and Appendix 17).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.128           | Frequencies in the bands 4 063-4 123 kHz and 4 130-4 438 kHz may be used exceptionally by stations in the fixed service, communicating only within the boundary of the country in which they are located, with a mean power not exceeding 50 W, on condition that harmful interference is not caused to the maritime mobile service. In addition, in Afghanistan, Argentina, Armenia, Belarus, Botswana, Burkina Faso, the Central African Rep., China, the Russian Federation, Georgia, India, Kazakhstan, Mali, Niger, Pakistan, Kyrgyzstan, Tajikistan, Chad, Turkmenistan and Ukraine, in the frequency bands 4 063-4 123 kHz, 4 130-4 133 kHz and 4 408-4 438 kHz, stations in the fixed service, with a mean power not exceeding 1 kW, can be operated on condition that they are situated at least 600 km from the coast and that harmful interference is not caused to the maritime mobile service. (WRC-19)                                                                                     |
| 5.130           | The conditions for the use of the carrier frequencies 4125 kHz and 6215 kHz are prescribed in Articles 31 and 52. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.131           | The frequency 4209.5 kHz is used exclusively for the transmission by coast stations of meteorological and navigational warnings and urgent information to ships by means of narrow-band direct-printing techniques. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.132           | The frequencies 4 210 kHz, 6 314 kHz, 8 416.5 kHz, 12 579 kHz, 16 806.5 kHz, 19 680.5 kHz, 22 376 kHz and 26 100.5 kHz are the international frequencies for the transmission of maritime safety information (MSI) (see Appendices 15 and 17). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.132A          | Stations in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the fixed or mobile services. Applications of the radiolocation service are limited to oceanographic radars operating in accordance with Resolution 612 (Rev.WRC-12) (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.132B          | Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 4 438- 4 488 kHz is allocated to the fixed and mobile, except aeronautical mobile (R), services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.133           | Different category of service: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kazakhstan, Latvia, Lithuania, Niger, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the allocation of the band 5130-5250 kHz to the mobile, except aeronautical mobile, service is on a primary basis (see No. 5.33). (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.133A          | Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency bands 5 250-5 275 kHz and 26 200-26 350 kHz are allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.133B          | Stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 15 W (e.i.r.p.). However, in Region 2 in Mexico, stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 20 W (e.i.r.p.). In the following Region 2 countries: Antigua and Barbuda, Argentina, Bahamas, Barbados, Belize, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Dominica, El Salvador, Ecuador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Nicaragua, Panama, Paraguay, Peru, Saint Lucia, Saint Kitts and Nevis, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, Uruguay, Venezuela, as well as the overseas countries and territories within the Kingdom of the Netherlands in Region 2, stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 25 W (e.i.r.p.). (WRC-19) |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 5.134           | The use of the frequency bands 5 900-5 950 kHz, 7 300-7 350 kHz, 9 400-9 500 kHz, 11 600-11 650 kHz, 12 050-12 100 kHz, 13 570-13 600 kHz, 13 800-13 870 kHz, 15 600-15 800 kHz, 17 480-17 550 kHz and 18 900-19 020 kHz by the broadcasting service is subject to the application of the procedure of Article 12. Administrations are encouraged to use these frequency bands to facilitate the introduction of digitally modulated emissions in accordance with the provisions of Resolution 517 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                          |
| 5.136           | Additional allocation: Frequencies in the band 5900-5950 kHz may be used by stations in the following services, communicating only within the boundary of the country in which they are located: fixed service (in all three Regions), land mobile service (in Region 1), mobile except aeronautical mobile (R) service (in Regions 2 and 3), on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-07)                                             |
| 5.137           | On condition that harmful interference is not caused to the maritime mobile service, the bands 6200-6213.5 kHz and 6220.5-6525 kHz may be used exceptionally by stations in the fixed service, communicating only within the boundary of the country in which they are located, with a mean power not exceeding 50 W. At the time of notification of these frequencies, the attention of the Bureau will be drawn to the above conditions.                                                                                                                                                                                                                                                                                         |
| 5.137A          | The frequencies 6 337.5 kHz, 8 443 kHz, 12 663.5 kHz, 16 909.5 kHz and 22 450.5 kHz are the regional frequencies for the transmission of maritime safety information (MSI) by means of the NAVDAT system (see Appendices 15 and 17). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.138           | The following bands: 6765-6795 kHz (centre frequency 6780 kHz), 433.05-434.79 MHz (centre frequency 433.92 MHz) in Region 1 except in the countries mentioned in No. 5.280, 61-61.5 GHz (centre frequency 61.25 GHz), 122-123 GHz (centre frequency 122.5 GHz), and 244-246 GHz (centre frequency 245 GHz) are designated for industrial, scientific and medical (ISM) applications. The use of these frequency bands for ISM applications shall be subject to special authorisation by the administration concerned, in agreement with other administrations whose radiocommunication services might be affected. In applying this provision, administrations shall have due regard to the latest relevant ITU-R Recommendations. |
| 5.140           | Additional allocation: in Angola, Iraq, Somalia and Togo, the frequency band 7000-7050 kHz is also allocated to the fixed service on a primary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.141           | Alternative allocation: in Egypt, Eritrea, Ethiopia, Guinea, Libya, Madagascar and Niger, the band 7000-7050 kHz is allocated to the fixed service on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.141A          | Additional allocation: in Uzbekistan and Kyrgyzstan, the bands 7000-7100 kHz and 7100-7200 kHz are also allocated to the fixed and land mobile services on a secondary basis. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.141B          | Additional allocation: in Algeria, Saudi Arabia, Australia, Bahrain, Botswana, Brunei Darussalam, China, Comoros, Korea (Rep. of), Diego Garcia, Djibouti, Egypt, United Arab Emirates, Eritrea, Guinea, Indonesia, Iran (Islamic Republic of), Japan, Jordan, Kuwait, Libya, Mali, Morocco, Mauritania, Niger, New Zealand, Oman, Papua New Guinea, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Sudan, South Sudan, Tunisia, Viet Nam and Yemen, the frequency band 7 100-7 200 kHz is also allocated to the fixed and the mobile, except aeronautical mobile (R), services on a primary basis. (WRC-19)                                                                                         |
| 5.143           | Additional allocation: frequencies in the band 7300-7350 kHz may be used by stations in the fixed service and in the land mobile service, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-07)                                                                                                                                                                |
| 5.143A          | In Region 3, frequencies in the band 7 350-7 450 kHz may be used by stations in the fixed service on a primary basis and land mobile service on a secondary basis, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-12)                                                                                                                                       |
| 5.143B          | In Region 1, frequencies in the band 7350-7450 kHz may be used by stations in the fixed and land mobile services communicating only within the boundary of the country in which they are located on condition that harmful interference is not caused to the broadcasting service. The total radiated power of each station shall not exceed 24 dBW. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                      |
| 5.143C          | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Comoros, Djibouti, Egypt, United Arab Emirates, Iran (Islamic Republic of), Jordan, Kuwait, Libya, Morocco, Mauritania, Niger, Oman, Qatar, the Syrian Arab Republic, Sudan, South Sudan, Tunisia and Yemen, the bands 7350-7400 kHz and 7400-7450 kHz are also allocated to the fixed service on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                              |
| 5.143D          | In Region 2, frequencies in the band 7 350-7 400 kHz may be used by stations in the fixed service and in the land mobile service, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies for these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-12)                                                                                                                                                                        |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 5.144           | In Region 3, the stations of those services to which the band 7 995-8 005 kHz is allocated may transmit standard frequency and time signals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.145           | The conditions for the use of the carrier frequencies 8291 kHz, 12290 kHz and 16420 kHz are prescribed in Articles 31 and 52. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.145A          | Stations in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the fixed service. Applications of the radiolocation service are limited to oceanographic radars operating in accordance with Resolution 612 (Rev.WRC-12) (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.145B          | Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency bands 9 305- 9 355 kHz and 16 100-16 200 kHz are allocated to the fixed service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.146           | Additional allocation: Frequencies in the bands 9400-9500 kHz, 11600-11650 kHz, 12050-12100 kHz, 15600-15800 kHz, 17480-17550 kHz and 18900-19020 kHz may be used by stations in the fixed service, communicating only within the boundary of the country in which they are located, on condition that harmful interference is not caused to the broadcasting service. When using frequencies in the fixed service, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.147           | On condition that harmful interference is not caused to the broadcasting service, frequencies in the bands 9775-9900 kHz, 11650-11700 kHz and 11975-12050 kHz may be used by stations in the fixed service communicating only within the boundary of the country in which they are located, each station using a total radiated power not exceeding 24 dBW.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.149           | In making assignments to stations of other services to which the bands: 13360-13410 kHz, 25550-25670 kHz, 37.5-38.25 MHz, 73-74.6 MHz in Regions 1 and 3, 150.05-153 MHz in Region 1, 322-328.6 MHz, 406.1-410 MHz, 608-614 MHz in Regions 1 and 3, 1330-1400 MHz, 1610.6-1613.8 MHz, 1660-1670 MHz, 1718.8-1722.2 MHz, 2655-2690 MHz, 3260-3267 MHz, 3332-3339 MHz, 3345.8-3352.5 MHz, 4825-4835 MHz, 4950-4990 MHz, 4990-5000 MHz, 6650-6675.2 MHz, 10.6-10.68 GHz, 14.47-14.5 GHz, 22.01-22.21 GHz, 22.21-22.5 GHz, 22.81-22.86 GHz, 23.07-23.12 GHz, 31.2-31.3 GHz, 31.5-31.8 GHz in Regions 1 and 3, 36.43-36.5 GHz, 42.5-43.5 GHz, 48.94-49.04 GHz, 76-86 GHz, 92-94 GHz, 94.1-100 GHz, 102-109.5 GHz, 111.8-114.25 GHz, 128.33-128.59 GHz, 129.23-129.49 GHz, 130-134 GHz, 136-148.5 GHz, 151.5-158.5 GHz, 168.59-168.93 GHz, 171.11-171.45 GHz, 172.31-172.65 GHz, 173.52-173.85 GHz, 195.75-196.15 GHz, 209-226 GHz, 241-250 GHz, 252-275 GHz are allocated, administrations are urged to take all practicable steps to protect the radio astronomy service from harmful interference. Emissions from spaceborne or airborne stations can be particularly serious sources of interference to the radio astronomy service (see Nos. 4.5 and 4.6 and Article 29). (WRC-07) |
| 5.149A          | Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 13 450-13 550 kHz is allocated to the fixed service on a primary basis and to the mobile, except aeronautical mobile (R), service on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.150           | The following bands: 13553-13567 kHz (centre frequency 13560 kHz), 26957-27283 kHz (centre frequency 27120 kHz), 40.66-40.70 MHz (centre frequency 40.68 MHz), 902-928 MHz in Region 2 (centre frequency 915 MHz), 2400-2500 MHz (centre frequency 2450 MHz), 5725-5875 MHz (centre frequency 5800 MHz), and 24-24.25 GHz (centre frequency 24.125 GHz) are also designated for industrial, scientific and medical (ISM) applications. Radiocommunication services operating within these bands must accept harmful interference which may be caused by these applications. ISM equipment operating in these bands is subject to the provisions of No. 15.13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.151           | Additional allocation: Frequencies in the bands 13570-13600 kHz and 13800-13870 kHz may be used by stations in the fixed service and in the mobile except aeronautical mobile (R) service, communicating only within the boundary of the country in which they are located, on the condition that harmful interference is not caused to the broadcasting service. When using frequencies in these services, administrations are urged to use the minimum power required and to take account of the seasonal use of frequencies by the broadcasting service published in accordance with the Radio Regulations.(WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.152           | Additional allocation: in Armenia, Azerbaijan, China, Ivory Coast, Georgia, Iran (Islamic Republic of), Kazakhstan, Uzbekistan, Kyrgyzstan, the Russian Federation, Tajikistan, Turkmenistan and Ukraine, the band 14250-14350 kHz is also allocated to the fixed service on a primary basis. Stations of the fixed service shall not use a radiated power exceeding 24 dBW. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.154           | Additional allocation: in Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, the Russian Federation, Tajikistan, Turkmenistan and Ukraine, the band 18068-18168 kHz is also allocated to the fixed service on a primary basis for use within their boundaries, with a peak envelope power not exceeding 1 kW. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.155           | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Moldova, Uzbekistan, Kyrgyzstan, Slovakia, Tajikistan, Turkmenistan and Ukraine, the frequency band 21 850-21 870 kHz is also allocated to the aeronautical mobile (R) service on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.155A          | In Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Moldova, Uzbekistan, Kyrgyzstan, Slovakia, Tajikistan, Turkmenistan and Ukraine, the use of the frequency band 21 850-21 870 kHz by the fixed service is limited to provision of services related to aircraft flight safety. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 5.155B          | The band 21870-21924 kHz is used by the fixed service for provision of services related to aircraft flight safety.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.156           | Additional allocation: in Nigeria, the band 22720-23200 kHz is also allocated to the meteorological aids service (radiosondes) on a primary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.156A          | The use of the band 23200-23350 kHz by the fixed service is limited to provision of services related to aircraft flight safety.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.157           | The use of the band 23350-24000 kHz by the maritime mobile service is limited to inter-ship radiotelegraphy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.158           | Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 24 450-24 600 kHz is allocated to the fixed and land mobile services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.159           | Alternative allocation: in Armenia, Belarus, Moldova and Kyrgyzstan, the frequency band 39-39.5 MHz is allocated to the fixed and mobile services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.159A          | The use of the frequency band 40-50 MHz by the Earth exploration-satellite service (active) shall be in accordance with the geographical area restrictions and the operational and technical conditions defined in Resolution 677 (WRC-23). The provisions of this footnote in no way diminish the obligation of the Earth exploration-satellite service (active) to operate as a secondary service in accordance with Nos. 5.29 and 5.30. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.160           | Additional allocation: in Botswana, Burundi, the Dem. Rep. of the Congo and Rwanda, the band 41-44 MHz is also allocated to the aeronautical radionavigation service on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.161           | Additional allocation: in Iran (Islamic Republic of) and Japan, the band 41-44 MHz is also allocated to the radiolocation service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.161A          | Additional allocation: in Korea (Rep. of), the United States and Mexico, the frequency bands 41.015-41.665 MHz and 43.35-44 MHz are also allocated to the radiolocation service on a primary basis. Stations in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the fixed or mobile services. Applications of the radiolocation service are limited to oceanographic radars operating in accordance with Resolution 612 (Rev.WRC-12). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.161B          | Alternative allocation: in Albania, Germany, Armenia, Austria, Belarus, Belgium, Bosnia and Herzegovina, Cyprus, Vatican, Croatia, Denmark, Spain, Estonia, Finland, France, Greece, Hungary, Ireland, Iceland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malta, Moldova, Monaco, Montenegro, Norway, Uzbekistan, Netherlands, Portugal, Kyrgyzstan, Slovakia, Czech Rep., Romania, United Kingdom, San Marino, Slovenia, Sweden, Switzerland, Turkey and Ukraine, the frequency band 42-42.5 MHz is allocated to the fixed and mobile services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.162           | Additional allocation: in Australia, the band 44-47 MHz is also allocated to the broadcasting service on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.162A          | Additional allocation: in Germany, Australia, Austria, Belgium, Bosnia and Herzegovina, China, Vatican, Korea (Rep. of), Denmark, Spain, Estonia, the Russian Federation, Finland, France, Indonesia, Ireland, Iceland, Italy, Japan, Latvia, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Monaco, Montenegro, Norway, the Netherlands, Poland, Portugal, the Dem. People's Rep. of Korea, the Czech Rep., the United Kingdom, Serbia, Slovenia, Sweden and Switzerland, the frequency band 46-68 MHz is also allocated to the radiolocation service on a secondary basis. This use is limited to the operation of wind profiler radars in accordance with Resolution 217 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.163           | Additional allocation: in Armenia, Belarus, the Russian Federation, Georgia, Kazakhstan, Latvia, Moldova, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency bands 47-48.5 MHz and 56.5-58 MHz are also allocated to the fixed and land mobile services on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.164           | Additional allocation: in Albania, Algeria, Germany, Austria, Belgium, Bosnia and Herzegovina, Botswana, Bulgaria, Ivory Coast, Croatia, Denmark, Spain, Estonia, Eswatini, Finland, France, Gabon, Greece, Hungary, Ireland, Israel, Italy, Jordan, Lebanon, Libya, Liechtenstein, Lithuania, Luxembourg, Madagascar, Mali, Malta, Morocco, Mauritania, Monaco, Montenegro, Nigeria, Norway, the Netherlands, Poland, Syrian Arab Republic, Slovakia, Czech Rep., Romania, the United Kingdom, Serbia, Slovenia, Sweden, Switzerland, Chad, Togo, Tunisia and Turkey, the frequency band 47-68 MHz, in South Africa the frequency band 47-50 MHz, and in Latvia the frequency bands 48.5-56.5 MHz and 58-68 MHz, are also allocated to the land mobile service on a primary basis. However, stations of the land mobile service in the countries mentioned in connection with each frequency band referred to in this footnote shall not cause harmful interference to, or claim protection from, existing or planned broadcasting stations of countries other than those mentioned in connection with the frequency band. (WRC-19) |
| 5.165           | Additional allocation: in Angola, Cameroon, Congo (Rep. of the), Egypt, Madagascar, Mozambique, Niger, Somalia, Sudan, South Sudan, Tanzania and Chad, the frequency band 47- 68 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.166A          | Different category of service: in Austria, Cyprus, the Vatican, Croatia, Denmark, Spain, Finland, Hungary, Latvia, the Netherlands, the Czech Republic, the United Kingdom, Slovakia and Slovenia, the frequency band 50.0-50.5 MHz is allocated to the amateur service on a primary basis. Stations in the amateur service in these countries shall not cause harmful interference to, or claim protection from, stations of the broadcasting, fixed and mobile services operating in accordance with the Radio Regulations in the frequency band 50.0-50.5 MHz in the countries not listed in this provision. For a station of these services, the protection criteria in No. 5.169B shall also apply. In Region 1, with the exception of those countries listed in No. 5.169, wind profiler radars operating in the radiolocation service under No. 5.162A are authorized to operate on the basis of equality with stations in the amateur service in the frequency band 50.0-50.5 MHz. (WRC-19)                                                                                                                                  |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 5.166B          | In Region 1, stations in the amateur service operating on a secondary basis shall not cause harmful interference to, or claim protection from, stations of the broadcasting service. The field strength generated by an amateur station in Region 1 in the frequency band 50-52 MHz shall not exceed a calculated value of +6 dB( $\mu$ V/m) at a height of 10 m above ground for more than 10% of time along the border of a country with operational analogue broadcasting stations in Region 1 and of neighbouring countries with broadcasting stations in Region 3 listed in Nos. 5.167 and 5.168. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.166C          | In Region 1, stations in the amateur service in the frequency band 50-52 MHz, with the exception of those countries listed in No. 5.169, shall not cause harmful interference to, or claim protection from, wind profiler radars operating in the radiolocation service under No. 5.162A. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.166E          | In the Russian Federation, only the frequency band 50.080-50.280 MHz is allocated to the amateur service on a secondary basis. The protection criteria for the other services in the countries not listed in this provision are specified in Nos. 5.166B and 5.169B. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.169A          | Alternative allocation: in the following countries in Region 1: Angola, Saudi Arabia, Bahrain, Burkina Faso, Burundi, the United Arab Emirates, Gambia, Jordan, Kenya, Kuwait, Mauritius, Mozambique, Oman, Uganda, Qatar, South Sudan and Tanzania, the frequency band 50- 54 MHz is allocated to the amateur service on a primary basis. In Guinea-Bissau, the frequency band 50.0-50.5 MHz is allocated to the amateur service on a primary basis. In Djibouti, the frequency band 50-52 MHz is allocated to the amateur service on a primary basis. With the exception of those countries listed in No. 5.169, stations in the amateur service operating in Region 1 under this footnote, in all or part of the frequency band 50-54 MHz, shall not cause harmful interference to, or claim protection from, stations of other services operating in accordance with the Radio Regulations in Algeria, Egypt, Iran (Islamic Republic of), Iraq, Israel, Libya, Palestine*, the Syrian Arab Republic, the Dem. People's Republic of Korea, Sudan and Tunisia. The field strength generated by an amateur station in the frequency band 50-54 MHz shall not exceed a value of +6 dB( $\mu$ V/m) at a height of 10 m above ground for more than 10% of time along the borders of listed countries requiring protection. (WRC-19) |
| 5.169B          | Except countries listed under No. 5.169, stations in the amateur service used in Region 1, in all or part of the 50-54 MHz frequency band, shall not cause harmful interference to, or claim protection from, stations of other services used in accordance with the Radio Regulations in Algeria, Armenia, Azerbaijan, Belarus, Egypt, Russian Federation, Iran (Islamic Republic of), Iraq, Kazakhstan, Kyrgyzstan, Libya, Uzbekistan, Palestine*, the Syrian Arab Republic, Sudan, Tunisia and Ukraine. The field strength generated by an amateur station in the frequency band 50-54 MHz shall not exceed a value of +6 dB( $\mu$ V/m) at a height of 10 m above ground for more than 10% of time along the borders of the countries listed in this provision. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.175           | Alternative allocation: in Armenia, Belarus, the Russian Federation, Kazakhstan, Moldova, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency bands 68-73 MHz and 76-87.5 MHz are allocated to the broadcasting service on a primary basis. In Latvia and Lithuania, the frequency bands 68-73 MHz and 76-87.5 MHz are allocated to the broadcasting and mobile, except aeronautical mobile, services on a primary basis. In Mongolia, the frequency band 76-87.5 MHz is allocated to the broadcasting service on a primary basis; the stations of the broadcasting service shall not cause harmful interference to, or claim protection from, existing or planned fixed and mobile stations in the neighbouring countries. The services to which these frequency bands are allocated in other countries and the broadcasting service in the countries listed above are subject to agreements with the neighbouring countries concerned. (WRC-23)                                                                                                                                                                                                                                                                                                                                                         |
| 5.177           | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the band 73-74 MHz is also allocated to the broadcasting service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.178           | Additional allocation: in Colombia, Cuba, El Salvador, Guatemala, Guyana, Honduras and Nicaragua, the band 73-74.6 MHz is also allocated to the fixed and mobile services on a secondary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.179           | Additional allocation: in Armenia, Azerbaijan, Belarus, China, the Russian Federation, Georgia, Kazakhstan, Lithuania, Mongolia, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the bands 74.6-74.8 MHz and 75.2-75.4 MHz are also allocated to the aeronautical radionavigation service, on a primary basis, for ground-based transmitters only. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.180           | The frequency 75 MHz is assigned to marker beacons. Administrations shall refrain from assigning frequencies close to the limits of the guardband to stations of other services which, because of their power or geographical position, might cause harmful interference or otherwise place a constraint on marker beacons. Every effort should be made to improve further the characteristics of airborne receivers and to limit the power of transmitting stations close to the limits 74.8 MHz and 75.2 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.187           | Alternative allocation: in Albania, the band 81-87.5 MHz is allocated to the broadcasting service on a primary basis and used in accordance with the decisions contained in the Final Acts of the Special Regional Conference (Geneva, 1960).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.190           | Additional allocation: in Monaco, the band 87.5-88 MHz is also allocated to the land mobile service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.192           | Additional allocation: in China and Korea (Rep. of), the band 100-108 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 5.194           | Additional allocation: in Kyrgyzstan, Somalia and Turkmenistan, the frequency band 104-108 MHz is also allocated to the mobile, except aeronautical mobile (R), service on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.197           | Additional allocation: in the Syrian Arab Republic, the band 108-111.975 MHz is also allocated to the mobile service on a secondary basis, subject to agreement obtained under No. 9.21. In order to ensure that harmful interference is not caused to stations of the aeronautical radionavigation service, stations of the mobile service shall not be introduced in the band until it is no longer required for the aeronautical radionavigation service by any administration which may be identified in the application of the procedures invoked under No. 9.21 (WRC-12)                                                                                            |
| 5.197A          | Additional allocation: the frequency band 108-117.975 MHz is also allocated on a primary basis to the aeronautical mobile (R) service, limited to systems operating in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 413 (Rev.WRC-23). The use of the frequency band 108-112 MHz by the aeronautical mobile (R) service shall be limited to systems composed of ground-based transmitters and associated receivers that provide navigational information in support of air navigation functions in accordance with recognized international aeronautical standards. (WRC-23)                           |
| 5.198A          | The use of the frequency band 117.975-137 MHz by the aeronautical mobile-satellite (R) service is subject to coordination under No. 9.11A. No. 9.16 does not apply. Such use shall be limited to non-geostationary-satellite systems operated in accordance with international aeronautical standards. Resolution 406 (WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                          |
| 5.198B          | The use of the frequency band 117.975-137 MHz by the aeronautical mobile (R) service shall have priority over use by the aeronautical mobile-satellite (R) service. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.200           | In the frequency band 117.975-137 MHz, the frequency 121.5 MHz is the aeronautical emergency frequency and, where required, the frequency 123.1 MHz is the aeronautical frequency auxiliary to 121.5 MHz. Mobile stations of the maritime mobile service may communicate on these frequencies under the conditions laid down in Article 31 for distress and safety purposes with stations of the aeronautical mobile service and the aeronautical mobile satellite service. (WRC-23)                                                                                                                                                                                      |
| 5.201           | Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Egypt, Estonia, the Russian Federation, Georgia, Hungary, Iran (Islamic Republic of), Iraq (Republic of), Japan, Kazakhstan, Mali, Mongolia, Mozambique, Uzbekistan, Papua New Guinea, Poland, Qatar, Kyrgyzstan, Romania, Senegal, Somalia, Tajikistan and Turkmenistan, the frequency band 132-136 MHz is also allocated to the aeronautical mobile (OR) service on a primary basis. In assigning frequencies to stations of the aeronautical mobile (OR) service, the administration shall take account of the frequencies assigned to stations in the aeronautical mobile (R) service. (WRC-23) |
| 5.202           | Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, the United Arab Emirates, the Russian Federation, Georgia, Iran (Islamic Republic of), Jordan, Mali, Oman, Uzbekistan, Poland, the Syrian Arab Republic, Kyrgyzstan, Romania, Senegal, Tajikistan and Turkmenistan, the frequency band 136-137 MHz is also allocated to the aeronautical mobile (OR) service on a primary basis. In assigning frequencies to stations of the aeronautical mobile (OR) service, the administration shall take account of the frequencies assigned to stations in the aeronautical mobile (R) service. (WRC-23)                                                       |
| 5.203C          | The use of the space operation service (space-to-Earth) with non-geostationary satellite short-duration mission systems in the frequency band 137-138 MHz is subject to Resolution 660 (WRC-19). Resolution 32 (WRC-19) applies. These systems shall not cause harmful interference to, or claim protection from, the existing services to which the frequency band is allocated on a primary basis. (WRC-19)                                                                                                                                                                                                                                                             |
| 5.204           | Different category of service: in Afghanistan, Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, China, Cuba, the United Arab Emirates, India, Indonesia, Iran (Islamic Republic of), Iraq, Kuwait, Montenegro, Oman, Pakistan, the Philippines, Qatar, Singapore, Thailand and Yemen, the frequency band 137-138 MHz is allocated to the fixed and mobile, except aeronautical mobile (R), services on a primary basis (see No. 5.33). (WRC-19)                                                                                                                                                                                                                      |
| 5.205           | Different category of service: in Israel and Jordan, the allocation of the band 137-138 MHz to the fixed and mobile, except aeronautical mobile, services is on a primary basis (see No. 5.33).                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.206           | Different category of service: in Armenia, Azerbaijan, Belarus, Bulgaria, Egypt, the Russian Federation, Finland, France, Georgia, Greece, Kazakhstan, Lebanon, Moldova, Mongolia, Uzbekistan, Poland, Kyrgyzstan, the Syrian Arab Republic, Slovakia, the Czech Republic, Romania, Tajikistan, Turkmenistan and Ukraine, the allocation of the band 137-138 MHz to the aeronautical mobile (OR) service is on a primary basis (see No. 5.33). (WRC-2000)                                                                                                                                                                                                                 |
| 5.207           | Additional allocation: in Australia, the band 137-144 MHz is also allocated to the broadcasting service on a primary basis until that service can be accommodated within regional broadcasting allocations.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.208           | The use of the band 137-138 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 5.208A          | In making assignments to space stations in the mobile-satellite service in the frequency bands 137-138 MHz, 387-390 MHz and 400.15-401 MHz and in the maritime mobile-satellite service (space-to-Earth) in the frequency bands 157.1875-157.3375 MHz and 161.7875-161.9375 MHz, administrations shall take all practicable steps to protect the radio astronomy service in the frequency bands 150.05-153 MHz, 322-328.6 MHz, 406.1-410 MHz and 608-614 MHz from harmful interference from unwanted emissions as shown in the most recent version of Recommendation ITU-R RA.769. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.208B          | In the frequency bands: 137-138 MHz, 157.1875-157.3375 MHz, 161.7875-161.9375 MHz, 387-390 MHz, 400.15-401 MHz, 1 452-1 492 MHz, 1 525-1 610 MHz, 1 613.8-1 626.5 MHz, 2 655-2 690 MHz, 21.4-22 GHz, Resolution 739 (Rev.WRC-19) applies. (WRC-19) *This provision was previously numbered as No. 5.347A. It was renumbered to preserve the sequential order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.209           | The use of the bands 137-138 MHz, 148-150.05 MHz, 399.9-400.05 MHz, 400.15-401 MHz, 454-456 MHz and 459-460 MHz by the mobile-satellite service is limited to non-geostationary-satellite systems. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.209A          | The use of the frequency band 137.175-137.825 MHz by non-geostationary satellite systems in the space operation service identified as short-duration mission in accordance with Appendix 4 is not subject to No. 9.11A. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.210           | Additional allocation: in Italy and the United Kingdom, the frequency bands 138-143.6 MHz and 143.65-144 MHz are also allocated to the space research service (space-to-Earth) on a secondary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.211           | Additional allocation: in Germany, Saudi Arabia, Austria, Bahrain, Belgium, Denmark, the United Arab Emirates, Spain, Finland, Greece, Guinea, Ireland, Israel, Kenya, Kuwait, Lebanon, Liechtenstein, Luxembourg, North Macedonia, Mali, Malta, Montenegro, Norway, the Netherlands, Qatar, Slovakia, the United Kingdom, Serbia, Slovenia, Somalia, Sweden, Switzerland, Tanzania, Tunisia and Turkey, the frequency band 138-144 MHz is also allocated to the maritime mobile and land mobile services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.212           | Alternative allocation: in Angola, Botswana, Cameroon, the Central African Rep., Congo (Rep. of the), Eswatini, Gabon, Gambia, Ghana, Guinea, Iraq, Jordan, Lesotho, Liberia, Libya, Malawi, Mozambique, Namibia, Niger, Oman, Uganda, Syrian Arab Republic, the Dem. Rep. of the Congo, Rwanda, Sierra Leone, South Africa, Chad, Togo, Zambia and Zimbabwe, the frequency band 138-144 MHz is allocated to the fixed and mobile services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.214           | Additional allocation: in Eritrea, Ethiopia, Kenya, North Macedonia, Montenegro, Serbia, Somalia, Sudan, South Sudan and Tanzania, the frequency band 138-144 MHz is also allocated to the fixed service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.216           | Additional allocation: in China, the band 144-146 MHz is also allocated to the aeronautical mobile (OR) service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.218           | Additional allocation: the band 148-149.9 MHz is also allocated to the space operation service (Earth-to-space) on a primary basis, subject to agreement obtained under No. 9.21. The bandwidth of any individual transmission shall not exceed $\pm 25$ kHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.218A          | The frequency band 148-149.9 MHz in the space operation service (Earth-to-space) may be used by non-geostationary satellite systems with short-duration missions. Non-geostationary satellite systems in the space operation service used for a short-duration mission in accordance with Resolution COM5/5 (WRC-19) of the Radio Regulations are not subject to agreement under No. 9.21. At the stage of coordination, the provisions of Nos. 9.17 and 9.18 also apply. In the frequency band 148-149.9 MHz, non-geostationary satellite systems with short-duration missions shall not cause unacceptable interference to, or claim protection from, existing primary services within this frequency band, or impose additional constraints on the space operation and mobile-satellite services. In addition, earth stations in non-geostationary satellite systems in the space operation service with short-duration missions in the frequency band 148-149.9 MHz shall ensure that the power flux-density does not exceed $-149$ dB(W/(m <sup>2</sup> $\square$ 4 kHz)) for more than 1% of time at the border of the territory of the following countries: Armenia, Azerbaijan, Belarus, China, Korea (Rep. of), Cuba, Russian Federation, India, Iran (Islamic Republic of), Japan, Kazakhstan, Malaysia, Uzbekistan, Kyrgyzstan, Thailand and Viet Nam. In case this power flux-density limit is exceeded, agreement under No. 9.21 is required to be obtained from countries mentioned in this footnote. (WRC-19)                                                                         |
| 5.219           | The use of the frequency band 148-149.9 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. The mobile-satellite service shall not constrain the development and use of the fixed, mobile and space operation services in the frequency band 148-149.9 MHz. The use of the frequency band 148-149.9 MHz by non-geostationary-satellite systems in the space operation service identified as short-duration mission is not subject to No. 9.11A. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.220           | The use of the bands 149.9-150.05 MHz and 399.9-400.05 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.221           | Stations of the mobile-satellite service in the frequency band 148-149.9 MHz shall not cause harmful interference to, or claim protection from, stations of the fixed or mobile services operating in accordance with the Table of Frequency Allocations in the following countries: Albania, Algeria, Germany, Saudi Arabia, Australia, Austria, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Benin, Bosnia and Herzegovina, Botswana, Brunei Darussalam, Bulgaria, Cameroon, China, Cyprus, Congo (Rep. of the), Korea (Rep. of), Côte d'Ivoire, Croatia, Cuba, Denmark, Djibouti, Egypt, the United Arab Emirates, Eritrea, Spain, Estonia, Eswatini, Ethiopia, the Russian Federation, Finland, France, Gabon, Georgia, Ghana, Greece, Guinea, Guinea Bissau, Hungary, India, Iran (Islamic Republic of), Ireland, Iceland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kuwait, Lesotho, Latvia, Lebanon, Libya, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malaysia, Mali, Malta, Mauritania, Moldova, Mongolia, Montenegro, Mozambique, Namibia, Norway, New Zealand, Oman, Uganda, Uzbekistan, Pakistan, Panama, Papua New Guinea, Paraguay, the Netherlands, the Philippines, Poland, Portugal, Qatar, the Syrian Arab Republic, Türkiye, Kyrgyzstan, Dem. People's Rep. of Korea, Slovakia, Romania, the United Kingdom, Senegal, Serbia, Sierra Leone, Singapore, Slovenia, Somalia, Sudan, Sri Lanka, South Africa, Sweden, Switzerland, Tanzania, Chad, Togo, Tonga, Trinidad and Tobago, Tunisia, Ukraine, Viet Nam, Yemen, Zambia and Zimbabwe. (WRC-23) |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.225A          | Additional allocation: in Algeria, Armenia, Azerbaijan, Belarus, China, France, Iran (Islamic Republic of), Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan, Ukraine and Viet Nam, the frequency band 154-156 MHz is also allocated to the radiolocation service on a primary basis. The usage of the frequency band 154-156 MHz by the radiolocation service shall be limited to space-object detection systems operating from terrestrial locations. The operation of stations in the radiolocation service in the frequency band 154-156 MHz shall be subject to agreement obtained under No. 9.21. For the identification of potentially affected administrations in Region 1, the instantaneous field-strength value of 12 dB( $\mu$ V/m) for 10% of the time produced at 10 m above ground level in the 25 kHz reference frequency band at the border of the territory of any other administration shall be used. For the identification of potentially affected administrations in Region 3, the interference-to-noise ratio (I/N) value of -6 dB (N = -161 dBW/4 kHz), or -10 dB for applications with greater protection requirements, such as public protection and disaster relief (PPDR (N = -161 dBW/4 kHz)), for 1% of the time produced at 60 m above ground level at the border of the territory of any other administration shall be used. In the frequency bands 156.7625-156.8375 MHz, 156.5125-156.5375 MHz, 161.9625-161.9875 MHz, 162.0125-162.0375 MHz, out-of-band e.i.r.p. of space surveillance radars shall not exceed -16 dBW. Frequency assignments to the radiolocation service under this allocation in Ukraine shall not be used without the agreement of Moldova. (WRC-12) |
| 5.226           | The frequency 156.8 MHz is the international distress, safety and calling frequency for the maritime mobile VHF radiotelephone service. The conditions for the use of this frequency and the band 156.7625-156.8375 MHz are contained in Article 31 and Appendix 18. The frequency 156.525 MHz is the international distress, safety and calling frequency for the maritime mobile VHF radiotelephone service using digital selective calling (DSC). The conditions for the use of this frequency and the band 156.4875-156.5625 MHz are contained in Articles 31 and 52, and in Appendix 18. In the bands 156-156.4875 MHz, 156.5625-156.7625 MHz, 156.8375-157.45 MHz, 160.6-160.975 MHz and 161.475-162.05 MHz, each administration shall give priority to the maritime mobile service on only such frequencies as are assigned to stations of the maritime mobile service by the administration (see Articles 31 and 52, and Appendix 18). Any use of frequencies in these bands by stations of other services to which they are allocated should be avoided in areas where such use might cause harmful interference to the maritime mobile VHF radiocommunication service. However, the frequencies 156.8 MHz and 156.525 MHz and the frequency bands in which priority is given to the maritime mobile service may be used for radiocommunications on inland waterways subject to agreement between interested and affected administrations and taking into account current frequency usage and existing agreements. (WRC-07)                                                                                                                                                                                     |
| 5.227           | Additional allocation: the bands 156.4875-156.5125 MHz and 156.5375-156.5625 MHz are also allocated to the fixed and land mobile services on a primary basis. The use of these bands by the fixed and land mobile services shall not cause harmful interference to nor claim protection from the maritime mobile VHF radiocommunication service. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.228           | The use of the frequency bands 156.7625-156.7875 MHz and 156.8125-156.8375 MHz by the mobile-satellite service (Earth-to-space) is limited to the reception of automatic identification system (AIS) emissions of long-range AIS broadcast messages (Message 27, see the most recent version of Recommendation ITU R M.1371). With the exception of AIS emissions, emissions in these frequency bands by systems operating in the maritime mobile service for communications shall not exceed 1 W. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.228A          | The frequency bands 161.9625-161.9875 MHz and 162.0125-162.0375 MHz may be used by aircraft stations for the purpose of search and rescue operations and other safety-related communications. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.228AA         | The use of the frequency bands 161.9375-161.9625 MHz and 161.9875-162.0125 MHz by the maritime mobile-satellite (Earth-to-space) service is limited to the systems which operate in accordance with Appendix 18. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.228AB         | The use of the frequency bands 157.1875-157.3375 MHz and 161.7875-161.9375 MHz by the maritime mobile-satellite service (Earth-to-space) is limited to non-GSO satellite systems operating in accordance with Appendix 18. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.228AC         | The use of the frequency bands 157.1875-157.3375 MHz and 161.7875-161.9375 MHz by the maritime mobile-satellite service (space-to-Earth) is limited to non-GSO satellite systems operating in accordance with Appendix 18. Such use is subject to agreement obtained under No. 9.21 with respect to the terrestrial services in Azerbaijan, Belarus, China, Korea (Rep. of), Cuba, the Russian Federation, the Syrian Arab Republic, the Dem. People's Rep. of Korea, South Africa and Viet Nam. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.228B          | The use of the frequency bands 161.9625-161.9875 MHz and 162.0125-162.0375 MHz by the fixed and land mobile services shall not cause harmful interference to, or claim protection from, the maritime mobile service. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.228F          | The use of the frequency bands 161.9625-161.9875 MHz and 162.0125-162.0375 MHz by the mobile-satellite service (Earth-to-space) is limited to the reception of automatic identification system emissions from stations operating in the maritime mobile service. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.235           | Additional allocation: in Germany, Austria, Belgium, Denmark, Spain, Finland, France, Israel, Italy, Liechtenstein, Malta, Monaco, Norway, the Netherlands, the United Kingdom, Sweden and Switzerland, the band 174 - 223 MHz is also allocated to the land mobile service on a primary basis. However, the stations of the land mobile service shall not cause harmful interference to, or claim protection from, broadcasting stations, existing or planned, in countries other than those listed in this footnote.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 5.243           | Additional allocation: in Somalia, the band 216-225 MHz is also allocated to the aeronautical radionavigation service on a primary basis, subject to not causing harmful interference to existing or planned broadcasting services in other countries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.246           | Alternative allocation: in Spain, France, Israel and Monaco, the band 223-230 MHz is allocated to the broadcasting and land mobile services on a primary basis (see No. 5.33) on the basis that, in the preparation of frequency plans, the broadcasting service shall have prior choice of frequencies; and allocated to the fixed and mobile, except land mobile, services on a secondary basis. However, the stations of the land mobile service shall not cause harmful interference to, or claim protection from, existing or planned broadcasting stations in Morocco and Algeria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.247           | Additional allocation: in Saudi Arabia, Bahrain, the United Arab Emirates, Jordan, Oman, Qatar and Syrian Arab Republic, the band 223-235 MHz is also allocated to the aeronautical radionavigation service on a primary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.251           | Additional allocation: in Nigeria, the band 230-235 MHz is also allocated to the aeronautical radionavigation service on a primary basis, subject to agreement obtained under No. 9.21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.252           | Alternative allocation: in Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia, South Africa, Zambia and Zimbabwe, the frequency bands 230-238 MHz and 246-254 MHz are allocated to the broadcasting service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.254           | The bands 235-322 MHz and 335.4-399.9 MHz may be used by the mobile-satellite service, subject to agreement obtained under No. 9.21, on condition that stations in this service do not cause harmful interference to those of other services operating or planned to be operated in accordance with the Table of Frequency Allocations except for the additional allocation made in footnote No. 5.256A. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.255           | The bands 312-315 MHz (Earth-to-space) and 387-390 MHz (space-to-Earth) in the mobile-satellite service may also be used by non-geostationary-satellite systems. Such use is subject to coordination under No. 9.11A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.256           | The frequency 243 MHz is the frequency in this band for use by survival craft stations and equipment used for survival purposes. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.256A          | Additional allocation: in China, the Russian Federation and Kazakhstan, the frequency band 258-261 MHz is also allocated to the space research service (Earth-to-space) and space operation service (Earth-to-space) on a primary basis. Stations in the space research service (Earth-to-space) and space operation service (Earth-to-space) shall not cause harmful interference to, or claim protection from, or constrain the use and development of the mobile service systems and mobile-satellite service systems operating in the frequency band. Stations in space research service (Earth-to-space) and space operation service (Earth-to-space) shall not constrain the future development of fixed service systems of other countries. (WRC-15)                                                                                                                                                                                                                                                                                                                                          |
| 5.257           | The band 267-272 MHz may be used by administrations for space telemetry in their countries on a primary basis, subject to agreement obtained under No. 9.21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.258           | The use of the band 328.6-335.4 MHz by the aeronautical radionavigation service is limited to Instrument Landing Systems (glide path).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.259           | Additional allocation: in Egypt and the Syrian Arab Republic, the band 328.6-335.4 MHz is also allocated to the mobile service on a secondary basis, subject to agreement obtained under No. 9.21. In order to ensure that harmful interference is not caused to stations of the aeronautical radionavigation service, stations of the mobile service shall not be introduced in the band until it is no longer required for the aeronautical radionavigation service by any administration which may be identified in the application of the procedure invoked under No. 9.21. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.260A          | In the frequency band 399.9-400.05 MHz, the maximum e.i.r.p. of any emission of earth stations in the mobile-satellite service shall not exceed 5 dBW in any 4 kHz band and the maximum e.i.r.p. of each earth station in the mobile-satellite service shall not exceed 5 dBW in the whole 399.9-400.05 MHz frequency band. Until 22 November 2022, this limit shall not apply to satellite systems for which complete notification information has been received by the Radiocommunication Bureau by 22 November 2019 and that have been brought into use by that date. After 22 November 2022, these limits shall apply to all systems within the mobile-satellite service operating in this frequency band. In the frequency band 399.99-400.02 MHz, the e.i.r.p. limits as specified above shall apply after 22 November 2022 to all systems within the mobile-satellite service. Administrations are requested that their mobile-satellite service satellite links in the 399.99-400.02 MHz frequency band comply with the e.i.r.p. limits as specified above, after 22 November 2019. (WRC-19) |
| 5.260B          | In the frequency band 400.02-400.05 MHz, the provisions of No. 5.260A are not applicable for telecommand uplinks within the mobile-satellite service. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.262           | Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Botswana, Colombia, Cuba, Egypt, the United Arab Emirates, Ecuador, the Russian Federation, Georgia, Hungary, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kazakhstan, Kuwait, Liberia, Malaysia, Moldova, Oman, Uzbekistan, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, Kyrgyzstan, Singapore, Somalia, Tajikistan, Chad, Turkmenistan and Ukraine, the band 400.05-401 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.263           | The band 400.15-401 MHz is also allocated to the space research service in the space-to-space direction for communications with manned space vehicles. In this application, the space research service will not be regarded as a safety service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5.264           | The use of the band 400.15-401 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. The power flux-density limit indicated in Annex 1 of Appendix 5 shall apply until such time as a competent world radiocommunication conference revises it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.264A          | In the frequency band 401-403 MHz, the maximum e.i.r.p. of any emission of each earth station in the meteorological-satellite service and the Earth exploration-satellite service shall not exceed 22 dBW in any 4 kHz band for geostationary systems and non-geostationary systems with an orbit of apogee equal or greater than 35 786 km. The maximum e.i.r.p. of any emission of each earth station in the meteorological satellite service and the Earth exploration-satellite service shall not exceed 7 dBW in any 4 kHz band for non-geostationary systems with an orbit of apogee lower than 35 786 km. The maximum e.i.r.p. of each earth station in the meteorological-satellite service and the Earth exploration-satellite service shall not exceed 22 dBW for geostationary systems and nongeostationary systems with an orbit of apogee equal or greater than 35 786 km in the whole 401-403 MHz frequency band. The maximum e.i.r.p. of each earth station in the meteorological-satellite service and the Earth exploration-satellite service shall not exceed 7 dBW for non-geostationary systems with an orbit of apogee lower than 35 786 km in the whole 401-403 MHz frequency band. Until 22 November 2029, these limits shall not apply to satellite systems for which complete notification information has been received by the Radiocommunication Bureau by 22 November 2019 and that have been brought into use by that date. After 22 November 2029, these limits shall apply to all systems within the meteorological-satellite service and the Earth exploration-satellite service operating in this frequency band. (WRC-19) |
| 5.264B          | Non-geostationary-satellite systems in the meteorological-satellite service and the Earth exploration-satellite service for which complete notification information has been received by the Radiocommunication Bureau no later than 28 April 2007 are exempt from provisions of No. 5.264A and may continue to operate in the frequency band 401.898-402.522 MHz on a primary basis without exceeding a maximum e.i.r.p. level of 12 dBW. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.265           | In the frequency band 403-410 MHz, Resolution 205 (Rev.WRC-19) applies. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.266           | The use of the band 406-406.1 MHz by the mobile-satellite service is limited to low power satellite emergency position-indicating radiobeacons (see also Article 31). (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.267           | Any emission capable of causing harmful interference to the authorised uses of the band 406-406.1 MHz is prohibited.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.268           | Use of the frequency band 410-420 MHz by the space research service is limited to space-to-space communications links with an orbiting, manned space vehicle. The power flux-density at the surface of the Earth produced by emissions from transmitting stations of the space research service (space-to-space) in the frequency band 410-420 MHz shall not exceed $-153 \text{ dB(W/m}^2\text{)}$ for $0^\circ \leq \delta \leq 5^\circ$ , $-153 + 0.077 (\delta - 5) \text{ dB(W/m}^2\text{)}$ for $5^\circ \leq \delta \leq 70^\circ$ and $-148 \text{ dB(W/m}^2\text{)}$ for $70^\circ \leq \delta \leq 90^\circ$ , where $\delta$ is the angle of arrival of the radio-frequency wave and the reference bandwidth is 4 kHz. In this frequency band, stations of the space research service (space-to-space) service shall not claim protection from, nor constrain the use and development of, stations of the fixed and mobile services. No. 4.10 does not apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.269           | Different category of service: in Australia, Brazil, the United States, India, Japan and the United Kingdom, the allocation of the frequency bands 420-430 MHz and 440-450 MHz to the radiolocation service is on a primary basis (see No. 5.33). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.270           | Additional allocation: in Australia, the United States, Jamaica and the Philippines, the bands 420-430 MHz and 440-450 MHz are also allocated to the amateur service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.271           | Additional allocation: in Belarus, China, India, Kyrgyzstan and Turkmenistan, the band 420-460 MHz is also allocated to the aeronautical radionavigation service (radio altimeters) on a secondary basis. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.274           | Alternative allocation: in Denmark, Norway, Sweden and Chad, the bands 430-432 MHz and 438-440 MHz are allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.275           | Additional allocation: in Croatia, Estonia, Finland, Libya, North Macedonia, Montenegro and Serbia, the frequency bands 430-432 MHz and 438-440 MHz are also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.276           | Additional allocation: in Afghanistan, Algeria, Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Burkina Faso, Djibouti, Egypt, the United Arab Emirates, Ecuador, Eritrea, Ethiopia, Greece, Guinea, India, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Italy, Jordan, Kenya, Kuwait, Libya, Malaysia, Niger, Nigeria, Oman, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, Switzerland, Thailand, Togo, Turkey and Yemen, the frequency band 430-440 MHz is also allocated to the fixed service on a primary basis and the frequency bands 430-435 MHz and 438-440 MHz are also allocated, except in Equador, to the mobile, except aeronautical mobile, service on a primary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.277           | Additional allocation: in Angola, Armenia, Azerbaijan, Belarus, Cameroon, Congo (Rep. of the), Djibouti, the Russian Federation, Georgia, Hungary, Israel, Kazakhstan, Mali, Uzbekistan, Poland, the Dem. Rep. of the Congo, Kyrgyzstan, Slovakia, Romania, Rwanda, Tajikistan, Chad, Turkmenistan and Ukraine, the frequency band 430-440 MHz is also allocated to the fixed service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 5.279A          | The use of the frequency band 432-438 MHz by sensors in the Earth exploration-satellite service (active) shall be in accordance with Recommendation ITU-R RS.1260-2. Additionally, the Earth exploration-satellite service (active) in the frequency band 432-438 MHz shall not cause harmful interference to the aeronautical radionavigation service in China. The provisions of this footnote in no way diminish the obligation of the Earth exploration-satellite service (active) to operate as a secondary service in accordance with Nos. 5.29 and 5.30. (WRC-19)                                                                          |
| 5.280           | In Germany, Austria, Bosnia and Herzegovina, Croatia, Liechtenstein, North Macedonia, Montenegro, Portugal, Serbia, Slovenia and Switzerland, the frequency band 433.05-434.79 MHz (centre frequency 433.92 MHz) is designated for industrial, scientific and medical (ISM) applications. Radiocommunication services of these countries operating within this frequency band must accept harmful interference which may be caused by these applications. ISM equipment operating in this frequency band is subject to the provisions of No. 15.13. (WRC-19)                                                                                      |
| 5.281           | Additional allocation: in the French Overseas Departments in Region 2 and India, the band 433.75-434.25 MHz is also allocated to the space operation service (Earth-to-space) on a primary basis. In France and in Brazil, the band is allocated to the same service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                        |
| 5.282           | In the bands 435-438 MHz, 1260-1270 MHz, 2400-2450 MHz, 3400-3410 MHz (in Regions 2 and 3 only) and 5650-5670 MHz, the amateur-satellite service may operate subject to not causing harmful interference to other services operating in accordance with the Table (see No. 5.43). Administrations authorising such use shall ensure that any harmful interference caused by emissions from a station in the amateur-satellite service is immediately eliminated in accordance with the provisions of No. 25.11. The use of the bands 1260-1270 MHz and 5650-5670 MHz by the amateur-satellite service is limited to the Earth-to-space direction. |
| 5.283           | Additional allocation: in Austria, the band 438-440 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.284           | Additional allocation: in Canada, the band 440-450 MHz is also allocated to the amateur service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.285           | Different category of service: in Canada, the allocation of the band 440-450 MHz to the radiolocation service is on a primary basis (see No. 5.33).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.286           | The band 449.75-450.25 MHz may be used for the space operation service (Earth-to-space) and the space research service (Earth-to-space), subject to agreement obtained under No. 9.21.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.286A          | The use of the bands 454-456 MHz and 459-460 MHz by the mobile-satellite service is subject to coordination under 9.11A. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.286AA         | The frequency band 450-470 MHz is identified for use by administrations wishing to implement International Mobile Telecommunications (IMT) - see Resolution 224 (Rev.WRC-19). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. (WRC-19)                                                                                                                                                                                                                                                                 |
| 5.286B          | The use of the band 454-455 MHz in the countries listed in No. 5.286D, 455-456 MHz and 459-460 MHz in Region 2, and 454-456 MHz and 459-460 MHz in the countries listed in No. 5.286E, by stations in the mobile-satellite service, shall not cause harmful interference to, or claim protection from, stations of the fixed or mobile services operating in accordance with the Table of Frequency Allocations. (WRC-97)                                                                                                                                                                                                                         |
| 5.286C          | The use of the band 454-455 MHz in the countries listed in No. 5.286D, 455-456 MHz and 459-460 MHz in Region 2, and 454-456 MHz and 459-460 MHz in the countries listed in No. 5.286E, by stations in the mobile-satellite service, shall not constrain the development and use of the fixed and mobile services operating in accordance with the Table of Frequency Allocations. (WRC-97)                                                                                                                                                                                                                                                        |
| 5.286D          | Additional allocation: in Canada, the United States and Panama, the band 454-455 MHz is also allocated to the mobile-satellite service (Earth-to-space) on a primary basis. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.286E          | Additional allocation: in Cabo Verde, Nepal and Nigeria, the bands 454-456 MHz and 459-460 MHz are also allocated to the mobile-satellite (Earth-to-space) service on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.287           | Use of the frequency bands 457.5125-457.5875 MHz and 467.5125-467.5875 MHz by the maritime mobile service is limited to on-board communication stations. The characteristics of the equipment and the channelling arrangement shall be in accordance with Recommendation ITU-R M.1174-4. The use of these frequency bands in territorial waters is subject to the national regulations of the administration concerned. (WRC-19)                                                                                                                                                                                                                  |
| 5.288           | In the territorial waters of the United States and the Philippines, the preferred frequencies for use by on-board communication stations shall be 457.525 MHz, 457.550 MHz, 457.575 MHz and 457.600 MHz paired, respectively, with 467.750 MHz, 467.775 MHz, 467.800 MHz and 467.825 MHz. The characteristics of the equipment used shall conform to those specified in Recommendation ITU-R M.1174-4. (WRC-19)                                                                                                                                                                                                                                   |
| 5.289           | Earth exploration-satellite service applications, other than the meteorological-satellite service, may also be used in the bands 460-470 MHz and 1690-1710 MHz for space-to-Earth transmissions subject to not causing harmful interference to stations operating in accordance with the Table.                                                                                                                                                                                                                                                                                                                                                   |
| 5.290           | Different category of service: in Afghanistan, Azerbaijan, Belarus, China, the Russian Federation, Kyrgyzstan, Tajikistan, and Turkmenistan, the allocation of the band 460-470 MHz to the meteorological-satellite service (space-to Earth) is on a primary basis (see No. 5.33), subject to agreement obtained under No. 9.21. (WRC-12)                                                                                                                                                                                                                                                                                                         |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 5.291A          | Additional allocation: in Germany, Austria, Denmark, Estonia, Liechtenstein, Serbia and Switzerland, the frequency band 470-494 MHz is also allocated to the radiolocation service on a secondary basis. This use is limited to the operation of wind profiler radars in accordance with Resolution 217 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.294           | Additional allocation: in Saudi Arabia, Cameroon, Ivory Coast, Egypt, Ethiopia, Israel, Libya, Palestine*, the Syrian Arab Republic, Chad and Yemen, the frequency band 470-582 MHz is also allocated to the fixed service on a secondary basis. (WRC-23) *Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.295A          | Additional allocation: in Albania, Germany, Andorra, Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Cyprus, Vatican, Croatia, Denmark, Estonia, Finland, France, Georgia, Greece, Hungary, Ireland, Iceland, Latvia, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malta, Moldova, Monaco, Montenegro, Norway, Uzbekistan, Kingdom of the Netherlands, Poland, Portugal, Türkiye, Slovakia, the Czech Republic, Romania, the United Kingdom, San Marino, Serbia, Slovenia, Sweden, Switzerland and Ukraine, the frequency band 470-694 MHz is allocated to the mobile, except aeronautical mobile, service on a secondary basis, subject to agreement obtained under No. 9.21. For the protection of the broadcasting service, stations in the mobile service shall not create a field strength for more than 1% of the time at the highest of the clutter height or 10 m above ground level at the border of the territory of any other administration that exceeds the field strength value as calculated using § 4.1.3.2 of Annex 2 to the GE06 Agreement with regard to allowance for multiple interference, Table A.1.10 and the methodology given in the GE06 Agreement. These limits may be exceeded on the territory of any country whose administration has so agreed. This allocation shall in no way adversely affect the broadcast development or undermine new entries of the broadcasting service to the GE06 Plan. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.296           | Additional allocation: in Albania, Algeria, Germany, Angola, Saudi Arabia, Austria, Bahrain, Belgium, Benin, Bosnia and Herzegovina, Botswana, Bulgaria, Burkina Faso, Burundi, Cameroon, Vatican, Congo (Rep. of the), Ivory Coast, Croatia, Denmark, Djibouti, Egypt, United Arab Emirates, Spain, Estonia, Eswatini, Finland, France, Gabon, Gambia, Georgia, Ghana, Hungary, Iraq, Ireland, Iceland, Israel, Italy, Jordan, Kenya, Kuwait, Lesotho, Latvia, Lebanon, Libya, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Malawi, Mali, Malta, Morocco, Mauritius, Mauritania, Moldova, Monaco, Mozambique, Namibia, Niger, Nigeria, Norway, Oman, Uganda, Palestine*, the Netherlands, Poland, Portugal, Qatar, the Syrian Arab Republic, Türkiye, Slovakia, the Czech Republic, Romania, the United Kingdom, Rwanda, San Marino, Senegal, Serbia, Sudan, South Africa, Sweden, Switzerland, Tanzania, Chad, Togo, Tunisia, Ukraine, Zambia and Zimbabwe, the frequency band 470-694 MHz is also allocated on a secondary basis to the land mobile service, intended for applications ancillary to broadcasting and programme-making. Stations of the land mobile service in the countries listed in this footnote shall not cause harmful interference to existing or planned stations operating in accordance with the Table in countries other than those listed in this footnote. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli Palestinian Interim Agreement of 28 September 1995                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.300           | Additional allocation: in Saudi Arabia, Cameroon, Egypt, the United Arab Emirates, Iraq, Israel, Jordan, Libya, Oman, Palestine*, Qatar, the Syrian Arab Republic and Sudan, the frequency band 582-790 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a secondary basis. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.304           | Additional allocation: in the African Broadcasting Area (see Nos. 5.10 to 5.13), the band 606-614 MHz is also allocated to the radio astronomy service on a primary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.306           | Additional allocation: in Region 1, except in the African Broadcasting Area (see Nos. 5.10 to 5.13), and in Region 3, the band 608-614 MHz is also allocated to the radio astronomy service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.307A          | Additional allocation: in Saudi Arabia, Bahrain, Egypt, the United Arab Emirates, Iraq, Jordan, Kuwait, Oman, Palestine*, Qatar and the Syrian Arab Republic, the frequency band 614-694 MHz is allocated to the mobile, except aeronautical mobile, service on a primary basis and identified for International Mobile Telecommunications (IMT) – see Resolution 224 (Rev.WRC-23) subject to the agreement obtained under No. 9.21. Stations in the mobile service shall not create a field strength for more than 1% of the time at the highest of the clutter height or 10 m above ground level at the border of the territory of any other administration that exceeds the field strength value as calculated using paragraph 4.1.3.2 of Annex 2 to the GE06 Agreement with regard to allowance for multiple interference, Table A.1.10 and the methodology given in the GE06 Agreement. Stations in the mobile service of the countries listed in this footnote shall not cause harmful interference to, or claim protection from the existing and future broadcasting stations of the neighbouring countries operating in accordance with the GE06 Plan. This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations and shall in no way adversely affect the development of the existing and future broadcasting service in accordance with the GE06 Agreement. For countries party to the GE06 Agreement, the use of stations in the mobile service is also subject to the successful application of the procedures of that Agreement. This allocation does not establish priority in the Radio Regulations and shall allow the implementation and development of the broadcasting service in accordance with the GE06 Agreement. The countries listed in this footnote and located in the African Broadcasting Area should ensure protection of the radio astronomy service within the frequency band 606-614 MHz, as allocated in No. 5.304, consistent with the most recent version of Recommendation ITU-R RA.769. The countries listed in this footnote, which are neighbouring to the countries listed in No. 5.312, should ensure the protection of the aeronautical radionavigation service in the frequency band 645-862 MHz. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995. |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 5.307B          | Additional allocation: in Gambia, Mauritania, Namibia, Nigeria, Senegal, Somalia, Tanzania and Chad, the frequency band 614-694 MHz is allocated to the mobile service on a secondary basis. For the protection of the broadcasting service, stations in the mobile service shall not create a field strength for more than 1% of the time at the highest of the clutter height or 10 m above ground level at the border of the territory of any other administration that exceeds the field strength value as calculated using paragraph 4.1.3.2 of Annex 2 to the GE06 Agreement with regard to allowance for multiple interference, Table A.1.10 and the methodology given in the GE06 Agreement. This allocation shall in no way adversely affect the broadcast development or undermine new entries of the broadcasting service to the GE06 Plan. Additional measures shall be used by administrations implementing stations in the mobile services to protect stations in the broadcasting service of neighbouring administrations such as a distance limitation from the border of a neighbouring country. (WRC-23) |
| 5.312           | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency band 645-862 MHz, and in Bulgaria the frequency bands 726-753 MHz, 778-811 MHz and 822-852 MHz, are also allocated to the aeronautical radionavigation service on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.312A          | In Region 1, the use of the frequency band 694-790 MHz by the mobile, except aeronautical mobile, service is subject to the provisions of Resolution 760 (Rev.WRC-23). See also Resolution 224 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.312B          | The frequency band 698-960 MHz, or portions thereof, in Region 2, and the frequency band 694-960 MHz, or portions thereof, in Region 1, are identified for use by high-altitude platform stations as International Mobile Telecommunications (IMT) base stations (HIBS). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 213 (WRC-23) shall apply. HIBS shall not claim protection from existing primary services. No. 5.43A does not apply, see resolves 2 of Resolution 213 (WRC-23). Such use of HIBS in the frequency bands 694-728 MHz, 830-835 MHz and 805.3-806.9 MHz is limited to reception by HIBS. (WRC-23)                                                                                                                                                                                                                                                                                                                                          |
| 5.316B          | In Region 1, the allocation to the mobile, except aeronautical mobile, service in the frequency band 790-862 MHz is subject to agreement obtained under No. 9.21 with respect to the aeronautical radionavigation service in countries mentioned in No. 5.312. For countries party to the GE06 Agreement, the use of stations of the mobile service is also subject to the successful application of the procedures of that Agreement. Resolutions 224 (Rev.WRC-23) and 749 (Rev.WRC-23) shall apply, as appropriate. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.317A          | The parts of the frequency band 698-960 MHz in Region 2 and the frequency bands 694-790 MHz in Region 1 and 790-960 MHz in Regions 1 and 3 which are allocated to the mobile service on a primary basis are identified for use by administrations wishing to implement International Mobile Telecommunications (IMT) – see Resolutions 224 (Rev.WRC-23), 760 (Rev.WRC-23) and 749 (Rev.WRC-23), where applicable. This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.319           | Additional Allocation: In Belarus, the Russian Federation and Ukraine, the bands 806-840 MHz (Earth-to-space) and 856-890 MHz (space-to-Earth) are also allocated to the mobile-satellite, except aeronautical mobile satellite (R), service. The use of these bands by this service shall not cause harmful interference to, or claim protection from, services in other countries operating in accordance with the Table of Frequency Allocations and is subject to special agreements between the administrations concerned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.322           | In Region 1, in the frequency band 862-960 MHz, stations of the broadcasting service shall be operated only in the African Broadcasting Area (see Nos. 5.10 to 5.13) excluding Algeria, Burundi, Djibouti, Egypt, Spain, Lesotho, Libya, Morocco, Malawi, Namibia, Nigeria, South Africa, Tanzania, Zimbabwe and Zambia, subject to agreement obtained under No. 9.21. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.323           | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan, Turkmenistan and Ukraine, the frequency band 862-960 MHz, in Bulgaria the frequency bands 862-880 MHz and 915-925 MHz, and in Romania the frequency bands 862-880 MHz and 915-925 MHz, are also allocated to the aeronautical radionavigation service on a primary basis. Such use is subject to agreement obtained under No. 9.21 with administrations concerned and limited to ground-based radiobeacons in operation on 27 October 1997 until the end of their lifetime. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.327A          | The use of the frequency band 960-1164 MHz by the aeronautical mobile (R) service is limited to systems that operate in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 417 (Rev. WRC-15). (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.328           | The use of the band 960-1215 MHz by the aeronautical radionavigation service is reserved on a worldwide basis for the operation and development of airborne electronic aids to air navigation and any directly associated ground-based facilities. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.328A          | Stations in the radionavigation-satellite service in the band 1164-1215 MHz shall operate in accordance with the provisions of Resolution 609 (Rev.WRC-07) and shall not claim protection from stations in the aeronautical radionavigation service in the band 960-1215 MHz. No. 5.43A does not apply. The provisions of No. 21.18 shall apply. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.328AA         | The frequency band 1 087.7-1 092.3 MHz is also allocated to the aeronautical mobile-satellite (R) service (Earth-to-space) on a primary basis, limited to the space station reception of Automatic Dependent Surveillance-Broadcast (ADS-B) emissions from aircraft transmitters that operate in accordance with recognized international aeronautical standards. Stations operating in the aeronautical mobile-satellite (R) service shall not claim protection from stations operating in the aeronautical radionavigation service. Resolution 425 (Rev.WRC-19) shall apply. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 5.328B          | The use of the bands 1164-1300 MHz, 1559-1610 MHz and 5010-5030 MHz by systems and networks in the radionavigation-satellite service for which complete coordination or notification information, as appropriate, is received by the Radiocommunication Bureau after 1 January 2005 is subject to the application of the provisions of Nos. 9.12, 9.12A and 9.13. Resolution 610 (WRC-03) shall also apply; however, in the case of radionavigation-satellite service (space-to-space) networks and systems, Resolution 610 (WRC-03) shall only apply to transmitting space stations. In accordance with No. 5.329A, for systems and networks in the radionavigation-satellite service (space-to-space) in the bands 1215-1300 MHz and 1559-1610 MHz, the provisions of Nos. 9.7, 9.12, 9.12A and 9.13 shall only apply with respect to other systems and networks in the radionavigation-satellite service (space-to-space). (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.329           | Use of the radionavigation-satellite service in the frequency band 1 215-1 300 MHz shall be subject to the condition that no harmful interference is caused to, and no protection is claimed from, the radionavigation service authorized under No. 5.331. Furthermore, the use of the radionavigation-satellite service in the frequency band 1 215- 1 300 MHz shall be subject to the condition that no harmful interference is caused to the radiolocation service. No. 5.43 shall not apply in respect of the radiolocation service. Resolution 608 (Rev.WRC-19) shall apply. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.329A          | Use of systems in the radionavigation-satellite service (space-to-space) operating in the bands 1215-1300 MHz and 1559-1610 MHz is not intended to provide safety service applications, and shall not impose any additional constraints on radionavigation-satellite service (space-to-Earth) systems or on other services operating in accordance with the Table of Frequency Allocations. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.330           | Additional allocation: in Angola, Saudi Arabia, Bahrain, Bangladesh, Cameroon, China, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Guyana, India, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Japan, Jordan, Kuwait, Nepal, Oman, Pakistan, Palestine*, the Philippines, Qatar, the Syrian Arab Republic, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the frequency band 1 215-1 300 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.331           | Additional allocation: in Algeria, Germany, Saudi Arabia, Australia, Austria, Bahrain, Belarus, Belgium, Benin, Bosnia and Herzegovina, Brazil, Burkina Faso, Burundi, Cameroon, China, Korea (Rep. of), Croatia, Denmark, Djibouti, Egypt, the United Arab Emirates, Estonia, the Russian Federation, Finland, France, Ghana, Greece, Guinea, Equatorial Guinea, Hungary, India, Indonesia, Iran (Islamic Republic of), Iraq, Ireland, Israel, Jordan, Kenya, Kuwait, Lesotho, Latvia, Lebanon, Liechtenstein, Lithuania, Luxembourg, North Macedonia, Madagascar, Mali, Mauritania, Montenegro, Nigeria, Norway, Oman, Pakistan, Palestine*, the Kingdom of the Netherlands, Poland, Portugal, Qatar, the Syrian Arab Republic, Turkiye, Dem. People's Rep. of Korea, Slovakia, the United Kingdom, Serbia, Slovenia, Somalia, Sudan, South Sudan, Sri Lanka, South Africa, Sweden, Switzerland, Thailand, Togo, Venezuela and Viet Nam, the frequency band 1 215-1 300 MHz is also allocated to the radionavigation service on a primary basis. In Canada and the United States, the frequency band 1 240-1 300 MHz is also allocated to the radionavigation service, and use of the radionavigation service shall be limited to the aeronautical radionavigation service. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995. |
| 5.332           | In the band 1215-1260 MHz, active spaceborne sensors in the earth exploration-satellite and space research services shall not cause harmful interference to, claim protection from, or otherwise impose constraints on operation or development of the radiolocation service, the radionavigation-satellite service and other services allocated on a primary basis. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.332A          | Administrations authorizing operation of the amateur and amateur-satellite services in the frequency band 1 240-1 300 MHz, or portions thereof, shall ensure that the amateur and amateur-satellite services do not cause harmful interference to radionavigation-satellite service (space-to-Earth) receivers in accordance with No. 5.29 (see the most recent version of Recommendation ITU-R M.2164). The authorizing administration, upon receipt of a report of harmful interference caused by a station of the amateur or amateur-satellite services, shall take all necessary steps to rapidly eliminate such interference. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.335           | In Canada and the United States in the band 1 240-1 300 MHz, active spaceborne sensors in the Earth exploration-satellite and space research services shall not cause interference to, claim protection from, or otherwise impose constraints on operation or development of the aeronautical radionavigation service. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.335A          | In the band 1260-1300 MHz, active spaceborne sensors in the Earth exploration-satellite and space research services shall not cause harmful interference to, claim protection from, or otherwise impose constraints on operation or development of the radiolocation service and other services allocated by footnotes on a primary basis. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.337           | The use of the bands 1300-1350 MHz, 2700-2900 MHz and 9000-9200 MHz by the aeronautical radionavigation service is restricted to ground-based radars and to associated airborne transponders which transmit only on frequencies in these bands and only when actuated by radars operating in the same band.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.337A          | The use of the band 1300-1350 MHz by earth stations in the radionavigation-satellite service and by stations in the radiolocation service shall not cause harmful interference to, nor constrain the operation and development of, the aeronautical-radionavigation service. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.338           | In Kyrgyzstan, Slovakia and Turkmenistan, existing installations of the radionavigation service may continue to operate in the band 1350-1400 MHz. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 5.338A          | In the frequency bands 1 350-1 400 MHz, 1 427-1 452 MHz, 22.55-23.55 GHz, 24.25-27.5 GHz, 30-31.3 GHz, 49.7-50.2 GHz, 50.4-50.9 GHz, 51.4-52.6 GHz, 81-86 GHz and 92-94 GHz, Resolution 750 (Rev.WRC-19) applies. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.339           | The bands 1370-1400 MHz, 2640-2655 MHz, 4950-4990 MHz and 15.20-15.35 GHz are also allocated to the space research (passive) and earth exploration-satellite (passive) services on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.340           | All emissions are prohibited in the following bands: 1400-1427 MHz 2690-2700 MHz, except those provided for by No. 5.422 10.68-10.7 GHz, except those provided for by No. 5.483 15.35-15.4 GHz, except those provided for by No. 5.511 23.6-24 GHz 31.3-31.5 GHz 31.5-31.8 GHz, in Region 2 48.94-49.04 GHz, from airborne stations 50.2-50.4 GHz (1) 52.6-54.25 GHz 86-92 GHz 100-102 GHz 109.5-111.8 GHz 114.25-116 GHz 148.5-151.5 GHz 164-167 GHz 182-185 GHz 190-191.8 GHz 200-209 GHz, 226-231.5 GHz 250-252 GHz. (WRC-03) / (1) 5.340 The allocation to the Earth exploration-satellite service (passive) and the space research service (passive) in the band 50.2-50.4 GHz should not impose undue constraints on the use of the adjacent bands by the primary allocated services in those bands. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.341           | In the bands 1400-1727 MHz, 101-120 GHz and 197-220 GHz, passive research is being conducted by some countries in a programme for the search for intentional emissions of extraterrestrial origin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.341A          | In Region 1, the frequency bands 1427-1452 MHz and 1492-1518 MHz are identified for use by administrations wishing to implement International Mobile Telecommunications (IMT) in accordance with Resolution 223 (Rev.WRC-15). This identification does not preclude the use of these frequency bands by any other application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of IMT stations is subject to agreement obtained under No. 9.21 with respect to the aeronautical mobile service used for aeronautical telemetry in accordance with No. 5.342. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.342           | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Uzbekistan, Kyrgyzstan and Ukraine, the frequency band 1429-1535 MHz also allocated to the aeronautical mobile service on a primary basis exclusively for the purposes of aeronautical telemetry within the national territory. As of 1 April 2007, the use of the band 1452-1492 MHz is subject to agreement between the administrations concerned. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.345           | Use of the frequency band 1 452-1 492 MHz by the broadcasting-satellite service, and by the broadcasting service, is limited to digital audio broadcasting and is subject to the provisions of Resolution 528 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.346           | In Algeria, Angola, Saudi Arabia, Bahrain, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Central African Republic, Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, United Arab Emirates, Eswatini, Gabon, Gambia, Ghana, Guinea, Iraq, Jordan, Kenya, Kuwait, Lesotho, Lebanon, Liberia, Madagascar, Malawi, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Uganda, Palestine**, Qatar, Dem. Rep. of the Congo, Rwanda, Senegal, Seychelles, Somalia, Sudan, South Sudan, South Africa, Tanzania, Chad, Togo, Tunisia, Zambia, and Zimbabwe, the frequency band 1 452-1 492 MHz is identified for use by administrations listed above wishing to implement International Mobile Telecommunications (IMT) in accordance with Resolution 223 (Rev.WRC-23). This identification does not preclude the use of this frequency band by any other application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of this frequency band for the implementation of IMT is subject to agreement obtained under No. 9.21 with respect to the aeronautical mobile service used for aeronautical telemetry in accordance with No. 5.342. See also Resolution 761 (Rev.WRC-19). (WRC-23) ** The use by Palestine of the allocation to the mobile service in the frequency band 1 452-1 492 MHz identified for IMT is noted, pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995. |
| 5.348           | The use of the band 1518-1525 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. In the band 1518-1525 MHz stations in the mobile-satellite service shall not claim protection from the stations in the fixed service. No. 5.43A does not apply. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.348A          | In the band 1518-1525 MHz, the coordination threshold in terms of the power flux-density levels at the surface of the Earth in application of No. 9.11A for space stations in the mobile-satellite (space-to-Earth) service, with respect to the land mobile service use for specialized mobile radios or used in conjunction with public switched telecommunication networks (PSTN) operating within the territory of Japan, shall be -150 dB(W/m2) in any 4 kHz band for all angles of arrival, instead of those given in Table 52 of Appendix 5. In the band 1518-1525 MHz stations in the mobile-satellite service shall not claim protection from stations in the mobile service in the territory of Japan. No. 5.43A does not apply. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.348B          | In the band 1518-1525 MHz, stations in the mobile-satellite service shall not claim protection from aeronautical mobile telemetry stations in the mobile service in the territory of the United States (see Nos. 5.343 and 5.344) and in the countries listed in No. 5.342. No. 5.43A does not apply. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.349           | Different category of service: in Saudi Arabia, Azerbaijan, Bahrain, Cameroon, Djibouti, Egypt, Iran (Islamic Republic of), Iraq, Israel, Kuwait, Lebanon, North Macedonia, Morocco, Qatar, Syrian Arab Republic, Kyrgyzstan, Turkmenistan and Yemen, the allocation of the frequency band 1 525-1 530 MHz to the mobile, except aeronautical mobile, service is on a primary basis (see No. 5.33). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 5.350           | Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 1 525-1 530 MHz is also allocated to the aeronautical mobile service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.351           | The bands 1525-1544 MHz, 1545-1559 MHz, 1626.5-1645.5 MHz and 1646.5-1660.5 MHz shall not be used for feeder links of any service. In exceptional circumstances, however, an earth station at a specified fixed point in any of the mobile-satellite services may be authorised by an administration to communicate via space stations using these bands.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.351A          | For the use of the frequency bands 1 518-1 544 MHz, 1 545-1 559 MHz, 1 610-1 645.5 MHz, 1 646.5-1 660.5 MHz, 1 668-1 675 MHz, 1 980-2 010 MHz, 2 170-2 200 MHz, 2 483.5-2 520 MHz and 2 670-2 690 MHz by the mobile-satellite service, see Resolutions 212 (Rev.WRC-23) and 225 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.352A          | In the frequency band 1 525-1 530 MHz, stations in the mobile-satellite service, except stations in the maritime mobile-satellite service, shall not cause harmful interference to, or claim protection from, stations of the fixed service in Algeria, Saudi Arabia, Egypt, Guinea, India, Israel, Italy, Jordan, Kuwait, Mali, Morocco, Mauritania, Nigeria, Oman, Pakistan, the Philippines, Qatar, Syrian Arab Republic, Viet Nam and Yemen notified prior to 1 April 1998. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.353A          | In applying the procedures of Section II of Article 9 to the mobile-satellite service in the frequency bands 1 530-1 544 MHz and 1 626.5-1 645.5 MHz, priority shall be given to accommodating the spectrum requirements for distress, urgency and safety communications of the global maritime distress and safety system (GMDSS). Maritime mobile-satellite distress, urgency and safety communications shall have priority access and immediate availability over all other mobile satellite communications operating within a network. Mobile-satellite systems shall not cause unacceptable interference to, or claim protection from, distress, urgency and safety communications of the GMDSS. Account shall be taken of the priority of safety-related communications in the other mobile-satellite services. The provisions of Resolution 222 (Rev.WRC-23) shall apply. (WRC-23)                                                                                                                                                                                                                                                                                                           |
| 5.354           | The use of the bands 1525-1559 MHz and 1626.5-1660.5 MHz by the mobile-satellite services is subject to coordination under No. 9.11A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.355           | Additional allocation: in Bahrain, Bangladesh, the Dem. Rep. of the Congo, Djibouti, Egypt, Eritrea, Iraq, Israel, Kuwait, Qatar, Syrian Arab Republic, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the bands 1540-1559 MHz, 1610-1645.5 MHz and 1646.5-1660 MHz are also allocated to the fixed service on a secondary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.356           | The use of the band 1544-1545 MHz by the mobile-satellite service (space-to-Earth) is limited to distress and safety communications (see Article 31).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.357           | Transmissions in the band 1545-1555 MHz from terrestrial aeronautical stations directly to aircraft stations, or between aircraft stations, in the aeronautical mobile (R) service are also authorised when such transmissions are used to extend or supplement the satellite-to-aircraft links.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.357A          | In applying the procedures of Section II of Article 9 to the mobile-satellite service in the frequency bands 1 545-1 555 MHz and 1 646.5-1 656.5 MHz, priority shall be given to accommodating the spectrum requirements of the aeronautical mobile-satellite (R) service providing transmission of messages with priority 1 to 6 in Article 44. Aeronautical mobile-satellite (R) service communications with priority 1 to 6 in Article 44 shall have priority access and immediate availability, by pre-emption if necessary, over all other mobile-satellite communications operating within a network. Mobile-satellite systems shall not cause unacceptable interference to, or claim protection from, aeronautical mobile-satellite (R) service communications with priority 1 to 6 in Article 44. Account shall be taken of the priority of safety-related communications in the other mobile-satellite services. The provisions of Resolution 222 (Rev.WRC-23) shall apply. (WRC-23)                                                                                                                                                                                                       |
| 5.359           | Additional allocation: in Germany, Saudi Arabia, Armenia, Azerbaijan, Belarus, Cameroon, the Russian Federation, Georgia, Guinea, Guinea-Bissau, Jordan, Kazakhstan, Kuwait, Lithuania, Mauritania, Uganda, Uzbekistan, Pakistan, Poland, the Syrian Arab Republic, Kyrgyzstan, the Dem. People's Rep. of Korea, Romania, Tajikistan, Tunisia and Turkmenistan, the frequency bands 1 550-1 559 MHz, 1 610-1 645.5 MHz and 1 646.5-1 660 MHz are also allocated to the fixed service on a primary basis. Administrations are urged to make all practicable efforts to avoid the implementation of new fixed-service stations in these frequency bands. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.364           | The use of the band 1610-1626.5 MHz by the mobile-satellite service (Earth-to-space) and by the radiodetermination-satellite service (Earth-to-space) is subject to coordination under No. 9.11A. A mobile earth station operating in either of the services in this band shall not produce a peak e.i.r.p. density in excess of -15 dB(W/4 kHz) in the part of the band used by systems operating in accordance with the provisions of No. 5.366 (to which No. 4.10 applies), unless otherwise agreed by the affected administrations. In the part of the band where such systems are not operating, the mean e.i.r.p. density of a mobile earth station shall not exceed -3 dB(W/4 kHz). Stations of the mobile-satellite service shall not claim protection from stations in the aeronautical radionavigation service, stations operating in accordance with the provisions of No. 5.366 and stations in the fixed service operating in accordance with the provisions of No. 5.359. Administrations responsible for the coordination of mobile-satellite networks shall make all practicable efforts to ensure protection of stations operating in accordance with the provisions of No. 5.366. |
| 5.365           | The use of the band 1613.8-1626.5 MHz by the mobile-satellite service (space-to-Earth) is subject to coordination under No. 9.11A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 5.366           | The band 1610-1626.5 MHz is reserved on a worldwide basis for the use and development of airborne electronic aids to air navigation and any directly associated ground-based or satellite-borne facilities. Such satellite use is subject to agreement obtained under No. 9.21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.367           | Additional allocation: the bands 1610-1626.5 MHz is also allocated to the aeronautical mobile-satellite (R) service on a primary basis, subject to agreement obtained under No. 9.21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.368           | The provisions of No. 4.10 do not apply with respect to the radiodetermination-satellite and mobile-satellite services in the frequency band 1 610-1 626.5 MHz. However, No. 4.10 applies in the frequency band 1 610-1 626.5 MHz with respect to the aeronautical radionavigation-satellite service when operating in accordance with No. 5.366, the aeronautical mobile-satellite (R) service when operating in accordance with No. 5.367, and in the frequency bands 1 614.4225-1 618.725 MHz or 1 616.3-1 620.38 MHz (Earth-to-space) (see resolves 5 of Resolution 365 (WRC-23)) and 1 621.35-1 626.5 MHz with respect to the maritime mobile-satellite service when used for the global maritime distress and safety system (GMDSS). In applying the procedure of Section II of Article 9, the provisions of No. 4.10 do not apply for the frequency bands 1 614.4225-1 618.725 MHz or 1 616.3-1 620.38 MHz (Earth-to-space) (see resolves 5 of Resolution 365 (WRC-23)) and 2 483.59-2 499.91 MHz (space-to-Earth) for the maritime mobile-satellite service when used for the GMDSS with satellite networks or systems for which complete coordination information has been received by the Radiocommunication Bureau before 20 November 2023. Resolution 365 (WRC-23) applies. (WRC-23) |
| 5.369           | Different category of service: in Angola, Australia, China, Eritrea, Ethiopia, India, Iran (Islamic Republic of), Israel, Lebanon, Liberia, Madagascar, Mali, Pakistan, Papua New Guinea, Syrian Arab Republic, the Dem. Rep. of the Congo, Sudan, South Sudan, Togo and Zambia, the allocation of the band 1610-1626.5 MHz to the radiodetermination-satellite service (Earth-to-space) is on a primary basis (see No. 5.33), subject to agreement obtained under No. 9.21 from countries not listed in this provision. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.371           | Additional allocation: in Region 1, the bands 1610-1626.5 MHz (Earth-to-space) is also allocated to the radiodetermination-satellite service on a secondary basis, subject to agreement obtained under No. 9.21. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.372           | Harmful interference shall not be caused to stations of the radio astronomy service using the frequency band 1 610.6-1 613.8 MHz by stations of the radiodetermination-satellite and mobile-satellite services (No. 29.13 applies). The equivalent power flux-density (epfd) produced in the frequency band 1 610.6-1 613.8 MHz by all space stations of a non-geostationary-satellite system in the mobile-satellite service (space-to-Earth) operating in frequency band 1 613.8-1 626.5 MHz shall be in compliance with the protection criteria provided in Recommendations ITU-R RA.769-2 and ITU-R RA.1513-2, using the methodology given in Recommendation ITU-R M.1583-1, and the radio astronomy antenna pattern described in Recommendation ITU-R RA.1631-0. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.372A          | The maritime mobile-satellite service in the frequency bands 1 614.4225-1 618.725 MHz or 1 616.3-1 620.38 MHz (Earth-to-space) (see resolves 5 of Resolution 365 (WRC-23)) and 2 483.59-2 499.91 MHz (space-to-Earth) when they are used for the global maritime distress and safety system (GMDSS) is limited to the geostationary-satellite networks identified in Resolution 365 (WRC-23) and their associated earth stations located within a service area from 75°E to 135°E longitude and from 10°N to 55°N latitude. Resolution 365 (WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.373           | Maritime mobile earth stations receiving in the frequency band 1 621.35-1 626.5 MHz shall not impose additional constraints on earth stations operating in the maritime mobile-satellite service or maritime earth stations of the radiodetermination-satellite service operating in accordance with the Radio Regulations in the frequency band 1 610-1 621.35 MHz or on earth stations operating in the maritime mobile-satellite service operating in accordance with the Radio Regulations in the frequency band 1 626.5-1 660.5 MHz, unless otherwise agreed between the notifying administrations. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.373A          | Maritime mobile earth stations receiving in the frequency band 1 621.35-1 626.5 MHz shall not impose constraints on the assignments of earth stations of the mobile-satellite service (Earth-to-space) and the radiodetermination-satellite service (Earth-to-space) in the frequency band 1 621.35-1 626.5 MHz in networks for which complete coordination information has been received by the Radiocommunication Bureau before 28 October 2019. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.374           | Mobile earth stations in the mobile-satellite service operating in the bands 1631.5-1634.5 MHz and 1656.5-1660 MHz shall not cause harmful interference to the stations in the fixed service operating in the countries listed in No. 5.359. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.375           | The use of the frequency band 1 645.5-1 646.5 MHz by the mobile-satellite service (Earth-to-space) and for inter-satellite links is limited to distress, urgency and safety communications (see Article 31). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.376           | Transmissions in the band 1646.5-1656.5 MHz from aircraft stations in the aeronautical mobile (R) service directly to terrestrial aeronautical stations, or between aircraft stations, are also authorised when such transmissions are used to extend or supplement the aircraft-to-satellite links.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.376A          | Mobile earth stations operating in the band 1660-1660.5 MHz shall not cause harmful interference to stations in the radio astronomy service. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.379           | Additional allocation: in Bangladesh, India, Indonesia, Nigeria and Pakistan, the band 1 660.5-1 668.4 MHz is also allocated to the meteorological aids service on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| 5.379A          | Administrations are urged to give all practicable protection in the band 1660.5-1668.4 MHz for future research in radio astronomy, particularly by eliminating air-to-ground transmissions in the meteorological aids service in the band 1664.4-1668.4 MHz as soon as practicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.379B          | The use of the frequency band 1 668-1 675 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.379C          | In order to protect the radio astronomy service in the band 1668-1670 MHz, the aggregate power flux-density values produced by mobile earth stations in a network of the mobile-satellite service operating in this band shall not exceed -181 dB(W/m <sup>2</sup> ) in 10 MHz and -194 dB(W/m <sup>2</sup> ) in any 20 kHz at any radio astronomy station recorded in the Master International Frequency Register, for more than 2% of integration periods of 2 000 s. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.379D          | For sharing of the frequency band 1 668.4-1 675 MHz between the mobile-satellite service and the fixed and mobile services, Resolution 744 (Rev.WRC-23) shall apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.379E          | In the band 1668.4-1675 MHz, stations in the mobile-satellite service shall not cause harmful interference to stations in the meteorological aids service in China, Iran (Islamic Republic of), Japan and Uzbekistan. In the band 1668.4-1675 MHz, administrations are urged not to implement new systems in the meteorological aids service and are encouraged to migrate existing meteorological aids service operations to other bands as soon as practicable. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.380A          | In the band 1670-1675 MHz, stations in the mobile-satellite service shall not cause harmful interference to, nor constrain the development of, existing earth stations in the meteorological-satellite service notified before 1 January 2004. Any new assignment to these earth stations in this band shall also be protected from harmful interference from stations in the mobile-satellite service. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.382           | Different category of service: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Congo (Rep. of the), Egypt, the United Arab Emirates, Eritrea, Ethiopia, the Russian Federation, Guinea, Iraq, Israel, Jordan, Kazakhstan, Kuwait, Lebanon, North Macedonia, Mauritania, Moldova, Mongolia, Oman, Uzbekistan, Poland, Qatar, the Syrian Arab Republic, Kyrgyzstan, Somalia, Tajikistan, Turkmenistan, Ukraine and Yemen, the allocation of the frequency band 1 690-1 700 MHz to the fixed and mobile, except aeronautical mobile, services is on a primary basis (see No. 5.33), and in the Dem. People's Rep. of Korea, the allocation of the frequency band 1 690-1 700 MHz to the fixed service is on a primary basis (see No. 5.33) and to the mobile, except aeronautical mobile, service on a secondary basis. (WRC-19)                                             |
| 5.384A          | The frequency bands 1710-1885 MHz, 2300-2400 MHz or 2500-2690 MHz, and portion thereof, are identified for use by administrations wishing to implement International Mobile Telecommunications(IMT)in accordance with Resolution 223 (Rev.WRC-15). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.385           | Additional allocation: the band 1718.8-1722.2 MHz is also allocated to the radio astronomy service on a secondary basis for spectral line observations. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.386           | Additional allocation: the band 1750-1850 MHz is also allocated to the space operation (Earth-to-space) and space research (Earth-to-space) services in Region 2, (except in Mexico) in Australia, Guam, India, Indonesia and Japan on a primary basis, subject to agreement obtained under No. 9.21, having particular regard to troposcatter systems. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.387           | Additional allocation: in Belarus, Georgia, Kyrgyzstan, Romania, Tajikistan and Turkmenistan, the frequency band 1 770-1 790 MHz is also allocated to the meteorological-satellite service on a primary basis, subject to agreement obtained under No. 9.21. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.388           | The frequency bands 1 885-2 025 MHz and 2 110-2 200 MHz are intended for use, on a worldwide basis, by administrations wishing to implement International Mobile Telecommunications (IMT). Such use does not preclude the use of these frequency bands by other services to which they are allocated. The frequency bands should be made available for IMT in accordance with Resolution 212 Rev.WRC-23) (see also Resolution 223 (Rev.WRC-23)). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.388A          | The frequency bands 1 710-1 980 MHz, 2 010-2 025 MHz and 2 110-2 170 MHz in Regions 1 and 3 and the frequency bands 1 710-1 980 MHz and 2 110-2 160 MHz in Region 2 are identified for the use by high altitude platform stations as International Mobile Telecommunications (IMT) base stations (HIBS). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 221 (Rev.WRC-23) shall apply. HIBS shall not claim protection from existing primary services. No. 5.43A does not apply. Such use of HIBS in the frequency bands 1 710-1 785 MHz in Regions 1 and 2, and 1 710-1 815 MHz in Region 3 is limited to reception by HIBS, and in the frequency band 2 110-2 170 MHz is limited to transmission from HIBS. (WRC-23) |
| 5.389A          | The use of the frequency bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service is subject to coordination under No. 9.11A and to the provisions of Resolution 716 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.389B          | The use of the frequency band 1 980-1 990 MHz by the mobile-satellite service shall not cause harmful interference to or constrain the development of the fixed and mobile services in Argentina, Brazil, Canada, Chile, Ecuador, the United States, Honduras, Jamaica, Mexico, Paraguay, Peru, Suriname, Trinidad and Tobago, Uruguay and Venezuela. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.389F          | In Algeria, Cape Verde, Egypt, Iran (Islamic Republic of), Mali, Syrian Arab Republic and Tunisia, the use of the frequency bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service shall neither cause harmful interference to the fixed and mobile services, nor hamper the development of those services prior to 1 January 2005, nor shall the former service request protection from the latter services. (WRC-19)                                                                                                                                                                                                                                                                                                                |
| 5.391           | In making assignments to the mobile service in the frequency bands 2025-2110 MHz and 2200-2290 MHz, administrations shall not introduce high-density mobile systems, as described in Recommendation ITU-R SA.1154-0, and shall take that Recommendation into account for the introduction of any other type of mobile system. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.392           | Administrations are urged to take all practicable measures to ensure that space-to-space transmissions between two or more non-geostationary satellites, in the space research, space operations and Earth exploration-satellite services in the bands 2025-2110 MHz and 2200-2290 MHz, shall not impose any constraints on Earth-to-space, space-to-Earth and other space-to-space transmissions of those services and in those bands between geostationary and non-geostationary satellites.                                                                                                                                                                                                                                                           |
| 5.395           | In France and Turkey, the use of the band 2310-2360 MHz by the aeronautical mobile service for telemetry has priority over other uses by the mobile service. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.398           | In respect of the radiodetermination-satellite service in the band 2483.5-2500 MHz, the provisions of No. 4.10 do not apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.398A          | Different category of service: In Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan and Ukraine, the band 2483.5-2500 MHz is allocated on a primary basis to the radiolocation service. The radiolocation stations in these countries shall not cause harmful interference to, or claim protection from, stations of the fixed, mobile and mobile-satellite services operating in accordance with the Radio Regulations in the frequency band 2483.5-2500 MHz. (WRC-12)                                                                                                                                                                                                                               |
| 5.399           | Except for cases referred to in No. 5.401, stations of the radiodetermination-satellite service operating in the frequency band 2483.5-2500 MHz for which notification information is received by the Bureau after 17 February 2012, and the service area of which includes Armenia, Azerbaijan, Belarus, the Russian Federation, Kazakhstan, Uzbekistan, Kyrgyzstan, Tajikistan and Ukraine, shall not cause harmful interference to, and shall not claim protection from stations of the radiolocation service operating in these countries in accordance with No.5.398A. (WRC-12)                                                                                                                                                                     |
| 5.401           | In Angola, Australia, Bangladesh, China, Eritrea, Eswatini, Ethiopia, India, Lebanon, Liberia, Libya, Madagascar, Mali, Pakistan, Papua New Guinea, Syrian Arab Republic, Dem. Rep. of the Congo, Sudan, Togo and Zambia, the frequency band 2 483.5-2 500 MHz was already allocated on a primary basis to the radiodetermination-satellite service before WRC-12, subject to agreement obtained under No. 9.21 from countries not listed in this provision. Systems in the radiodetermination-satellite service for which complete coordination information has been received by the Radiocommunication Bureau before 18 February 2012 will retain their regulatory status, as of the date of receipt of the coordination request information. (WRC-19) |
| 5.402           | The use of the band 2483.5-2500 MHz by the mobile-satellite and the radiodetermination-satellite services is subject to the coordination under No. 9.11A. Administrations are urged to take all practicable steps to prevent harmful interference to the radio astronomy service from emissions in the 2483.5-2500 MHz band, especially those caused by second-harmonic radiation that would fall into the 4990-5000 MHz band allocated to the radio astronomy service worldwide.                                                                                                                                                                                                                                                                        |
| 5.409A          | The frequency band 2 500-2 690 MHz in Regions 1 and 2, and the frequency band 2 500-2 655 MHz in Region 3 are identified for use by high-altitude platform stations as International Mobile Telecommunications (IMT) base stations (HIBS). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 218 (WRC-23) shall apply. HIBS shall not claim protection from existing primary services. No. 5.43A does not apply. Such use of HIBS in the frequency bands 2 500-2 510 MHz in Regions 1 and 2, and 2 500-2 535 MHz in Region 3 is limited to reception by HIBS. (WRC-23)                          |
| 5.410           | The band 2500-2690 MHz may be used for tropospheric scatter systems in Region 1, subject to agreement obtained under No. 9.21. No. 9.21 does not apply to tropospheric scatter links situated entirely outside Region 1. Administrations shall make all practicable efforts to avoid developing new tropospheric scatter systems in this band. When planning new tropospheric scatter radio-relay links in this band, all possible measures shall be taken to avoid directing the antennas of these links towards the geostationary-satellite orbit.(WRC-12)                                                                                                                                                                                             |
| 5.412           | Alternative allocation:in Kyrgyzstan and Turkmenistan, the band 2 500-2 690 MHz is allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.413           | In the design of systems in the broadcasting-satellite service in the bands between 2500 MHz and 2690 MHz, administrations are urged to take all necessary steps to protect the radio astronomy service in the band 2690-2700 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.416           | The use of the band 2520-2670 MHz by the broadcasting-satellite service is limited to national and regional systems for community reception, subject to agreement obtained under No. 9.21. The provisions of No. 9.19 shall be applied by administrations in this band in their bilateral and multilateral negotiations. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.418B          | Use of the band 2630-2655 MHz by non geostationary-satellite systems in the broadcasting-satellite service (sound), pursuant to No. 5.418, for which complete Appendix 4 coordination information, or notification information, has been received after 2 June 2000, is subject to the application of the provisions of No. 9.12. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                               |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5.418C          | Use of the band 2630-2655 MHz by geostationary-satellite networks for which complete Appendix 4 coordination information, or notification information, has been received after 2 June 2000 is subject to the application of the provisions of No. 9.13 with respect to non geostationary-satellite systems in the broadcasting-satellite service (sound), pursuant to No. 5.418 and No. 22.2 does not apply. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.422           | Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Brunei Darussalam, Congo (Rep. of the), Ivory Coast, Cuba, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Gabon, Georgia, Guinea, Guinea-Bissau, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kuwait, Lebanon, Mauritania, Mongolia, Montenegro, Nigeria, Oman, Pakistan, the Philippines, Qatar, Syrian Arab Republic, Kyrgyzstan, the Dem. Rep. of the Congo, Romania, Somalia, Tajikistan, Tunisia, Turkmenistan, Ukraine and Yemen, the band 2690-2700 MHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. Such use is limited to equipment in operation by 1 January 1985. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.423           | In the band 2700-2900 MHz, ground-based radars used for meteorological purposes are authorised to operate on a basis of equality with stations of the aeronautical radionavigation service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.424A          | In the band 2900-3100 MHz, stations in the radiolocation service shall not cause harmful interference to, nor claim protection from, radar systems in the radionavigation service. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.425           | In the band 2900-3100 MHz, the use of the shipborne interrogator-transponder system (SIT) shall be confined to the sub-band 2930-2950 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.426           | The use of the band 2900-3100 MHz by the aeronautical radionavigation service is limited to ground-based radars.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.427           | In the bands 2900-3100 MHz and 9300-9500 MHz, the response from radar transponders shall not be capable of being confused with the response from radar beacons (racons) and shall not cause interference to ship or aeronautical radars in the radionavigation service, having regard, however, to No. 4.9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.428           | Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 3 100-3 300 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.429           | Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Benin, Brunei Darussalam, Cambodia, Cameroon, China, Congo (Rep. of the), Korea (Rep. of), Ivory Coast, Djibouti, Egypt, the United Arab Emirates, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kenya, Kuwait, Lao P.D.R., Lebanon, Libya, Malaysia, Mongolia, Myanmar, New Zealand, Oman, Uganda, Pakistan, Palestine*, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, Thailand, Viet Nam and Yemen, the frequency band 3 300-3 400 MHz is also allocated to the fixed and mobile services on a primary basis. Mongolia, New Zealand and the countries bordering the Mediterranean shall not claim protection for their fixed and mobile services from the radiolocation service. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.429A          | Additional allocation: in Angola, Benin, Botswana, Burkina Faso, Burundi, Djibouti, Eswatini, Ghana, Guinea, Guinea-Bissau, Lesotho, Liberia, Malawi, Mauritania, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sudan, South Sudan, South Africa, Tanzania, Chad, Togo, Zambia and Zimbabwe, the frequency band 3 300-3 400 MHz is allocated to the mobile, except aeronautical mobile, service on a primary basis. Stations in the mobile service operating in the frequency band 3 300-3 400 MHz shall not cause harmful interference to, or claim protection from, stations operating in the radiolocation service. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.429B          | In the following countries of Region 1: Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Comoros, Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, Eritrea, Eswatini, Ethiopia, Gambia, Ghana, Guinea, Guinea-Bissau, Equatorial Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mauritius, Mauritania, Mongolia, Mozambique, Namibia, Niger, Nigeria, Uganda, the Dem. Rep. of the Congo, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, Sudan, South Sudan, South Africa, Tanzania, Chad, Togo, Zambia and Zimbabwe, the frequency band 3 300-3 400 MHz is identified for the implementation of International Mobile Telecommunications (IMT). The use of this frequency band shall be in accordance with Resolution 223 (Rev.WRC-23). The use of the frequency band 3 300-3 400 MHz by IMT stations in the mobile service shall not cause harmful interference to, or claim protection from, systems in the radiolocation service, and administrations wishing to implement IMT shall obtain the agreement of neighbouring countries to protect operations within the radiolocation service. This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.430           | Additional allocation: in Kyrgyzstan and Turkmenistan, the frequency band 3 300-3 400 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.430A          | The allocation of the frequency band 3400-3600 MHz to the mobile, except aeronautical mobile, service subject to agreement obtained under No. 9.21. This frequency band is identified for International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The provisions of Nos. 9.17 and 9.18 shall also apply in the coordination phase. Before an administration brings into use a (base or mobile) station of the mobile service in this frequency band it shall ensure that the power flux-density (pfd) produced at 3 m above ground does not exceed -154.5 dBW/(m <sup>2</sup> · 4 kHz) for more than 20% of time at the border of the territory of any other administration. This limit may be exceeded on the territory of any country whose administration has so agreed. In order to ensure that the pfd limit at the border of the territory of any other administration is met, the calculations and verification shall be made, taking into account all relevant information, with the mutual agreement of both administrations (the administration responsible for the terrestrial station and the administration responsible for the earth station), and with the assistance of the Bureau if so requested. In case of disagreement, the calculation and verification of the pfd shall be made by the Bureau, taking into account the information referred to above. Stations of the mobile service in the frequency band 3400-3600 MHz shall not claim more protection from space stations than that provided in Table 21-4 of the Radio Regulations (Edition of 2004). (WRC-15) |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5.431           | Additional allocation: in Germany, the frequency band 3 400-3 475 MHz is also allocated to the amateur service on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.433B          | In Angola, Botswana, Guinea, Lesotho, Malawi and South Sudan, the frequency band 3 600-3 700 MHz is identified for International Mobile Telecommunications (IMT). This identification does not preclude the use of the frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The conditions of No. 5.434A shall apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.434A          | The use of the frequency band 3 600-3 800 MHz by the mobile, except aeronautical mobile, service on a primary basis in Region 1 is subject to agreement obtained under No. 9.21 if the power flux-density (pfd) limit below is exceeded. The provisions of Nos. 9.17 and 9.18 shall also apply in the coordination phase. Before an administration in Region 1 brings into use a station in the mobile service in the frequency band 3 600-3 800 MHz, for the protection of stations in the fixed and fixed-satellite services, it shall ensure that the pfd produced at 3 m above ground does not exceed $-154.5 \text{ dB(W/(m}^2 \text{ 4 kHz))}$ for more than 20% of the time at the border of the territory of any other administration. Stations in the mobile service operating in the frequency band 3 600-3 800 MHz shall not claim more protection from space stations than that provided in Table 21-4 of the Radio Regulations. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.434B          | In Algeria, Saudi Arabia, Azerbaijan, Bahrain, Belarus, Benin, Burkina Faso, Burundi, Cameroon, Central African Rep., Comoros, Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, United Arab Emirates, Eswatini, Gabon, Gambia, Ghana, Guinea, Iraq, Jordan, Kazakhstan, Kenya, Kuwait, Lebanon, Liberia, Libya, Madagascar, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Uganda, Uzbekistan, Palestine*, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Rwanda, Sao Tome and Principe, Senegal, Sierra Leone, Somalia, Sudan, South Africa, Tanzania, Chad, Togo, Tunisia, Yemen, Zambia and Zimbabwe, the frequency band 3 600-3 800 MHz is identified for International Mobile Telecommunications (IMT). This identification does not preclude the use of the frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The conditions of No. 5.434A shall apply. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.435A          | Different category of service: In Angola, Botswana, Guinea, Lesotho, Malawi and South Sudan, the frequency band 3 700-3 800 MHz is allocated to the mobile service on a secondary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.436           | Use of the frequency band 4 200-4 400 MHz by stations in the aeronautical mobile (R) service is reserved exclusively for wireless avionics intra-communication systems that operate in accordance with recognized international aeronautical standards. Such use shall be in accordance with Resolution 424 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.437           | Passive sensing in the Earth exploration-satellite and space research services may be authorized in the frequency band 4200-4400 MHz on a secondary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.438           | Use of the frequency band 4200-4400 MHz by the aeronautical radionavigation service is reserved exclusively for radio altimeters installed on board aircraft and for the associated transponders on the ground. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.439           | Additional allocation: in Iran (Islamic Republic of), the band 4200-4400 MHz is also allocated to the fixed service on a secondary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.440           | The standard frequency and time signal-satellite service may be authorised to use the frequency 4202 MHz for space-to-Earth transmissions and the frequency 6427 MHz for Earth-to-space transmissions. Such transmissions shall be confined within the limits of $\pm 2 \text{ MHz}$ of these frequencies, subject to agreement obtained under No. 9.21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.440A          | In Region 2 (except Brazil, Cuba, French overseas departments and communities, Guatemala, Paraguay, Uruguay and Venezuela), and in Australia, the band 4400-4940 MHz may be used for aeronautical mobile telemetry for flight testing by aircraft stations (see No. 1.83). Such use shall be in accordance with Resolution 416 (WRC-07) and shall not cause harmful interference to, nor claim protection from, the fixed-satellite and fixed service. Any such use does not preclude the use of this band by other mobile service applications or by other services to which this band is allocated on a co-primary basis and does not establish priority in the Radio Regulations. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.441           | The use of the bands 4500-4800 MHz (space-to-Earth), 6725-7025 MHz (Earth-to-space) by the fixed-satellite service shall be in accordance with the provisions of Appendix 30B. The use of the bands 10.7-10.95 GHz (space-to-Earth), 11.2-11.45 GHz (space-to-Earth) and 12.75-13.25 GHz (Earth-to-space) by geostationary-satellite systems in the fixed-satellite service shall be in accordance with the provisions of Appendix 30B. The use of the bands 10.7-10.95 GHz (space-to Earth), 11.2-11.45 GHz (space-to-Earth) and 12.75-13.25 GHz (Earth-to-space) by a non-geostationary-satellite system in the fixed-satellite service is subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non-geostationary-satellite system in the fixed-satellite service shall not claim protection from geostationary-satellite networks in the fixed-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite system in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. (WRC-2000) |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.441A          | In Brazil, Paraguay and Uruguay, the frequency band 4 800-4 900 MHz, or portions thereof, is identified for the implementation of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of this frequency band for the implementation of IMT is subject to agreement obtained with neighbouring countries, and IMT stations shall not claim protection from stations of other applications of the mobile service. Such use shall be in accordance with Resolution 223 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.441B          | In Angola, Argentina, Armenia, Azerbaijan, Benin, Botswana, Brazil, Burkina Faso, Burundi, Cabo Verde, Cambodia, Cameroon, Chile, China, Colombia, Congo (Rep. of the), Ivory Coast, Djibouti, Eswatini, Russian Federation, Gabon, Ghana, Guinea, Iran (Islamic Republic of), Iraq, Kazakhstan, Lao P.D.R., Lesotho, Liberia, Madagascar, Malawi, Mali, Mongolia, Namibia, Niger, Uganda, Uzbekistan, the Dem. Rep. of the Congo, Kyrgyzstan, the Dem. People's Rep. of Korea, South Sudan, South Africa, Chad, Togo, Viet Nam, Zambia and Zimbabwe, the frequency band 4 800-4 990 MHz, or portions thereof, is identified for use by administrations wishing to implement International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. The use of IMT stations is subject to agreement obtained under No. 9.21 with concerned administrations, and IMT stations shall not claim protection from stations of other applications of the mobile service. In addition, before an administration brings into use an IMT station in the mobile service, it shall ensure that the power flux-density (pfd) produced by this station does not exceed $-155 \text{ dB(W/(m}^2 \cdot 1 \text{ MHz))}$ produced up to 19 km above sea level at 20 km from the coast, defined as the low-water mark, as officially recognized by the coastal State. Resolution 223 (Rev.WRC-23) applies. (WRC-23) |
| 5.442           | In the bands 4825-4835 MHz and 4950-4990 MHz, the allocation to the mobile service is restricted to the mobile, except aeronautical mobile, service. In Region 2 (except Brazil, Cuba, Guatemala, Mexico, Paraguay, Uruguay and Venezuela), and in Australia, the band 4825-4835 MHz is also allocated to the aeronautical mobile service, limited to aeronautical mobile telemetry for flight testing by aircraft stations. Such use shall be in accordance with Resolution 416 (WRC-07) and shall not cause harmful interference to the fixed service. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.443           | Different category of service: in Argentina, Australia and Canada, the allocation of the bands 4 825-4 835 MHz and 4 950-4 990 MHz to the radio astronomy service is on a primary basis (see No. 5.33).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.443AA         | In the frequency bands 5000-5030 MHz and 5091-5150 MHz, the aeronautical mobile-satellite (R) service is subject to agreement obtained under No. 9.21. The use of these bands by the aeronautical mobile-satellite (R) service is limited to internationally standardized aeronautical systems. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.443B          | In order not to cause harmful interference to the microwave landing system operating above 5030 MHz, the aggregate power flux-density produced at the Earth's surface in the frequency band 5030-5150 MHz by all the space stations within any radionavigation-satellite service system (space-to-Earth) operating in the frequency band 5 010-5 030 MHz shall not exceed $-124.5 \text{ dB(W/m}^2)$ in a 150 kHz band. In order not to cause harmful interference to the radio astronomy service in the frequency band 4990-5000 MHz, radionavigation-satellite service systems operating in the frequency band 5 010-5 030 MHz shall comply with the limits in the frequency band 4990-5000 MHz defined in Resolution 741 (Rev.WRC-15). (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.443C          | The use of the frequency band 5030-5091 MHz by the aeronautical mobile (R) service is limited to internationally standardized aeronautical systems. Unwanted emissions from the aeronautical mobile (R) service in the frequency band 5030-5091 MHz shall be limited to protect RNSS system downlinks in the adjacent 5010-5030 MHz band. Until such time that an appropriate value is established in a relevant ITU-R Recommendation, the e.i.r.p. density limit of $-75 \text{ dBW/MHz}$ in the frequency band 5010-5030 MHz for any AM(R)S station unwanted emission should be used. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.443D          | In the frequency band 5030-5091 MHz, the aeronautical mobile-satellite (R) service is subject to coordination under No. 9.11A. The use of this frequency band by the aeronautical mobile-satellite (R) service is limited to internationally standardized aeronautical systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.444           | The frequency band 5030-5150 MHz is to be used for the operation of the international standard system (microwave landing system) for precision approach and landing. In the frequency band 5030-5091 MHz, the requirements of this system shall take precedence over other uses of this frequency band. For the use of the frequency band 5091-5150 MHz, No. 5.444A and Resolution 114 (Rev.WRC-15) apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.444A          | The use of allocation to the fixed-satellite service (Earth-to-space) in the frequency band 5091-5150 MHz is limited to feeder links of non-geostationary satellite systems in the mobile-satellite service and is subject to coordination under No. 9.11A. The use of the frequency band 5091-5150 MHz by feeder links of non-geostationary satellite systems in the mobile-satellite service shall be subject to application of Resolution 114 (Rev. WRC-15). Moreover, to ensure that the aeronautical radionavigation service is protected from harmful interference, coordination is required for feeder-link earth stations of the non geostationary satellite systems in the mobile-satellite service which are separated by less than 450 km from the territory of an administration operating ground stations in the aeronautical radionavigation service. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.444B          | The use of the frequency band 5 091-5 150 MHz by the aeronautical mobile service is limited to: – systems operating in the aeronautical mobile (R) service and in accordance with international aeronautical standards, limited to surface applications at airports. Such use shall be in accordance with Resolution 748 (Rev.WRC-19); – aeronautical telemetry transmissions from aircraft stations (see No. 1.83) in accordance with Resolution 418 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 5.446           | Additional allocation: in the countries listed in No. 5.369, the frequency band 5150-5216 MHz is also allocated to the radiodetermination-satellite service (space-to-Earth) on a primary basis, subject to agreement obtained under No. 9.21. In Region 2 (except in Mexico), the frequency band is also allocated to the radiodetermination-satellite service (space-to-Earth) on a primary basis. In Regions 1 and 3, except those countries listed in No. 5.369 and Bangladesh, the frequency band is also allocated to the radiodetermination-satellite service (space-to-Earth) on a secondary basis. The use by the radiodetermination-satellite service is limited to feeder links in conjunction with the radiodetermination-satellite service operating in the frequency bands 1610-1626.5 MHz and/or 2483.5-2500 MHz. The total power flux-density at the Earth's surface shall in no case exceed $-159$ dB(W/m <sup>2</sup> ) in any 4 kHz band for all angles of arrival. (WRC-15)                                                                                                                                                              |
| 5.446A          | The use of the frequency bands 5 150-5 350 MHz and 5 470-5 725 MHz by the stations in the mobile, except aeronautical mobile, service shall be in accordance with Resolution 229 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.446B          | In the band 5150-5250 MHz, stations in the mobile service shall not claim protection from earth stations in the fixed-satellite service. No. 5.43A does not apply to the mobile service with respect to fixed-satellite service earth stations. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.446C          | Additional allocation: in Region 1 (except in Algeria, Saudi Arabia, Bahrain, Egypt, United Arab Emirates, Iraq, Jordan, Kuwait, Lebanon, Morocco, Oman, Qatar, Syrian Arab Republic, Sudan, South Sudan and Tunisia), the frequency band 5 150-5 250 MHz is also allocated to the aeronautical mobile service on a primary basis, limited to aeronautical telemetry transmissions from aircraft stations (see No. 1.83), in accordance with Resolution 418 (Rev.WRC-19). These stations shall not claim protection from other stations operating in accordance with Article 5. No. 5.43A does not apply. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.446D          | Additional allocation: in Brazil, the band 5 150-5 250 MHz is also allocated to the aeronautical mobile service on a primary basis, limited to aeronautical telemetry transmissions from aircraft stations (see No. 1.83), in accordance with Resolution 418 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.447           | Additional allocation: in Ivory Coast, Egypt, Lebanon, the Syrian Arab Republic and Tunisia, the frequency band 5 150-5 250 MHz is also allocated to the mobile service, on a primary basis, subject to agreement obtained under No. 9.21. In this case, the provisions of Resolution 229 (Rev.WRC-23) do not apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.447A          | The allocation to the fixed-satellite service (Earth-to-space) is limited to feeder links of non-geostationary-satellite systems in the mobile-satellite service and is subject to coordination under No. 9.11A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.447B          | Additional allocation: the band 5150-5216 MHz is also allocated to the fixed-satellite service (space-to-Earth) on a primary basis. This allocation is limited to feeder links of non-geostationary-satellite systems in the mobile-satellite service and is subject to provisions of No. 9.11A. The power flux-density at the Earth's surface produced by space stations of the fixed-satellite service operating in the space-to-Earth direction in the band 5150-5216 MHz shall in no case exceed $-164$ dB(W/m <sup>2</sup> ) in any 4 kHz band for all angles of arrival.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.447C          | Administrations responsible for fixed-satellite service networks in the band 5150-5250 MHz operated under Nos. 5.447A and 5.447B shall coordinate on an equal basis in accordance with No. 9.11A with administrations responsible for non-geostationary-satellite networks operated under No. 5.446 and brought into use prior to 17 November 1995. Satellite networks operated under No. 5.446 brought into use after 17 November 1995 shall not claim protection from, and shall not cause harmful interference to, stations of the fixed-satellite service operated under Nos. 5.447A and 5.447B.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.447D          | The allocation of the band 5250-5255 MHz to the space research service on a primary basis is limited to active spaceborne sensors. Other uses of the band by the space research service are on a secondary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.447E          | Additional allocation: The frequency band 5 250-5 350 MHz is also allocated to the fixed service on a primary basis in the following countries in Region 3: Australia, Korea (Rep. of), India, Indonesia, Iran (Islamic Republic of), Japan, Malaysia, Papua New Guinea, the Philippines, Dem. People's Rep. of Korea, Sri Lanka, Thailand and Viet Nam. The use of this frequency band by the fixed service is intended for the implementation of fixed wireless access systems and shall comply with Recommendation ITU-R F.1613-0. In addition, the fixed service shall not claim protection from the radiodetermination, Earth exploration-satellite (active) and space research (active) services, but the provisions of No. 5.43A do not apply to the fixed service with respect to the Earth exploration-satellite (active) and space research (active) services. After implementation of fixed wireless access systems in the fixed service with protection for the existing radiodetermination systems, no more stringent constraints should be imposed on the fixed wireless access systems by future radiodetermination implementations. (WRC-15) |
| 5.447F          | In the frequency band 5 250-5 350 MHz, stations in the mobile service shall not claim protection from the radiolocation service, the Earth exploration-satellite service (active) and the space research service (active). The radiolocation service, the Earth exploration-satellite service (active) and the space research service (active) shall not impose more stringent conditions upon the mobile service than those stipulated in Resolution 229 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.448           | Additional allocation: in Kyrgyzstan, Romania and Turkmenistan, the frequency band 5 250-5 350 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 5.448A          | The Earth exploration-satellite (active) and space research (active) services in the frequency band 5 250-5 350 MHz shall not claim protection from the radiolocation service. No. 5.43A does not apply. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.448B          | The Earth exploration-satellite service (active) operating in the band 5350-5570 MHz and space research service (active) operating in the band 5460-5570 MHz shall not cause harmful interference to the aeronautical radionavigation service in the band 5350-5460 MHz, the radionavigation service in the band 5460-5470 MHz and the maritime radionavigation service in the band 5470-5570 MHz. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.448C          | The space research service (active) operating in the band 5350-5460 MHz shall not cause harmful interference to nor claim protection from other services to which this band is allocated. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.448D          | In the frequency band 5350-5470 MHz, stations in the radiolocation service shall not cause harmful interference to, nor claim protection from, radar systems in the aeronautical radionavigation service operating in accordance with No. 5.449. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.449           | The use of the band 5350-5470 MHz by the aeronautical radionavigation service is limited to airborne radars and associated airborne beacons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.450           | Additional allocation: in Austria, Azerbaijan, Iran (Islamic Republic of), Kyrgyzstan, Romania, Turkmenistan and Ukraine, the band 5470-5650 MHz is also allocated to the aeronautical radionavigation service on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.450A          | In the frequency band 5 470-5 725 MHz, stations in the mobile service shall not claim protection from radiodetermination services. The radiodetermination services shall not impose more stringent conditions upon the mobile service than those stipulated in Resolution 229 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.450B          | In the frequency band 5470-5650 MHz, stations in the radiolocation service, except ground-based radars used for meteorological purposes in the band 5 600-5 650 MHz, shall not cause harmful interference to, nor claim protection from, radar systems in the maritime radionavigation service. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.451           | Additional allocation: in the United Kingdom, the band 5470-5850 MHz is also allocated to the land mobile service on a secondary basis. The power limits specified in Nos. 21.2, 21.3, 21.4 and 21.5 shall apply in the band 5725-5850 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.452           | Between 5600 MHz and 5650 MHz, ground-based radars used for meteorological purposes are authorised to operate on a basis of equality with stations of the maritime radionavigation service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.453           | Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Korea (Rep. of), Ivory Coast, Djibouti, Egypt, the United Arab Emirates, Eswatini, Gabon, Guinea, Equatorial Guinea, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kenya, Kuwait, Lebanon, Libya, Madagascar, Malaysia, Niger, Nigeria, Oman, Uganda, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sri Lanka, Tanzania, Chad, Thailand, Togo, Viet Nam and Yemen, the frequency band 5 650-5 850 MHz is also allocated to the fixed and mobile services on a primary basis. In this case, the provisions of Resolution 229 (Rev.WRC-23) do not apply. In addition, in Afghanistan, Angola, Benin, Bhutan, Botswana, Burkina Faso, Burundi, Dem. Rep. of the Congo, Fiji, Ghana, Kiribati, Lesotho, Malawi, Maldives, Mauritius, Micronesia, Mongolia, Mozambique, Myanmar, Namibia, Nauru, New Zealand, Papua New Guinea, Rwanda, Solomon Islands, South Sudan, South Africa, Tonga, Vanuatu, Zambia and Zimbabwe, the frequency band 5 725-5 850 MHz is allocated to the fixed service on a primary basis, and stations operating in the fixed service shall not cause harmful interference to and shall not claim protection from other primary services in the frequency band. (WRC-23) |
| 5.454           | Different category of service: in Azerbaijan, the Russian Federation, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the allocation of the band 5670-5725 MHz to the space research service is on a primary basis (see No. 5.33). (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.455           | Additional allocation: in Armenia, Azerbaijan, Belarus, Cuba, the Russian Federation, Georgia, Hungary, Kazakhstan, Moldova, Uzbekistan, Kyrgyzstan, Romania, Tajikistan, Turkmenistan and Ukraine, the frequency band 5 670-5 850 MHz is also allocated to the fixed service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.457           | In Australia, Burkina Faso, Ivory Coast, Mali and Nigeria, the allocation to the fixed service in the bands 6440-6520 MHz (HAPS-to-ground direction) and 6560-6640 MHz (ground-to-HAPS direction) may also be used by gateway links for high-altitude platform stations (HAPS) within the territory of these countries. Such use is limited to operation in HAPS gateway links and shall not cause harmful interference to, and shall not claim protection from, existing services, and shall be in compliance with Resolution 150 (WRC-12). Existing services shall not be constrained in future development by HAPS gateway links. The use of HAPS gateway links in these bands requires explicit agreement with other administrations whose territories are located within 1000 kilometres from the border of an administration intending to use the HAPS gateway links. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.457A          | In the frequency bands 5 925-6 425 MHz and 14-14.5 GHz, earth stations located on board vessels may communicate with space stations of the fixed-satellite service. Such use shall be in accordance with Resolution 902 (Rev.WRC-23). In the frequency band 5 925-6 425 MHz, earth stations located on board vessels and communicating with space stations of the fixed-satellite service may employ transmit antennas with minimum diameter of 1.2 m and operate without prior agreement of any administration if located at least 330 km away from the low-water mark as officially recognized by the coastal State. All other provisions of Resolution 902 (Rev.WRC-23) shall apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 5.457B          | In the frequency bands 5 925-6 425 MHz and 14-14.5 GHz, earth stations located on board vessels may operate with the characteristics and under the conditions contained in Resolution 902 (Rev.WRC-23) in Algeria, Saudi Arabia, Bahrain, Comoros, Djibouti, Egypt, United Arab Emirates, Jordan, Kuwait, Libya, Morocco, Mauritania, Oman, Qatar, the Syrian Arab Republic, Sudan, Tunisia and Yemen, in the maritime mobile-satellite service on a secondary basis. Such use shall be in accordance with Resolution 902 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.457C          | In Region 2 (except Brazil, Cuba, French overseas departments and communities, Guatemala, Mexico, Paraguay, Uruguay and Venezuela), the frequency band 5925-6700 MHz may be used for aeronautical mobile telemetry for flight testing by aircraft stations (see No. 1.83). Such use shall be in accordance with Resolution 416 (WRC-07) and shall not cause harmful interference to, or claim protection from, the fixed-satellite and fixed services. Any such use does not preclude the use of this frequency band by other mobile service applications or by other services to which this frequency band is allocated on a co-primary basis and does not establish priority in the Radio Regulations. (WRC-15)                                                                                                                                                                                                                                                                                                                                             |
| 5.457E          | The frequency bands 6 425-7 125 MHz in Region 1 and 7 025-7 125 MHz in Region 3 are identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of these frequency bands by any application of the services to which they are allocated and does not establish priority in the Radio Regulations. Resolution 220 (WRC-23) applies. The frequency bands are also used for the implementation of wireless access systems (WAS), including radio local area networks (RLANs). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.458           | In the band 6425-7075 MHz, passive microwave sensor measurements are carried out over the oceans. In the band 7075-7250 MHz, passive microwave sensor measurements are carried out. Administrations should bear in mind the needs of the Earth exploration-satellite (passive) and space research (passive) services in their future planning of the bands 6425-7025 MHz and 7075-7250 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.458A          | In making assignments in the band 6700-7075 MHz to space stations of the fixed-satellite service, administrations are urged to take all practicable steps to protect spectral line observations of the radio astronomy service in the band 6650-6675.2 MHz from harmful interference from unwanted emissions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.458B          | The space-to-Earth allocation to the fixed-satellite service in the band 6700-7075 MHz is limited to feeder links for non-geostationary satellite systems of the mobile-satellite service and is subject to coordination under No. 9.11A. The use of the band 6700-7075 MHz (space-to-Earth) by feeder links for non-geostationary satellite systems in the mobile-satellite service is not subject to No. 22.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.460           | No emissions from space research service (Earth-to-space) systems intended for deep space shall be effected in the frequency band 7190-7235 MHz. Geostationary satellites in the space research service operating in the frequency band 7190-7235 MHz shall not claim protection from existing and future stations of the fixed and mobile services and No. 5.43A does not apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.460A          | The use of the frequency band 7190-7250 MHz (Earth-to-space) by the Earth exploration-satellite service shall be limited to tracking, telemetry and command for the operation of spacecraft. Space stations operating in the Earth exploration-satellite service (Earth-to-space) in the frequency band 7190-7250 MHz shall not claim protection from existing and future stations in the fixed and mobile services, and No. 5.43A does not apply. No. 9.17 applies. Additionally, to ensure protection of the existing and future deployment of fixed and mobile services, the location of earth stations supporting spacecraft in the Earth exploration-satellite service in non-geostationary orbits or geostationary orbit shall maintain a separation distance of at least 10 km and 50 km, respectively, from the respective border(s) of neighbouring countries, unless a shorter distance is otherwise agreed between the corresponding administrations. (WRC-15)                                                                                     |
| 5.460B          | Space stations on the geostationary orbit operating in the Earth exploration-satellite service (Earth-to-space) in the frequency band 7190-7235 MHz shall not claim protection from existing and future stations of the space research service, and No. 5.43A does not apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.461           | Additional allocation: the frequency bands 7 250-7 375 MHz (space-to-Earth) and 7 900-8 025 MHz (Earth-to-space) are also allocated to the mobile-satellite service on a primary basis, subject to agreement obtained under No. 9.21, with the exception that No. 9.21 shall not apply to the geostationary-satellite networks in the mobile-satellite service for which complete coordination information is received by the Bureau as of 1 January 2025 with respect to non-geostationary-satellite systems for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025. Non-geostationary-satellite systems for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025 shall not cause unacceptable interference to and shall not claim protection from geostationary-satellite networks in the mobile-satellite service operating in accordance with these Regulations. No. 5.43A does not apply. (WRC-23) |
| 5.461A          | The use of the band 7450-7550 MHz by the meteorological-satellite service (space-to-Earth) is limited to geostationary-satellite systems. Non-geostationary meteorological-satellite systems in this band notified before 30 November 1997 may continue to operate on a primary basis until the end of their lifetime. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.461AA         | The use of the frequency band 7375-7750 MHz by the maritime mobile-satellite service is limited to geostationary-satellite networks. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.461AB         | In the frequency band 7375-7750 MHz, earth stations in the maritime mobile-satellite service shall not claim protection from, nor constrain the use and development of, stations in the fixed and mobile, except aeronautical mobile, services. No. 5.43A does not apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 5.461AC         | In the frequency band 7 375-7 750 MHz, non-geostationary-satellite systems operating in the fixed-satellite service for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025 shall not cause unacceptable interference to and shall not claim protection from geostationary-satellite networks in the maritime mobile-satellite service operating in accordance with these Regulations. No. 5.43A does not apply. (WRC-23)                                                                                                                                                                                                                                                                                              |
| 5.461B          | The use of the band 7750-7900 MHz by the meteorological-satellite service (space-to-Earth) is limited to non-geostationary satellite systems. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.462A          | In Regions 1 and 3 (except for Japan), in the band 8025-8400 MHz, the Earth exploration-satellite service using geostationary satellites shall not produce a power flux-density in excess of the following provisional values for angles of arrival ( $\theta$ ), without the consent of the affected administration: - 135 dB(W/m <sup>2</sup> ) in a 1 MHz band for $0^\circ \leq \theta < 5^\circ$ - 135 + 0.5 ( $\theta - 5$ ) dB(W/m <sup>2</sup> ) in a 1 MHz band for $5^\circ \leq \theta < 25^\circ$ - 125 dB(W/m <sup>2</sup> ) in a 1 MHz band for $25^\circ \leq \theta < 90^\circ$ (WRC-12)                                                                                                                                                                                          |
| 5.463           | Aircraft stations are not permitted to transmit in the band 8025-8400 MHz. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.465           | In the space research service, the use of the band 8400-8450 MHz is limited to deep space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.466           | Different category of service: in Singapore and Sri Lanka, the allocation of the band 8400-8500 MHz to the space research service is on a secondary basis (see No. 5.32). (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.468           | Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Burundi, Cameroon, China, Congo (Rep. of the), Djibouti, Egypt, the United Arab Emirates, Eswatini, Gabon, Guyana, Indonesia, Iran (Islamic Republic of), Iraq, Jamaica, Jordan, Kenya, Kuwait, Lebanon, Libya, Malaysia, Mali, Morocco, Mauritania, Nepal, Nigeria, Oman, Uganda, Pakistan, Qatar, Syrian Arab Republic, the Dem. People's Rep. of Korea, Senegal, Singapore, Somalia, Sudan, Chad, Togo, Tunisia and Yemen, the frequency band 8 500-8 750 MHz is also allocated to the fixed and mobile services on a primary basis. (WRC-19)                                                                                                                                                                  |
| 5.469           | Additional allocation: in Armenia, Azerbaijan, Belarus, the Russian Federation, Georgia, Hungary, Lithuania, Uzbekistan, Poland, Kyrgyzstan, the Czech Rep., Romania, Tajikistan, Turkmenistan and Ukraine, the frequency band 8 500-8 750 MHz is also allocated to the land mobile and radionavigation services on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.469A          | In the band 8550-8650 MHz, stations in the earth exploration-satellite service (active) and space research service (active) shall not cause harmful interference to, or constrain the use and development of, stations of the radiolocation service. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.470           | The use of the band 8750-8850 MHz by the aeronautical radionavigation service is limited to airborne Doppler navigation aids on a centre frequency of 8800 MHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.471           | Additional allocation: in Algeria, Germany, Bahrain, Belgium, China, Egypt, the United Arab Emirates, France, Greece, Indonesia, Iran (Islamic Republic of), Libya, the Netherlands, Qatar, and Sudan, the frequency bands 8825-8850 MHz and 9000-9200 MHz are also allocated to the maritime radionavigation service, on a primary basis, for use by shore-based radars only. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.472           | In the bands 8850-9000 MHz and 9200-9225 MHz, the maritime radionavigation service is limited to shore-based radars.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.473           | Additional allocation: in Armenia, Austria, Azerbaijan, Belarus, Cuba, the Russian Federation, Georgia, Hungary, Uzbekistan, Poland, Kyrgyzstan, Romania, Tajikistan, Turkmenistan and Ukraine, the frequency bands 8 850-9 000 MHz and 9 200-9 300 MHz are also allocated to the radionavigation service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.473A          | In the band 9000-9200 MHz, stations operating in the radiolocation service shall not cause harmful interference to, nor claim protection from, systems identified in No. 5.337 operating in the aeronautical radionavigation service, or radar systems in the maritime radionavigation service operating in this band on a primary basis in the countries listed in No. 5.471. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.474           | In the band 9200-9500 MHz, search and rescue transponders (SART) may be used, having due regard to the appropriate ITU-R Recommendation (see also Article 31).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.474A          | The use of the frequency bands 9 200-9 300 MHz and 9 900-10 400 MHz by the Earth exploration-satellite service (active) is limited to systems requiring necessary bandwidth greater than 600 MHz that cannot be fully accommodated within the frequency band 9 300-9 900 MHz. Such use is subject to agreement to be obtained under No. 9.21 from Algeria, Saudi Arabia, Bahrain, Egypt, Indonesia, Iran (Islamic Republic of), Lebanon and Tunisia. An administration that has not replied under No. 9.52 is considered as not having agreed to the coordination request. In this case, the notifying administration of the satellite system operating in the Earth exploration-satellite service (active) may request the assistance of the Bureau under Sub-Section IID of Article 9. (WRC-15) |
| 5.474B          | Stations operating in the Earth exploration-satellite (active) service shall comply with Recommendation ITU-R RS.2066-0. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.474C          | Stations operating in the Earth exploration-satellite (active) service shall comply with Recommendation ITU-R RS.2065-0. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.474D          | Stations operating in the Earth exploration-satellite service (active) shall not cause harmful interference to, or claim protection from, stations of the maritime radionavigation and radiolocation services in the frequency band 9 200-9 300 MHz, the radionavigation and radiolocation services in the frequency band 9 900-10 000 MHz and the radiolocation service in the frequency band 10.0-10.4 GHz. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                            |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 5.475           | The use of the band 9300-9500 MHz by the aeronautical radionavigation service is limited to airborne weather radars and ground-based radars. In addition, ground-based radar beacons in the aeronautical radionavigation service are permitted in the band 9300-9320 MHz on condition that harmful interference is not caused to the maritime radionavigation service. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.475A          | The use of the band 9300-9500 MHz by the Earth exploration-satellite service (active) and the space research service (active) is limited to systems requiring necessary bandwidth greater than 300 MHz that cannot be fully accommodated within the 9500-9800 MHz band. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.475B          | In the band 9300-9500 MHz, stations operating in the radiolocation service shall not cause harmful interference to, nor claim protection from, radars operating in the radionavigation service in conformity with the Radio Regulations. Ground-based radars used for meteorological purposes have priority over other radiolocation uses. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.476A          | In the band 9300-9800 MHz, stations in the Earth exploration-satellite service (active) and space research service (active) shall not cause harmful interference to, nor claim protection from, stations of the radionavigation and radiolocation services. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.477           | Different category of service: in Algeria, Saudi Arabia, Bahrain, Bangladesh, Brunei Darussalam, Cameroon, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Guyana, India, Indonesia, Iran (Islamic Republic of), Iraq, Jamaica, Japan, Jordan, Kuwait, Lebanon, Liberia, Malaysia, Nigeria, Oman, Pakistan, Qatar, Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, South Sudan, Trinidad and Tobago, Uganda and Yemen, the allocation of the band 9800-10000 MHz to the fixed service is on a primary basis (see No. 5.33). (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.478           | Additional allocation: in Azerbaijan, Kyrgyzstan, Romania, Turkmenistan and Ukraine, the frequency band 9 800-10 000 MHz is also allocated to the radionavigation service on a primary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.478A          | The use of the band 9800-9900 MHz by the Earth exploration-satellite service (active) and space research service (active) is limited to systems requiring necessary bandwidth greater than 500 MHz that cannot be fully accommodated within the 9300-9800 MHz band. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.478B          | In the band 9800-9900 MHz, stations in the Earth exploration-satellite service (active) and the space research service (active) shall not cause harmful interference to, nor claim protection from stations of the fixed service to which this band is allocated on a secondary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.479           | The band 9975-10025 MHz is also allocated to the meteorological-satellite service on a secondary basis for use by weather radars.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.481           | Additional allocation: in Algeria, Germany, Angola, Brazil, China, Colombia, Costa Rica, Ivory Coast, Cuba, Djibouti, the Dominican Republic, Egypt, El Salvador, Ecuador, Spain, Guatemala, Hungary, Jamaica, Japan, Kenya, Morocco, Mexico, Nigeria, Oman, Uzbekistan, Pakistan, Palestine*, Paraguay, Peru, the Dem. People's Rep. of Korea, Romania, Somalia, Suriname, Tunisia and Uruguay, the frequency band 10.45-10.5 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.482           | In the band 10.6-10.68 GHz, the power delivered to the antenna of stations of the fixed and mobile, except aeronautical mobile, services shall not exceed -3 dBW. This limit may be exceeded, subject to agreement obtained under No. 9.21. However, in Algeria, Saudi Arabia, Armenia, Azerbaijan, Bahrain, Bangladesh, Belarus, Egypt, United Arab Emirates, Georgia, India, Indonesia, Iran (Islamic Republic of), Iraq, Jordan, Libyan Arab Jamahiriya, Kazakhstan, Kuwait, Lebanon, Morocco, Mauritania, Moldova, Nigeria, Oman, Uzbekistan, Pakistan, Philippines, Qatar, Syrian Arab Republic, Kyrgyzstan, Singapore, Tajikistan, Tunisia, Turkmenistan and Viet Nam, this restriction on the fixed and mobile, except aeronautical mobile, service is not applicable. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.482A          | For sharing of the band 10.6-10.68 GHz between the Earth exploration-satellite (passive) service and the fixed and mobile, except aeronautical mobile, services, Resolution 751 (WRC-07) applies. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.483           | Additional allocation: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, China, Colombia, Korea (Rep. of), Egypt, the United Arab Emirates, Georgia, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kazakhstan, Kuwait, Lebanon, Mongolia, Qatar, Kyrgyzstan, the Dem. People's Rep. of Korea, Tajikistan, Turkmenistan and Yemen, the frequency band 10.68-10.7 GHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. Such use is limited to equipment in operation by 1 January 1985. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.484           | In Region 1, the use of the band 10.7-11.7 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links for the broadcasting-satellite service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.484A          | The use of the frequency bands 10.95-11.2 GHz (space-to-Earth), 11.45-11.7 GHz (space-to-Earth), 11.7-12.2 GHz (space-to-Earth) in Region 2, 12.2-12.75 GHz (space-to-Earth) in Region 3, 12.5-12.75 GHz (space-to-Earth) in Region 1, 13.75-14.5 GHz (Earth-to-space), 17.3-17.7 GHz (space-to-Earth) in Region 2, 17.8-18.6 GHz (space-to-Earth), 19.7-20.2 GHz (space-to-Earth), 27.5-28.6 GHz (Earth-to-space), 29.5-30 GHz (Earth-to-space) by a non-geostationary-satellite system in the fixed-satellite service is subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non-geostationary-satellite systems in the fixed-satellite service shall not claim protection from geostationary-satellite networks in the fixed-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite systems in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. In Region 2, No. 22.2 shall continue to apply in the frequency band 17.3-17.7 GHz. (WRC-23) |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 5.484B          | Resolution 155 (WRC-15) shall apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.487           | In the band 11.7-12.5 GHz in Regions 1 and 3, the fixed, fixed-satellite, mobile, except aeronautical mobile, and broadcasting services, in accordance with their respective allocations, shall not cause harmful interference to, or claim protection from, broadcasting-satellite stations operating in accordance with the Regions 1 and 3 Plan in Appendix 30. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.487A          | Additional allocation: in Region 1, the band 11.7-12.5 GHz, in Region 2, the band 12.2-12.7 GHz and, in Region 3, the band 11.7-12.2 GHz, are also allocated to the fixed-satellite service (space-to-Earth) on a primary basis, limited to non geostationary systems and subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non-geostationary-satellite systems in the fixed satellite service shall not claim protection from geostationary-satellite networks in the broadcasting-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite systems in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. (WRC-03) |
| 5.492           | Assignments to stations of the broadcasting-satellite service which are in conformity with the appropriate regional Plan or included in the Regions 1 and 3 List in Appendix 30 may also be used for transmissions in the fixed-satellite service (space-to-Earth), provided that such transmissions do not cause more interference, or require more protection from interference, than the broadcasting-satellite service transmissions operating in conformity with the Plan or the List, as appropriate. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.494           | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Cameroon, the Central African Rep., Congo (Rep. of the), Ivory Coast, Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Gabon, Ghana, Guinea, Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Madagascar, Mali, Morocco, Mongolia, Nigeria, Oman, Palestine*, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the frequency band 12.5-12.75 GHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a primary basis. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference, and taking into account the Israeli Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.495           | Additional allocation: in Greece, Monaco, Montenegro, Uganda and Tunisia, the frequency band 12.5- 12.75 GHz is also allocated to the fixed and mobile, except aeronautical mobile, services on a secondary basis. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.496           | Additional allocation: in Austria, Azerbaijan, Kyrgyzstan and Turkmenistan, the band 12.5-12.75 GHz is also allocated to the fixed service and the mobile, except aeronautical mobile, service on a primary basis. However, stations in these services shall not cause harmful interference to fixed-satellite service earth stations of countries in Region 1 other than those listed in this footnote. Coordination of these earth stations is not required with stations of the fixed and mobile services of the countries listed in this footnote. The power flux-density limit at the Earth's surface given in Table 21-4 of Article 21, for the fixed-satellite service shall apply on the territory of the countries listed in this footnote. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.496A          | The frequency band 12.75-13.25 GHz (Earth-to-space) may be used by earth stations in motion, limited to earth stations on aircraft and vessels, communicating with geostationary space stations in the fixed-satellite service. Resolution 121 (WRC-23) shall apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.497           | The use of the band 13.25-13.4 GHz by the aeronautical radionavigation service is limited to Doppler navigation aids.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.498A          | The Earth exploration-satellite (active) and space research (active) services operating in the band 13.25-13.4 GHz shall not cause harmful interference to, or constrain the use and development of, the aeronautical radionavigation service. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.499           | Additional allocation: in Bangladesh and India, the band 13.25-14 GHz is also allocated to the fixed service on a primary basis. In Pakistan, the band 13.25-13.75 GHz is allocated to the fixed service on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.499A          | The use of the frequency band 13.4-13.65 GHz by the fixed-satellite service (space-to-Earth) is limited to geostationary-satellite systems and is subject to agreement obtained under No. 9.21 with respect to satellite systems operating in the space research service (space-to-space) to relay data from space stations in the geostationary-satellite orbit to associated space stations in non-geostationary satellite orbits for which advance publication information has been received by the Bureau by 27 November 2015. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.499B          | Administrations shall not preclude the deployment and operation of transmitting earth stations in the standard frequency and time signal-satellite service (Earth-to-space) allocated on a secondary basis in the frequency band 13.4-13.65 GHz due to the primary allocation to FSS (space-to-Earth). (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.499C          | The allocation of the frequency band 13.4-13.65 GHz to the space research service on a primary basis is limited to: - satellite systems operating in the space research service (space-to-space) to relay data from space stations in the geostationary-satellite orbit to associated space stations in non-geostationary satellite orbits for which advance publication information has been received by the Bureau by 27 November 2015, - active spaceborne sensors, - satellite systems operating in the space research service (space-to-Earth) to relay data from space stations in the geostationary-satellite orbit to associated earth stations. Other uses of the band by the space research service are on a secondary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 5.499D          | In the frequency band 13.4-13.65 GHz, satellite systems in the space research service (space-to-Earth) and/or the space research service (space-to-space) shall not cause harmful interference to, nor claim protection from, stations in the fixed, mobile, radiolocation and Earth exploration-satellite (active) services. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.499E          | In the frequency band 13.4-13.65 GHz, geostationary-satellite networks in the fixed-satellite service (space-to-Earth) shall not claim protection from space stations in the Earth exploration-satellite service (active) operating in accordance with these Regulations, and No. 5.43A does not apply. The provisions of No. 22.2 do not apply to the Earth exploration-satellite service (active) with respect to the fixed-satellite service (space-to-Earth) in this band. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.500           | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Brunei Darussalam, Cameroon, Djibouti, Egypt, the United Arab Emirates, Gabon, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kuwait, Lebanon, Madagascar, Malaysia, Mali, Morocco, Mauritania, Niger, Nigeria, Oman, Qatar, the Syrian Arab Republic, Singapore, Somalia, Sudan, South Sudan, Chad and Tunisia, the frequency band 13.4-14 GHz is also allocated to the fixed and mobile services on a primary basis. In Pakistan, the frequency band 13.4-13.75 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.501           | Additional allocation: in Hungary, Japan, Kyrgyzstan, Romania and Turkmenistan, the frequency band 13.4-14 GHz is also allocated to the radionavigation service on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.501A          | The allocation of the frequency band 13.65-13.75 GHz to the space research service on a primary basis is limited to active spaceborne sensors. Other uses of the frequency band by the space research service are on a secondary basis. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.501B          | In the band 13.4-13.75 GHz, the earth exploration-satellite (active) and space research (active) services shall not cause harmful interference to, or constrain the use and development of, the radiolocation service. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.502           | In the band 13.75-14 GHz, an earth station of a geostationary fixed-satellite service network shall have a minimum antenna diameter of 1.2 m and an earth station of a non-geostationary fixed-satellite service system shall have a minimum antenna diameter of 4.5 m. In addition, the e.i.r.p., averaged over one second, radiated by a station in the radiolocation or radionavigation services shall not exceed 59 dBW for elevation angles above 2° and 65 dBW at lower angles. Before an administration brings into use an earth station in a geostationary-satellite network in the fixed-satellite service in this band with an antenna size smaller than 4.5 m, it shall ensure that the power flux-density produced by this earth station does not exceed: - 115 dB(W/(m <sup>2</sup> · 10 MHz)) for more than 1% of the time produced at 36 m above sea level at the low water mark, as officially recognized by the coastal state; - 115 dB(W/(m <sup>2</sup> · 10 MHz)) for more than 1% of the time produced 3 m above ground at the border of the territory of an administration deploying or planning to deploy land mobile radars in this band, unless prior agreement has been obtained. For earth stations within the fixed-satellite service having an antenna diameter greater than or equal to 4.5 m, the e.i.r.p. of any emission should be at least 68 dBW and should not exceed 85 dBW. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.503           | In the band 13.75-14 GHz, geostationary space stations in the space research service for which information for advance publication has been received by the Bureau prior to 31 January 1992 shall operate on an equal basis with stations in the fixed-satellite service; after that date, new geostationary space stations in the space research service will operate on a secondary basis. Until those geostationary space stations in the space research service for which information for advance publication has been received by the Bureau prior to 31 January 1992 cease to operate in this band: - in the band 13.77-13.78 GHz, the e.i.r.p. density of emissions from any earth station in the fixed-satellite service operating with a space station in geostationary-satellite orbit shall not exceed: i) 4.7D + 28 dB(W/40 kHz), where D is the fixed-satellite service earth station antenna diameter (m) for antenna diameters equal to or greater than 1.2 m and less than 4.5 m; ii) 49.2 + 20 log(D/4.5) dB(W/40 kHz), where D is the fixed-satellite service earth station antenna diameter (m) for antenna diameters equal to or greater than 4.5 m and less than 31.9 m; iii) 66.2 dB(W/40 kHz) for any fixed-satellite service earth station for antenna diameters (m) equal to or greater than 31.9 m; iv) 56.2 dB(W/4 kHz) for narrow-band (less than 40 kHz of necessary bandwidth) fixed-satellite service earth station emissions from any fixed-satellite service earth station having an antenna diameter of 4.5 m or greater; - the e.i.r.p. density of emissions from any earth station in the fixed-satellite service operating with a space station in non-geostationary-satellite orbit shall not exceed 51 dBW in the 6 MHz band from 13.772 to 13.778 GHz. Automatic power control may be used to increase the e.i.r.p. density in these frequency ranges to compensate for rain attenuation, to the extent that the power flux-density at the fixed-satellite service space station does not exceed the value resulting from use by an earth station of an e.i.r.p. meeting the above limits in clear-sky conditions. (WRC-03) |
| 5.504           | The use of the band 14-14.3 GHz by the radionavigation service shall be such as to provide sufficient protection to space stations of the fixed-satellite service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.504A          | In the band 14-14.5 GHz, aircraft earth stations in the secondary aeronautical mobile-satellite service may also communicate with space stations in the fixed-satellite service. The provisions of Nos. 5.29, 5.30 and 5.31 apply. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.504B          | Aircraft earth stations operating in the aeronautical mobile-satellite service in the frequency band 14-14.5 GHz shall comply with the provisions of Annex 1, Part C of Recommendation ITU-R M.1643-0, with respect to any radio astronomy station performing observations in the 14.47-14.5 GHz frequency band located on the territory of Spain, France, India, Italy, the United Kingdom and South Africa. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.504C          | In the frequency band 14-14.25 GHz, the power flux-density produced on the territory of the countries of Saudi Arabia, Bahrain, Botswana, Ivory Coast, Egypt, Guinea, India, Iran (Islamic Republic of), Kuwait, Nigeria, Oman, the Syrian Arab Republic and Tunisia by any aircraft earth station in the aeronautical mobile-satellite service shall not exceed the limits given in Annex 1, Part B of Recommendation ITU-R M.1643-0, unless otherwise specifically agreed by the affected administration(s). The provisions of this footnote in no way derogate the obligations of the aeronautical mobile-satellite service to operate as a secondary service in accordance with No. 5.29. (WRC-15)                                                                                   |
| 5.505           | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Botswana, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Korea (Rep. of), Djibouti, Egypt, the United Arab Emirates, Eswatini, Gabon, Guinea, India, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Japan, Jordan, Kuwait, Lebanon, Malaysia, Mali, Morocco, Mauritania, Oman, the Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, South Sudan, Chad, Viet Nam and Yemen, the frequency band 14-14.3 GHz is also allocated to the fixed service on a primary basis. (WRC-19)                                                                                                                                                                         |
| 5.506           | The band 14-14.5 GHz may be used, within the fixed-satellite service (Earth-to-space), for feeder links for the broadcasting-satellite service, subject to coordination with other networks in the fixed-satellite service. Such use of feeder links is reserved for countries outside Europe.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.506A          | In the frequency band 14-14.5 GHz, ship earth stations with an equivalent isotropically radiated power (e.i.r.p.) greater than 21 dBW shall operate under the same conditions as earth stations located on board vessels, as provided in Resolution 902 (Rev.WRC-23). This footnote shall not apply to ship earth stations for which the complete Appendix 4 information has been received by the Bureau prior to 5 July 2003. (WRC-23)                                                                                                                                                                                                                                                                                                                                                  |
| 5.506B          | Earth stations located on board vessels communicating with space stations in the fixed-satellite service may operate in the frequency band 14-14.5 GHz without the need for prior agreement from Cyprus and Malta, within the minimum distance given in Resolution 902 (Rev.WRC-23) from these countries. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.508           | Additional allocation: in Germany, Italy, Libya, North Macedonia and the United Kingdom, the frequency band 14.25-14.3 GHz is also allocated to the fixed service on a primary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.508A          | In the frequency band 14.25-14.3 GHz, the power flux-density produced on the territory of the countries of Saudi Arabia, Bahrain, Botswana, China, Ivory Coast, Egypt, Guinea, India, Iran (Islamic Republic of), Italy, Kuwait, Nigeria, Oman, the Syrian Arab Republic, the United Kingdom and Tunisia by any aircraft earth station in the aeronautical mobile-satellite service shall not exceed the limits given in Annex 1, Part B of Recommendation ITU-R M.1643-0, unless otherwise specifically agreed by the affected administration(s). The provisions of this footnote in no way derogate the obligations of the aeronautical mobile-satellite service to operate as a secondary service in accordance with No. 5.29. (WRC-23)                                               |
| 5.509A          | In the frequency band 14.3-14.5 GHz, the power flux-density produced on the territory of the countries of Saudi Arabia, Bahrain, Botswana, Cameroon, China, Ivory Coast, Egypt, Gabon, Guinea, India, Iran (Islamic Republic of), Italy, Kuwait, Morocco, Nigeria, Oman, the Syrian Arab Republic, the United Kingdom, Sri Lanka, Tunisia and Viet Nam by any aircraft earth station in the aeronautical mobile-satellite service shall not exceed the limits given in Annex 1, Part B of Recommendation ITU-R M.1643-0, unless otherwise specifically agreed by the affected administration(s). The provisions of this footnote in no way derogate the obligations of the aeronautical mobile-satellite service to operate as a secondary service in accordance with No. 5.29. (WRC-23) |
| 5.509B          | The use of the frequency bands 14.5-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.5-14.8 GHz in countries listed in Resolution 164 (WRC-15) by the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service is limited to geostationary-satellites. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.509C          | For the use of the frequency bands 14.5-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.5-14.8 GHz in countries listed in Resolution 164 (WRC-15) by the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service, the fixed-satellite service earth stations shall have a minimum antenna diameter of 6 m and a maximum power spectral density of -44.5 dBW/Hz at the input of the antenna. The earth stations shall be notified at known locations on land. (WRC-15)                                                                                                                                                                                                                                                       |
| 5.509D          | Before an administration brings into use an earth station in the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service in the frequency bands 14.5-14.75 GHz (in countries listed in Resolution 163 (WRC-15)) and 14.5-14.8 GHz (in countries listed in Resolution 164 (WRC-15)), it shall ensure that the power flux-density produced by this earth station does not exceed -151.5 dB(W/(m <sup>2</sup> · 4 kHz)) produced at all altitudes from 0 m to 19 000 m above sea level at 22 km seaward from all coasts, defined as the low-water mark, as officially recognized by each coastal State. (WRC-15)                                                                                                                               |
| 5.509E          | In the frequency bands 14.50-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.50-14.8 GHz in countries listed in Resolution 164 (WRC-15), the location of earth stations in the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service shall maintain a separation distance of at least 500 km from the border(s) of other countries unless shorter distances are explicitly agreed by those administrations. No. 9.17 does not apply. When applying this provision, administrations should consider the relevant parts of these Regulations and the latest relevant ITU-R Recommendations. (WRC-15)                                                                                                                        |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.509F          | In the frequency bands 14.50-14.75 GHz in countries listed in Resolution 163 (WRC-15) and 14.50-14.8 GHz in countries listed in Resolution 164 (WRC-15), earth stations in the fixed-satellite service (Earth-to-space) not for feeder links for the broadcasting-satellite service shall not constrain the future deployment of the fixed and mobile services. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.509G          | The frequency band 14.5-14.8 GHz is also allocated to the space research service on a primary basis. However, such use is limited to the satellite systems operating in the space research service (Earth-to-space) to relay data to space stations in the geostationary-satellite orbit from associated earth stations. Stations in the space research service shall not cause harmful interference to, or claim protection from, stations in the fixed and mobile services and in the fixed-satellite service limited to feeder links for the broadcasting-satellite service and associated space operations functions using the guardbands under Appendix 30A and feeder links for the broadcasting-satellite service in Region 2. Other uses of this frequency band by the space research service are on a secondary basis. (WRC-15) |
| 5.510           | Except for use in accordance with Resolution 163 (WRC-15) and Resolution 164 (WRC-15), the use of the frequency band 14.5-14.8 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links for the broadcasting-satellite service. This use is reserved for countries outside Europe. Uses other than feeder links for the broadcasting-satellite service are not authorized in Regions 1 and 2 in the frequency band 14.75-14.8 GHz. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                        |
| 5.510A          | The allocation of the frequency band 14.8-15.35 GHz to the space research service on a primary basis is limited to satellite systems operating in the space-to-space, space-to-Earth and Earth-to-space directions at distances from the Earth of less than $2 \times 10^6$ km in accordance with Resolution 678 (WRC-23). Other uses of the frequency band by the space research service are on a secondary basis. The use of the frequency band 14.8-15.35 GHz by the space research service (space-to-Earth) (Earth-to-space) is on a secondary basis with respect to the terrestrial services in Algeria, Saudi Arabia, Bahrain, Korea (Rep. of), Egypt, the United Arab Emirates, the United States, India, Iraq, Japan, Kuwait, Libya, Morocco, Mauritania, Oman, Qatar, the Syrian Arab Republic, Tunisia and Yemen. (WRC-23)     |
| 5.511           | Additional allocation: in Saudi Arabia, Bahrain, Cameroon, Djibouti, Egypt, the United Arab Emirates, Guinea, Iran (Islamic Republic of), Iraq, Israel, Kuwait, Lebanon, Oman, Pakistan, Qatar, the Syrian Arab Republic and Somalia, the frequency band 15.35-15.4 GHz is also allocated to the fixed and mobile services on a secondary basis. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.511A          | Use of the band 15.43-15.63 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links of non-geostationary systems in the mobile-satellite service, subject to coordination under No. 9.11A. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.511C          | Stations operating in the aeronautical radionavigation service shall limit the effective e.i.r.p. in accordance with Recommendation ITU-R S.1340-0. The minimum coordination distance required to protect the aeronautical radionavigation stations (No. 4.10 applies) from harmful interference from feeder link earth stations and the maximum e.i.r.p. transmitted towards the local horizontal plane by a feeder link earth station shall be in accordance with Recommendation ITU-R S.1340-0. (WRC-15)                                                                                                                                                                                                                                                                                                                              |
| 5.511E          | In the frequency band 15.4-15.7 GHz, stations operating in the radiolocation service shall not cause harmful interference to, or claim protection from, stations operating in the aeronautical radionavigation service. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.511F          | In order to protect the radio astronomy service in the frequency band 15.35-15.4 GHz, radiolocation stations operating in the frequency band 15.4-15.7 GHz shall not exceed the power flux-density level of $-156$ dB(W/m <sup>2</sup> ) in a 50 MHz bandwidth in the frequency band 15.35-15.4 GHz, at any radio astronomy observatory site for more than 2 per cent of the time. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.511G          | Stations in the aeronautical mobile (OR) service operating in the frequency band 15.41-15.7 GHz shall not cause harmful interference to the radio astronomy service operating in the frequency band 15.35-15.4 GHz. The aggregate power flux-density (pfd) received from stations in the aeronautical mobile (OR) service operating in the frequency band 15.41-15.7 GHz at any radio astronomy station operating in the frequency band 15.35-15.4 GHz shall be in compliance with the protection criteria provided in Recommendations ITU-R RA.769-2 and ITU-R RA.1513-2, unless specifically agreed by the affected administration(s). (WRC-23)                                                                                                                                                                                        |
| 5.512           | Additional allocation: in Algeria, Saudi Arabia, Austria, Bahrain, Bangladesh, Brunei Darussalam, Cameroon, Congo (Rep. of the), Egypt, El Salvador, the United Arab Emirates, Eritrea, Finland, Guatemala, India, Indonesia, Iran (Islamic Republic of), Jordan, Kenya, Kuwait, Lebanon, Libya, Malaysia, Mali, Morocco, Mauritania, Montenegro, Nepal, Nicaragua, Niger, Oman, Pakistan, Qatar, Syrian Arab Republic, the Dem. Rep. of the Congo, Singapore, Somalia, Sudan, South Sudan, Chad, Togo and Yemen, the frequency band 15.7-17.3 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-15)                                                                                                                                                                                                       |
| 5.513           | Additional allocation: in Israel, the band 15.7-17.3 GHz is also allocated to the fixed and mobile services on a primary basis. These services shall not claim protection from or cause harmful interference to services operating in accordance with the Table in countries other than those included in No. 5512.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.513A          | Spaceborne active sensors operating in the band 17.2-17.3 GHz shall not cause harmful interference to, or constrain the development of, the radiolocation and other services allocated on a primary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 5.514           | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Bangladesh, Cameroon, Djibouti, El Salvador, the United Arab Emirates, Guatemala, India, Iran (Islamic Republic of), Iraq, Israel, Italy, Japan, Jordan, Kuwait, Libya, Lithuania, Nepal, Nicaragua, Nigeria, Oman, Uzbekistan, Pakistan, Qatar, Kyrgyzstan, Somalia, Sudan and South Sudan, the frequency band 17.3-17.7 GHz is also allocated to the fixed and mobile services on a secondary basis. The power limits given in Nos. 21.3 and 21.5 shall apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.516           | The use of the band 17.3-18.1 GHz by geostationary-satellite systems in the fixed-satellite service (Earth-to-space) is limited to feeder links for the broadcasting-satellite service. The use of the band 17.3-17.8 GHz in Region 2 by systems in the fixed-satellite service (Earth-to-space) is limited to geostationary satellites. For the use of the band 17.3-17.8 GHz in Region 2 by feeder links for the broadcasting satellite service in the band 12.2-12.7 GHz, see Article 11. The use of the bands 17.3-18.1 GHz (Earth-to-space) in Regions 1 and 3 and 17.8-18.1 GHz (Earth-to-space) in Region 2 by non geostationary-satellite systems in the fixed-satellite service is subject to application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service. Non geostationary-satellite systems in the fixed satellite service shall not claim protection from geostationary-satellite networks in the fixed-satellite service operating in accordance with the Radio Regulations, irrespective of the dates of receipt by the Bureau of the complete coordination or notification information, as appropriate, for the non-geostationary-satellite systems in the fixed-satellite service and of the complete coordination or notification information, as appropriate, for the geostationary-satellite networks, and No. 5.43A does not apply. Non-geostationary-satellite systems in the fixed-satellite service in the above bands shall be operated in such a way that any unacceptable interference that may occur during their operation shall be rapidly eliminated. (WRC-2000) |
| 5.516A          | In the band 17.3-17.7 GHz, earth stations of the fixed-satellite service (space-to-Earth) in Region 1 shall not claim protection from the broadcasting-satellite service feeder-link earth stations operating under Appendix 30A, nor put any limitations or restrictions on the locations of the broadcasting-satellite service feeder-link earth stations anywhere within the service area of the feeder link. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.516B          | The following bands are identified for use by high-density applications in the fixed-satellite service: 17.3-17.7 GHz (space-to-Earth) in Region 1, 18.3-19.3 GHz (space-to-Earth) in Region 2, 19.7-20.2 GHz (space-to-Earth) in all Regions, 39.5-40 GHz (space-to-Earth) in Region 1, 40-40.5 GHz (space-to-Earth) in all Regions, 40.5-42 GHz (space-to-Earth) in Region 2, 47.5-47.9 GHz (space-to-Earth) in Region 1, 48.2-48.54 GHz (space-to-Earth) in Region 1, 49.44-50.2 GHz (space-to-Earth) in Region 1, and 27.5-27.82 GHz (Earth-to-space) in Region 1, 28.35-28.45 GHz (Earth-to-space) in Region 2, 28.45-28.94 GHz (Earth-to-space) in all Regions, 28.94-29.1 GHz (Earth-to-space) in Region 2 and 3, 29.25-29.46 GHz (Earth-to-space) in Region 2, 29.46-30 GHz (Earth-to-space) in all Regions, 48.2-50.2 GHz (Earth-to-space) in Region 2. ART5 – 37 – This identification does not preclude the use of these frequency bands by other fixed-satellite service applications or by other services to which these frequency bands are allocated on a co-primary basis and does not establish priority in these Radio Regulations among users of the frequency bands. Administrations should take this into account when considering regulatory provisions in relation to these frequency bands. See Resolution 143 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                |
| 5.517A          | The operation of earth stations in motion communicating with geostationary fixed-satellite service space stations within the frequency bands 17.7-19.7 GHz (space-to-Earth) and 27.5-29.5 GHz (Earth-to-space) shall be subject to the application of Resolution 169 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.517B          | The operation of aeronautical and maritime earth stations in motion communicating with non-geostationary space stations in the fixed-satellite service in the frequency bands 17.7-18.6 GHz, 18.8-19.3 GHz and 19.7-20.2 GHz (space-to-Earth) and 27.5-29.1 GHz and 29.5-30 GHz (Earth-to-space) shall be subject to the application of Resolution 123 (WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.519           | Additional allocation: the bands 18.0-18.3 GHz in Region 2 and 18.1-18.4 GHz in Regions 1 and 3 are also allocated to the meteorological-satellite service (space-to-Earth) on a primary basis. Their use is limited to geostationary satellites. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.520           | The use of the band 18.1-18.4 GHz by the fixed-satellite service (Earth-to-space) is limited to feeder links of geostationary-satellite systems in the broadcasting-satellite service. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.521           | Alternative allocation: in the United Arab Emirates, the frequency band 18.1-18.4 GHz is allocated to the fixed, fixed-satellite (space-to-Earth) and mobile services on a primary basis (see No. 5.33). The provisions of No. 5.519 also apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.521A          | For use of the frequency bands 18.1-18.6 GHz, 18.8-20.2 GHz and 27.5-30 GHz, or parts thereof, by space stations in the inter-satellite service, Resolution 679 (WRC-23) shall apply. Such use is limited to space research, space operation and/or Earth exploration-satellite applications, and also transmissions of data originating from industrial and medical activities in space. When using these frequencies, administrations shall ensure that this inter-satellite service is used only for the aforementioned purposes and is not subject to coordination under No. 9.11A. For use of the frequency bands 18.1-18.6 GHz, 18.8-20.2 GHz, 27.5-29.1 GHz and 29.5-30 GHz by space stations, the allocation is limited to inter-satellite links between non-geostationary satellites or between non-geostationary satellites and geostationary satellites. For use of the frequency band 29.1-29.5 GHz by space stations, the allocation is limited to inter-satellite links between non-geostationary satellites and geostationary satellites. No. 4.10 does not apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.522A          | The emissions of the fixed service and the fixed-satellite service in the band 18.6-18.8 GHz are limited to the values given in Nos. 21.5A and 21.16.2, respectively. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 5.522B          | The use of the band 18.6-18.8 GHz by the fixed-satellite service is limited to geostationary systems and systems with an orbit of apogee greater than 20 000 km. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.522C          | In the band 18.6-18.8 GHz, in Algeria, Saudi Arabia, Bahrain, Egypt, the United Arab Emirates, the Libyan Arab Jamahiriya, Jordan, Lebanon, Morocco, Oman, Qatar, the Syrian Arab Republic, Tunisia and Yemen, fixed-service systems in operation at the date of entry into force of the Final Acts of WRC-2000 are not subject to the limits of No. 21.5A. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.523A          | The use of the bands 18.8-19.3 GHz (space-to-Earth) and 28.6-29.1 GHz (Earth-to-space) by geostationary and non-geostationary fixed-satellite service networks is subject to the application of the provisions of No. 9.11A and No. 22.2 does not apply. Administrations having geostationary-satellite networks under coordination prior to 18 November 1995 shall cooperate to the maximum extent possible to coordinate pursuant to No. 9.11A with non-geostationary-satellite networks for which notification information has been received by the Bureau prior to that date, with a view to reaching results acceptable to all the parties concerned. Non-geostationary-satellite networks shall not cause unacceptable interference to geostationary fixed-satellite service networks for which complete Appendix 4 notification information is considered as having been received by the Bureau prior to 18 November 1995. (WRC-97)                                                                                                                                                                                                                                                                                       |
| 5.523B          | The use of the band 19.3-19.6 GHz (Earth-to-space) by the Fixed-satellite service is limited to feeder links for non-geostationary-satellite systems in the mobile-satellite service. Such use is subject to the application of the provisions of No. 9.11A, and No. 22.2 does not apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.523C          | No. 22.2 of the Radio Regulations shall continue to apply in the bands 19.3-19.6 GHz and 29.1-29.4 GHz, between feeder links of non-geostationary mobile-satellite service networks and those fixed-satellite service networks for which complete Appendix 4 coordination information, or notification information, is considered as having been received by the Bureau prior to 18 November 1995. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.523D          | The use of the band 19.3-19.7 GHz (space-to-Earth) by geostationary fixed-satellite service systems and by feeder links for non-geostationary-satellite systems in the mobile-satellite service is subject to the application of the provisions of No. 9.11A, but not subject to the provisions of No. 22.2. The use of this band for other non-geostationary fixed-satellite service systems, or for the cases indicated in Nos. 5.523C and 5.523E, is not subject to the provisions of No. 9.11A and shall continue to be subject to Articles 9 (except No. 9.11A) and 11 procedures, and to the provisions of No. 22.2. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.523DA         | In order to protect feeder links of non-geostationary networks in the mobile-satellite service in the frequency band 19.3-19.7 GHz, the power flux-density values produced at the surface of the Earth for all angles of arrival by a space station in the inter-satellite service operating in this band in accordance with Resolution 679 (WRC-23) shall not exceed -140 dB(W/m <sup>2</sup> ) in any 1 MHz within 150 km of any of the above feeder-link earth stations recorded in the Master International Frequency Register. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.523E          | No. 22.2 shall continue to apply in the bands 19.6-19.7 GHz and 29.4-29.5 GHz, between feeder links of non-geostationary mobile-satellite service networks and those fixed-satellite service networks for which complete Appendix 4 coordination information, or notification information, is considered as having been received by the Bureau by 21 November 1997. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.524           | Additional allocation: in Afghanistan, Algeria, Saudi Arabia, Bahrain, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Costa Rica, Djibouti, Egypt, the United Arab Emirates, Gabon, Guatemala, Guinea, India, Iran (Islamic Republic of), Iraq, Israel, Japan, Jordan, Kuwait, Lebanon, Malaysia, Mali, Morocco, Mauritania, Nepal, Nigeria, Oman, Pakistan, Palestine*, the Philippines, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, the Dem. People's Rep. of Korea, Singapore, Somalia, Sudan, South Sudan, Chad, Togo and Tunisia, the frequency band 19.7-21.2 GHz is also allocated to the fixed and mobile services on a primary basis. This additional use shall not impose any limitation on the power flux-density of space stations in the fixed-satellite service in the frequency band 19.7-21.2 GHz and of space stations in the mobile-satellite service in the frequency band 19.7-20.2 GHz where the allocation to the mobile-satellite service is on a primary basis in the latter frequency band. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995 |
| 5.525           | In order to facilitate interregional coordination between networks in the mobile-satellite and fixed-satellite services, carriers in the mobile-satellite service that are most susceptible to interference shall, to the extent practicable, be located in the higher parts of the bands 19.7-20.2 GHz and 29.5-30 GHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.526           | In the bands 19.7-20.2 GHz and 29.5-30 GHz in Region 2, and in the bands 20.1-20.2 GHz and 29.9-30 GHz in Regions 1 and 3, networks which are both in the fixed-satellite service and in the mobile-satellite service may include links between earth stations at specified or unspecified points or while in motion, through one or more satellites for point-to-point and point-to-multipoint communications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.527           | In the bands 19.7-20.2 GHz and 29.5-30 GHz, the provisions of No 4.10 do not apply with respect to the mobile-satellite service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.527A          | The operation of earth stations in motion communicating with the FSS is subject to Resolution 156 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.528           | The allocation to the mobile-satellite service is intended for use by networks which use narrow spot-beam antennas and other advanced technology at the space stations. Administrations operating systems in the mobile-satellite service in the band 19.7 - 20.1 GHz in Region 2 and in the band 20.1 - 20.2 GHz shall take all practicable steps to ensure the continued availability of these bands for administrations operating fixed and mobile systems in accordance with the provisions of No. 5.524.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5.529A          | In the frequency bands 20.2-21.2 GHz and 30-31 GHz, non-geostationary-satellite systems for which complete coordination or notification information, according to the case, is received by the Bureau as of 1 January 2025 shall not cause unacceptable interference to and shall not claim protection from geostationary-satellite networks in the mobile-satellite service operating in accordance with these Regulations. No. 5.43A does not apply. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.530A          | Unless otherwise agreed between the administrations concerned, any station in the fixed or mobile services of an administration shall not produce a power flux-density in excess of -120.4 dB(W/(m <sup>2</sup> · MHz)) at 3 m above the ground of any point of the territory of any other administration in Regions 1 and 3 for more than 20% of the time. In conducting the calculations, administrations should use the most recent version of Recommendation ITU-R P.452 (see also the most recent version of Recommendation ITU-R BO.1898). (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.530B          | In the band 21.4-22 GHz, in order to facilitate the development of the broadcasting-satellite service, administrations in Regions 1 and 3 are encouraged not to deploy stations in the mobile service and are encouraged to limit the deployment of stations in the fixed service to point-to-point links. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.531A          | The use of the aeronautical mobile (OR) service in the frequency band 22-22.2 GHz is limited to non-safety applications. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.531B          | Aircraft stations in the aeronautical mobile (OR) service operating in the frequency band 22-22.2 GHz are subject to agreement obtained under No. 9.21 with respect to the fixed service and shall not cause harmful interference to, nor claim protection from, the fixed service. The following power flux-density values shall be used as a threshold for coordination under No. 9.21: -110 dB(W/(m <sup>2</sup> · MHz)) for 0° ≤ θ ≤ 12.6° 2.86 θ - 146 dB(W/(m <sup>2</sup> · MHz)) for 12.6° < θ ≤ 15° 0.87 θ - 116 dB(W/(m <sup>2</sup> · MHz)) for 15° < θ ≤ 30° 0.067 θ - 92 dB(W/(m <sup>2</sup> · MHz)) for 30° < θ ≤ 90° where θ is the angle of arrival of the incident wave above the horizontal plane, in degrees. This criterion should be applied at the border of the territory of another administration for any aircraft station located at an altitude of up to 15 km above the ground. In conducting the calculations, the most recent version of Recommendation ITU-R P.525 should be used. (WRC-23) |
| 5.531C          | Stations in the aeronautical mobile (OR) service operating in the frequency band 22-22.2 GHz shall not cause harmful interference to the radio astronomy service operating in the frequency band 22.21-22.5 GHz. The aggregate power flux-density (pfd) received from these stations at any radio astronomy station operating in the frequency band 22.21-22.5 GHz shall be in compliance with the protection criteria provided in Recommendations ITU-R RA.769-2 and ITU-R RA.1513-2, unless specifically agreed by the affected administration(s). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.531D          | The use of the aeronautical mobile (OR) service in the frequency band 22-22.2 GHz outside national boundaries shall not cause harmful interference to, or claim protection from, services in other countries operating in accordance with the Table of Frequency Allocations. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5.531F          | In order to protect stations of the Earth exploration-satellite service (passive) operating in the frequency band 22.21-22.5 GHz, the unwanted equivalent isotropically radiated power (e.i.r.p.) of stations operating in the aeronautical mobile (OR) service shall not exceed -23 dBW in any 100 MHz band in the frequency band 22.21-22.5 GHz. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.532           | The use of the band 22.21-22.5 GHz by the earth exploration-satellite (passive) and space research (passive) services shall not impose constraints upon the fixed and mobile, except aeronautical mobile, services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.532A          | The location of earth stations in the space research service shall maintain a separation distance of at least 54 km from the respective border(s) of neighbouring countries to protect the existing and future deployment of fixed and mobile services unless a shorter distance is otherwise agreed between the corresponding administrations. Nos. 9.17 and 9.18 do not apply. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.532AB         | The frequency band 24.25-27.5 GHz is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. Resolution 242 (Rev.WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.532B          | Use of the band 24.65-25.25 GHz in Region 1 and the band 24.65-24.75 GHz in Region 3 by the fixed-satellite service (Earth-to-space) is limited to earth stations using a minimum antenna diameter of 4.5 m. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.534A          | The allocation to the fixed service in the frequency band 25.25-27.5 GHz is identified in Region 2 for use by high-altitude platform stations (HAPS) in accordance with the provisions of Resolution 166 (Rev.WRC-23). Such use of the fixed-service allocation by HAPS shall be limited to the ground-to-HAPS direction in the frequency band 25.25-27.0 GHz and to the HAPS-to-ground direction in the frequency band 27.0-27.5 GHz. Furthermore, the use of the frequency band 25.5-27.0 GHz by HAPS shall be limited to gateway links. This identification does not preclude the use of this frequency band by other fixed-service applications or by other services to which this band is allocated on a co-primary basis, and does not establish priority in the Radio Regulations. (WRC-23)                                                                                                                                                                                                                          |
| 5.535A          | The use of the band 29.1-29.5 GHz (Earth-to-space) by the fixed-satellite service is limited to geostationary-satellite systems and feeder links to non-geostationary-satellite systems in the mobile-satellite service. Such use is subject to the application of the provisions of No. 9.11A, but not subject to the provisions of No. 22.2, except as indicated in Nos. 5.523C and 5.523E where such use is not subject to the provisions of No. 9.11A and shall continue to be subject to Articles 9 (except No. 9.11A) and 11 procedures, and to the provisions of No. 22.2. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 5.536           | Use of the 25.25-27.5 GHz band by the inter-satellite service is limited to space research and Earth exploration-satellite applications, and also transmissions of data originating from industrial and medical activities in space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.536A          | Administrations operating earth stations in the Earth exploration-satellite service or the space research service shall not claim protection from stations in the fixed and mobile services operated by other administrations. In addition, earth stations in the Earth exploration-satellite service or in the space research service should be operated taking into account the most recent version of Recommendation ITU-R SA.1862. Resolution 242 (Rev.WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.536B          | In Algeria, Saudi Arabia, Austria, Bahrain, Belgium, Brazil, China, Korea (Rep. of), Denmark, Egypt, United Arab Emirates, Estonia, Finland, Hungary, India, Iran (Islamic Republic of), Iraq, Ireland, Israel, Italy, Jordan, Kenya, Kuwait, Lebanon, Libya, Lithuania, Moldova, Norway, Oman, Uganda, Pakistan, the Philippines, Poland, Portugal, Qatar, the Syrian Arab Republic, Turkiye, Dem. People's Rep. of Korea, Slovakia, the Czech Rep., Romania, the United Kingdom, Singapore, Slovenia, Somalia, Sudan, Sweden, Tanzania, Viet Nam and Zimbabwe, earth stations operating in the Earth exploration-satellite service in the frequency band 25.5-27 GHz shall not claim protection from, or constrain the use and deployment of, stations of the fixed and mobile services. Resolution 242 (Rev.WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.536C          | In Algeria, Saudi Arabia, Bahrain, Botswana, Brazil, Cameroon, Comoros, Cuba, Djibouti, Egypt, United Arab Emirates, Estonia, Finland, Iran (Islamic Rep. of), Israel, Jordan, Kenya, Kuwait, Lithuania, Malaysia, Morocco, Nigeria, Oman, Qatar, Syrian Arab Republic, Somalia, Sudan, South Sudan, Tanzania, Tunisia, Uruguay, Zambia and Zimbabwe, earth stations operating in the space research service in the band 25.5-27 GHz shall not claim protection from, or constrain the use and deployment of, stations of the fixed and mobile services. (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.537A          | In Bhutan, Cameroon, China, Korea (Rep. of), the Russian Federation, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Kazakhstan, Malaysia, Maldives, Mongolia, Myanmar, Uzbekistan, Pakistan, the Philippines, Kyrgyzstan, the Dem. People's Rep. of Korea, Sudan, Sri Lanka, Thailand and Viet Nam, the allocation to the fixed service in the frequency band 27.9-28.2 GHz may also be used by high altitude platform stations (HAPS) within the territory of these countries. Such use of 300 MHz of the fixed-service allocation by HAPS in the above countries is further limited to operation in the HAPS-to-ground direction and shall not cause harmful interference to, nor claim protection from, other types of fixed-service systems or other co-primary services. Furthermore, the development of these other services shall not be constrained by HAPS. See Resolution 145 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.538           | Additional allocation: the bands 27.500-27.501 GHz and 29.999-30.000 GHz are also allocated to the fixed-satellite service (space to Earth) on a primary basis for the beacon transmissions intended for up-link power control. Such space-to-Earth transmissions shall not exceed an equivalent isotropically radiated power (e.i.r.p.) of +10 dBW in the direction of adjacent satellites on the geostationary-satellite orbit. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.539           | The band 27.5-30 GHz may be used by the fixed-satellite service (Earth-to-space) for the provision of feeder links for the broadcasting-satellite service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.540           | Additional allocation: the band 27.501-29.999 GHz is also allocated to the fixed-satellite service (space-to-Earth) on a secondary basis for beacon transmissions intended for up-link power control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.541           | In the band 28.5-30 GHz, the earth exploration-satellite service is limited to the transfer of data between stations and not to the primary collection of information by means of active or passive sensors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.541A          | Feeder links of non-geostationary networks in the mobile-satellite service and geostationary networks in the fixed-satellite service operating in the band 29.1-29.5 GHz (Earth-to-space) shall employ uplink adaptive power control or other methods of fade compensation, such that the earth station transmissions shall be conducted at the power level required to meet the desired link performance while reducing the level of mutual interference between both networks. These methods shall apply to networks for which Appendix 4 coordination information is considered as having been received by the Bureau after 17 May 1996 and until they are changed by a future competent world radiocommunication conference. Administrations submitting Appendix 4 information for coordination before this date are encouraged to utilize these techniques to the extent practicable. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.542           | Additional allocation: in Algeria, Saudi Arabia, Bahrain, Brunei Darussalam, Cameroon, China, Congo (Rep. of the), Djibouti, Egypt, the United Arab Emirates, Eritrea, Ethiopia, Guinea, India, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kuwait, Lebanon, Malaysia, Mali, Morocco, Mauritania, Nepal, Oman, Pakistan, Palestine*, Philippines, Qatar, the Syrian Arab Republic, the Dem. People's Rep. of Korea, Somalia, Sudan, South Sudan, Sri Lanka and Chad, the frequency band 29.5-31 GHz is also allocated to the fixed and mobile services on a secondary basis. The power limits specified in Nos. 21.3 and 21.5 shall apply. (WRC-23) * Pursuant to Resolution 99 (Rev. Dubai, 2018) of the Plenipotentiary Conference and taking into account the Israeli-Palestinian Interim Agreement of 28 September 1995.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.543           | The band 29.95-30 GHz may be used for space-to-space links in the earth exploration-satellite service for telemetry, tracking, and control purposes, on a secondary basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.543A          | In Bhutan, Cameroon, Korea (Rep. of), the Russian Federation, India, Indonesia, Iran (Islamic Republic of), Iraq, Japan, Kazakhstan, Malaysia, Maldives, Mongolia, Myanmar, Uzbekistan, Pakistan, the Philippines, Kyrgyzstan, the Dem. People's Rep. of Korea, Sudan, Sri Lanka, Thailand and Viet Nam, the allocation to the fixed service in the frequency band 31-31.3 GHz may also be used by systems using high altitude platform stations (HAPS) in the ground-to-HAPS direction. The use of the frequency band 31-31.3 GHz by systems using HAPS is limited to the territory of the countries listed above and shall not cause harmful interference to, nor claim protection from, other types of fixed-service systems, systems in the mobile service and systems operated under No. 5.545. Furthermore, the development of these services shall not be constrained by HAPS. Systems using HAPS in the frequency band 31-31.3 GHz shall not cause harmful interference to the radio astronomy service having a primary allocation in the frequency band 31.3-31.8 GHz, taking into account the protection criterion as given in the most recent version of Recommendation ITU-R RA.769. In order to ensure the protection of satellite passive services, the level of unwanted power density into a HAPS ground station antenna in the frequency band 31.3-31.8 GHz shall be limited to -106 dB(W/MHz) under clear-sky conditions, and may be increased up to -100 dB(W/MHz) under rainy conditions to mitigate fading due to rain, provided the effective impact on the passive satellite does not exceed the impact under clear-sky conditions. See Resolution 145 (Rev.WRC-12). (WRC-15) |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 5.543B          | The allocation to the fixed service in the frequency band 31-31.3 GHz is identified for worldwide use by high-altitude platform stations (HAPS). This identification does not preclude the use of this frequency band by other fixed-service applications or by other services to which this frequency band is allocated on a co-primary basis, and does not establish priority in the Radio Regulations. Such use of the fixed-service allocation by HAPS shall be in accordance with the provisions of Resolution 167 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                           |
| 5.544           | In the band 31-31.3 GHz the power flux-density limits specified in Article 21, Table 21-4 shall apply to the space research service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.545           | Different category of service: in Armenia, Georgia, Kyrgyzstan, Tajikistan and Turkmenistan, the allocation of the band 31-31.3 GHz to the space research service is on a primary basis (see No. 5.33). (WRC-12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.546           | Different category of service: in Saudi Arabia, Armenia, Azerbaijan, Bahrain, Belarus, Djibouti, Egypt, the United Arab Emirates, Spain, Estonia, the Russian Federation, Georgia, Hungary, Iran (Islamic Republic of), Israel, Jordan, Lebanon, Moldova, Mongolia, Oman, Uzbekistan, Poland, the Syrian Arab Republic, Turkiye, Kyrgyzstan, Romania, the United Kingdom, Somalia, South Africa, Tajikistan and Turkmenistan, the allocation of the frequency band 31.5-31.8 GHz to the fixed and mobile, except aeronautical mobile, services is on a primary basis (see No. 5.33). (WRC-23)                                                                                                                                                                                                                                                            |
| 5.547           | The frequency bands 31.8-33.4 GHz, 37-40 GHz, 40.5-43.5 GHz, 51.4-52.6 GHz, 55.78-59 GHz and 64-66 GHz are available for high-density applications in the fixed service. Administrations should take this into account when considering regulatory provisions in relation to these bands. Because of the potential deployment of high-density applications in the fixed-satellite service in the frequency bands 39.5-40 GHz and 40.5-42 GHz (see No. 5.516B), administrations should further take into account potential constraints to high-density applications in the fixed service, as appropriate. (WRC-23)                                                                                                                                                                                                                                        |
| 5.547A          | Administrations should take practical measures to minimize the potential interference between stations in the fixed service and airborne stations in the radionavigation service in the 31.8-33.4 GHz band, taking into account the operational needs of the airborne radar systems. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.547B          | Alternative allocation: in the United States, the band 31.8-32 GHz is allocated to the radionavigation and space research (deep space) (space-to-Earth) services on a primary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.547C          | Alternative allocation: in the United States, the band 32-32.3 GHz is allocated to the radionavigation and space research (deep space) (space-to-Earth) services on a primary basis. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.547D          | Alternative allocation: in the United States, the band 32.3-33 GHz is allocated to the inter-satellite and radionavigation services on a primary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.547E          | Alternative allocation: in the United States, the band 33-33.4 GHz is allocated to the radionavigation service on a primary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.548           | In designing systems for the inter-satellite service in the frequency band 32.3-33 GHz, for the radionavigation service in the frequency band 32-33 GHz, and for the space research service (deep space) in the frequency band 31.8-32.3 GHz, administrations shall take all necessary measures to prevent harmful interference between these services, bearing in mind the safety aspects of the radionavigation service (see Recommendation 707 (Rev.WRC-23)). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                |
| 5.549           | Additional allocation: in Saudi Arabia, Bahrain, Bangladesh, Egypt, the United Arab Emirates, Gabon, Indonesia, Iran (Islamic Republic of), Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Malaysia, Mali, Morocco, Mauritania, Nepal, Nigeria, Oman, Pakistan, the Philippines, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Singapore, Somalia, Sudan, South Sudan, Sri Lanka, Togo, Tunisia and Yemen, the band 33.4-36 GHz is also allocated to the fixed and mobile services on a primary basis. (WRC-12)                                                                                                                                                                                                                                                                                                                            |
| 5.549A          | In the band 35.5-36.0 GHz, the mean power flux-density at the Earth's surface, generated by any spaceborne sensor in the Earth exploration-satellite service (active) or space research service (active), for any angle greater than 0.8° from the beam centre shall not exceed -73.3 dB(W/m²) in this band. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.550A          | For sharing of the band 36-37 GHz between the Earth exploration-satellite (passive) service and the fixed and mobile services, Resolution 752 (WRC-07) shall apply. (WRC-07)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.550B          | The frequency band 37-43.5 GHz, or portions thereof, is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. Because of the potential deployment of FSS earth stations within the frequency range 37.5-42.5 GHz and high-density applications in the fixed-satellite service in the frequency bands 39.5-40 GHz in Region 1, 40-40.5 GHz in all Regions and 40.5-42 GHz in Region 2 (see No. 5.516B), administrations should further take into account potential constraints to IMT in these frequency bands, as appropriate. Resolution 243 (Rev.WRC-23) applies. (WRC-23) |
| 5.550C          | The use of the frequency bands 37.5-39.5 GHz (space-to-Earth), 39.5-42.5 GHz (space-to-Earth), 47.2- 50.2 GHz (Earth-to-space) and 50.4-51.4 GHz (Earth-to-space) by a non-geostationary-satellite system in the fixed-satellite service is subject to the application of the provisions of No. 9.12 for coordination with other non-geostationary-satellite systems in the fixed-satellite service but not with non-geostationary-satellite systems in other services. Resolution 770 (WRC-19) shall also apply, and No. 22.2 shall continue to apply. (WRC-19)                                                                                                                                                                                                                                                                                         |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.550CA         | Non-geostationary-satellite systems in the fixed-satellite service operating with an apogee altitude above 407 km and below 2 000 km in the frequency band 37.5-38 GHz shall not exceed an unwanted emission e.i.r.p. density of $-21 \text{ dB(W/100 MHz)}$ per space station for angles greater than $65.0^\circ$ from nadir relative to the space station in the fixed-satellite service in the frequency band 36-37 GHz in order to protect the Earth exploration-satellite service (passive) operating in the latter frequency band. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.550D          | The allocation to the fixed service in the frequency band 38-39.5 GHz is identified for worldwide use by administrations wishing to implement high-altitude platform stations (HAPS). In the HAPS-to-ground direction, the HAPS ground station shall not claim protection from stations in the fixed, mobile and fixed-satellite services; and No. 5.43A does not apply. This identification does not preclude the use of this frequency band by other fixed-service applications or by other services to which this frequency band is allocated on a co-primary basis and does not establish priority in the Radio Regulations. Furthermore, the development of the fixed-satellite, fixed and mobile services shall not be unduly constrained by HAPS. Such use of the fixed-service allocation by HAPS shall be in accordance with the provisions of Resolution 168 (Rev.WRC-23). (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.550E          | The use of the frequency bands 39.5-40 GHz and 40-40.5 GHz by non-geostationary-satellite systems in the mobile-satellite service (space-to-Earth) and by non-geostationary-satellite systems in the fixed-satellite service (space-to-Earth) is subject to the application of the provisions of No. 9.12 for coordination with other non-geostationarysatellite systems in the fixed-satellite and mobile-satellite services but not with non-geostationary-satellite systems in other services. No. 22.2 shall continue to apply for non-geostationary-satellite-systems. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.551H          | The equivalent power flux-density (epfd) produced in the frequency band 42.5-43.5 GHz by all space stations in any non-geostationary-satellite system in the fixed-satellite service, or in the broadcasting-satellite service operating in the frequency band 42-42.5 GHz, shall not exceed the following values at the site of any radio astronomy station for more than 2% of the time: $-230 \text{ dB(W/m}^2\text{)}$ in 1 GHz and $-246 \text{ dB(W/m}^2\text{)}$ in any 500 kHz of the frequency band 42.5-43.5 GHz at the site of any radio astronomy station registered as a single-dish telescope; and $-209 \text{ dB(W/m}^2\text{)}$ in any 500 kHz of the frequency band 42.5-43.5 GHz at the site of any radio astronomy station registered as a very long baseline interferometry station. These epfd values shall be evaluated using the methodology given in Recommendation ITU-R S.1586-1 and the reference antenna pattern and the maximum gain of an antenna in the radio astronomy service given in Recommendation ITU-R RA.1631-0 and shall apply over the whole sky and for elevation angles higher than the minimum operating angle $\theta_{\text{min}}$ of the radiotelescope (for which a default value of $5^\circ$ should be adopted in the absence of notified information). These values shall apply at any radio astronomy station that either: - was in operation prior to 5 July 2003 and has been notified to the Bureau before 4 January 2004; or - was notified before the date of receipt of the complete Appendix 4 information for coordination or notification, as appropriate, for the space station to which the limits apply. Other radio astronomy stations notified after these dates may seek an agreement with administrations that have authorized the space stations. In Region 2, Resolution 743 (WRC-03) shall apply. The limits in this footnote may be exceeded at the site of a radio astronomy station of any country whose administration so agreed. (WRC-15) |
| 5.551I          | The power flux-density in the band 42.5-43.5 GHz produced by any geostationary space station in the fixed-satellite service (space-to-Earth), or the broadcasting-satellite service (space-to-Earth) operating in the 42-42.5 GHz band, shall not exceed the following values at the site of any radio astronomy station: $-137 \text{ dB(W/m}^2\text{)}$ in 1 GHz and $-153 \text{ dB(W/m}^2\text{)}$ in any 500 kHz of the 42.5-43.5 GHz band at the site of any radio astronomy station registered as a single-dish telescope; and $-116 \text{ dB(W/m}^2\text{)}$ in any 500 kHz of the 42.5-43.5 GHz band at the site of any radio astronomy station registered as a very long baseline interferometry station. These values shall apply at the site of any radio astronomy station that either: - was in operation prior to 5 July 2003 and has been notified to the Bureau before 4 January 2004; or - was notified before the date of receipt of the complete Appendix 4 information for coordination or notification, as appropriate, for the space station to which the limits apply. Other radio astronomy stations notified after these dates may seek an agreement with administrations that have authorized the space stations. In Region 2, Resolution 743 (WRC-03) shall apply. The limits in this footnote may be exceeded at the site of a radio astronomy station of any country whose administration so agreed. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.552           | The allocation of the spectrum for the fixed-satellite service in the bands 42.5-43.5 GHz and 47.2-50.2 GHz for Earth-to-space transmission is greater than that in the band 37.5-39.5 GHz for space-to-Earth transmission in order to accommodate feeder links to broadcasting satellites. Administrations are urged to take all practicable steps to reserve the band 47.2-49.2 GHz for feeder links for the broadcasting-satellite service operating in the band 40.5-42.5 GHz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.552A          | The allocation to the fixed service in the frequency bands 47.2-47.5 GHz and 47.9-48.2 GHz is identified for use by high-altitude platform stations (HAPS). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated on a co-primary basis, and does not establish priority in the Radio Regulations. Such use of the fixed-service allocation in the frequency bands 47.2-47.5 GHz and 47.9-48.2 GHz by HAPS shall be in accordance with the provisions of Resolution 122 (Rev.WRC-19). (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5.553           | In the bands 43.5-47 GHz and 66-71 GHz, stations in the land mobile service may be operated subject to not causing harmful interference to the space radiocommunication services to which these bands are allocated (see No. 5.43). (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.553A          | In Algeria, Angola, Bahrain, Belarus, Benin, Botswana, Brazil, Burkina Faso, Cabo Verde, Korea (Rep. of), Ivory Coast, Croatia, Djibouti, Egypt, United Arab Emirates, Estonia, Eswatini, Gabon, Gambia, Ghana, Greece, Guinea, Guinea-Bissau, Hungary, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lesotho, Latvia, Liberia, Lithuania, Madagascar, Malawi, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Qatar, Senegal, Seychelles, Sierra Leone, Slovenia, Somalia, Sudan, South Africa, Sweden, Tanzania, Togo, Tunisia, Zambia and Zimbabwe, the frequency band 45.5-47 GHz is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT), taking into account No. 5.553. With respect to the aeronautical mobile service and radionavigation service, the use of this frequency band for the implementation of IMT is subject to agreement obtained under No. 9.21 with concerned administrations and shall not cause harmful interference to, or claim protection from these services. This identification does not preclude the use of this frequency band by any application of the services to which it is allocated and does not establish priority in the Radio Regulations. Resolution 244 (Rev.WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 5.553B          | In Region 2 and Algeria, Angola, Saudi Arabia, Australia, Bahrain, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Central African Rep., Comoros, Congo (Rep. of the), Korea (Rep. of), Ivory Coast, Djibouti, Egypt, United Arab Emirates, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Equatorial Guinea, India, Iran (Islamic Republic of), Iraq, Japan, Jordan, Kenya, Kuwait, Lesotho, Liberia, Libya, Lithuania, Madagascar, Malaysia, Malawi, Mali, Morocco, Mauritius, Mauritania, Mozambique, Namibia, Niger, Nigeria, Oman, Uganda, Qatar, the Syrian Arab Republic, the Dem. Rep. of the Congo, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Singapore, Slovenia, Somalia, Sudan, South Sudan, South Africa, Sweden, Tanzania, Chad, Togo, Tunisia, Zambia and Zimbabwe, the frequency band 47.2-48.2 GHz is identified for use by administrations wishing to implement International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which it is allocated, and does not establish any priority in the Radio Regulations. Resolution 243 (Rev.WRC-23) applies. (WRC-23) |
| 5.554           | In the bands 43.5-47 GHz, 66-71 GHz, 95-100 GHz, 123-130 GHz, 191.8-200 GHz and 252-265 GHz, satellite links connecting land stations at specified fixed points are also authorized when used in conjunction with the mobile-satellite service or the radionavigation-satellite service. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.554A          | The use of the bands 47.5-47.9 GHz, 48.2-48.54 GHz and 49.44-50.2 GHz by the fixed-satellite service (space-to-Earth) is limited to geostationary satellites. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.555           | Additional allocation: the band 48.94-49.04 GHz is also allocated to the radio astronomy service on a primary basis. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.555B          | The power flux-density in the band 48.94-49.04 GHz produced by any geostationary space station in the fixed-satellite service (space-to-Earth) operating in the bands 48.2-48.54 GHz and 49.44-50.2 GHz shall not exceed -151.8 dB(W/m <sup>2</sup> ) in any 500 kHz band at the site of any radio astronomy station. (WRC-03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.555C          | The use of the frequency band 51.4-52.4 GHz by the fixed-satellite service (Earth-to-space) is limited to geostationary-satellite networks. The earth stations shall be limited to gateway earth stations with a minimum antenna diameter of 2.4 metres. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.556           | In the bands 51.4-54.25 GHz, 58.2-59 GHz and 64-65 GHz, radio astronomy observations may be carried out under national arrangements. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.556A          | Use of the bands 54.25-56.9 GHz, 57-58.2 GHz and 59-59.3 GHz by the inter-satellite service is limited to satellites in the geostationary-satellite orbit. The single-entry power flux-density at all altitudes from 0 km to 1 000 km above the Earth's surface produced by a station in the inter-satellite service, for all conditions and for all methods of modulation, shall not exceed -147 dB(W/m <sup>2</sup> /100 MHz) for all angles of arrival. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.556B          | Additional allocation: in Japan, the band 54.25-55.78 GHz is also allocated to the mobile service on a primary basis for low-density use. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.557           | Additional allocation: in Japan, the band 55.78-58.2 GHz is also allocated to the radiolocation service on a primary basis. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.557A          | In the band 55.78-56.26 GHz, in order to protect stations in the Earth exploration-satellite service (passive), the maximum power density delivered by a transmitter to the antenna of a fixed service station is limited to -26 dB(W/MHz). (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.558           | In the bands 55.78-58.2 GHz, 59-64 GHz, 66-71 GHz, 122.25-123 GHz, 130-134 GHz, 167-174.8 GHz and 191.8-200 GHz, stations in the aeronautical mobile service may be operated subject to not causing harmful interference to the inter-satellite service (see No. 5.43). (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.558A          | Use of the band 56.9-57 GHz by inter-satellite systems is limited to links between satellites in geostationary-satellite orbit and to transmissions from non-geostationary satellites in high-Earth orbit to those in low Earth orbit. For links between satellites in the geostationary-satellite orbit, the single entry power flux-density at all altitudes from 0 km to 1 000 km above the Earth's surface, for all conditions and for all methods of modulation, shall not exceed -147 dB(W/(m <sup>2</sup> □ 100 MHz)) for all angles of arrival. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.559           | In the band 59-64 GHz, airborne radars in the radiolocation service may be operated subject to not causing harmful interference to the inter-satellite service (see No. 5.43). (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.559AA         | The frequency band 66-71 GHz is identified for use by administrations wishing to implement the terrestrial component of International Mobile Telecommunications (IMT). This identification does not preclude the use of this frequency band by any application of the services to which this frequency band is allocated and does not establish priority in the Radio Regulations. Resolution 241 (Rev.WRC-23) applies. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.559B          | The use of the frequency band 77.5-78 GHz by the radiolocation service shall be limited to short-range radar for ground-based applications, including automotive radars. The technical characteristics of these radars are provided in the most recent version of Recommendation ITU-R.M.2057. The provisions of No. 4.10 do not apply. (WRC-15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.560           | In the band 78-79 GHz radars located on space stations may be operated on a primary basis in the earth exploration-satellite service and in the space research service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.561           | In the band 74-76 GHz, stations in the fixed, mobile and broadcasting services shall not cause harmful interference to stations of the fixed-satellite service or stations of the broadcasting-satellite service operating in accordance with the decisions of the appropriate frequency assignment planning conference for the broadcasting-satellite service. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Footnote Number | Footnote Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 5.561A          | The 81-81.5 GHz band is also allocated to the amateur and amateur-satellite services on a secondary basis. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.562           | The use of the band 94-94.1 GHz by the Earth exploration-satellite (active) and space research (active) services is limited to spaceborne cloud radars. (WRC-97)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.562A          | In the bands 94-94.1 GHz and 130-134 GHz, transmissions from space stations of the Earth exploration-satellite service (active) that are directed into the main beam of a radio astronomy antenna have the potential to damage some radio astronomy receivers. Space agencies operating the transmitters and the radio astronomy stations concerned should mutually plan their operations so as to avoid such occurrences to the maximum extent possible. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.562B          | In the frequency bands 105-109.5 GHz, 111.8-114.25 GHz and 217-226 GHz, the use of this allocation is limited to space-based radio astronomy only. (WRC-19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.562C          | Use of the band 116-122.25 GHz by the inter-satellite service is limited to satellites in the geostationary-satellite orbit. The single-entry power flux-density produced by a station in the inter-satellite service, for all conditions and for all methods of modulation, at all altitudes from 0 km to 1 000 km above the Earth's surface and in the vicinity of all geostationary orbital positions occupied by passive sensors, shall not exceed -148 dB(W/(m <sup>2</sup> · MHz)) for all angles of arrival. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.562E          | The allocation to the Earth exploration-satellite service (active) is limited to the band 133.5-134 GHz. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.562H          | Use of the bands 174.8-182 GHz and 185-190 GHz by the inter-satellite service is limited to satellites in the geostationary-satellite orbit. The single-entry power flux-density produced by a station in the inter-satellite service, for all conditions and for all methods of modulation, at all altitudes from 0 km to 1000 km above the Earth's surface and in the vicinity of all geostationary orbital positions occupied by passive sensors, shall not exceed -144 dB(W/(m <sup>2</sup> · MHz)) for all angles of arrival. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5.563A          | In the bands 200-209 GHz, 235-238 GHz, 250-252 GHz and 265-275 GHz, ground-based passive atmospheric sensing is carried out to monitor atmospheric constituents. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.563AA         | In the frequency band 235-238 GHz, stations in the Earth exploration-satellite service (passive) shall not claim protection from stations in the fixed and mobile services. (WRC-23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.563B          | The band 237.9-238 GHz is also allocated to the Earth exploration-satellite service (active) and the space research service (active) for spaceborne cloud radars only. (WRC-2000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.564A          | For the operation of fixed and land mobile service applications in frequency bands in the range 275-450 GHz: The frequency bands 275-296 GHz, 306-313 GHz, 318-333 GHz and 356-450 GHz are identified for use by administrations for the implementation of land mobile and fixed service applications where no specific conditions are necessary to protect Earth exploration-satellite service (passive) applications. The frequency bands 296-306 GHz, 313-318 GHz and 333-356 GHz may only be used by fixed and land mobile service applications when specific conditions to ensure the protection of Earth exploration-satellite service (passive) applications are determined in accordance with Resolution 731 (Rev.WRC-23). In those portions of the frequency range 275-450 GHz where radio astronomy applications are used, specific conditions (e.g. minimum separation distances and/or avoidance angles) may be necessary to ensure protection of radio astronomy sites from land mobile and/or fixed service applications, on a case-by-case basis, in accordance with Resolution 731 (Rev.WRC-23). The use of the above-mentioned frequency bands by land mobile and fixed service applications does not preclude use by, and does not establish priority over, any other applications of radio services in the range of 275-450 GHz. (WRC-23) |
| 5.565           | The following frequency bands in the range 275-1000 GHz are identified for use by administrations for passive services applications: - radio astronomy service: 275-323 GHz, 327-371 GHz, 388-424 GHz, 426-442 GHz, 453-510 GHz, 623-711 GHz, 795-909 GHz and 926-945 GHz; - Earth exploration-satellite service (passive) and space research service (passive): 275-286 GHz, 296-306 GHz, 313-356 GHz, 361-365 GHz, 369-392 GHz, 397-399 GHz, 409-411 GHz, 416-434 GHz, 439-467 GHz, 477-502 GHz, 523-527 GHz, 538-581 GHz, 611-630 GHz, 634-654 GHz, 657-692 GHz, 713-718 GHz, 729-733 GHz, 750-754 GHz, 771-776 GHz, 823-846 GHz, 850-854 GHz, 857-862 GHz, 866-882 GHz, 905-928 GHz, 951-956 GHz, 968-973 GHz and 985-990 GHz. The use of the range 275-1000 GHz by the passive services does not preclude use of this range by active services. Administrations wishing to make frequencies in the 275-1000 GHz range available for active service applications are urged to take all practicable steps to protect these passive services from harmful interference until the date when the Table of Frequency Allocations is established in the above-mentioned 275-1000 GHz frequency range. All frequencies in the range 1000-3000 GHz may be used by both active and passive services. (WRC-12)                                                     |

## CEPT Deliverables

| Document       | Description                                                                                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECC/DEC/(25)01 | Designation of the frequency bands used by the Global Navigation Satellite System, Galileo, and Technical and operational measures for the use of the frequency band 1258-1300 MHz by the amateur and amateur-satellite services in order to protect the radionavigation-satellite service (space-to-Earth) |
| ECC/DEC/(24)01 | Harmonised technical conditions for the shared use of the 3800-4200 MHz frequency band by low/medium power terrestrial wireless broadband systems (WBB LMP) providing local-area network connectivity                                                                                                       |
| ECC/DEC/(23)01 | On the use of the band 40.5-42.5 GHz by earth stations in the fixed-satellite service (space-to-Earth) and broadcasting-satellite service and on the use of the band 42.5-43.5 GHz by earth stations in the fixed-satellite service (Earth-to-space)                                                        |
| ECC/DEC/(22)07 | Harmonised technical conditions for the usage of aerial UE for communications based on LTE and 5G NR in the bands 703-733 MHz, 832-862 MHz, 880-915 MHz, 1710-1785 MHz, 1920-1980 MHz, 2500-2570 MHz and 2570-2620 MHz harmonised for MFCN                                                                  |
| ECC/DEC/(22)06 | Harmonised technical conditions for Mobile/Fixed Communications Networks (MFCN) in the band 40.5-43.5 GHz                                                                                                                                                                                                   |
| ECC/DEC/(22)03 | Technical characteristics, exemption from individual licensing and free circulation and use of specific radiodetermination applications in the frequency range 116-260 GHz                                                                                                                                  |
| ECC/DEC/(22)02 | Regulation to operate Autonomous Maritime Radio Devices (AMRD) in CEPT                                                                                                                                                                                                                                      |
| ECC/DEC/(22)01 | Free circulation and use of Mobile/Fixed Communication Networks (MFCN) terminals operating under the control of terrestrial networks                                                                                                                                                                        |
| ECC/DEC/(21)02 | The harmonised frequency band 76-77 GHz, technical characteristics, exemption from individual licensing and free circulation and use of High Definition Ground Based Synthetic Aperture Radar (HD-GBSAR)                                                                                                    |
| ECC/DEC/(21)01 | The use of the bands 47.2-50.2 GHz and 50.4-52.4 GHz by the fixed-satellite service (Earth-to-space)                                                                                                                                                                                                        |
| ECC/DEC/(20)02 | Harmonised use of the paired frequency bands 874.4-880.0 MHz and 919.4-925.0 MHz and of the unpaired frequency band 1900-1910 MHz for Railway Mobile Radio (RMR)                                                                                                                                            |
| ECC/DEC/(20)01 | On the harmonised use of the frequency band 5945-6425 MHz for Wireless Access Systems including Radio Local Area Networks (WAS/RLAN)                                                                                                                                                                        |
| ECC/DEC/(19)04 | The harmonised use of spectrum, free circulation and use of earth stations on-board aircraft operating with GSO FSS networks and NGSO FSS systems in the frequency bands 12.75-13.25 GHz (Earth-to-space) and 10.7-12.75 GHz (space-to-Earth)                                                               |
| ECC/DEC/(19)03 | Harmonised usage of the channels of the Radio Regulations Appendix 18 (Table of transmitting frequencies in the VHF maritime mobile band)                                                                                                                                                                   |
| ECC/DEC/(19)02 | Land mobile systems in the frequency ranges 68-87.5 MHz, 146-174 MHz, 406.1-410 MHz, 410-430 MHz, 440-450 MHz and 450-470 MHz                                                                                                                                                                               |
| ECC/DEC/(18)06 | The harmonised technical conditions for Mobile/Fixed Communications Networks (MFCN) in the band 24.25-27.5 GHz                                                                                                                                                                                              |
| ECC/DEC/(18)05 | The harmonised use, exemption from individual licensing and free circulation and use of Earth Stations In-Motion (ESIM) operating with NGSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz                                                                                   |
| ECC/DEC/(18)04 | The harmonised use, exemption from individual licensing and free circulation and use of land based Earth Stations In-Motion (ESIM) operating with GSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz                                                                         |
| ECC/DEC/(17)06 | The harmonised use of the frequency bands 1427-1452 MHz and 1492-1518 MHz for Mobile/Fixed Communications Networks Supplemental Downlink (MFCN SDL)                                                                                                                                                         |
| ECC/DEC/(17)04 | The harmonised use and exemption from individual licensing of fixed earth stations operating with NGSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz                                                                                                                        |
| ECC/DEC/(16)02 | Harmonised technical conditions and frequency bands for the implementation of Broadband Public Protection and Disaster Relief (BB-PPDR) systems                                                                                                                                                             |
| ECC/DEC/(16)01 | The harmonised frequency band 76-77 GHz, technical characteristics, exemption from individual licensing and free carriage and use of obstacle detection radars for rotorcraft use                                                                                                                           |
| ECC/DEC/(15)05 | The harmonised frequency range 446.0-446.2 MHz, technical characteristics, exemption from individual licensing and free carriage and use of analogue and digital PMR 446 applications                                                                                                                       |

| Document       | Description                                                                                                                                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECC/DEC/(15)04 | The harmonised use, free circulation and exemption from individual licensing of Land, Maritime and Aeronautical Earth Stations On Mobile Platforms (ESOMPs) operating with NGSO FSS satellite systems in the frequency ranges 17.3-20.2 GHz, 27.5-29.1 GHz and 29.5-30.0 GHz |
| ECC/DEC/(15)01 | The harmonised technical conditions for mobile/fixed communications networks (MFCN) in the band 694-790 MHz including a paired frequency arrangement (Frequency Division Duplex 2x30 MHz) and an optional unpaired frequency arrangement (Supplemental Downlink)             |
| ECC/DEC/(14)02 | The harmonised technical and regulatory conditions for the use of the band 2300-2400 MHz for Mobile/Fixed Communications Networks (MFCN)                                                                                                                                     |
| ECC/DEC/(13)03 | The harmonised use of the frequency band 1452-1492 MHz for Mobile/Fixed Communications Networks Supplemental Downlink (MFCN SDL)                                                                                                                                             |
| ECC/DEC/(13)01 | The use, free circulation, and exemption from individual licensing of Earth stations on mobile platforms (ESOMPs) in the frequency bands available for use by uncoordinated FSS Earth stations within the ranges 17.3-20.2 GHz and 27.5-30.0 GHz                             |
| ECC/DEC/(12)03 | The harmonised conditions for UWB applications onboard aircraft                                                                                                                                                                                                              |
| ECC/DEC/(12)01 | Exemption from individual licensing and free circulation and use of satellite mobile terminals operating under the control of networks in the range 1 to 3 GHz                                                                                                               |
| ECC/DEC/(11)06 | The harmonised frequency arrangements and Least Restrictive Technical Conditions (LRTCs) for Mobile/Fixed Communications Networks (MFCN) operating in the band 3400-3800 MHz                                                                                                 |
| ECC/DEC/(11)03 | The harmonised use of frequencies for Citizen' Band (CB) radio equipment                                                                                                                                                                                                     |
| ECC/DEC/(11)02 | Industrial Level Probing Radars (LPR) operating in frequency bands 6 - 8.5 GHz, 24.05 - 26.5 GHz, 57 - 64 GHz and 75 - 85 GHz                                                                                                                                                |
| ECC/DEC/(11)01 | The protection of the Earth exploration satellite service (passive) in the 1400-1427 MHz band                                                                                                                                                                                |
| ECC/DEC/(10)02 | Compatibility between the fixed satellite service in the 30-31 GHz band and the Earth exploration satellite service (passive) in the 31.3-31.5 GHz band                                                                                                                      |
| ECC/DEC/(10)01 | Sharing conditions in the 10.6-10.68 GHz band between the fixed service, mobile service and Earth exploration satellite service (passive)                                                                                                                                    |
| ECC/DEC/(09)04 | Exemption from individual licensing and the free circulation and use of transmit-only mobile satellite terminals operating in the Mobile-Satellite Service allocations in the 1613.8-1626.5 MHz band                                                                         |
| ECC/DEC/(09)03 | Harmonised conditions for Mobile/Fixed Communications Networks (MFCN) operating in the band 791-862 MHz                                                                                                                                                                      |
| ECC/DEC/(09)02 | The harmonisation of the bands 1610-1626.5 MHz and 2483.5-2500 MHz for use by systems in the Mobile-Satellite Service                                                                                                                                                        |
| ECC/DEC/(09)01 | Harmonised use of the 63.72-65.88 GHz frequency band for Intelligent Transport Systems (ITS)                                                                                                                                                                                 |
| ECC/DEC/(08)08 | The harmonised use of GSM systems in the 900 MHz and 1800 MHz bands, UMTS systems in the 2 GHz band and LTE and 5G NR non-AAS systems in the 1800 MHz and 2.6 GHz (FDD) bands on board vessels                                                                               |
| ECC/DEC/(08)05 | The harmonisation of frequency bands for the implementation of digital Public Protection and Disaster Relief (PPDR) radio applications in bands within the 380-470 MHz range                                                                                                 |
| ECC/DEC/(08)01 | The harmonised use of Safety-Related Intelligent Transport Systems (ITS) in the 5875-5935 MHz frequency band                                                                                                                                                                 |
| ECC/DEC/(07)01 | The harmonised use, exemption from individual licensing and free circulation of Material Sensing Devices using Ultra-Wideband (UWB) technology                                                                                                                               |
| ECC/DEC/(06)13 | Harmonised technical conditions for mobile/fixed communications networks (MFCN) including terrestrial IMT systems, other than GSM and EC-GSM IoT, in the bands 880-915/925-960 MHz and 1710-1785/1805-1880 MHz                                                               |
| ECC/DEC/(06)10 | Transition of terrestrial service operations from the Bands 1980-2010 MHz and 2170-2200 MHz in order to facilitate the Harmonised Introduction and Development of Systems in the mobile-satellite service including those supplemented by a Complementary Ground Component   |
| ECC/DEC/(06)09 | The designation of the bands 1980-2010 MHz and 2170-2200 MHz for use by systems in the Mobile-Satellite Service including those supplemented by a Complementary Ground Component (CGC)                                                                                       |
| ECC/DEC/(06)07 | The harmonised use of airborne GSM and LTE systems in the frequency bands 1710-1785 and 1805-1880 MHz, and airborne UMTS systems in the frequency bands 1920-1980 MHz and 2110-2170 MHz                                                                                      |
| ECC/DEC/(06)05 | The harmonised frequency bands to be designated for Air-Ground-Air operation (AGA) of the Digital Land Mobile Systems for the Emergency Services                                                                                                                             |

| Document       | Description                                                                                                                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECC/DEC/(06)04 | The harmonised use, exemption from individual licensing and free circulation of devices using Ultra-Wideband (UWB) technology in bands below 10.6 GHz                                                                                                      |
| ECC/DEC/(06)03 | Exemption from Individual Licensing of high e.i.r.p. satellite terminals (HEST) operating with geostationary satellites and in the frequency bands 10.70-12.75 GHz or 19.70-20.20 GHz space-to-Earth and 14.00-14.25 GHz or 29.50-30.00 GHz Earth-to-space |
| ECC/DEC/(06)01 | The harmonised utilisation of the bands 1920-1980 MHz and 2110-2170 MHz for mobile/fixed communications networks (MFCN) including terrestrial IMT systems                                                                                                  |
| ECC/DEC/(05)11 | The free circulation and use of Aircraft Earth Stations (AES) in the frequency bands 14-14.5 GHz (Earth-to-space), 10.7-11.7GHz (space-to-Earth) and 12.5-12.75 GHz (space-to-Earth)                                                                       |
| ECC/DEC/(05)10 | The free circulation and use of Earth Stations on board Vessels operating in fixed satellite service networks in the frequency bands 14-14.5 GHz (Earth-to-space), 10.7-11.7 GHz (space-to-Earth) and 12.5-12.75 GHz (space-to-Earth)                      |
| ECC/DEC/(05)09 | The free circulation and use of Earth Stations on board Vessels operating in Fixed Satellite service networks in the frequency bands 5925-6425 MHz (Earth-to-space) and 3700-4200 MHz (space-to-Earth)                                                     |
| ECC/DEC/(05)08 | The availability of frequency bands for high density applications in the Fixed-Satellite Service (space-to-Earth and Earth-to-space)                                                                                                                       |
| ECC/DEC/(05)05 | Harmonised utilisation of spectrum for Mobile/Fixed Communications Networks (MFCN) operating within the band 2500-2690 MHz                                                                                                                                 |
| ECC/DEC/(05)02 | A harmonised frequency plan for the use of the band 169.4-169.8125 MHz                                                                                                                                                                                     |
| ECC/DEC/(05)01 | The use of the band 27.5-29.5 GHz by the Fixed Service and uncoordinated Earth stations of the Fixed-Satellite Service (Earth-to-space)                                                                                                                    |
| ECC/DEC/(04)10 | The frequency bands to be designated for the temporary introduction of Automotive Short Range Radars (SRR)                                                                                                                                                 |
| ECC/DEC/(04)09 | Designation of the bands 1518-1525 MHz and 1670-1675 MHz for the Mobile Satellite Service                                                                                                                                                                  |
| ECC/DEC/(04)08 | The harmonised use of the 5 GHz frequency bands for Wireless Access Systems including Radio Local Area Networks (WAS/RLAN)                                                                                                                                 |
| ECC/DEC/(04)03 | The frequency band 77-81 GHz to be designated for the use of Automotive Short Range Radars                                                                                                                                                                 |
| ECC/DEC/(03)04 | The Exemption from Individual Licensing of Very Small Aperture Terminals (VSAT) operating in the frequency bands 14.25 - 14.50 GHz Earth-to-space and 10.70-11.70 GHz space-to-Earth                                                                       |
| ERC/DEC/(99)06 | The harmonised introduction of satellite personal communication systems operating in the bands below 1 GHz (S-PCS<1GHz)                                                                                                                                    |
| ERC/DEC/(99)05 | Free Circulation, Use and Exemption from Individual Licensing of Mobile Earth Stations.(S-PCS < 1GHz)                                                                                                                                                      |
| ERC/DEC/(98)22 | Exemption from individual licensing and free circulation and use of DECT equipment                                                                                                                                                                         |
| ERC/DEC/(97)02 | The extended frequency bands to be used for the GSM Digital Pan-European Communications system                                                                                                                                                             |
| ERC/DEC/(95)03 | The frequency bands to be designated for the introduction of DCS 1800                                                                                                                                                                                      |
| ERC/DEC/(94)03 | The frequency band to be designated for the coordinated introduction of the Digital European Cordless Telecommunications system                                                                                                                            |
| ERC/DEC/(94)01 | The frequency bands to be designated for the coordinated introduction of the GSM digital pan-European communications system                                                                                                                                |
| ERC/DEC/(01)19 | Harmonised frequency bands to be designated for the Direct Mode Operation (DMO) of the Digital Land Mobile Systems for the Emergency Services                                                                                                              |
| ERC/DEC/(01)17 | Harmonised frequencies, technical characteristics and exemption from individual licensing of Ultra Low Power Active Medical Implant (ULP-AMI) communication systems operating in the frequency band 401 - 406 MHz on a secondary basis                     |
| ERC/DEC/(01)12 | Harmonised frequencies, technical characteristics and exemption from individual licensing of Short Range Devices used for Model control operating in the frequencies 40.665, 40.675, 40.685 and 40.695 MHz                                                 |
| ERC/DEC/(01)11 | Harmonised frequencies, technical characteristics and exemption from individual licensing of Short Range Devices used for Flying Model control operating in the frequency band 34.995 - 35.225 MHz                                                         |

| Document       | Description                                                                                                                                                                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ERC/DEC/(00)08 | The use of the band 10.7 - 12.5 GHz by the fixed service and Earth stations of the broadcasting-satellite and fixed-satellite Service (space-to-Earth)                                                                                                                                                                                   |
| ERC/DEC/(00)07 | The shared use of the band 17.7 - 19.7 GHz by the fixed service and Earth stations of the fixed-satellite service (space-to-Earth)                                                                                                                                                                                                       |
| ERC/DEC/(00)02 | Use of the band 37.5-39.5 GHz by the fixed service and by earth stations of the fixed-satellite service (space-to-Earth) and use of the band 39.5-40.5 GHz by earth stations of the fixed-satellite service and the mobile-satellite service (space-to-Earth)                                                                            |
| ECC/REC/(25)02 | Cross-border coordination for Railway Mobile Radio (RMR) in the 900 MHz frequency band excluding GSM-R vs. GSM-R                                                                                                                                                                                                                         |
| ECC/REC/(24)03 | Licensing of earth stations for space tracking, space telemetry and space telecommand in the bands 2025-2110 MHz (Earth-to-space) and 2200-2290 MHz (space-to-Earth)                                                                                                                                                                     |
| ECC/REC/(24)02 | Guidance for the use of governmental UAS operating within the frequency bands 1880-1900 MHz and 1910-1920 MHz                                                                                                                                                                                                                            |
| ECC/REC/(23)02 | Cross-border coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency band 24.25-27.5 GHz                                                                                                                                                                                                                           |
| ECC/REC/(23)01 | Cross-border coordination for Railway Mobile Radio (RMR) in the 1900-1910 MHz TDD frequency band                                                                                                                                                                                                                                         |
| ECC/REC/(22)02 | Guidelines on measures to facilitate compatibility between MFCN operating in 40.5-43.5 GHz and FSS earth stations receiving in 39.5-40.5 GHz and to prevent and/or resolve interference issues                                                                                                                                           |
| ECC/REC/(22)01 | Guidelines to support the introduction of MFCN in 40.5-43.5 GHz while ensuring, in a proportionate way, the use of FSS receiving earth stations in the frequency band 40.5-42.5 GHz and the use of FSS transmitting earth stations in the frequency band 42.5-43.5 GHz and the possibility for future deployment of these earth stations |
| ECC/REC/(21)02 | Guidance on the application of the least restrictive technical conditions (LRTC) in ECC Decision (11)06 (amended 26 October 2018) to ensure protection of the military radiolocation systems operating below 3400 MHz from indoor non-AAS small cells operating in the band 3400-3800 MHz                                                |
| ECC/REC/(20)03 | Frame structures to facilitate cross-border coordination of TDD MFCN in the frequency band 3400-3800 MHz                                                                                                                                                                                                                                 |
| ECC/REC/(20)01 | Guidelines to support the introduction of 5G while ensuring, in a proportionate way, the use of existing and planned FSS transmitting earth stations in the frequency band 24.65-25.25 GHz and the possibility for future deployment of these earth stations                                                                             |
| ECC/REC/(19)01 | Technical toolkit to support the introduction of 5G while ensuring, in a proportionate way, the use of existing and planned EESS/SRS receiving earth stations in the 26 GHz band and the possibility for future deployment of these earth stations                                                                                       |
| ECC/REC/(18)02 | Radio frequency channel/block arrangements for fixed service systems operating in the bands 92-94 GHz, 94.1-100 GHz, 102-109.5 GHz and 111.8-114.25 GHz                                                                                                                                                                                  |
| ECC/REC/(18)01 | Radio frequency channel/block arrangements for Fixed Service systems operating in the bands 130 - 134 GHz, 141-148.5 GHz, 151.5-164 GHz and 167 - 174.8 GHz                                                                                                                                                                              |
| ECC/REC/(17)03 | Guidance for the harmonised use and coordination of Maritime Broadband Radio (MBR) systems on board ships and off-shore platforms operating within the frequency bands 5852-5872 MHz and 5880-5900 MHz                                                                                                                                   |
| ECC/REC/(16)03 | Cross-border coordination for Broadband Public Protection and Disaster Relief (BB-PPDR) systems in the frequency band 698 to 791 MHz                                                                                                                                                                                                     |
| ECC/REC/(15)04 | Guidance for the implementation of a sharing framework between MFCN and PMSE within 2300-2400 MHz                                                                                                                                                                                                                                        |
| ECC/REC/(15)01 | Cross-border coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency bands: 694-790 MHz, 1427-1518 MHz and 3400-3800 MHz                                                                                                                                                                                           |
| ECC/REC/(14)06 | Implementation of Fixed Service Point-to-Point narrow channels (3.5 MHz, 1.75 MHz, 0.5 MHz, 0.25 MHz, 0.025 MHz) in the guard bands and center gaps of the lower 6 GHz (5925-6425 MHz) and upper 6 GHz (6425-7125 MHz) bands                                                                                                             |
| ECC/REC/(14)04 | Cross-border coordination for mobile/fixed communications networks (MFCN) and between MFCN and other systems in the frequency band 2300-2400 MHz                                                                                                                                                                                         |
| ECC/REC/(14)01 | Radio frequency channel arrangements for fixed service systems operating in the band 92-95 GHz                                                                                                                                                                                                                                           |
| ECC/REC/(11)10 | Location Tracking Application for emergency and disaster situations                                                                                                                                                                                                                                                                      |
| ECC/REC/(11)09 | UWB Location Tracking Systems Type 2 (LT2)                                                                                                                                                                                                                                                                                               |

| Document       | Description                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ECC/REC/(11)08 | Framework for authorisation regime of indoor global navigation satellite system (GNSS) pseudolites in the band 1559-1610 MHz                                                       |
| ECC/REC/(11)05 | Cross-border Coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency band 2500-2690 MHz                                                                      |
| ECC/REC/(11)04 | Cross-border Coordination for Mobile/Fixed Communications Networks (MFCN) in the frequency band 791-862 MHz                                                                        |
| ECC/REC/(11)01 | Guidelines for assignment of frequency blocks for Fixed Wireless Systems in the bands 24.5-26.5 GHz, 27.5-29.5 GHz and 31.8-33.4 GHz                                               |
| ECC/REC/(10)02 | A framework for authorisation regime of Global Navigation Satellite System (GNSS) repeaters                                                                                        |
| ECC/REC/(10)01 | Guidelines for compatibility between Complementary Ground Components (CGC) operating in the band 2170-2200 MHz and EESS/SOS/SRS earth stations operating in the band 2200-2290 MHz |
| ECC/REC/(08)04 | The identification of frequency bands for the implementation of Broad Band Disaster Relief (BBDR) radio applications in the 5 GHz frequency range                                  |
| ECC/REC/(08)02 | Frequency planning and frequency coordination for GSM / UMTS / LTE / WiMAX Land Mobile systems operating within the 900 and 1800 MHz bands                                         |
| ECC/REC/(08)01 | Use of the band 5855-5875 MHz for non-safety related Road Intelligent Transport Systems (ITS)                                                                                      |
| ECC/REC/(06)04 | Use of the band 5725-5875 MHz for Broadband Fixed Wireless Access (BFWA)                                                                                                           |
| ECC/REC/(05)08 | Frequency planning and cross-border coordination between GSM Land Mobile Systems (GSM 900, GSM 1800 and GSM-R)                                                                     |
| ECC/REC/(05)07 | Radio frequency channel arrangements for Fixed Service Systems operating in the bands 71-76 GHz and 81-86 GHz                                                                      |
| ECC/REC/(02)09 | Protection of Aeronautical Radio Navigation Service in the band 2700-2900 MHz from interference caused by the operation of Digital Cordless Cameras                                |
| ECC/REC/(02)06 | Preferred channel arrangements for digital Fixed Service Systems operating in the frequency range 7125-8500 MHz                                                                    |
| ECC/REC/(02)02 | Preferred channel arrangements for fixed service systems (point-to-point and point-to-multipoint) operating in the frequency band 31.0-31.3 GHz                                    |
| ECC/REC/(01)04 | Radio frequency channel arrangements for point-to-point (P-P) fixed wireless systems in the frequency band 40.5 - 43.5 GHz                                                         |
| ERC/REC/(01)02 | Preferred channel arrangement for digital FS systems operating in the band 31.8-33.4 GHz                                                                                           |
| ERC/REC/(00)04 | Harmonised frequencies and free circulation and use for meteor scatter applications                                                                                                |
| ERC/REC 70-03  | Relating to the Use of Short Range Devices (SRD)                                                                                                                                   |
| ERC/REC 62-02  | Harmonised frequency band for civil and military airborne telemetry applications                                                                                                   |
| ERC/REC 25-10  | Frequency ranges for the use of terrestrial audio and video Programme Making and Special Events (PMSE) applications                                                                |
| ERC/REC 14-02  | Radio-frequency channel arrangements for high, medium and low capacity digital Fixed Service systems operating in the band 6425-7125 MHz                                           |
| ERC/REC 14-01  | Radio-frequency channel arrangements for high capacity analogue and digital radio-relay systems operating in the band 5925 to 6425 MHz                                             |
| ERC/REC 13-03  | The use of the band 14.0 - 14.5 GHz for Very Small Aperture Terminals (VSAT) and Satellite News Gathering (SNG)                                                                    |
| ERC/REC 12-12  | Radio frequency channel arrangement for fixed service systems operating in the band 55.78-57.0 GHz (as amended in 2015)                                                            |
| ERC/REC 12-11  | Radio frequency channel arrangements for Fixed Service systems operating in the bands 48.5-50.2 / 50.9-52.6 GHz                                                                    |
| ERC/REC 12-08  | Harmonised radio frequency channel arrangements and block allocations for low, medium and high capacity systems in the band 3600 MHz to 4200 MHz                                   |
| ERC/REC 12-07  | Harmonised radio frequency channel arrangements for digital terrestrial fixed systems operating in the band 14.5 - 14.62 GHz paired with 15.23 - 15.35 GHz                         |
| ERC/REC 12-06  | Preferred channel arrangements for fixed service systems operating in the frequency band 10.7-11.7 GHz                                                                             |
| ERC/REC 12-05  | Harmonised radio frequency channel arrangements for digital terrestrial fixed systems operating in the band 10.0 - 10.68 GHz                                                       |
| ERC/REC 12-03  | Harmonised radio frequency channel arrangements for digital terrestrial fixed systems operating in the band 17.7 GHz to 19.7 GHz                                                   |

| Document      | Description                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ERC/REC 12-02 | Harmonised radio frequency channel arrangements for analogue and digital terrestrial fixed systems operating in the band 12.75 GHz to 13.25 GHz |
| ERC/REC 01-01 | Cross-border coordination for mobile/fixed communications networks (MFCN) in the frequency bands: 1920-1980 MHz and 2110-2170 MHz               |
| T/R 25-08     | Planning criteria and cross-border coordination of frequencies for land mobile systems in the range 29.7-470 MHz                                |
| T/R 13-02     | Preferred channel arrangements for fixed service systems in the frequency range 22.0-29.5 GHz                                                   |
| T/R 13-01     | Preferred channel arrangements for fixed service systems operating in the frequency range 1-2-3 GHz                                             |
| T/R 12-01     | Harmonised radio frequency channel arrangements for analogue/digital terrestrial FS operating in 37-39.5 GHz                                    |

## European Standards

| Document   | Description                                                                                                                                                                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 300 065 | Narrow-band direct-printing telegraph equipment for receiving meteorological or navigational information (NAVTEX)                                                                                                                                                                      |
| EN 300 066 | Float-free maritime satellite Emergency Position Indicating Radio Beacons (EPIRBs) operating in the 406,0 MHz to 406,1 MHz frequency band                                                                                                                                              |
| EN 300 086 | Land Mobile Service; Radio equipment with an internal or external RF connector intended primarily for analogue speech                                                                                                                                                                  |
| EN 300 113 | Land Mobile Service; Radio equipment intended for the transmission of data (and speech) and having an antenna connector                                                                                                                                                                |
| EN 300 152 | Maritime Emergency Position Indicating Radio Beacons (EPIRBs) intended for use on the frequency 121.5 MHz or the frequencies 121.5 MHz and 243 MHz for homing purposes only                                                                                                            |
| EN 300 162 | Radiotelephone transmitters and receivers for the maritime mobile service operating in VHF bands                                                                                                                                                                                       |
| EN 300 219 | Land Mobile Service; Radio equipment transmitting signals to initiate a specific response in the receiver                                                                                                                                                                              |
| EN 300 220 | Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz                                                                                                                                                                                                         |
| EN 300 224 | On-site paging service                                                                                                                                                                                                                                                                 |
| EN 300 296 | Land Mobile Service; Radio equipment using integral antennas intended primarily for analogue speech                                                                                                                                                                                    |
| EN 300 328 | Wideband Transmission systems; Data transmission equipment operating in the 2.4 GHz ISM band and using spread spectrum modulation techniques                                                                                                                                           |
| EN 300 330 | SRD; Radio equipment in the frequency range 9 kHz to 25 MHz and inductive loop systems in the frequency range 9 kHz to 30 MHz                                                                                                                                                          |
| EN 300 338 | Radio equipment for generation, transmission and reception of Digital Selective Calling (DSC) in the maritime MF, MF/HF and/or VHF mobile service                                                                                                                                      |
| EN 300 341 | Land Mobile Service; Radio equipment using an integral antenna transmitting signals to initiate a specific response in the receiver                                                                                                                                                    |
| EN 300 390 | Land Mobile Service; Radio equipment intended for the transmission of data (and speech) and using an integral antenna                                                                                                                                                                  |
| EN 300 422 | Wireless Microphones; Audio PMSE up to 3 GHz; Audio PMSE Equipment up to 3 GHz; part 1 and 4                                                                                                                                                                                           |
| EN 300 433 | Citizens' Band (CB) radio equipment                                                                                                                                                                                                                                                    |
| EN 300 440 | Radio equipment to be used in the 1 to 40 GHz frequency range                                                                                                                                                                                                                          |
| EN 300 454 | Wide band audio links                                                                                                                                                                                                                                                                  |
| EN 300 471 | Rules for Access and the Sharing of common used channels by equipment complying with EN 300 113                                                                                                                                                                                        |
| EN 300 674 | Transport and Traffic Telematics (TTT); Dedicated Short Range Communication (DSRC) transmission equipment (500 kbit/s / 250 kbit/s) operating in the 5 795 MHz to 5 815 MHz frequency band; Part 2: Harmonised Standard for access to radio spectrum; Sub-part 2: On-Board Units (OBU) |
| EN 300 676 | Ground-based VHF hand-held, mobile and fixed radio transmitters, receivers and transceivers for the VHF aeronautical mobile service using amplitude modulation                                                                                                                         |
| EN 300 698 | Radio telephone transmitters and receivers for the maritime mobile service operating in the VHF bands used on inland waterways                                                                                                                                                         |
| EN 300 718 | Avalanche Beacons; Transmitter-receiver systems                                                                                                                                                                                                                                        |
| EN 300 720 | Ultra-High Frequency (UHF) on-board vessels communications systems and equipment                                                                                                                                                                                                       |
| EN 301 025 | VHF radiotelephone equipment for general communications and associated equipment for Class "D" Digital Selective Calling (DSC)                                                                                                                                                         |
| EN 301 091 | Radar equipment operating in the 76 GHz to 77 GHz range                                                                                                                                                                                                                                |
| EN 301 166 | Land Mobile Service; Radio equipment for analogue and/or digital communication (speech and/or data) and operating on narrow band channels and having an antenna connector                                                                                                              |

| Document   | Description                                                                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 301 178 | Portable Very High Frequency (VHF) radiotelephone equipment for the maritime mobile service operating in the VHF bands (for non-GMDSS applications only)                                                                                                                                      |
| EN 301 357 | Cordless audio devices in the range 25 MHz to 2000 MHz                                                                                                                                                                                                                                        |
| EN 301 406 | Digital Enhanced Cordless Telecommunications (DECT)                                                                                                                                                                                                                                           |
| EN 301 426 | Low data rate Land Mobile satellite Earth Stations (LMES) and Maritime Mobile satellite Earth Stations (MMES) not intended for distress and safety communications operating in the 1.5/1.6 GHz frequency bands                                                                                |
| EN 301 427 | Low data rate Mobile satellite Earth Stations (MESs) except aeronautical mobile satellite earth stations, operating in the 11/12/14 GHz frequency bands                                                                                                                                       |
| EN 301 428 | Transmit-only, transmit/receive or receive-only satellite earth stations operating in the 11/12/14 GHz frequency bands                                                                                                                                                                        |
| EN 301 430 | Satellite News Gathering Transportable Earth Stations (SNG TES) operating in the 11-12/13-14 GHz frequency bands                                                                                                                                                                              |
| EN 301 441 | Handheld earth stations, for Satellite Personal Communications Networks (S-PCN) in the 1.6/2,4 GHz bands under the Mobile Satellite Service (MSS)                                                                                                                                             |
| EN 301 442 | Handheld earth stations, for Satellite Personal Communications Networks (S-PCN) in the 2.0 GHz bands under the Mobile Satellite Service (MSS)                                                                                                                                                 |
| EN 301 443 | Harmonised Standard for Very Small Aperture Terminal (VSAT); Transmit-only, transmit-and-receive, receive-only satellite earth stations operating in the 4 GHz and 6 GHz frequency bands                                                                                                      |
| EN 301 444 | LMES operating in the 1.5 GHz and 1.6 GHz bands providing voice and/or data communications                                                                                                                                                                                                    |
| EN 301 447 | Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 4/6 GHz frequency bands allocated to the Fixed Satellite Service (FSS) covering the essential requirements of article 3.2 of the Directive 2014/53/EU                                               |
| EN 301 449 | Electromagnetic compatibility and Radio spectrum Matters (ERM); Harmonized EN for CDMA spread spectrum base stations operating in the 450 MHz cellular band (CDMA 450) and 410, 450 and 870 MHz PAMR bands (CDMA-PAMR) covering essential requirements of article 3.2 of the R&TTE; Directive |
| EN 301 459 | SIT and SUT transmitting towards satellites in geostationary orbit in the 29.5 to 30.0 GHz frequency bands                                                                                                                                                                                    |
| EN 301 473 | Aircraft Earth Stations (AES) operating below 3 GHz under the Aeronautical Mobile Satellite Service (AMSS)/Mobile Satellite Service (MSS) and/or the Aeronautical Mobile Satellite on Route Service (AMS(R)S)/Mobile Satellite Service (MSS)                                                  |
| EN 301 502 | Global System for Mobile communications (GSM); Base Station and Repeater equipment                                                                                                                                                                                                            |
| EN 301 511 | Mobile stations in the GSM 900 and GSM 1800 bands                                                                                                                                                                                                                                             |
| EN 301 526 | Harmonized EN for CDMA spread spectrum mobile stations operating in the 450 MHz cellular band (CDMA 450) and 410, 450 and 870 MHz PAMR bands (CDMA-PAMR) covering essential requirements of article 3.2 of the R&TTE; Directive                                                               |
| EN 301 559 | Low Power Active Medical Implants (LP-AMI) and associated Peripherals (LP-AMI-P) operating in the frequency range 2 483,5 MHz to 2 500 MHz; Harmonised Standard covering the essential requirements of article 3.2 of the Directive 2014/53/EU                                                |
| EN 301 681 | Geostationary mobile satellite systems, including handheld earth stations, for Satellite Personal Communications Networks (S-PCN) in the 1.5/1.6 GHz bands under the Mobile Satellite Service (MSS)                                                                                           |
| EN 301 721 | Providing Low Bit Rate Data Communications (LBRDC) using Low Earth Orbiting (LEO) satellites operating below 1 GHz                                                                                                                                                                            |
| EN 301 783 | Land Mobile Service; Commercially available amateur radio equipment                                                                                                                                                                                                                           |
| EN 301 839 | Ultra Low Power Active Medical Implants (ULP-AMI) and Peripherals (ULP-AMI-P) operating in the frequency range 402 MHz to 405 MHz                                                                                                                                                             |
| EN 301 841 | VHF air-ground Digital Link (VDL) Mode 2; Technical characteristics and methods of measurement for ground-based equipment; Part 3                                                                                                                                                             |
| EN 301 842 | VHF air-ground Digital Link (VDL) Mode 4 radio equipment; Technical characteristics and methods of measurement for ground-based equipment; Part 5                                                                                                                                             |
| EN 301 893 | 5 GHz WAS/RLAN Harmonised Standard for access to radio spectrum                                                                                                                                                                                                                               |

| Document   | Description                                                                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 301 908 | IMT cellular networks                                                                                                                                                                                                                                                                                     |
| EN 301 929 | VHF transmitters and receivers as Coast Stations for GMDSS and other appls in the maritime mobile service                                                                                                                                                                                                 |
| EN 302 017 | Transmitting equipment for the Amplitude Modulated (AM) sound broadcasting service                                                                                                                                                                                                                        |
| EN 302 018 | Transmitting equipment for the Frequency Modulated (FM) sound broadcasting service                                                                                                                                                                                                                        |
| EN 302 054 | Meteorological Aids (Met Aids); Radiosondes to be used in the 400.15 to 406 MHz frequency range with power levels ranging up to 200 mW                                                                                                                                                                    |
| EN 302 064 | Wireless Digital Video Links operating in the 1,3 GHz to 50 GHz frequency band                                                                                                                                                                                                                            |
| EN 302 065 | Ultra Wide Band (UWB) technologies (multiple parts)                                                                                                                                                                                                                                                       |
| EN 302 077 | Transmitting equipment for the Terrestrial - Digital Audio Broadcasting (T-DAB) service                                                                                                                                                                                                                   |
| EN 302 152 | Satellite Personal Locator Beacons (PLBs) operating in the 406.0 MHz to 406.1 MHz frequency band                                                                                                                                                                                                          |
| EN 302 186 | Satellite mobile Aircraft Earth Stations (AESs) operating in the 11/12/14 GHz frequency bands                                                                                                                                                                                                             |
| EN 302 194 | Electromagnetic compatibility and Radio spectrum Matters (ERM); Navigation radar used on inland waterways                                                                                                                                                                                                 |
| EN 302 195 | Radio equipment in the frequency range 9 kHz to 315 kHz for ULP-AMI and accessories                                                                                                                                                                                                                       |
| EN 302 208 | Radio Frequency Identification Equipment operating in the band 865 to 868 MHz with power levels up to 2 W and in the band 915 MHz to 921 MHz with power levels up to 4 W                                                                                                                                  |
| EN 302 217 | Characteristics and requirements for point-to-point equipment and antennas                                                                                                                                                                                                                                |
| EN 302 245 | Transmitting equipment for the Digital Radio Mondiale (DRM) broadcasting service                                                                                                                                                                                                                          |
| EN 302 248 | Navigation radar for use on non-SOLAS vessels                                                                                                                                                                                                                                                             |
| EN 302 264 | Short Range Radar equipment operating in the 77 GHz to 81 GHz band                                                                                                                                                                                                                                        |
| EN 302 288 | Short range radar equipment operating in the 24 GHz range                                                                                                                                                                                                                                                 |
| EN 302 296 | Transmitting equipment for the digital television broadcast service, Terrestrial (DVB-T)                                                                                                                                                                                                                  |
| EN 302 326 | Multipoint Equipment and Antennas                                                                                                                                                                                                                                                                         |
| EN 302 340 | Harmonised Standard for satellite Earth Stations on board Vessels (ESVs) operating in the 11/12/14 GHz frequency bands allocated to the Fixed Satellite Service (FSS) covering the essential requirements of article 3.2 of the Directive 2014/53/EU                                                      |
| EN 302 372 | Short Range Devices (SRD); Tank Level Probing Radar (TLPR) equipment operating in the frequency ranges 4,5 GHz to 7 GHz, 8,5 GHz to 10,6 GHz, 24,05 GHz to 27 GHz, 57 GHz to 64 GHz, 75 GHz to 85 GHz; Harmonised Standard covering the essential requirements of article 3.2 of the Directive 2014/53/EU |
| EN 302 426 | Harmonized EN for CDMA spread spectrum Repeaters operating in the 450 MHz cellular band (CDMA450) and the 410 MHz, 450 MHz and 870 MHz PAMR bands (CDMA-PAMR) covering essential requirements of article 3.2 of the R&TTE; Directive                                                                      |
| EN 302 448 | Earth Stations on Trains (ESTs) operating in the 14/12 GHz frequency bands                                                                                                                                                                                                                                |
| EN 302 454 | Radiosondes to be used in the 1 668.4 MHz to 1 690 MHz frequency range                                                                                                                                                                                                                                    |
| EN 302 480 | GSM onboard aircraft system                                                                                                                                                                                                                                                                               |
| EN 302 502 | Broadband Radio Access Networks (BRAN); 5800 MHz fixed broadband data transmitting systems                                                                                                                                                                                                                |
| EN 302 510 | Radio equipment in the range 30-37.5 MHz for Ultra Low Power Active Medical Membrane Implants and Accessories                                                                                                                                                                                             |

| Document   | Description                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 302 536 | Radio equipment operating in the frequency range 315 kHz to 600 kHz for Ultra Low Power Animal Implantable Devices (ULP-AID) and associated peripherals                                                                    |
| EN 302 537 | Ultra Low Power Medical Data Service Systems operating in the frequency range 401-402 MHz and 405-406 MHz                                                                                                                  |
| EN 302 561 | Radio equipment using constant or non-constant envelope modulation operating in a channel bandwidth of 25 kHz, 50 kHz, 100 kHz or 150 kHz                                                                                  |
| EN 302 567 | 60 GHz Multiple-Gigabit WAS/RLAN Systems                                                                                                                                                                                   |
| EN 302 571 | Intelligent Transport Systems (ITS); Radiocommunications equipment operating in the 5855 MHz to 5925 MHz frequency band                                                                                                    |
| EN 302 574 | Satellite earth station for MSS operating in 1980-2010 MHz (E/s) and 2170-2200 MHz (s/E) frequency bands                                                                                                                   |
| EN 302 608 | Radio equipment for Eurobalise railway systems                                                                                                                                                                             |
| EN 302 609 | Radio equipment for Euroloop communication systems                                                                                                                                                                         |
| EN 302 617 | Ground-based UHF radio transmitters, receivers and transceivers for the UHF aeronautical mobile service using amplitude modulation                                                                                         |
| EN 302 625 | 5 GHz BroadBand Disaster Relief applications (BBDR)                                                                                                                                                                        |
| EN 302 645 | Global Navigation Satellite Systems (GNSS) Repeaters                                                                                                                                                                       |
| EN 302 686 | Intelligent Transport Systems (ITS); Radiocommunications equipment operating in the 63 GHz to 64 GHz frequency band                                                                                                        |
| EN 302 729 | LPR equipment operating in the frequency ranges 6.0 GHz to 8.5 GHz, 24.05 GHz to 26.5 GHz, 57 GHz to 64 GHz, 75 GHz to 85 GHz                                                                                              |
| EN 302 752 | Active Radar Target Enhancers                                                                                                                                                                                              |
| EN 302 858 | Automotive radar equipment operating in the 24.05 GHz up to 24.25 GHz or 24.50 GHz frequency range                                                                                                                         |
| EN 302 885 | VHF radiotelephone equipment for the maritime mobile service                                                                                                                                                               |
| EN 302 961 | Maritime Personal Homing Beacon for search and rescue purposes intended for use on the frequency 121.5 MHz for search and rescue purposes only                                                                             |
| EN 302 977 | Vehicle-Mounted Earth stations (VMES) operating 14/12 GHz frequency bands                                                                                                                                                  |
| EN 303 039 | Land Mobile Service; Multichannel transmitter specification for the PMR Service                                                                                                                                            |
| EN 303 064 | Primary Surveillance Radar (PSR);                                                                                                                                                                                          |
| EN 303 084 | Technical characteristics and methods of measurement for ground-based equipment                                                                                                                                            |
| EN 303 098 | Maritime low power personal locating devices employing AIS                                                                                                                                                                 |
| EN 303 132 | Maritime low power VHF personal locating beacons employing Digital Selective Calling (DSC)                                                                                                                                 |
| EN 303 135 | Coastal Surveillance, Vessel Traffic Systems and Harbour Radars (CS/VTS/HR)                                                                                                                                                |
| EN 303 203 | Medical Body Area Network Systems (MBANS) operating in the 2483.5 MHz to 2500 MHz range                                                                                                                                    |
| EN 303 204 | Fixed Short Range Devices (SRD) in data networks; Radio equipment to be used in the 870 MHz to 876 MHz frequency range with power levels ranging up to 500 mW e.r.p.; Harmonised Standard for access to the radio spectrum |
| EN 303 213 | Advanced Surface Movement Guidance and Control System (A-SMGCS)                                                                                                                                                            |
| EN 303 258 | Wireless Industrial Applications (WIA); Equipment operating in the 5 725 MHz to 5 875 MHz frequency range with power levels ranging up to 400 mW                                                                           |
| EN 303 276 | Maritime Broadband Radio (MBR) links for ships and fixed installations engaged in off-shore activities                                                                                                                     |
| EN 303 340 | Digital Terrestrial TV Broadcast Receivers; Harmonised Standard for access to radio spectrum                                                                                                                               |

| Document   | Description                                                                                                                                                                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 303 345 | Broadcast Sound Receivers; Part 3: FM broadcast sound service; Harmonised Standard for access to radio spectrum                                                                                                                                                                     |
| EN 303 347 | Meteorological Radars; Harmonised Standard for access to radio spectrum; Part 1: Meteorological Radar Sensor operating in the frequency band 2 700 MHz to 2 900 MHz (S band)                                                                                                        |
| EN 303 360 | Transport and Traffic Telematics (TTT); for heliborne obstacle detection radars operating in the 76-77 GHz range                                                                                                                                                                    |
| EN 303 402 | Maritime mobile transmitters and receivers for use in the MF and HF bands                                                                                                                                                                                                           |
| EN 303 405 | Analogue and Digital PMR446 Equipment                                                                                                                                                                                                                                               |
| EN 303 406 | Social Alarms Equipment operating in the frequency range 25 MHz to 1000 MHz;                                                                                                                                                                                                        |
| EN 303 413 | Global Navigation Satellite System (GNSS) receivers; Radio equipment operating in the 1 164 MHz to 1 300 MHz and 1 559 MHz to 1 610 MHz frequency bands                                                                                                                             |
| EN 303 447 | Short Range Devices (SRD); Inductive loop systems for robotic mowers in the frequency range 0 Hz to 148,5 kHz                                                                                                                                                                       |
| EN 303 454 | Short Range Devices (SRD); Metal and object detection sensors in the frequency range 1 kHz to 148,5 kHz                                                                                                                                                                             |
| EN 303 520 | Ultra Low Power (ULP) wireless medical capsule endoscopy devices operating in the band 430 MHz to 440 MHz                                                                                                                                                                           |
| EN 303 609 | GSM Repeaters                                                                                                                                                                                                                                                                       |
| EN 303 659 | Short Range Devices (SRD) in Data Networks; Radio equipment to be used in the frequency ranges 865 MHz to 868 MHz and 915 MHz to 919,4 MHz; Harmonised Standard for access to radio spectrum                                                                                        |
| EN 303 661 | Short Range Devices (SRD); Ground Based Synthetic Aperture Radar (GBSAR) in the frequency range 17,1 GHz to 17,3 GHz and High Definition Ground Based Synthetic Aperture Radar (HD-GBSAR) in the frequency range 76 GHz to 77 GHz; Harmonised Standard for access to radio spectrum |
| EN 303 687 | 6 GHz WAS/RLAN Harmonised Standard for access to radio spectrum                                                                                                                                                                                                                     |
| EN 303 699 | Satellite Earth Stations and Systems (SES); Fixed earth stations communicating with non-geostationary satellite systems in the 20 GHz and 30 GHz FSS bands; Harmonised Standard for access to radio spectrum                                                                        |
| EN 303 722 | Wideband Data Transmission Systems (WDTS) for Fixed Network Radio Equipment operating in the 57 GHz to 71 GHz band                                                                                                                                                                  |
| EN 303 753 | Wideband Data Transmission Systems (WDTS) for Mobile and Fixed Radio Equipment operating in the 57 - 71 GHz band                                                                                                                                                                    |
| EN 303 978 | Earth Stations on Mobile Platforms ESOMP transmitting towards satellites in geostationary orbit in the 27.5-30.0 GHz frequency bands                                                                                                                                                |
| EN 303 979 | Fixed Earth Stations and Earth Stations on Mobile Platforms (ESOMPs) transmitting towards satellites in non-geostationary orbit in the 27.5 GHz to 29.1 GHz and 29.5 GHz to 30.0 GHz bands                                                                                          |
| EN 303 980 | Fixed and in-motion Earth Stations communicating with non-geostationary satellite systems in the 11 GHz to 14 GHz frequency bands                                                                                                                                                   |
| EN 303 981 | Satellite Earth Stations and Systems (SES); Fixed and in-motion Wide Band Earth Stations communicating with non-geostationary satellite systems (WBES) in the 11 GHz to 14 GHz frequency bands; Harmonised Standard for access to radio spectrum                                    |
| EN 304 220 | Wideband data transmission SRD; Harmonised Standard for access to radio spectrum; Part 1: Wideband data transmission devices: network access points operating in the frequency bands 863 MHz to 868 MHz and 915,8 MHz to 919,4 MHz                                                  |
| EN 305 550 | Short Range Devices (SRD); Radio equipment to be used in the 40 GHz to 246 GHz frequency range                                                                                                                                                                                      |

# European Standards for Receive-Only Equipment

| Document   | Description                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 300 487 | Satellite Earth Stations and Systems (SES); Harmonised Standard for Receive-Only Mobile Earth Stations (ROMES) providing data communications operating in the 1,5 GHz frequency band |
| EN 303 372 | Satellite Earth Stations and Systems (SES); Satellite broadcast reception equipment. Part 1: Outdoor unit receiving in the 10,7 GHz to 12,75 GHz frequency band                      |
| EN 303 345 | Broadcast Sound Receivers                                                                                                                                                            |

## Abbreviations

| Abbreviation | Description                                                          |
|--------------|----------------------------------------------------------------------|
| (OR)         | Off-Route                                                            |
| (R)          | Route                                                                |
| 1800         | Global System for Mobile Communications using 1800 MHz band          |
| ADS          | Automatic Dependant Surveillance (Aeronautical)                      |
| AES          | Aircraft Earth Stations                                              |
| AGA          | Air Ground Air                                                       |
| AIS          | Automatic Identification System                                      |
| ALS          | Assistive Listening Systems                                          |
| AM           | Amplitude Modulation                                                 |
| AMRD         | Autonomous Maritime Radio Device                                     |
| AMS(R)S      | Aeronautical Mobile Satellite (Route) Services                       |
| APP          | Appendix of the ITU Radio Regulations                                |
| AVI          | Automatic Vehicle Identification                                     |
| BBDR         | Broad Band Disaster Relief                                           |
| BFWA         | Broadband Fixed Wireless Access                                      |
| BMA          | Building Material Analysis                                           |
| BSS          | Broadcasting Satellite Service                                       |
| CB           | Citizen Band                                                         |
| CEPT         | European Conference of Postal and Telecommunications Administrations |
| CGC          | Complementary Ground Component                                       |
| CRS          | Central Radio Station                                                |
| CT           | Cordless Telephone                                                   |
| DEC          | Decision                                                             |
| DECT         | Digital Enhanced Cordless Telecommunication                          |
| D-GPS        | Differential Global Positioning System                               |
| DME          | Distance Measuring Equipment                                         |
| DMO          | Direct Mode Operation                                                |
| DRM          | Digital Radio Mondiale                                               |
| DSC          | Digital Selective Calling                                            |

| Abbreviation | Description                                 |
|--------------|---------------------------------------------|
| DSI          | Detailed Spectrum Investigation             |
| DVB-T        | Terrestrial Digital Video Broadcasting      |
| E/s          | Earth-to-space direction                    |
| ECA          | European Common Allocation                  |
| ECC          | Electronic Communications Committee         |
| ECM          | Electronic Countermeasures                  |
| ECP          | European Common Proposal                    |
| EESS         | Earth Exploration-Satellite Service         |
| EFIS         | European Frequency Information System       |
| EGSM         | Extended GSM                                |
| EISCAT       | European Incoherent SCATter facility        |
| ELT          | Emergency locator transmitter               |
| ENG          | Electronic News Gathering                   |
| EPIRB        | Emergency Position-Indicating Radiobeacon   |
| ERC          | European Radiocommunications Committee      |
| ERO          | European Radiocommunications Office         |
| ESIM         | Earth Stations In Motion                    |
| ESOMPs       | Earth Stations On Mobile Platforms          |
| EST          | Earth Stations on Trains                    |
| ESV          | Earth Stations on-board Vessels             |
| EU           | European footnote                           |
| FDD          | Frequency Division Duplex                   |
| FM           | Frequency Modulation                        |
| FSS          | Fixed-Satellite Service                     |
| FWA          | Fixed Wireless Access                       |
| GALILEO      | European Global Navigation Satellite System |
| GBAS         | Ground Based Augmentation System            |
| GBSAR        | Ground Based Synthetic Aperture Radar       |
| GE06         | Geneva 2006 Agreement                       |
| GE75         | Geneva 1975 Agreement                       |
| GE85         | Geneva 1985 Agreement                       |

| Abbreviation | Description                                                 |
|--------------|-------------------------------------------------------------|
| GMDSS        | Global Maritime Distress and Safety System                  |
| GNSS         | Global Navigation Satellite System                          |
| GPR/WPR      | Ground Probing Radar / Wall Probing Radar                   |
| GPS          | Global Positioning System                                   |
| GSM          | Global System for Mobile Communications                     |
| GSM 1800     | Global System for Mobile Communications using 1800 MHz band |
| GSM-R        | GSM for Railways                                            |
| GSO          | GeoStationary Orbit                                         |
| HAPS         | High Altitude Platform Systems                              |
| HDFS         | High Density Fixed Service                                  |
| HDFSS        | High Density Fixed-Satellite Service                        |
| HDTV         | High Definition Television                                  |
| HEST         | High E.i.r.p. Satellite Terminals                           |
| HF           | High Frequency                                              |
| HIPERLAN     | High Performance Radio Local Area Network                   |
| IALA         | International Association of Lighthouse Authorities         |
| IBCN         | Integrated Broadband Communications Network                 |
| IFF          | Identification Friend or Foe                                |
| ILS          | Instrument Landing System                                   |
| IMO          | International Maritime Organisation                         |
| IMT          | International Mobile Telecommunications                     |
| IMT-2000     | International Mobile Telecommunications-2000                |
| IMT-Advanced | Systems beyond IMT-2000                                     |
| IoT          | Internet of Things                                          |
| ISM          | Industrial, Scientific and Medical                          |
| ITS          | Intelligent Transport Systems                               |
| ITU          | International Telecommunication Union                       |
| JTIDS        | Joint Tactical Information Distribution System              |
| LAES         | Location Application for Emergency Services                 |
| LANs         | Local Area Networks                                         |
| LDC          | Low Duty Cycle                                              |

| Abbreviation | Description                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| LP-AMI       | Low Power Active Medical Implants                                                                                                          |
| LPD-S        | Low Power Devices communicating with Satellites                                                                                            |
| LPR          | Level Probing Radar                                                                                                                        |
| LT2          | Location Tracking Type 2                                                                                                                   |
| MBANS        | Medical Body Area Network Systems                                                                                                          |
| MBR          | Maritime Broadband Radio Links                                                                                                             |
| MCA          | Mobile Communications Services on Board Aircraft                                                                                           |
| MCV          | Mobile Communication Services on Board Vessels                                                                                             |
| MES          | Mobile Earth Stations                                                                                                                      |
| MFCN         | Mobile/Fixed Communications Networks                                                                                                       |
| MIDS         | Multifunctional Information Distribution System                                                                                            |
| MLS          | Microwave Landing System                                                                                                                   |
| MSI          | Maritime Safety Information                                                                                                                |
| MSS          | Mobile-Satellite Service                                                                                                                   |
| MWS          | Multimedia Wireless System                                                                                                                 |
| NATO         | North Atlantic Treaty Organisation                                                                                                         |
| NAVTEX       | Narrow-band direct-printing telegraphy system for transmission of navigational and meteorological warnings and urgent information to ships |
| NDB          | Non-Directional Beacon                                                                                                                     |
| NGSO         | Non-GeoStationary Orbit                                                                                                                    |
| NJFA         | NATO Joint Civil/Military Frequency Agreement                                                                                              |
| NMR          | Nuclear Magnetic Resonance                                                                                                                 |
| NTFA         | National Table of Frequency Allocation                                                                                                     |
| OB           | Outside Broadcasting                                                                                                                       |
| PAMR         | Public Access Mobile Radio                                                                                                                 |
| PKO          | Peace Keeping Operations                                                                                                                   |
| PLB          | Personal Locator Beacons                                                                                                                   |
| PMR          | Professional Mobile Radio, Private Mobile Radio                                                                                            |
| PMSE         | Programme Making and Special Events                                                                                                        |
| POCSAG       | Post Office Code Standards Advisory Group                                                                                                  |
| PPDR         | Public Protection and Disaster Relief                                                                                                      |
| PWAP         | Private Wide Area Paging                                                                                                                   |

| Abbreviation        | Description                             |
|---------------------|-----------------------------------------|
| RA                  | Radio Astronomy                         |
| REC                 | Recommendation                          |
| RFID                | Radio Frequency Identification          |
| RLAN                | Radio Local Area Network System         |
| RR                  | ITU Radio Regulations                   |
| RTE                 | Radar Target Enhancer                   |
| RTTT                | Road Transport & Traffic Telematics     |
| s/E                 | space-to-Earth direction                |
| SAB                 | Services Ancillary to Broadcasting      |
| SAP                 | Services Ancillary to Programming       |
| SAR(communciations) | Search and Rescue                       |
| SIT                 | Satellite Interactive Terminal          |
| SNG                 | Satellite News Gathering                |
| S-PCS               | Satellite Personal Communication System |
| SRD                 | Short Range Device                      |
| SRR                 | Short Range Radar                       |
| SRS                 | Space Research Service                  |
| SSR                 | Secondary Surveillance Radar            |
| SUT                 | Satellite User Terminal                 |
| TACAN               | Tactical Air Navigation                 |
| T-DAB               | Terrestrial Digital Audio Broadcasting  |
| TDD                 | Time Division Duplex                    |
| TETRA               | Terrestrial Trunked Radio               |
| TLPR                | Tank Level Probing Radar                |
| TRR                 | Tactical Radio Relays                   |
| TS                  | Terminal Station                        |
| TTT                 | Transport and Traffic Telematics        |
| TV                  | Television                              |
| UIC                 | International Union for Railways        |
| ULP-AID             | Ultra Low Power Animal Implants Devices |
| ULP-AMI             | Ultra Low Power Active Medical Implants |

| Abbreviation | Description                                         |
|--------------|-----------------------------------------------------|
| ULP-MMI      | Ultra Low Power Medical Membrane Implants           |
| ULP-WMCE     | Ultra-Low Power Wireless Medical Capsule Endoscopy  |
| UMTS         | Universal Mobile Telecommunications System          |
| UWB          | Ultra – Wideband                                    |
| VDB          | VHF ground-air Data Broadcast                       |
| VLBI         | Very Long Baseline Interferometry (Radio Astronomy) |
| VOR          | VHF Omni-directional Range                          |
| VSAT         | Very Small Aperture Terminal                        |
| VTs          | Vessel Traffic System (radar)                       |
| WAIC         | Wireless Avionics Intra-Communication systems       |
| WARC         | World Administrative Radio Conference               |
| WAS          | Wireless Access System                              |
| WIA          | Wireless Industrial Applications                    |